

they lack some portion of the phage genome, but they are capable of transducing and integrating into a new host, carrying the original host genes with them (C). This establishes a merodiploid situation. At this point, the defective λ cannot produce more phage particles on its own. Nevertheless, a high-frequency transducing lysate can be made following an LFT event. This happens because a cell doubly lysogenized with a defective λ (carrying donor DNA) and a normal λ can be induced to yield new phage progeny (D). The normal copy of λ provides replication proteins missing from the defective λ . Thus, an HFT lysate is produced in which approximately half of the particles will be specialized transducing particles.

As with conjugation, transduction usually occurs most readily between closely related species of the same bacterial genus (intrageneric). This is due to the need for specific cell surface receptors for the phage. However, intergeneric transduction has been demonstrated between closely related members of the enteric group of organisms—for example, between *E. coli* and *Salmonella* or *Shigella* species. Various genetic traits such as fermentation capabilities, antigenic structure, and resistance to chemotherapeutic agents are transducible. Keep in mind that transduction is not limited to the bacterial chromosome. Genetic information residing on plasmids may also be transferred by transduction.

Lysogenic conversion is dependent on the establishment of lysogeny between a bacteriophage and the host bacterial cell. Lysogeny occurs when a bacteriophage coexists with its bacterial host without lysing it. In lysogenic conversion, a new phenotypic trait acquired by the host cell is due to a phage gene. Because the gene is part of the normal phage genome, every cell in a population that has been lysogenized acquires the genetic property. This mass conversion of the cell population distinguishes lysogenic conversion from transduction and other genetic transfer mechanisms.

One of the most interesting and thoroughly investigated examples of lysogenic conversion is the relationship of lysogeny to the production of toxin by the diphtheria bacillus, *Corynebacterium diphtheriae*. Cells that are lysogenized by β phage are designated *tox*⁺. The ability to produce toxin is inherent in the genome of the bacteriophage rather than in any trait that may have been transduced from the original host cell. Thus, cells that are “cured” of the lysogenic state no longer produce toxin. The *tox*⁺ gene is considered to be a part of the prophage genome; however, it is not essential for any known phage function and may be modified or eliminated without any effect on β phage replication. The production of *erythrogenic toxin* by members of the group A streptococci and toxins produced by *Clostridium botulinum* types C and D have also been shown to be the result of lysogenization by bacteriophage.

RECOMBINATION

The success of bacterial genetics has been due, in great measure, to the fortuitous ability of bacteria to integrate donor DNA into their genomes. Without this ability, we would be considerably more ignorant of life's secrets. Of course, recombination is not really a philanthropic activity of bacteria toward science but a rather important component of their survival and evolution (see “DNA Repair Systems”). There are basically two types of recombination that can occur in *E. coli* and many other bacteria. These are **RecA-dependent** general recombination (or homologous recombination) and **RecA-independent** or nonhomologous recombination. General recombination requires a large

region of homology between the donor and recipient DNAs but does not produce a net gain of DNA by the recipient genome. Homologous sequences are merely exchanged. In contrast, the nonhomologous recombinational mechanisms require very little sequence homology (as little as 5 or 6 bp) but actually produce a net gain of DNA by the recipient. Nonhomologous recombination may be divided into two types: **site specific**, where exchange occurs only at specific sites located on one or both participating DNAs (see “ λ Phage” in Chapter 6), and **illegitimate**, which includes other RecA-independent events such as transposition (see “Insertion Sequences and Transposable Elements”). Because these types of recombination increase the length of the recipient’s DNA, they have been referred to as **additive**.

General Recombination

Most recombination events that occur in *E. coli* following DNA exchanges are mediated by RecA-dependent pathways that require large regions of homology between donor and recipient DNA. The *recA* product is a synaptase that facilitates the alignment of homologous DNA sequences. Loss of *recA* through mutation reduces recombination frequencies by 99.9%. The process of recombination can be viewed in six steps: (1) strand breakage, (2) strand pairing, (3) strand invasion/assimilation, (4) chiasma or crossover formation, (5) breakage and reunion, and (6) mismatch repair.

Recombination events in *E. coli*, or in any other organism for that matter, are not really a random process. A specific DNA sequence called chi (χ) is the primary target for initial recombination events. χ sequences, which are naturally present in *E. coli*, clearly enhance recombination and define recombinational hotspots. The χ -consensus sequence is an octamer, 5'-GCTGGTGG-3', and occurs every 5 to 10 kb (there are 1009 sites in the *E. coli* genome).

χ sequences are important for accomplishing the first goal in the process of recombination, which is to take a double-stranded donor molecule and produce a single-strand from it that can invade and displace one strand of the recipient DNA (see Fig. 3-13). The process begins with the *recBCD* or *recJ* products entering at the ends of a double-stranded DNA molecule and unwinding the DNA using their helicase activities. RecJ is part of an alternative recombination pathway called the RecF pathway. When the sliding RecBCD complex nears a χ site, it introduces a single-strand nick a few nucleotides from the 3' side of χ . RecBCD then degrades the 3'-terminated strand during further DNA unwinding. When RecBCD reaches χ , the enzyme is altered and degradation is reduced.

Once the enzyme is altered by interaction with χ , it does not recognize other χ sites on the same molecule even though it continues to unwind the DNA. RecBCD then loads the new single-stranded 3' end with the RecA synaptase. RecD actually inhibits recombination at sites other than χ by inhibiting the loading of RecA on single-stranded DNA. It has been proposed that upon encountering χ , RecD undergoes an as-yet undocumented change that either ejects RecD from the complex or exposes a domain for RecA loading. Once the resultant single-strand tail is bound to RecA, the complex is poised to invade the donor DNA and begin the recombination process. (Figs. 3-13C and 3-14c). The RecA protein (active species is a tetramer of 38,000 dalton monomers) is known to promote rapid renaturation of complementary single strands, hydrolyzing ATP in the process. It is believed that the RecA synaptase binds to ssDNA, which spontaneously increases RecA ability to bind dsDNA. RecA protein bound to ssDNA then aids in a search for homology between the donor strand and the recipient molecule.

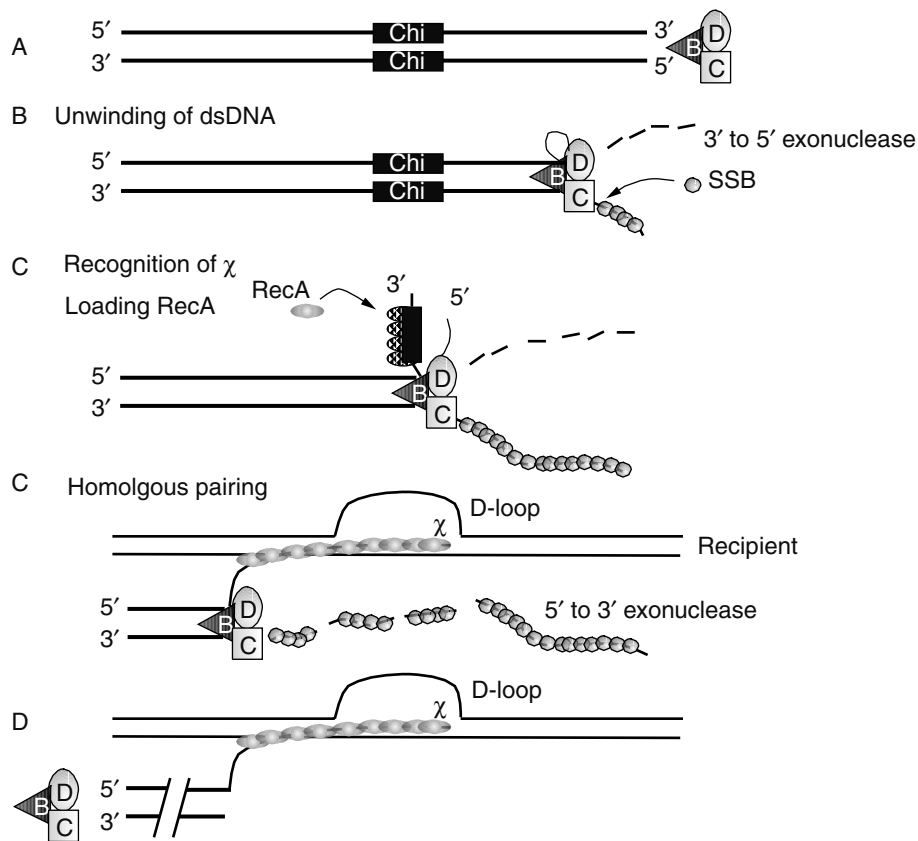
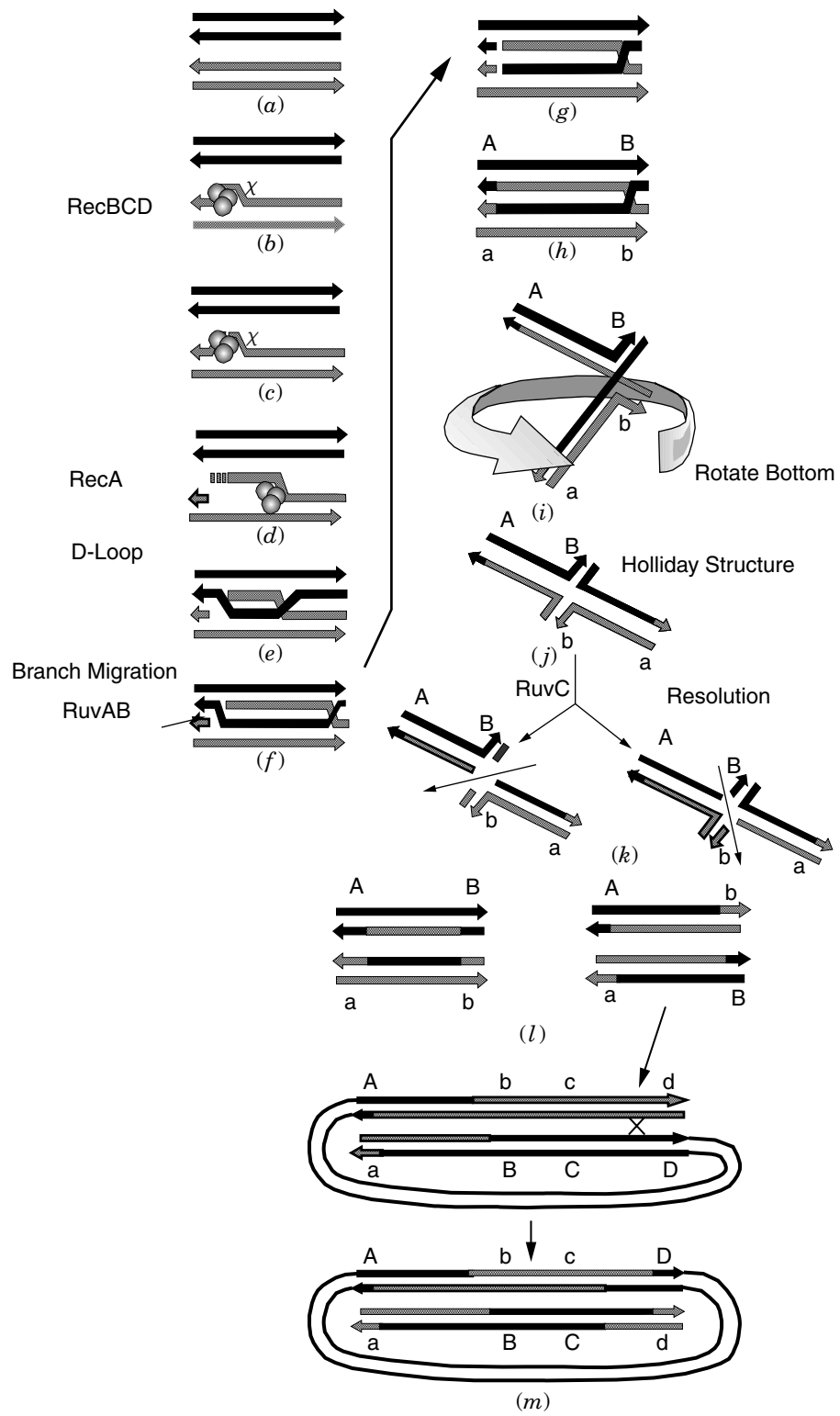


Fig. 3-13. Recombination: D-loop formation.

Experiments suggest a mechanism in which the RecA-mediated interaction between the searching single strand and the recipient double helix occurs randomly at regions of nonhomology, forming nonspecific triple-stranded complexes. The search for homology continues as the RecA protein either reiteratively forms such complexes or translates the two DNA molecules relative to each other until areas of homology are found. Once homologous regions are encountered and the single- and double-stranded DNAs are complexed, a stable displacement loop (D loop) is formed (see Figs. 3-13C and D, 3-14e).

Next, strand assimilation occurs where the donor strand progressively displaces the recipient strand (also called branch migration). Some data suggest that RecA protein will partially denature the recipient DNA duplex and allow the assimilating strand to track in with a 5' to 3' polarity. Extensive assimilation seems to involve RecA binding cooperatively to additional areas in the crossover region to form a "protein-DNA filament." Subsequent to RecA binding, the RuvAB complex is responsible for the branch migration phase of recombination that occurs after the crossover site, known as the **Holliday junction**, has formed (Fig. 3-14g and h). RuvA targets the complex to the Holliday junction, enabling RuvB to assemble as a hexamer around two opposite arms of the Holliday junction. The DNA is then pumped out of the RuvAB complex in opposite directions to drive branch migration (animation at



<http://www.sdsc.edu/journals/mbb/ruva.html>). The RuvAB complex has the ability to promote branch migration for at least 1 kb.

D-loop formation can be coupled with replication of the unpaired donor strand. The fact that certain *dna* mutations cause a decrease in recombination frequencies tends to support the involvement of replication in genetic exchange. Alternatively, a simultaneous assimilation of the sister strand may occur as depicted in Figure 3-14e. While strand assimilation occurs, it is proposed that an endonuclease such as *exoV* (*recBCD* gene product) attacks the D loop. *ExoV* possesses both *exo*- and *endo*nuclease activity. D loops are known to be susceptible to *ExoV*. A general model that accounts for most of the experimental evidence is shown in Figure 3-14.

The resolution of recombination intermediates is catalyzed by RuvC protein, which binds the Holliday junction as a dimer. This process effects the crossover. Cleavage requires homology and occurs at a preferred sequence [5'(A/T)TT ↓(G/C)3']. Nicks are introduced symmetrically across the junction to give ligatable products. Mentally rotating the bottom half of the molecule in Figure 3-14i to make the molecule depicted in Figure 3-14j helps to visualize the final crossover event. Breakage and reunion are required to resolve this structure and can occur in either the vertical or horizontal plane. Note that the exchange of markers A/a and B/b occur following vertical but not horizontal cleavage. Figure 3-14l illustrates the end products of the right-hand resolution pathway as it would appear in a circular DNA molecule (e.g., chromosome). Note that to retain continuity of the circular DNA molecule following resolution event 2, a second crossover event is required (Fig. 3-14m). A short animation of recombination may be found at <http://www.wisc.edu/genetics/Holliday/holliday3.html>.

Genetics of Recombination

Many genes that participate in recombination have been identified in *E. coli*. Table 3-1 provides a list of the known loci associated with recombination in this organism.

Fig. 3-14. Model for the recombination of genetic material. (a) The RecBCD enzyme moves along a donor DNA duplex until it encounters a χ sequence (b) at which point a single-stranded nick (c) is introduced, creating a single-stranded DNA tail (d). (e) RecA protein, along with single-stranded-binding proteins, can bind to the ssDNA region and subsequently bind to recipient duplex DNA searching for regions of homology. (f) Upon encountering a region of homology, D-loop formation occurs. Concomitant to D-loop formation, replication may occur on the unpaired donor strand. Some branch migration of the D loop occurs by cooperative binding of additional RecA tetramers to the donor ssDNA. At some point, a nuclease cleaves and partially degrades the D loop (f). The single-stranded tail remaining from the D loop is trimmed (g) and the exchanged strands ligated (by DNA ligase) to their recipient molecules (h). This structure as shown in (i) is referred to as the Holliday intermediate or chiasma formation and has been observed in electron micrographs. Further branch migration at this point occurs by the action of RuvAB. Mentally rotating the bottom half of the molecule while keeping the top half stationary (i) helps to visualize the final crossover event (j). Subsequent breakage and reunion, catalyzed by RuvC or RecG, is required to resolve this structure and can occur either in the vertical or horizontal plane (k). Note the exchange of markers A/a and B/b occur following vertical but not horizontal cleavage. The products of the *ruvC* and *recG* loci are thought to be alternative endonucleases specific for Holliday structures. In (m), the second crossover event is shown that maintains the circularity of the bacterial chromosome following the vertical resolution event in (l).

TABLE 3-1. Recombination Genes

Gene	Map Location ^a	Function of Gene, Distinguishing Characteristics of Mutants, or Other Pertinent Information
<i>recA</i>	60.8	Complete recombination deficiency and many other phenotype defects including suppression of <i>tifl</i> , DNA-dependent ATPase, 38 kDa
<i>recB</i>	63.6	Structural gene of exonuclease V, 135 kDa, couples ATP hydrolysis to DNA unwinding
<i>recC</i>	63.7	Structural gene of exonuclease V, 125 kDa
<i>recD</i>	63.6	α subunit of <i>exo V</i> , 67 kDa
<i>recE (sbcA)</i>	30.5	Exonuclease VIII (140 kDa), 5' $\rightarrow$ 3' dsDNA
<i>recF</i>	83.6	ATPase stimulated by RecR
<i>recJ</i>	64.6	5' $\rightarrow$ 3' exonuclease
<i>recG</i>	82.4	ATPase, disrupts Holliday structures, 76 kDa
<i>recN</i>	59.3	Unknown function, 60 kDa
<i>recO</i>	58.2	Promotes renaturation of complimentary ssDNA, 31 kDa (interacts with RecR)
<i>recR</i>	10.6	Helps RecA utilize SSB-SSDNA complexes as substrates, 22 kDa
<i>recT (sbcA)</i>	30.4	DNA annealing, in <i>rac</i> phage, activated by <i>sbcA</i> mutation
<i>ruvA</i>	41.9	Complexes with Holliday junctions, branch migration
<i>ruvB</i>	41.9	ATPase, branch migration at Holliday junctions
<i>ruvC</i>	41.9	Endonuclease, Holliday junction, resolvase
<i>recQ</i>	86.3	DNA helicase, 74 kDa
<i>recL (uvrD)</i>	86.1	Recombination deficiency of <i>recB</i> ⁻ <i>recC</i> ⁻ <i>sbcB</i> ⁻ strain
<i>rus</i>	12.3	Holliday junction resolvase
<i>sbcB (xon)</i>	44.8	The structural gene for exonuclease I; <i>recB</i> ⁻ <i>recC</i> ⁻ <i>sbcB</i> ⁻
<i>sbcC</i>	8.9	ATP-dependent dsDNA exonuclease

^aMap positions are derived from the *E. coli* K-12 linkage map at <http://ecocyc.pangeasystems.com> or <http://genolist.pasteur.fr/Colibri/>.

The identification of these genes has led to the discovery of multiple recombinational pathways. Mutations in *recA* reduce recombination frequencies to 0.1% of normal levels commensurate with its central role in many homologous recombination systems. One of these pathways was defined by mutations in *recBC*, which reduce the frequency to 1%. Other pathways were discovered following a search for revertants of these *recBC* mutants. Among the revertants found were two additional genes that can suppress the Rec⁻ phenotype. The suppressors of *recB* and *recC* (*sbc*) include *sbcA* and *sbcB*. The *sbcA*⁺ locus negatively controls the expression of *exoVIII*, the *recE*⁺ gene product. The *sbcA* mutations increase *exoVIII* activity, which presumably can substitute for *exoV* (*recBCD*) in the RecBC pathway. It turns out that both *sbcA* (*recT*) and *recE* are genes present on a lambdoid prophage called *rac* that maps near the replication terminus.

The fact that *recBC* mutants retain some recombination ability suggested there was at least one RecBCD-independent recombinational pathway. Support for this theory came from another suppressor of *recBC* mutations, *sbcB*, which encodes exonuclease I. Mutations in *sbcB* decrease exonuclease I activity. Exo I is proposed to shuttle a DNA recombinational intermediate from an alternative pathway to the RecBCD

pathway. However, since *recBC* mutants cannot use the RecBC intermediate, very little recombination would occur because the important intermediate in the alternate RecBCD-independent pathway would be siphoned off by Exo I. Exo I mutants (*sbcB*) would accumulate the alternative intermediate rather than siphon it off, allowing the intermediate to serve as a substrate for the RecBCD-independent system, thereby increasing recombination through this pathway. Actually, the *sbcB* mutants contain a second mutation in the unlinked *sbcC* gene. Apparently both mutations are required to suppress the *recBC* mutant phenotype, although the reason for this is unclear. The RecBCD-independent pathway is now called the RecF pathway, since mutations in *recF* eliminate this alternative recombinational route. The *recF* pathway also requires the participation of the *recJ*, *N*, *R*, *O*, *Q* and *ruv* gene products.

Mismatch Repair. Following recombination, the initial duplex product may contain unpaired regions resulting from genetic differences. These regions are unstable and susceptible to mismatch repair. The mechanism involves the excision of one or the other mismatched base along with up to 3000 nucleotides (see below). RecFJO is also involved in short patch mismatch repair and thus is involved not only in the initial stages of the recombinational process but also in its ultimate resolution.

Restriction and Modification

Although bacteria have elaborate methods to exchange DNA, they have also developed a means to protect themselves from potentially harmful foreign DNA (e.g., bacteriophage DNA). Restriction and modification of prokaryotic DNA were discovered when bacteriophage λ was grown on one strain of *E. coli* (*E. coli* K-12) and then used to infect a different strain (*E. coli* B). It was noticed that the plating efficiency of the virus was less on the strain B relative to the K strain. Plating efficiency is determined by diluting the phage lysate (λ , K) and adding aliquots of diluted phage to tubes containing soft agar seeded with either *E. coli* K-12 or B. The soft agar is then poured onto a normal nutrient agar plate and incubated. After incubation, the plate will be confluent with growth of *E. coli* except where a phage particle has infected a cell. This area will appear clear and is called a plaque. The plaque is visible to the naked eye because progeny phage from the initial infected cell will progressively infect and lyse adjacent cells.

The unusual observation was that equivalent amounts of λ , K yielded more plaques on *E. coli* K-12 than on *E. coli* B. However, if λ was isolated from the plaques on *E. coli* B and propagated on *E. coli* B, this lysate (λ , B) produced more plaques on *E. coli* B than on *E. coli* K, the exact opposite of the original observation. The underlying mechanism for this phenomenon is the presence of specific endodeoxyribonucleases known as restriction endonucleases in each strain of *E. coli*. Invading bacteriophage DNA originating from the K-12 strain of *E. coli* will undergo cleavage (restriction) by the B endonuclease and then will be degraded rapidly to nucleotides by subsequent exonuclease action. Bacterial host DNA is not degraded because the nucleotide sequence recognized by the restriction endonuclease has been modified by methylation. The few molecules of bacteriophage DNA that survive the initial infection do so because they were modified by the host methylase before the restriction enzyme had a chance to cleave the DNA. The lucky methylated phage is thus protected by methylation.

TABLE 3-2. Characteristics of Types I, II, and III Restriction-Modification Systems

Characteristics	Type I	Type II	Type III	Type IV
Protein structure	Three different subunits	One or two identical subunits	Two different subunits	Two peptides
Endonuclease and methylase activities performed by:	One multifunctional enzyme	Separate methylase and endonuclease enzymes	One multifunctional enzyme	Bifunctional endonuclease/separate methylase
Cofactor requirements for endonuclease	ATP, SAM, Mg^{2+}	Mg^{2+}	ATP	Mg^{2+}
Stimulatory cofactors (not required) for endonuclease	—	—	SAM, Mg^{2+}	SAM
Cofactor requirements for methylase	SAM	SAM	SAM	SAM
Stimulatory cofactors (not required) for methylase	ATP, Mg^{2+}	—	ATP, Mg^{2+}	Ca^{2+}
Cleavage and modification sites	Random, from 1000 bp from recognition site	At or near recognition site	24–26 bp from recognition site	—
Recognition site	<i>Eco</i> K: AACN ⁶ GTGC <i>Eco</i> B: TGAN ⁸ TGCT	Mostly at sites with dyad symmetry	<i>Eco</i> P1: AGACC <i>Hinf</i> III: CAGCAG CGAAT	<i>Eco</i> 571 CTGAAG
DNA translocation	Yes	No	No	—

λ DNA that has been modified in *E. coli* K-12 carries the K modification and is referred to as λ , K while lambda grown on *E. coli* B carries the B modification and is referred to as λ , B. The genetics of this system is dealt with in more detail below. There are four classes of restriction endonucleases (Table 3-2). The enzymes in the *E. coli* K-12 and B systems belong to class I while most of the other known restriction enzymes belong to class II.

Class I enzymes (e.g., *EcoB*, *EcoK*, and *EcoPI*) are complex multisubunit enzymes that cleave unmodified DNA in the presence of S-adenosylmethionine (SAM), ATP, and Mg^{2+} . These enzymes are also methylases. Each contains two α subunits (mwt 135,000), two β subunits (mwt 60,000), and one γ subunit (mwt 55,000). Prior to binding to DNA, *EcoK* enzyme binds rapidly to SAM. This initiates an allosteric conformation of the enzyme to an active form that will interact with DNA at random, nonspecific sites. The enzyme then moves to the recognition site with subsequent events depending on the state of the site (Fig. 3-15).

The recognition sequence for both *EcoK* and *EcoB* includes a group of three bases and a group of four bases separated by six (*EcoK*) or eight (*EcoB*) nonspecific bases. Four of the specific bases are conserved (boxes in Fig. 3-15). The adenines with an asterisk are methylated by the methylase activity. If both strands of the recognition site are methylated when encountered, the enzyme does not recognize it. If one strand is methylated, as would be found immediately upon replication, the enzyme binds and methylates the second strand. However, if both strands are unmethylated, the restriction capability is activated. Restriction does not occur directly at the binding site. Rather, the DNA loops past the enzyme, which does not leave the recognition site (DNA translocation), forming supercoils that are cleaved at nonrandom sites approximately 100 bp from the recognition site.

Elegant genetic experiments have demonstrated the existence of three genes whose products are required for these systems. These include *hsdM* (β subunit, methylase activity), *hsdR* (α subunit, restriction endonuclease activity), and *hsdS* (γ subunit, site recognition). Whether *EcoK* recognizes the K recognition site appears to be dependent on the *hsdS* gene product. Hybrid proteins consisting of HsdM and HsdR from *EcoK* and HsdS from *EcoB* have B-site specificity. When characterizing the phenotype of strains carrying mutations in one or more of these genes, r^+m^+ indicates both restriction and modification activities are functioning, r^-m^+ indicates a restriction-less strain that can still modify its DNA, and r^-m^- indicates the loss of both activities. An r^-m^- phenotype can be the result of *hsdR hsdM hsdS*⁺

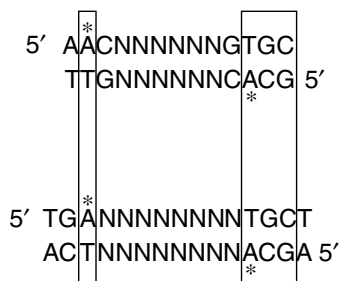


Fig. 3-15. Recognition sequences for the *EcoK* and *EcoB* restriction modification enzymes. Asterisks mark sites of methylation.

or *hsdR*⁺ *hsdM*⁺ *hsdS* genotypes, since the *hsdS* product is required for site recognition.

Class II restriction-modification enzymes are less complex and only require Mg²⁺ for activity. These enzymes typically recognize palindromic sequences four to eight base pairs in length, depending on the enzyme. Where the type I enzymes function as a complex including endonuclease and methylase activities, the type II endonucleases and methylases are distinct enzymes. Type II restriction enzymes, some of which are listed in Table 3-3, will only cleave DNA that is unmodified by the cognate type II methylase. If even one of the strands of the sequence is modified, the endonuclease will not recognize it. As opposed to type I enzymes, these endonucleases cleave at the recognition site, forming either blunt ends (e.g., *Hae*III) by cutting both strands in the center of the site or cohesive ends by introducing staggered, single-strand nicks. An example of cohesive ends is as follows:

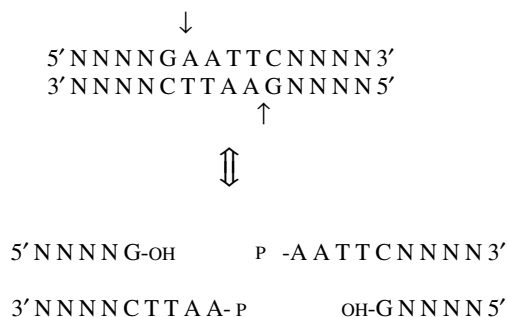


TABLE 3-3. Examples of Class II DNA Restriction and Modification Enzymes

Enzyme	Restriction and Modification Site ^a	Bacterial Strain
<i>Eco</i> RI	G↓A*ATTC	<i>E. coli</i> RY13
<i>Eco</i> RII	↓C*CTGG	<i>E. coli</i> R245
<i>Ava</i> I	C↓Py°CGRG	<i>Anabaena variabilis</i>
<i>Bam</i> HI	G↓A*ATTC	<i>Bacillus amyloliquefaciens</i> II
<i>Bg</i> III	A↓GATCT	<i>Bacillus globigii</i>
<i>Hah</i> I	G°CG↓C	<i>Haemophilus haemolyticus</i>
<i>Hae</i> III	GG↓CC	<i>Haemophilus aegyptius</i>
<i>Hind</i> III	*A↓AGCTT	<i>Haemophilus influenzae</i> Rd
<i>Hin</i> fl	G↓ANTC	<i>Haemophilus influenzae</i> Rf
<i>Hpa</i> II	C↓*CGG	<i>Haemophilus parainfluenzae</i>
<i>Hga</i> I	GACGCNNNN↓	<i>Haemophilus gallinarum</i>
<i>Sma</i> I	CCC↓GGG	<i>Serratia marcescens</i> Sb
<i>Dpn</i>	*AC↓TC	<i>Diplococcus pneumoniae</i>

^a ↓Shows the site of cleavage and *shows the site of methylation of the corresponding methylase where known. °shows the site of action of a presumed methylase—that is, methylation at this site blocks restriction, but the methylase has not yet been isolated.

Note that the two cohesive ends can reanneal to each other and, in the presence of DNA ligase, will form an intact molecule. These enzymes are heavily responsible for the current explosion in molecular biology in that they are essential in forming recombinant DNA molecules (cloning). The reason is that DNA from two unrelated organisms will both contain sequences that can be recognized by these enzymes *in vitro*. Thus, DNA from evolutionarily distinct organisms (e.g., humans and *E. coli*) can be cut with EcoRI, for example, their DNAs mixed and ligated to each other because of the identical cohesive ends. This forms a recombinant molecule. Figure 3-16 illustrates how a recombinant molecule can be formed *in vitro*. This molecule, because it is a plasmid, can be used to transform *E. coli*.

Class III restriction-modification systems include those from prophage P1, the *E. coli* plasmid P15, and *Haemophilus influenzae* RF. Unlike the type I systems, the type III enzymes do not require SAM for restriction and cleave target DNA to fragments of distinct size, typically 25 to 27 bp away from the recognition sequence.

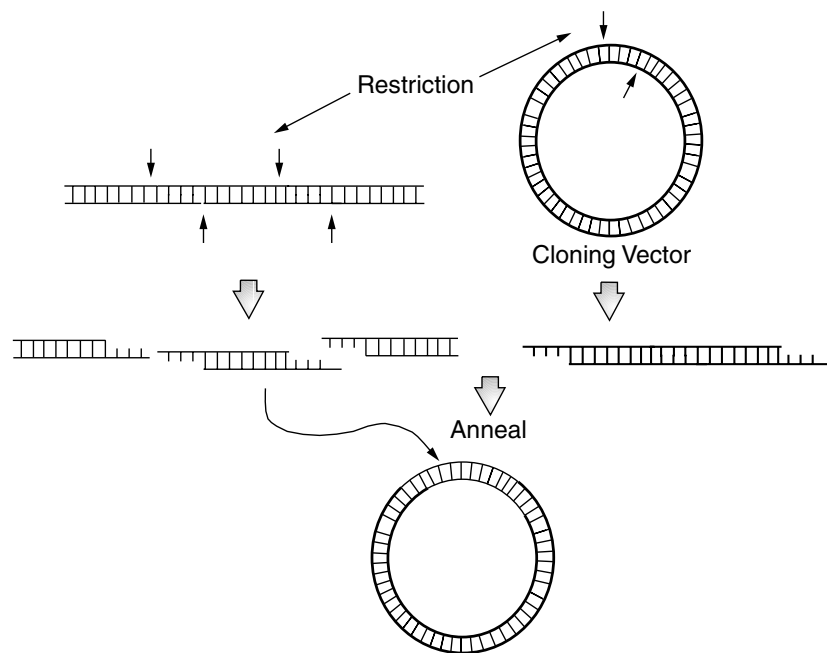


Fig. 3-16. Cloning a fragment of DNA in a plasmid vector. DNA to be cloned (thin lines) and cloning vector (thick lines) are treated with the same type II restriction enzyme to generate molecules with complementary “sticky” end sequences. The cloning vector should only contain a single restriction site for the enzyme being used as is the case in this example. A recombinant molecule is formed by *in vitro* annealing and ligation with a DNA ligase (usually phage T4 DNA ligase). The recombinant molecules are then transformed into suitable *E. coli* hosts (usually $r^- m^+$ to avoid degrading the foreign DNA insert). An antibiotic resistance gene on the vector will allow selection of cells that received the vector. Clones carrying recombinants may be selected when the inserted DNA complements a mutant defect (e.g., cloning of *arg* genes), insertionally inactivates a readily selectable marker (e.g., tetracycline resistance), or can be detected using labeled DNA probes.

Class IV restriction-modification systems (e.g., *Eco571*) include a large bifunctional polypeptide with endonuclease and methylase activities and a separate methyltransferase polypeptide. The large protein methylates only one strand of the asymmetric recognition site, whereas the smaller methyltransferase methylates both strands. Cleavage has an absolute requirement for Mg^{2+} and is stimulated by SAM. A key distinction between type III and type IV endonucleases is that the latter does not depend on ATP.

INSERTION SEQUENCES AND TRANSPOSABLE ELEMENTS

Transposable elements are discrete sequences of DNA that encode functions to catalyze the movement (**translocation**) of the transposable element from one DNA site to a second, *target* site (Fig. 3-17A). The target site is duplicated during the transposition event with a copy found to either side of the transposed element. There are essentially two types of transposition. Replicative transposition involves both replication and recombination with a copy of the element remaining at the original site. Conservative

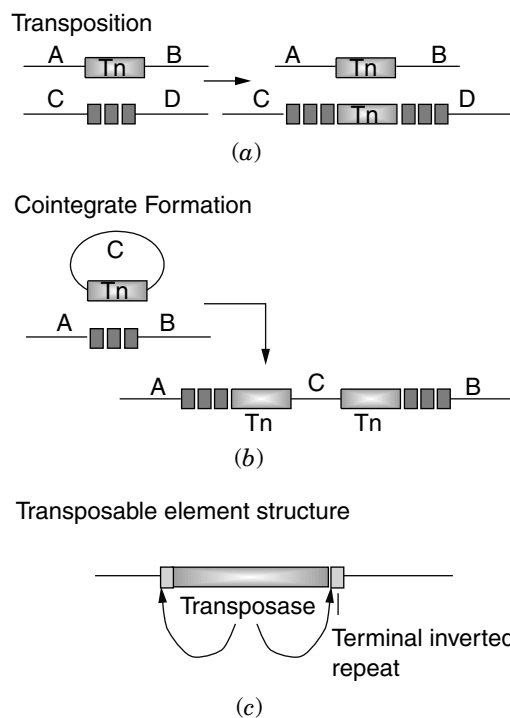


Fig. 3-17. Transposition, cointegrate formation, and a transposable element. (a) Transposition. A schematic presentation of replicative transposition in which a transposable element (Tn) translocates to a new site (■ ■ ■) with concomitant duplication of the site. (b) Cointegrate formation. Replicon fusion mediated by a transposable element. (c) A transposable element. The minimal components of a transposable element include a gene encoding a transposase and terminal target sites for the transposase. These sites are typically the same sequence in inverted orientation.

transposition does not involve replication. The element is simply moved to a new location. When the target site occurs within a gene, either type of transposition will generate insertion mutations. In addition to causing insertion mutations, transposition of these elements can cause deletions, inversions, and cointegrate formation in which two distinct replicons (e.g., plasmids) are joined (replicon fusion). Cointegrate formation is illustrated in Figure 3-17B.

There are essentially three classes of transposable elements in *E. coli*:

1. **Insertion sequences (IS)**, which encode no function other than transposition, and can be simply diagrammed as possessing inverted repeats at either of its ends and a gene encoding the “transposase” responsible for recognizing the terminal repeats and catalyzing the transposition process.
2. **Transposons**, which possess additional genetic information encoding properties such as drug resistance unrelated to the transposition process. There are two subclasses of transposons. **Class I transposons** are composite elements in which two insertion sequence elements flank an intervening sequence encoding antibiotic resistance or some other function (e.g., *Tn10*). **Class II transposons** are complex elements that do not appear to be related to IS elements (e.g., *Tn3*).
3. **Bacteriophage Mu**, a lysogenic bacteriophage that employs transposition as a way of life.

Table 3-4 lists the characteristics of several transposable elements.

Before discussing specific examples of transposable elements, it is useful to discuss a model for transposition. The diagram in Figure 3-18 shows the origin of the short, duplicated regions of DNA that flank the insertion element. The target DNA is cleaved with staggered cuts by the transposase, the transposon is attached to the protruding

TABLE 3-4. Examples of Transposable Elements

Element	Size	Terminal Repeat		
		Element	Target (bp)	Drug Resistance
<i>Insertion Sequences</i>				
IS1	768	30 bp inverted	9	None
IS2	1,327	32 bp inverted	5	None
IS3	1,400	32 bp inverted	3–4	None
<i>Transposons</i>				
Tn1	4,957	38 bp inverted	5	Ap ^R
Tn5	5,400	1,450 bp (IS50) inverted	9	Km ^R
Tn9	2,638	768 bp (IS1 direct)	9	Cm ^R
Tn10	9,300	1,400 (IS10) inverted	9	Tet ^R
Tn3	4,957	38 bp inverted	5	Ap ^R
<i>Bacteriophages</i>				
Mu	38,000	11 bp inverted	5	None

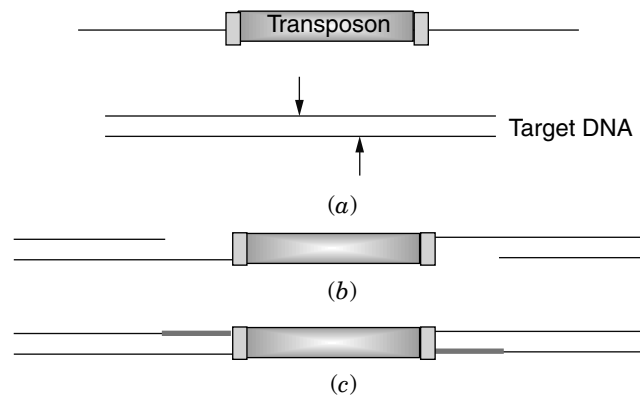


Fig. 3-18. Origin of the short, flanking duplications of target DNA. The target is cleaved with staggered cuts and the extended single strands of the target are then joined to the transposon termini. Repair of the single-stranded gaps completes the duplication.

single-stranded ends, and the short ends are filled in by repair synthesis, leading to the flanking duplications.

A model that explains the transposition process is presented in Figure 3-19. A **transpososome** complex forms between the transposase, the paired ends of the transposon, and the DNA possessing the intended target site (A). The transposase hydrolytically nicks one strand at each end of the transposon (B). In replicative transposition, the 3'-OH ends of the nicked element attack the target DNA, resulting in a transesterification reaction covalently joining the transposon ends to the target DNA (C). Attachment of both ends of the transposon to the target immediately forms two replication forks (D). The transposon is completely replicated, with final sealing of the replicated DNA to flanking sequences generating a cointegrate (E). Resolution of the cointegrate by genetic exchange between the two transposon copies results in a simple insertion and regeneration of the donor replicon (F). While this model appears adequate for Tn3-like transposons, it does not completely hold for IS elements or Mu.

Transposons such as Tn10 or Tn5 generate simple insertions by a **nonreplicative** cut-and-paste mechanism that does not involve cointegrate formation. The model can be seen in Figure 3-19G, H, and I. Nonreplicative transposition involves, at each end of the transposon, the 3'-OH end of the nicked strand attacking the opposite unnicked strand (G). The resulting transesterification reaction yields a hairpin at the transposon termini and a double-strand cleaved flanking DNA (H). The hairpins are then nicked to regenerate the 3'-OH ends, which then carry out a second transesterification reaction with DNA strands at the target site (I). The result is movement ("jumping") of the transposon from one site to another without making a copy of the transposon.

Transposon Tn10

While there are several examples of transposons that could be discussed in greater detail, the one most clearly understood is Tn10. Tn10 is actually a composite element in which two IS-like sequences cooperate to mediate the transposition of the entire element. The basic structure of Tn10 can be seen in Figure 3-20. There are two IS10 elements that flank a central region containing a tetracycline-resistant locus. The two

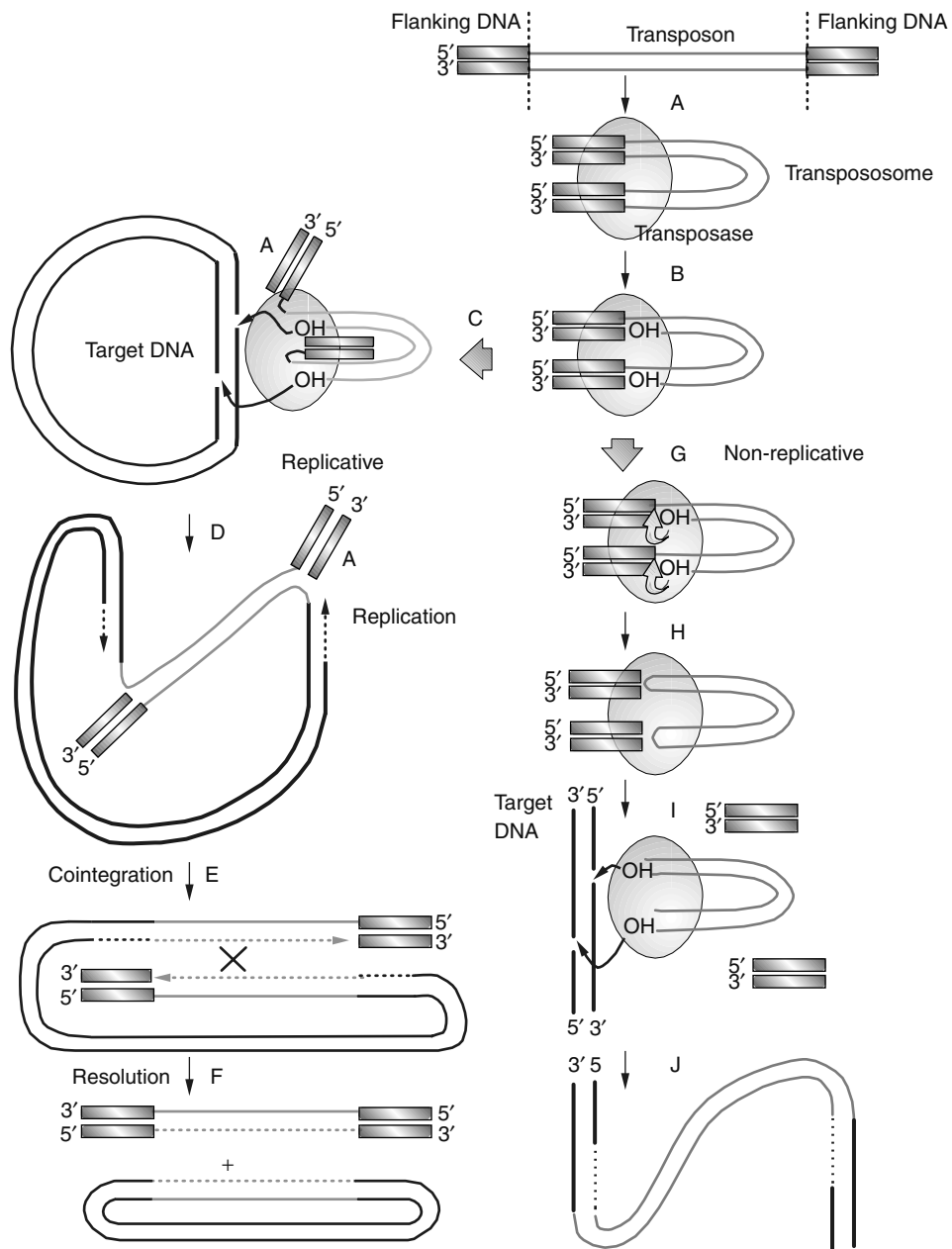


Fig. 3-19. Symmetric transposition model.

IS10 elements are not identical, however. It is believed that both started out the same, but IS10-left has evolved to the point where its ability to mediate transposition is very low relative to IS10-right.

IS10-right produces a trans-acting function required for transposition (the transposase). There is an inward promoter called p-IN that is required for transcription of the

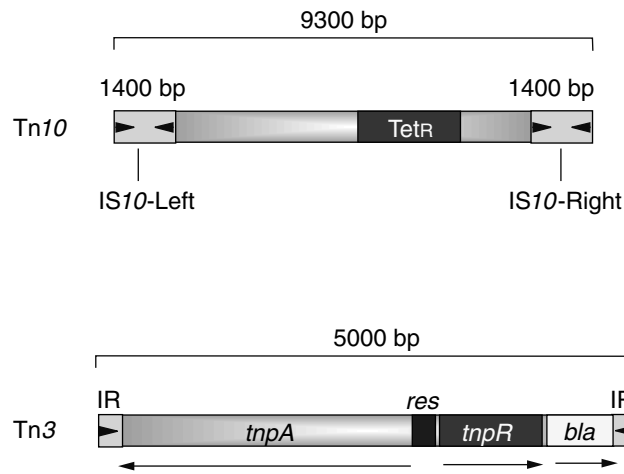


Fig. 3-20. Basic structure of transposons Tn10 and Tn3. IR represents the inverted repeats located at either end; *tnpR*, resolvase locus; *bla*, β -lactamase locus; *res*, site-specific resolution site; transcripts are represented by arrows.

transposase, but there is also an outward promoter (p-OUT) that produces an antisense RNA that regulates transposition. This 69 nucleotide RNA binds to the ribosome-binding site of the transposase message, preventing the overproduction of transposase. The p-OUT promoter can also be used to activate genes adjacent to where a Tn10 has inserted. The IS10-right transposase acts on sites located within the outermost 70 bp of both IS10 elements. This was determined by showing that deletion of this region prevented transposition even when the transposase function was supplied in trans.

The activity of IS10 is regulated by *dam* methylation (see “Initiation of DNA Replication” in Chapter 2). IS10 has two GATC methylation sites: one overlapping the pIN -10 region and the other located at the inner end of IS10 where transposase might bind. Methylation of the *dam* site within pIN reduces promoter efficiency of the transposase gene. The reduction in transposase will lower Tn10 transposition frequency. Methylation of the second *dam* site will lower independent transposition of the constituent IS10 elements, thereby increasing the cohesiveness of the composite element. Consequently, the element can only transpose after replication by the host machinery and before the newly synthesized strand is methylated. The transposition process for Tn10 is nonreplicative.

Although Tn10 can insert at many different sites in both *E. coli* and *S. enterica* chromosomes, the sites of insertion are not totally random. There is approximately one Tn10 insertion “hot spot” per thousand base pairs of DNA. The specificity sequence is a 6 bp consensus sequence (GCTNAGC). This consensus specificity is located within the 9 bp target site, which is cleaved by staggered nicks during Tn10 insertion. As originally illustrated in Figure 3-18, the 9 bp target site is duplicated during the insertion process.

As noted earlier, in addition to promoting the movement of a transposon from one site to another, transpositional recombination can result in a variety of genetic rearrangements such as inversions or deletions. Figure 3-21 illustrates how Tn10 can generate either deletions (A) or inversions (B). Note in both cases that the central

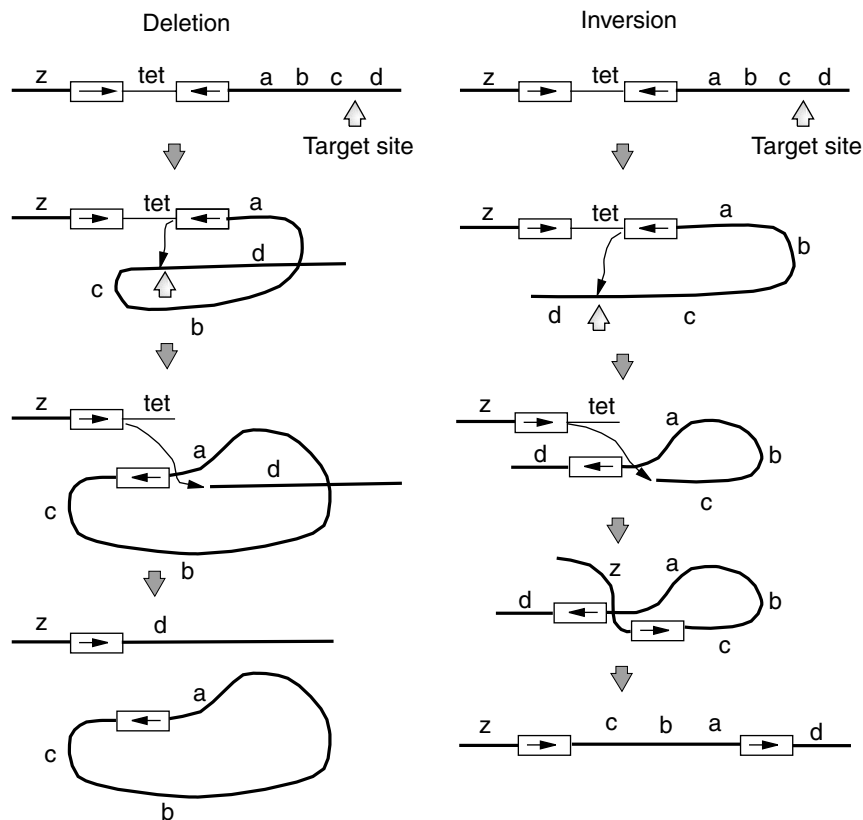


Fig. 3-21. Mechanisms of transposon Tn10-promoted rearrangements.

portion of Tn10 containing the Tet^R determinant is lost after the rearrangement. In both situations, the inner aspects of the IS10 elements are utilized rather than the outermost sequences, as is the case for transposition. The only difference for whether an inversion or a deletion occurs is the orientation of the target site relative to the transposon.

Transposon Tn3

Tn3 is an example of a complex transposon. It does not have a modular structure like Tn10, does not appear to be based on IS elements, and has no obvious evolutionary link to them. But even though Tn10 and Tn3 move by different mechanisms, they both go through the same intermediates as illustrated in Figure 3-19. Figure 3-20 illustrates the basic structure of Tn3. The *tnpA* locus codes for the transposase, which works at the inverted repeats located at either end of the element. TnpR is the repressor for *tnpA* but also acts as a site-specific resolvase. The replicative model for transposition presented in Figure 3-19 involves cointegrate formation in which the transposon has duplicated, joining the donor and recipient molecules. Resolution of this structure requires a recombinational event between the two transposons (Fig. 3-19E). For Tn3, the res site is the point where recombination, carried out by the TnpR resolvase, occurs.

Conjugative Transposition

Conjugative transposons are characterized by their ability to move between bacterial cells by a process that requires cell-to-cell contact. This phenomenon was first observed in plasmid-free strains of the gram-positive organism *Enterococcus hirae* (formerly *E. faecalis*). This organism contains an 18 kb Tet^R element designated Tn916 that mediates mating and self-transfer to other Enterococci. Now known to be more widespread in nature, conjugative transposons can excise from the chromosome, form a circular intermediate, and be transferred to a new cell via conjugation. Thus, conjugative transposons are plasmid-like in that they have a covalently closed circular transfer intermediate and they are transferred by conjugation, but unlike plasmids, the circular intermediate of a conjugative transposon does not appear to replicate. However, the possibility that circular forms might undergo autonomous replication in some hosts cannot be ruled out.

The conjugative transposons are also phage-like in that their excision and integration resembles excision/integration of temperate bacteriophages such as λ (see Chapter 6). An interesting feature of Tn916 is that the excision process involves dissimilar sequences at either end of the transposon. During excision, staggered cuts are made at both ends of the element, leaving six nucleotide single-strand overhangs called coupling sequences. The Tn916 circle is formed by covalently coupling the ends, but since the coupling sequences are not homologous, they do not base pair but form a bubble. The mismatched joint region undergoes some form of repair, since all the joint regions of circular intermediates end up as homoduplexes, with 80% of the joints reflecting the left coupling sequence and 20% reflecting the right coupling sequence.

Little is known about the actual process of cell-to-cell transfer of conjugative transposons and their establishment in the recipient. The precise excision of Tn916 suggests a conservative mechanism of transposition in which replication does not leave a copy of the transposon at the donor site. However, following mating, the recipient chromosome will end up with multiple copies of the transposon, suggesting the possibility of replicative transposition.

Although initially discovered in gram-positive organisms, examples of conjugative transposons are also found in the gram-negative genera *Bacteroides* and *Neisseria*. The transfer of many conjugative plasmids is regulated, but only in a few cases is the signal known. The signal for a *Bacteroides* conjugative plasmid named Tc^R Em^R Dot is tetracycline. Like the bacterial equivalent of an aphrodisiac, exposure to a low level of tetracycline stimulates mating 1000-fold. Curiously, *Bacteroides* conjugative transposons transfer only when the donor and recipient reside on solid surfaces. Mating in liquid culture does not occur. How mating became regulated by tetracycline is an evolutionary mystery. Tet has only been in use as an antibiotic for a short time—not long enough for such a complex system to develop in response. One theory is that *Bacteroides* ancestors encountered tetracycline produced in nature by the actinomycetes. Alternatively, tetracycline may bear a resemblance to the true original inducer, perhaps a plant phenolic compound.

Evolutionary Consideration

Insertion sequences and transposons are thought to have been, and probably continue to be, an important component of evolution, since they can catalyze major rearrangements of chromosomes as well as serve to introduce new genes into a given organism. For

example, there is evidence suggesting that the chromosome of *E. coli*, as we know it, is the result of perhaps two major duplications (which IS elements can also initiate). It only needs to be noted that genes with related biochemical functions often reside in positions 90° to 180° apart from each other for this hypothesis to be appreciated. Also, even though *S. enterica* and *E. coli* are closely related organisms, a comparison of their genetic maps reveals several inversions and deletions. Consequently, it would appear that insertion elements may be an important key to understanding evolutionary processes.

Integrans

One of the consequences of the microbial genome sequencing era is the realization that bacteria are promiscuous gene swappers. This is not restricted to passing antibiotic resistance genes around. There are numerous unexplained resemblances between DNAs of evolutionarily distinct species. Genetic elements called integrans are the reason for this high level of gene trafficking. Integrans are elements that encode a site-specific recombination system that recognizes and captures mobile gene cassettes. Integrans include an integrase (Int) and an adjacent recombination site (*attI*).

A glimpse into how this type of gene trafficking occurs comes from studies with *Vibrio cholerae*, the cholera agent. This organism has a versatile integran acquisition system embedded in its own genome that appears capable of capturing genes serendipitously encountered from other organisms. The proposed integran contains an integrase gene and adjacent gene(s) flanked by DNA repeats that are targets for the integrase. The integrase, however, can act on similar DNA repeats that may be found flanking genes in DNA from heterologous organisms. *Vibrio* may encounter this heterologous DNA as a result of conjugation or transformation. The integrase can then “capture” this new DNA by recognizing these DNA repeats and can integrate the heterologous gene at one of *Vibrio*’s own DNA repeats. Although not proven, there is plenty of evidence that many pathogens have acquired new virulence genes using this strategy.

MUTAGENESIS

During growth of an organism, DNA can become damaged by a variety of conditions. Any heritable change in the nucleotide sequence of a gene is called a mutation regardless of whether there is any observable change in the characteristics (**phenotype**) of the organism. We will now discuss the various mechanisms by which mutations are introduced and repaired. You may be surprised to learn that it is not usually a chemical agent that causes heritable mutations but rather the bacterial attempt to repair chemically damaged DNA.

But before delving into the molecular details of this process, the terminology of mutations must be defined. For example, a bacterial strain that contains all of the genetic information required to grow on a minimal salts medium is called wild-type or **prototrophic**, whereas a mutant strain requiring one or more additional nutrients is **auxotrophic**. Mutations themselves come in a variety of different forms. A change in a single base is a **point mutation**. The most common type of point mutations are **transition mutations** that involve changing a purine to a different purine (A ↔ G) or

a pyrimidine to a different pyrimidine. A **transversion mutation** occurs when a purine is replaced by a pyrimidine or vice versa.

A point mutation can change a specific codon, resulting in an incorrect amino acid being incorporated into a protein. The result is a **missense mutation**. The codon change could also result in a translational stop codon being inserted into the middle of a gene. Since a mutation of this sort does not code for an amino acid, it is a **nonsense mutation**. A missense mutation allows the formation of a complete polypeptide, whereas a nonsense mutation results in an incomplete (truncated) protein. Nonsense mutations (also referred to as amber, UAG; ochre, UGA; or opal, UAA mutations), if they occur in an operon involving several genes transcribed from a single promoter, may have polar (distal) effects on the expression of genes downstream from the mutation. A mutation of this type is a **polar mutation**. Figure 3-22 compares a polar nonsense mutation with a nonpolar nonsense mutation.

Ordinarily, you would not expect a translational stop codon to interfere with the transcription or translation of downstream members of an operon, since each gene member of the operon has its own ribosome-binding site. The key to understanding polar mutations is that some nonsense mutations will cause premature transcription termination, lowering the amount of downstream message produced and available for translation. Premature transcription termination occurs when the nontranslated mRNA downstream from the nonsense codon possesses a secondary structure (i.e., stem-loop) that mimics transcription termination signals.

If a process causes the removal of a series of bases in a sequence, the result is a **deletion mutation**. Likewise, the addition of extra bases into a sequence is an addition or **insertion mutation**. Both additions and deletions can result in changing the translational reading frame, causing all of the amino acids situated downstream of the mutation to be incorrect. The offending mutation in this event is a **frameshift mutation**.

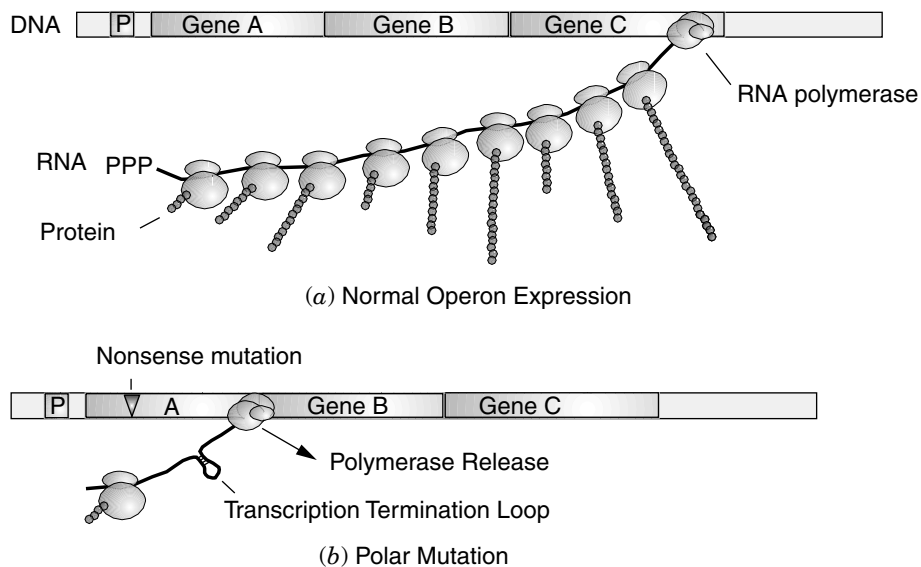


Fig. 3-22. Mechanism of transcription termination by a polar mutation.

One type of mutation that has proven extremely useful is the **conditional mutation**, an example of which is the temperature-sensitive (ts) mutant. These mutants grow normally at a low, permissive temperature (30 °C) but exhibit a mutant phenotype at a higher, nonpermissive temperature (42 °C). An amino acid replacement altering the conformational stability of the gene product is usually thought to be the cause. The mutant protein, being less stable, unfolds at the nonpermissive temperature and becomes inactive.

Spontaneous Mutations

In a population of cells, mutations can arise spontaneously without overt treatment with a mutagen. Spontaneous mutations are rare, ranging from 10^{-6} to 10^{-8} per generation depending on the gene and organism. **Mutation rate** is calculated as the number of mutations formed per cell doubling according to the formula $a = m/\text{cell generations} = m \ln 2 / n - n_0$, where a = mutation rate and m is the number of mutations that occur as the number of cells increases from n_0 to n .

There are many causes of spontaneous mutations. Even though the replication apparatus is very accurate with various “proofreading” and repair functions, such as the 3′ → 5′ exonuclease activity of DNA polymerase III, mistakes do occur at a very low rate. One reason that misincorporation by DNA polymerase can go undetected involves tautomeric considerations discussed below. Besides misincorporation mistakes, endogenously occurring DNA damage is an important source of spontaneous mutations. For example, cytosine deaminates spontaneously to yield uracil (G : C → A : T). In addition, purines are particularly susceptible to spontaneous loss from DNA by breakage of the glycosidic bond between the base and sugar. The result is the formation of an **apurinic site** in the DNA.

Metabolic activities can yield DNA-damaging intermediates (e.g., H_2O_2 , O_2^- , $\cdot\text{OH}$), and endogenous methylation agents (e.g., S-adenosylmethionine) can spontaneously methylate DNA bases. The spontaneous methylation of the N7 position of guanine, for example, weakens the glycosidic bond, resulting in spontaneous loss of the base (apurinic site) or opening of the imidazole ring forming a methyl formamide pyrimidine. In addition to mispairing, some of these spontaneous events can lead to major chromosomal rearrangements such as duplications, inversions, and deletions. Mutagens tend to increase the mutation rate by increasing the number of mistakes in a DNA molecule, as well as by inducing repair pathways that themselves introduce mutations.

THE NATURE OF MUTATIONAL EVENTS

Mutations can arise by a number of molecular events depending in part on the nature of the mutagen. Table 3-5 lists various mutagens, the type of mutation(s) that can arise from each, as well as an indication of the molecular event involved. This section discusses some of the commonly employed mutagens and their mechanisms for causing mutations. Ultraviolet (UV) irradiation at 254 nm causes the production of pyrimidine dimers between thymine-thymine (Fig. 3-23) or thymine-cytosine or cytosine-cytosine pairs. Ultraviolet irradiation may also result in distortion of the backbone of the

TABLE 3-5. Mutagen Action

Mutagen	Specificity	Mechanism ^a
Spontaneous	Substitution	Mispairing
	Frameshift	Slipping
	Multisite	Recombination
	All types	Misrepair
	All types	Misrepair
UV radiation		
Base analogs		
5-Bromouracil	A-T ↔ G-C ^b	Mispairing
2-Aminopurine	A-T ↔ G-C ^c	Mispairing
Base modifiers		
Nitrous acid	A-T ↔ G-C	Mispairing
Hydroxylamine	G-C → A-T	Mispairing
Alkylating agents	Mainly transitions ^d	Mispairing
	All types	Misrepair
Intercalators	Frameshift	Slipping
	All types	Misrepair

^aMispairing—that is, nonstandard base pairing—may arise spontaneously or through the presence of nucleotide derivatives. Slipping refers to imperfect pairing between complementary strands due to base sequence redundancy. All types of mutations may be induced indirectly by faulty repair mechanisms—that is, misrepair.

^bG-C → A-T is favored.

^cA-T → G-C transitions occur 10–20-fold more frequently.

^dEMS (ethylmethane sulfonate) and MNNG (*N*-methyl-*N'*-nitro-*N*-nitrosoguanidine).

^eThe wide range of lesions induced by ICR compounds may stem from their alkylating side chain.

DNA helix, causing replication errors. Ionizing radiations cause instability in the DNA molecule, resulting in single-strand breaks.

Chemical mutagens may directly modify the purine or pyrimidine bases, causing errors in base pairing. Local distortions in the helix may result in replication or recombination errors. Compounds such as nitrous acid cause the deamination of adenine and cytosine (Fig. 3-23B), and alkylating agents such as ethylethane sulfonate produce base analogs that result in transitions in base pairing in the nucleic acid structure. Acridine dyes may intercalate between the stacked bases (Fig. 3-24), distorting the structure of the DNA and causing frameshift errors during replication. Abnormal base pairing may also result from tautomeric shifts in the chemical structure of the bases as shown in Figure 3-25.

Normally, the amino and keto forms predominate (ca. 85%). Tautomeric shifts occurring during the replication of DNA will increase the number of mutational events. Certain analogs of the purine or pyrimidine bases may be incorporated into DNA in place of the normal base. For example, 5-bromouracil, upon incorporation into DNA, may pair with either adenine or guanine, as shown in Figure 3-26, resulting in base transitions in the DNA. Figure 3-27 illustrates the target sites on bases that yield common modified bases. It should also be recognized that some modified bases are mutagenic (altered base pairing) while others are lethal (stop replication).

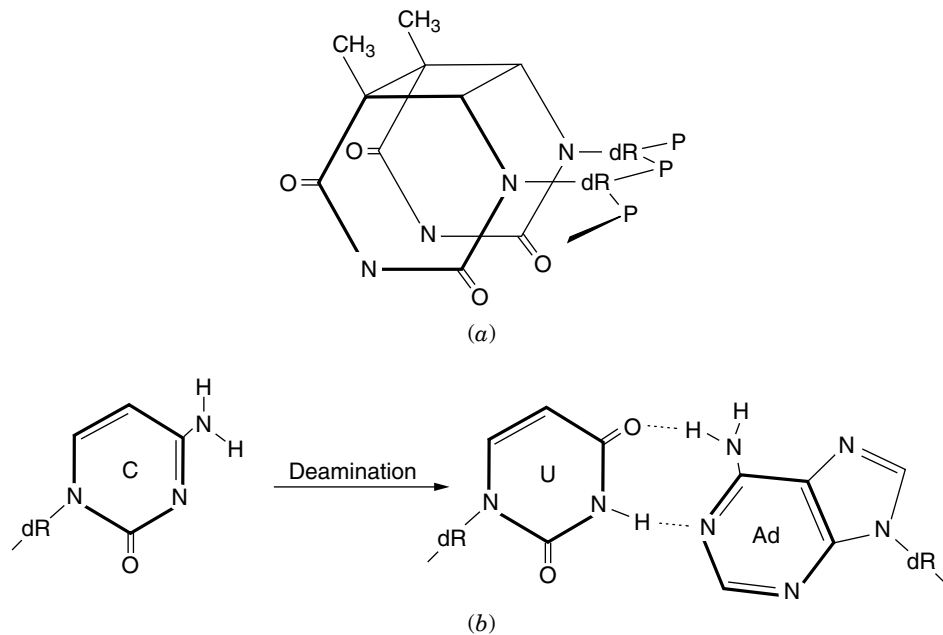


Fig. 3-23. (a) Photodimer of thymine found in ultraviolet-irradiated DNA. Of four possible stereoisomeric photodimers of thymine, only the *cis* 5,5':6,6' isomer shown is obtained by ultraviolet irradiation of DNA. Thymine-cytosine and cytosine-cytosine dimers are also produced. (b) Altered base-pairing as a result of deamination by nitrous acid.

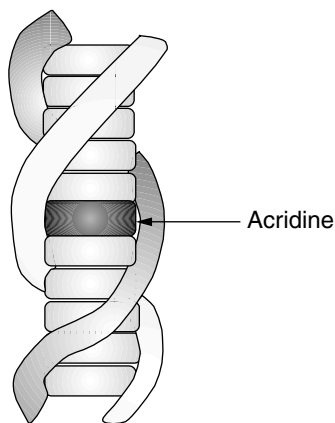


Fig. 3-24. Diagrammatic view of the manner in which acridine orange and ethidium bromide intercalate between the stacked base pairs in the double-stranded DNA molecule.

Suppressor Mutations

Suppression is the reversal of a mutant phenotype as a result of another, secondary mutation. The second mutation may occur in the same gene as the original mutation (**intragenic suppressor**) or in a different gene (**extragenic suppressor**). Intragenic

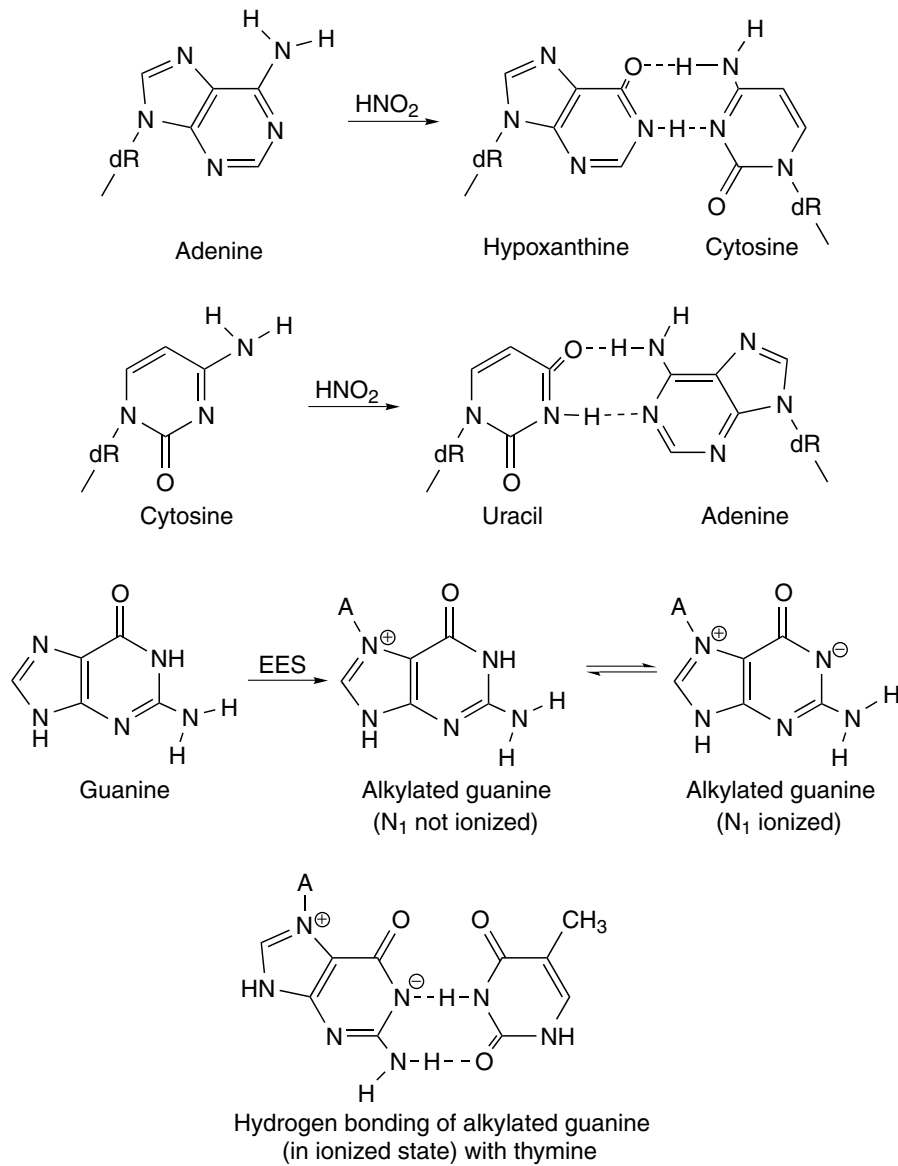
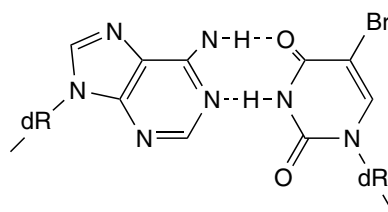
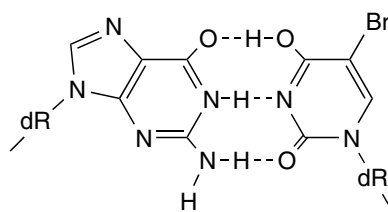


Fig. 3-25. Normal ($\text{A}=\text{T}$; $\text{G}=\text{C}$) and abnormal ($\text{A}=\text{C}$; $\text{G}=\text{T}$) base pairing as a result of tautomeric shifting.

suppressors may cause an amino acid substitution that compensates for the primary missense mutation, thereby partially restoring the lost function, or they may be the result of an insertion or deletion that compensates for the original frameshift mutation. Extragenic nonsense suppressors are usually mutations that alter the genes encoding tRNAs. The revertant phenotypes may be found to contain tRNAs that recognize nonsense codons (e.g., UAA and UAG) and result in the insertion of a particular amino acid at the site of the nonsense mutation.



Adenine: 5-bromouracil
(keto state)



Guanine: 5-bromouracil
(enol state)

Fig. 3-26. Base pairing of 5-bromouracil (keto state) with adenine and 5-bromouracil (enol state) with guanine.

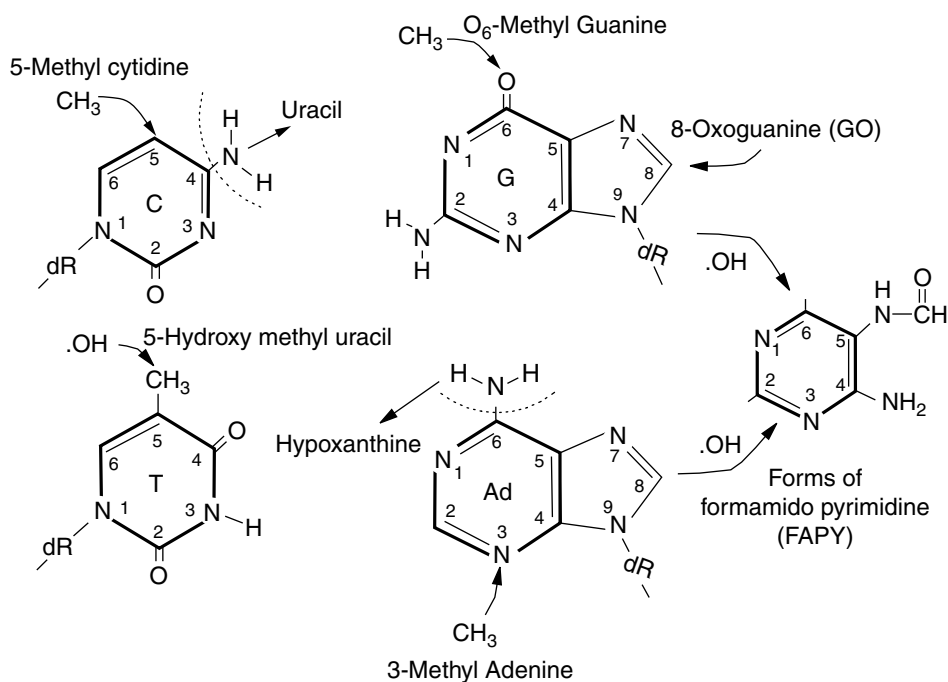


Fig. 3-27. Common chemical modifications of DNA bases.

Revertants that are due to suppressor mutations are usually less efficient than the wild type. For example, in the reversion of an auxotroph to wild type as a result of a tRNA suppressor mutation (*sup*), the revertants usually grow more slowly and produce smaller colonies on minimal agar than the wild type. The direct demonstration of a suppressor mutation can be accomplished by genetic crosses, since a suppressor mutation does not result in any alteration of the original mutation. The original mutant type can be recovered among the progeny of a cross between wild type and the revertant. Also, by means of appropriate crosses, suppressor mutations can be introduced into other mutants in order to assess their effect on other alleles.

DNA REPAIR SYSTEMS

Preventing an unacceptably high mutation rate is of extreme importance to the cell. Consequently, a variety of mechanisms have evolved to repair misincorporated residues or bases altered by exposure to radiation or chemical mutagens. Repair pathways can be viewed as either **prereplicative** or **postreplicative** and as **error-proof** or **error-prone**. The error-prone pathways are often responsible for producing heritable mutations. Table 3-6 lists the genes associated with various aspects of DNA repair in *E. coli*.

Photoreactivation

The cyclobutane ring structure in pyrimidine dimers produced during UV irradiation can be removed enzymatically by the product of the *phr* locus. The *E. coli phr*⁺ gene product (photolyase) will bind to pyrimidine dimers in the dark. However, the reaction that monomerizes the dimer requires activation by visible light (340–400 nm); hence the term *photoreactivation*. The absorption responsible for photoreactivation is developed only while the enzyme is bound to UV-damaged DNA. The reaction does not require the removal of any bases, just the monomerization of dimers (Fig. 3-23). This activity is, therefore, considered an error-proof repair pathway.

Nucleotide Excision Repair

Bulky lesions such as UV-induced pyrimidine dimers can be excised by a complex exonuclease encoded by the *uvrA*, *uvrB*, and *uvrC* genes of *E. coli*. The UvrABC nuclease recognizes distortions created by cyclobutane rings (Fig. 3-28). Two single-strand incisions are made: one at the eighth phosphodiester bond 5' to the dimer and the other at the fourth or fifth phosphodiester bond 3' to the dimer. The net result is the release of a 12–13 nucleotide DNA fragment that contains the site of damage. Subsequently, DNA polymerase I uses the 3'-OH end of the gapped DNA to synthesize a new stretch of DNA containing the correct nucleotide sequence. Finally, DNA ligase seals the remaining single-strand nick.

The *uvrA* gene product is a damage-recognition DNA-binding protein that forms a UvrA₂B complex. UvrA initially recognizes DNA lesions, binds, and delivers UvrB to the damaged sites. UvrC protein then binds to the UvrB-bound damaged DNA to induce dual incisions as indicated above. The *uvrD* product (DNA helicase II) is required for release of the UvrABC nuclease. This *uvr*⁺-dependent excision produces a relatively short patch of repair and is very accurate, since UV treatment is far more

TABLE 3-6. Selected Genetic Loci Associated with Repair

Gene	Map Location	Type of Mutation	Size	Name and/or Function
<i>ada</i>	49.7	G : C $\rightleftharpoons$ A : T	39,166	Regulates adaptive response, O ⁶ -alkylguanine-DNA alkyl transferase
<i>alkA</i>	46.2		31,241	3-methyladenine glycosylase II; hypoxanthine DNA glycosylase; sensitive to methylmethanesulfonate
<i>dcm</i>	43.7	GC $\rightleftharpoons$ AT	53,300	DNA cytosine methylase, 5'-CC(AT)GG target site
<i>dam</i>	74		31,947	DNA adenine methylase (mismatch correction), 5'-GATC target site
<i>endA</i>	66.6		26,560	DNA-specific endonuclease I
<i>fpg (mutM)</i>	82		30,139	Formamidopyrimidine DNA glycosylase, 8-hydroxyguanine, 8-oxodeoxyguanine glycosylase
<i>lexA</i>	91.7		21,212	Repressor of SOS mutations stimulate transversions; tRNA ^{gly} inserts at ASP codons; mutant induces unknown error-prone polymerase
<i>mutC (glyY)</i>	42.9			Pol III subunit ϵ
<i>mutD (dnaQ)</i>	5.1	Base substitutions; frameshifts	26,949	
<i>mutH</i>	63.9	G : C $\rightleftharpoons$ A : T frameshifts	25,380	Increased rate of frameshifts (methyl-directed mismatch repair)
<i>mutL</i>	95	G : C $\rightleftharpoons$ A : T frameshifts	67,750	Increased rate of AT $\leftrightarrow$ GC transitions, methyl-directed mismatch repair
<i>mutS</i>	61.5	G : C $\rightleftharpoons$ A : T frameshifts	95,050	Increased rate of AT $\leftrightarrow$ GC transversions, methyl-directed mismatch repair
<i>mutT</i>	2.4	A : T G : C	14,786	Hydrolyzes 8-oxo dGTP; prevents incorporation into DNA
<i>mutY</i>	66.8	G : C $\rightarrow$ T : A	38,992	Adenine glycosylase excises A from G-A mispair; mutants stimulate G-C $\rightarrow$ T-A transversions
<i>mutR (topB)</i>	39.7	Deletions	73,038	Topoisomerase III, increase in spontaneous deletions

(continued overleaf)

TABLE 3-6. (continued)

Gene	Map Location	Type of Mutation	Size	Name and/or Function
<i>nei</i>	16.0	G : C $\rightleftharpoons$ A : T	29,695	Endo VIII, 8-oxoguanine DNA glycosylase; oxidatively damaged pyr.
<i>nfo</i>	48.4	Base substitutions	31,326	EndoIV; AP-specific (BER)
<i>nth</i>	36.9	G : C $\rightleftharpoons$ A : T	23,415	Endo III (BER), oxidatively damaged pyr.
<i>ogt</i>	30.1	G : C $\rightleftharpoons$ A : T	19,035	O ⁶ -alkylguanine-DNA transferase
<i>phrB</i>	15.9	CC, TT dimers	53,501	Photolyase
<i>recA</i>	60.8		22,212	Recombination, effector of SOS regulon
<i>recBCD</i>	63.7		Table 3-1	Postreplication repair, Exo V
<i>recF</i>	83.6		40,355	Post replication repair, replication restart
<i>ruvA</i>	41.9		21,939	See Table 3-1
<i>ruvB</i>	41.9		37,017	See Table 3-1
<i>ruvC</i>	41.9		18,603	See Table 3-1
<i>tag</i>	79.9		20,955	3-methyladenine glycosylase I
<i>ung</i>	58.5	G : C $\rightleftharpoons$ A : T	25,540	Uracil DNA glycosylase
<i>uvrA</i>	92.0	CC or TT dimers	103,668	
<i>uvrB</i>	17.5	CC or TT dimers	76,044	ATP-dependent endonuclease (NER)
<i>uvrC</i>	42.9	CC or TT dimers	68,013	
<i>uvrD</i>	86.1	G : C $\rightleftharpoons$ A : T, frameshifts	81,803	Helicase II (mutU, uvrE, recL)
<i>umuDC</i>	26		14,922/47,516	Error-prone repair; DNA pol V
<i>vsr</i>	43.7		17,872	Endonuclease, cleaves 5' of the T in T/G mismatches [5'C(T/G)(A/T)GG]
<i>xseA</i>	56.7	Base substitutions	51,669	ExoVII; 5' $\rightarrow$ 3' exo ss specific
<i>xthA</i>	39.5	Base substitutions	30,818	ExoIII, EndoII (AP endonuclease)

BER, involved in base excision repair; NER, involved in nucleotide excision repair.

Source: Data derived from: <http://genolist.pasteur.fr/Colibri/>.

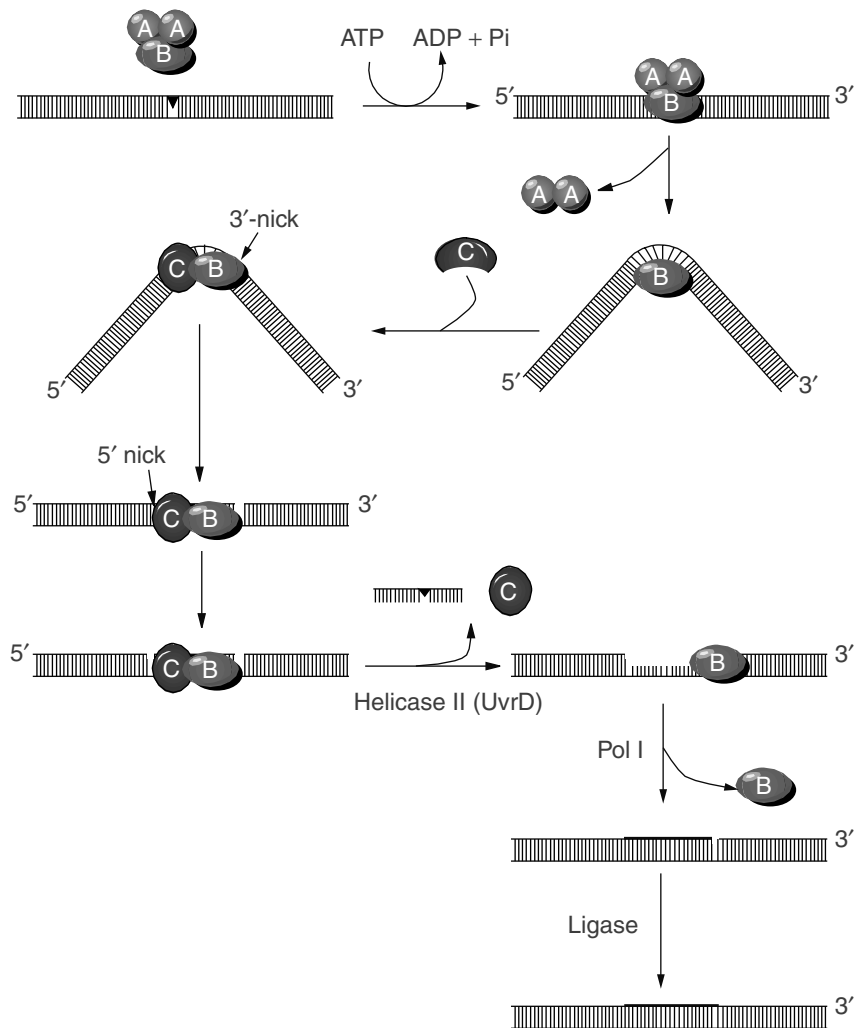


Fig. 3-28. Mechanism of the base excision repair.

mutagenic in *uvr* mutants. Thus, nucleotide excision repair (NER) is often referred to as **short-patch repair** and is error-proof, since a proofreading DNA polymerase (Pol I) fills in the gap. Short patch-type repair systems can also recognize other structural alterations such as missing bases (AP sites, see “DNA Glycosylases and Base Excision Repair”).

Transcription-Coupled Repair

An ability by bacteria and mammalian cells to repair actively transcribing genes more rapidly than nontranscribed (silent) genes can be described as a form of targeted NER. A current model to explain this phenomenon is based on the observation that the movement of RNA polymerase is blocked by lesions in the template but not in the coding strand of a gene. The product of the *mfd* (mutation frequency

decline) gene, called transcription repair coupling factor (TRCF), binds to the RNA pol–mRNA–DNA complex, causing release of RNA pol and the truncated transcript. TRCF is thought to replace RNA polymerase at the lesion site and attract the UvrABC complex by its affinity for UvrA. TRCF and UvrA then simultaneously disassociate, leaving the preincision UvrB–DNA complex to bind with UvrC, which makes the dual incisions.

Another model for transcription-coupled repair (TCR) is based on the observation that UvrA can associate with the β subunit of RNA polymerase. The resulting complex may act as a lesion sensor in which recognition of a lesion by UvrA could stall RNA polymerase. The stalled complex is proposed to localize to the inner membrane, where TRCF coaxes RNA polymerase off of the lesion site, allowing the Uvr repair proteins to function.

Transcription-coupled repair in *E. coli* is very important for gene expression in cells that have been exposed to DNA-damaging agents. For example, expression of *lacZ* (encoding β -galactosidase; see Chapter 4) during UV irradiation is dramatically enhanced in cells with a functional TCR. Consequently, targeting DNA repair enzymes to transcriptionally active genes is clearly advantageous for cell survival, since it permits the selective repair of genes deemed essential by virtue of their need to be transcribed. The TCR process may be the only means of repairing transcription-blocking damage at active genes.

Methyl-Directed Mismatch Repair

Methyl-directed mismatch repair (long-patch repair) is a postreplicative DNA repair system that recognizes mismatched bases produced in newly synthesized DNA. These misincorporated bases have eluded the 3' $\rightarrow$ 5' proofreading function of DNA polymerase (DnaQ subunit). Correction of this sort of mismatch must occur prior to a subsequent round of replication in order to prevent a mutation. The mismatch repair system requires the products of the *mutH*, *mutL*, *mutS*, and *uvrD* (*mutU*) genes. Mutations in these genes cause a tremendous increase in the spontaneous mutation rate (10^4 times higher than wild type) referred to as a **mutator phenotype**. To assure that only the newly misincorporated base is removed, this system must discriminate newly synthesized strands from parental strands. Discrimination is accomplished by recognizing methylation at the N6 position of adenine at GATC sequences, the target sequence for DNA adenine methylase (*dam*). Newly synthesized DNA is hemimethylated (methylated on only one strand) while parental DNA is methylated on both strands. The methyl-directed mismatch repair system recognizes hemimethylated DNA and repairs the mismatch on the unmethylated strand. The system works as represented in Figure 3-29.

The *mutS* gene (so named because mutations in these genes cause a mutator phenotype) encodes a 97 kDa protein that specifically binds to single base-pair mismatches and directs subsequent repair. The resulting conformational change in DNA-bound MutS allows a homodimer of MutL to bind. DNA is then translocated through the MutS–MutL recognition complex to form a loop. This translocation may allow the MutSL complex to interact with the nearest hemimethylated GATC site. The result is association and activation of the latent MutH endonuclease, which incises at unmethylated GATC sites and initiates the excision process. DNA helicase II (UvrD) then unwinds the DNA, exposing ssDNA to attack by exonuclease I (*xon*),

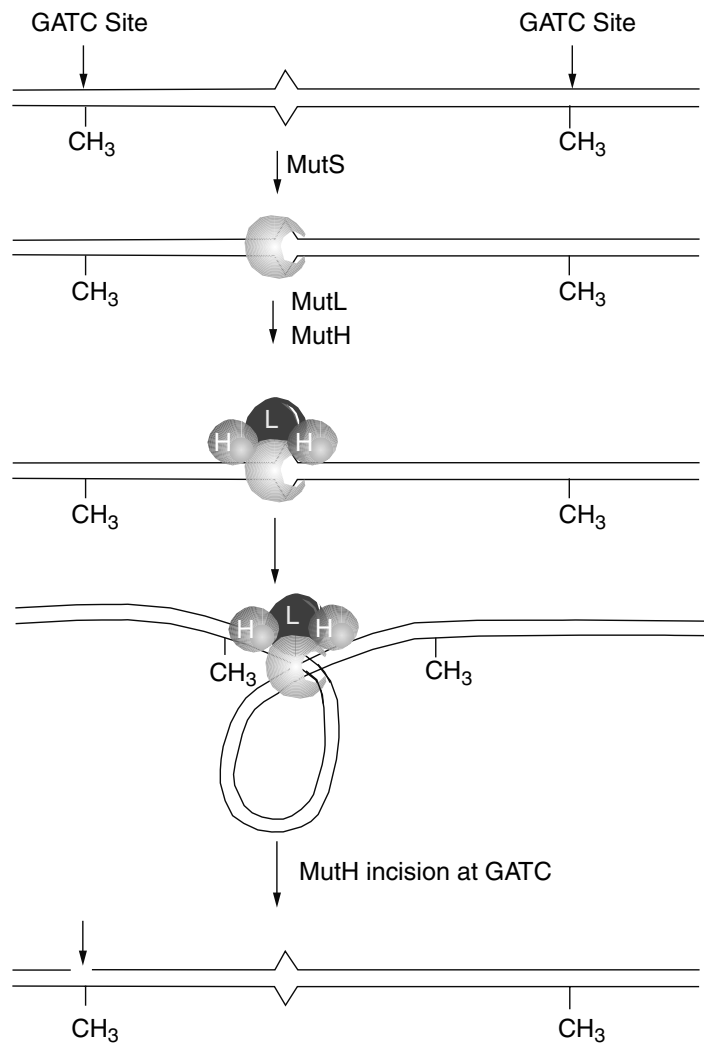


Fig. 3-29. Methyl-directed mismatch repair.

exonuclease VII (*xseA*), or the RecJ exonuclease. A strand break 3' to the mismatch requires Exo I for repair while either RecJ or Exo VII will suffice for processing strand breaks 5' to a mismatch. Once excised, DNA polymerase III will fill in the gap. The methyl-directed mismatch repair system is very effective in preventing transition and frameshift mutations but is less efficient in preventing transversions. Thus, methyl-directed mismatch repair is well suited to correct replication errors because the replicative polymerase Pol III produces primarily frameshifts and transitions (75% of all point mutations).

As you might predict, mutants lacking the Dam methylase also have a mutator phenotype—that is, since there is no discrimination between unmethylated parent and daughter strands, either the incorrect or correct base may be excised. If the correct base is repaired, then a mutation occurs.

Very Short-Patch Mismatch Repair

This system in *E. coli* efficiently corrects the T in G:T mismatches that occur within the Dcm methylase recognition sequence CC(A/T)GG. Dcm methylates the internal C's to generate 5-methyl cytosine. Dcm recognition sites are hot spots for mutations because of elevated spontaneous deamination of 5-methyl cytosine over that of cytosine. Deamination of 5-methyl cytosine in a G:5MeC base pair will lead to a G:T mispair. Very short-patch (VSP) mismatch repair involves MutS and MutL (but not MutH or UvrD) and requires the *vsr* product as well as DNA Pol I. Vsr (18 kDa) is a strand-specific mismatch endonuclease. It recognizes G:T mismatches in 5'CT(A/T)GG and NT(A/T)GG, making incisions 5' of the underlined T. DNA polymerase I removes the T with its 5' → 3' exonuclease activity and then carries out repair synthesis. MutS and MutL, though not required, stimulate Vsr function, enhancing very short-patch repair.

DNA Glycosylases and Base Excision Repair

DNA glycosylases catalyze the cleavage of sugar-base bonds in DNA and act only on altered or damaged nucleotide residues. The products of these glycosylase reactions are apurinic or apyrimidinic sites (AP sites) whose phosphodiester backbone can be cleaved by AP-specific endonucleases. The AP endonucleases cleave either at the 3' (class I) or the 5' (class II) side of the AP site (Fig. 3-30). Some glycosylases are monofunctional and require a separate endonuclease, whereas others are bifunctional, possessing both glycosylase and AP lyase activities. The bifunctional glycosylases remove the base and cleave the phosphodiester backbone. AP-specific endonucleases in *E. coli* include the following: (1) exonuclease III (*xth*), although initially identified as an exonuclease, it is also a class II endonuclease, in which mutations result in an increased sensitivity to hydrogen peroxide; (2) endonuclease IV (*nfo*), a class II endonuclease with no associated exonuclease activity; and (3) endonuclease V, a class I endonuclease with an unclear role.

There are a variety of DNA glycosylases that recognize a broad range of damaged bases. **Uracil–DNA glycosylase**, the product of the *ung* locus, removes deaminated cytosine residues and is responsible for repairing misincorporated uridine residues during replication. **Hypoxanthine–DNA glycosylase** removes spontaneously deaminated adenine residues. No mutant has been found lacking this enzyme, which suggests a critical role in maintaining cell viability. A major alkylation product in DNA treated with methylating agents such as methyl methane sulfonate or MNNG is 3-methyladenine. This modified base is very lethal because it interferes with DNA replication. The enzyme **3-methyladenine–DNA glycosylase I** (*tag*) removes 3-methyladenine from DNA. There is a second, inducible enzyme, **3-methyladenine–DNA glycosylase II** (*alkA*), produced during the adaptive response to methylating agents (see below). 3-Methyladenine glycosylases can release 3-methylguanine, 7-methylguanine, 7-methyladenine, hypoxanthine, 8-oxoguanine, and 5 hydroxymethylurea as well as 3-methyladenine from methylated DNA.

A major advance in understanding how 3-methyladenine DNA glycosylases operate was achieved by solving the crystal structure of AlkA. AlkA protein has three domains bordering on an adjustable cleft that contacts DNA substrates. The enzyme is thought to flip the nucleotide out of the DNA helix and into an active site lined with amino acids that interact with the target base. Although the AlkA substrates differ in their

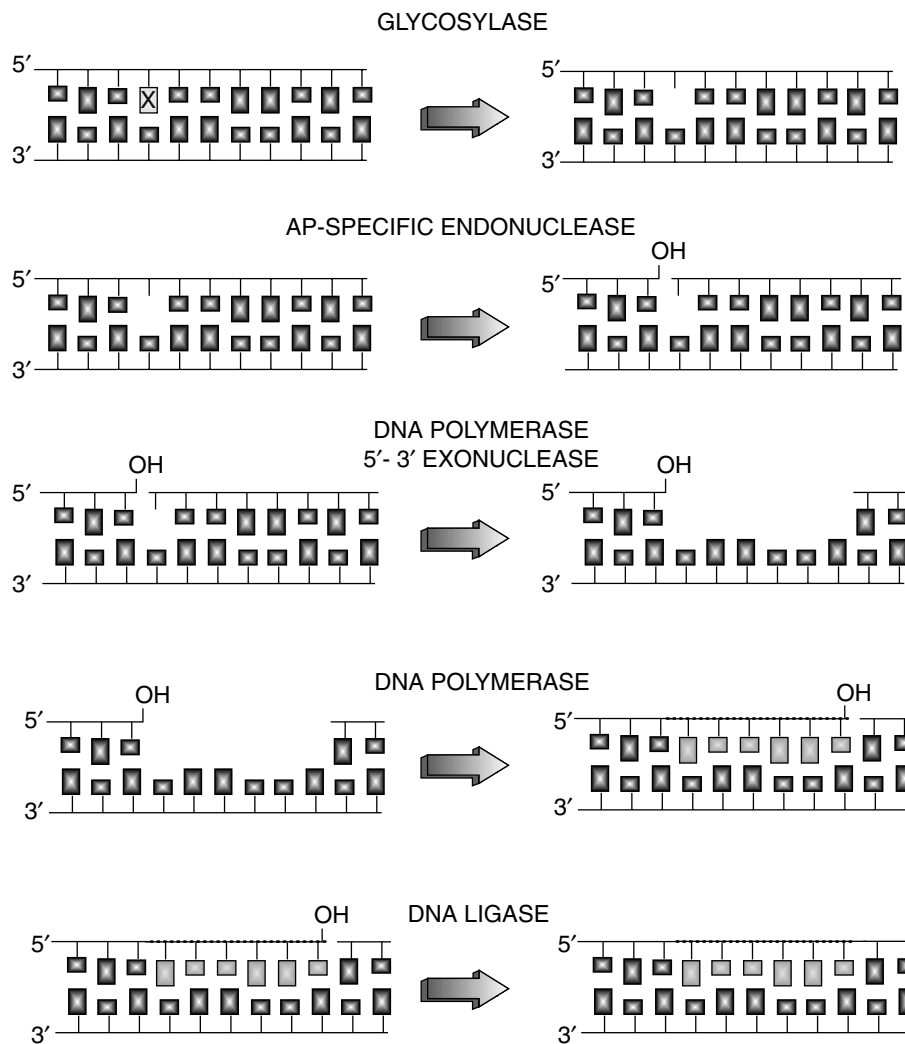


Fig. 3-30. Enzyme activities implicated in DNA repair. These functions are presented in the sequence in which they are thought to act. Resynthesis (DNA polymerase $5' \rightarrow 3'$ activity) may or may not take place in conjunction with nucleotide excision (associated $5' \rightarrow 3'$ exonuclease activity of DNA polymerase I). DNA ligase would, of course, be required to complete repair.

chemical structures, they share the trait of being electron-poor. The AlkA active site pocket is lined with hydrophobic tryptophan and tyrosine residues with electron-rich conjugated rings that stack against the target bases. These electron-rich amino acids may be the sensors for electron-deficient AlkA substrates.

A common DNA lesion, 7, 8 dihydro-8-oxoguanine (8-oxoG, GO), results from attack by various oxidizing chemical species (see Fig. 3-27). **MutY** is a bifunctional adenine DNA glycosylase that removes adenines misincorporated across from 8-oxoG as well as from AG and AC mismatches. 8-OxoG itself, along with other derivatives of A and G, are removed by another bifunctional enzyme, **FAPY glycosylase** (*fpg*).

Adaptive Response to Methylating and Ethylating Agents

The major mutagenic lesion in cells exposed to alkylating agents such as MNNG is O⁶-methylguanine. Such lesions give rise to mutations by mispairing during replication. However, two **O⁶-alkylguanine-DNA alkyltransferases**, Ada (O⁶-MGT I) and Ogt (O⁶-MGT II), specifically remove the methyl group from the O⁶ position of guanine. The adaptive response occurs when *E. coli* is exposed to low concentrations of methylating and ethylating agents (e.g., nitrosoguanidine). Following this exposure, the cells become resistant to the mutagenic and lethal effects of higher doses of these agents. The *ada*⁺ gene product, one of the methyl transferases already discussed, is also an AraC-like regulatory protein that transcriptionally controls this adaptive response.

Ada is able to transfer methyl groups from DNA to two of its Cys residues: Cys-69 and Cys-321 (Fig. 3-31). Methyl and/or ethyl groups from either the O⁶ position of guanine or the O⁴ position of thymine are transferred to Ada cysteine residue 321, a reaction that irreversibly inactivates the methyltransferase activity, making Ada an example of a suicide enzyme. However, Ada has a second methyltransferase activity that removes methyl groups from DNA methylphosphotriesters and transfers them to residue Cys-69. Methylation at Cys-69 results in a conformational change that converts Ada into a strong transcriptional activator. The residue Cys-69 is important in binding Zn²⁺. Once Cys-69 is methylated, the Zn²⁺ is released. This form of Ada stimulates transcription of the other adaptive response genes including *alkA* (3-methyladenine-DNA glycosylase II), *alkB* (unknown function), *aid* (similarity to acyl coenzyme A dehydrogenases, function unknown), as well as *ada* itself.

An interesting regulatory aspect of this system is that Ada regulates *ada* and *alkA* differently! Ada-Me binds to a sequence called an Ada box located upstream of the *ada* -35 recognition site as well as to a sequence that overlaps the *alkA* -35 region. Ada is classified as a class I transcription factor because it requires interaction with the carboxyl terminal domain of the RNA polymerase α subunit to transcriptionally activate *ada*. Truncated forms of Ada missing 10 to 20% of the C terminus constitutively activate *ada* but require methylation to induce *alkA*. In addition, the amino terminal half of Ada will not activate *ada* expression but is still capable of inducing *alkA* in the presence of methylating agents, indicating that elements in the carboxyl terminal half of Ada are required for activation of *ada* but not *alkA* transcription.

The mechanism by which the adaptive response shuts off appears to reflect a repressor effect of unmethylated Ada at the *ada*, but not the *alkA*, promoters. Once all the repairable methylphosphotriesters have been repaired, the unmethylated Ada that accumulates shuts down *ada* expression (Fig. 3-31).

Postreplication Daughter Strand Gap Repair

Although replication after the introduction of DNA damage often results in a heritable mutation, some DNA damage can be repaired after replication. Figure 3-32 illustrates that repair of DNA damage can be accomplished following replication by two basic routes: an error-proof recombinational route or an error-prone DNA synthesis route. If a pyrimidine dimer, for example, is not repaired prior to replication, a gap occurs opposite the dimer where normal DNA Pol III cannot synthesize DNA because it does not recognize the dimer as a template. The gap can be repaired by recombining the appropriate region from the intact sister chain (Fig. 3-32). This pathway is error-proof but requires the *recA*⁺, *recBC*⁺, *polA*⁺, and *uvr* systems. Following recombination,

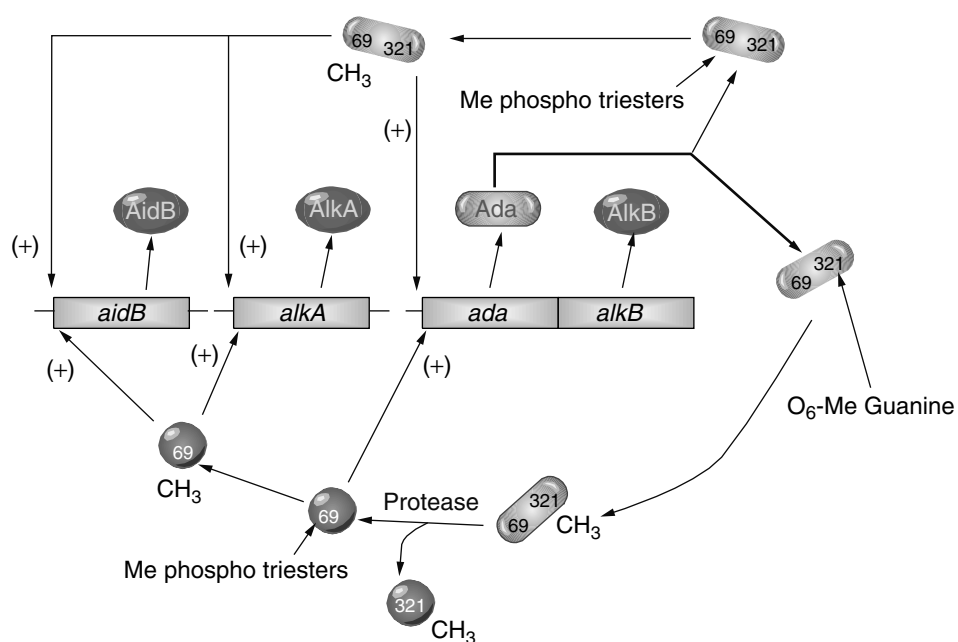


Fig. 3-31. Outline of the adaptive response.

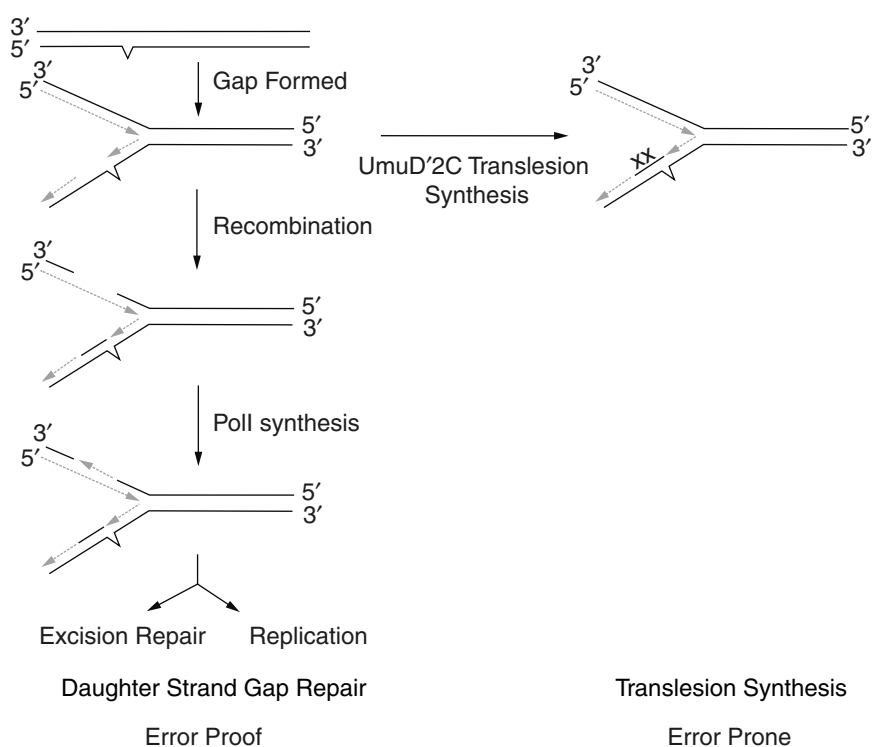


Fig. 3-32. Postreplication repair mechanisms.

the dimer can either be removed by excision repair or can undergo another round of replication. If the mutation still manages to elude error-proof repair, an alternative polymerase produced as a result of SOS induction (see below) can be used to synthesize DNA across the dimer. The bases inserted opposite the dimer are randomly chosen—that is, there is a good chance one or another of the inserted bases will represent a mutation.

SOS-Inducible Repair

The “Save-Our-Ship” Damage Response. Conditions that either damage DNA or interfere with DNA replication result in the increased expression of genes that are members of the SOS regulatory network. This repair system profoundly affects the mutagenic consequences (i.e., whether a heritable mutation is produced) of a number of DNA-damaging treatments. The SOS system is under the coordinate control of *recA* and *lexA*. Aspects of this regulation are outlined in Figure 3-33. In an uninduced cell, the *lexA*⁺ gene product acts as a repressor for a number of unlinked genes including *recA* and *lexA*. Several of these genes, such as *recA* and *lexA*, are significantly expressed even in the repressed state relative to the other genes under LexA control.

The RecA protein, along with its synaptase activity, possesses a coprotease activity. Cells sense they have experienced DNA damage when RecA forms nucleoprotein filaments with ssDNA produced as a consequence of damage. The coprotease activity of RecA is activated, and, following an interaction with LexA, triggers the cleavage of the LexA repressor (as well as a few other proteins such as λ phage cI repressor; see Chapter 6) at an ala-gly peptide bond, yielding two LexA fragments. RecA actually activates an autocatalytic cleavage of LexA and cI. In any case, as the quantitative levels of LexA decrease in the cell, the expression of the various SOS genes increases. Genes with operators that bind LexA weakly are the first to become induced. During recovery of the cell from the inducing treatment, RecA molecules return to their inactive state, helped by interaction with DNA damage–inducible DinI. A quiescent RecA allows LexA repressor to accumulate and restore repression of the SOS system.

Halting Cell Division. One of the physiological responses that occurs following UV irradiation is filamentation due to a cessation of cell division. The cell continues to grow but fails to divide. A halt in cell division is desirable for DNA-damaged cells in order to allow sufficient time for DNA repair systems to fix damage before replication makes the changes permanent. In addition, replication and cell division of heavily UV-damaged cells is often lethal. Consequently, the SOS response has been engineered to stop cell division by regulating expression of the cell division inhibitor gene *sulA*. The LexA protein normally represses the *sulA* gene, but, following mutagenesis, when LexA is cleaved by RecA, SulA protein is synthesized and binds to the *ftsZ* gene product, inhibiting its activity. The *ftsZ* gene product plays a critical role in defining the cell division plane. However, when SulA protein is produced, cell division does not occur. The result is long, filamentous cells. As the cells recover from irradiation, RecA stops degrading LexA, LexA represses *sulA* expression, and the *lon* gene product, a protease whose general function is to degrade abnormal cellular proteins (see Chapter 2), degrades SulA protein, allowing cell division to resume.

Translesion Synthesis. The major genetic response following SOS induction is the production of mutations. Two systems have been identified that are responsible

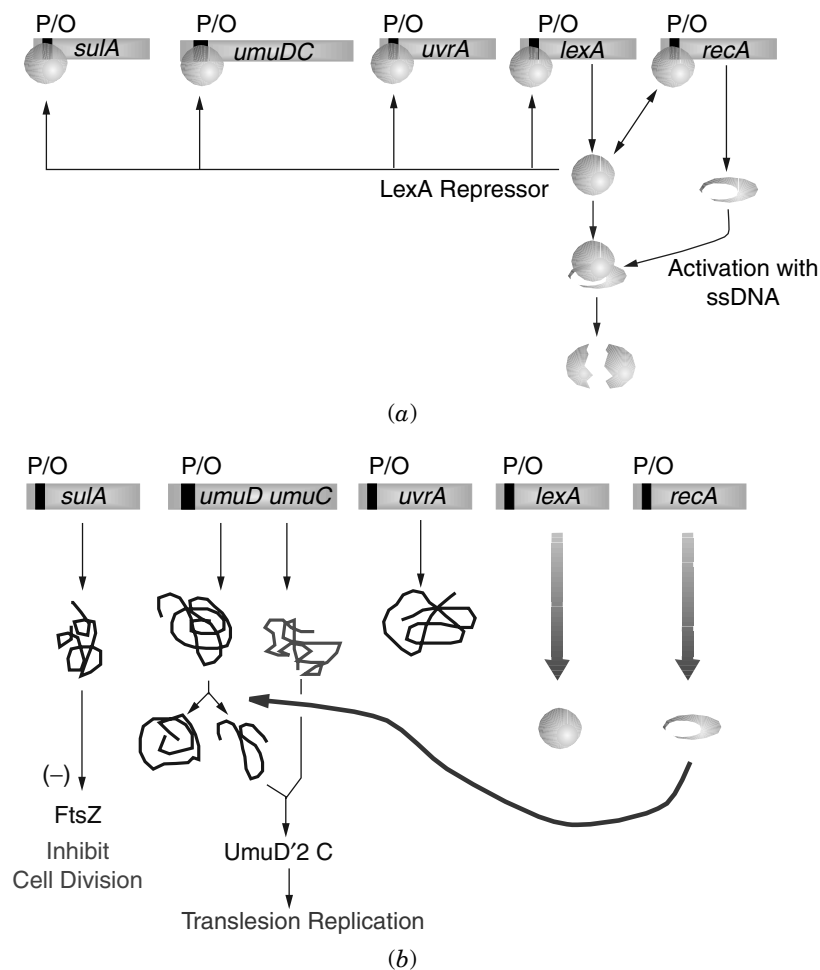


Fig. 3-33. Model of the SOS regulatory system. (a) The LexA repressor binds to the operator regions of 20 SOS genes (4 examples are given including *recA*), inhibiting their expression. The coprotease activity of RecA is activated by DNA damage. Activated RecA will trigger autocleavage of LexA. (b) The removal of LexA allows the induction of the 20 SOS genes. When DNA damage is repaired, the RecA coprotease is deactivated and LexA can reestablish control. As the cell is recovering from irradiation, the *lon* gene product, a protease whose general function is to degrade abnormal cellular proteins (see Chapter 2), will degrade SulA protein, allowing cell division to resume.

for mutability. The major system requires two genes, *umuD* and *umuC*, which map at 25 minutes and produce 16,000 and 45,000 dalton proteins, respectively. The genes are organized in an operon controlled by the LexA protein with *umuD* located upstream of *umuC*. The UmuD and UmuC proteins are uniquely required for SOS processing and mutagenesis, since mutations in either locus result in a nonmutable phenotype. While other SOS functions are induced in a *umuC* mutant, the mutation rate does not increase. However, elevated levels of UmuCD alone are not sufficient to promote mutagenesis. UmuD must be posttranslationally modified to an active form (UmuD').

This activation occurs by an autocatalytic cleavage triggered once again by activated RecA and therefore occurs only in the face of DNA damage. The active form is actually a dimer of UmuD'.

The Umu mutagenesis model places a UmuD₂C complex at the site of translesion synthesis (Fig. 3-32). UmuC is a member of a new family of DNA polymerases specialized for lesion bypass replication by virtue of a relaxed proofreading capability. UmuC is now referred to as Pol V. When a replication fork is blocked at a DNA lesion, Pol V is recruited to the site, becomes activated by UmuD', and catalyzes translesion DNA synthesis. Translesion synthesis is the nonspecific insertion of a base opposite a lesion in the template that blocks replication by Pol III. A mutation may result immediately across from the lesion, but additional mutations may occur downstream of the lesion as Pol V continues synthesizing DNA. The *dinB* gene encodes another lesion bypass DNA polymerase (Pol IV) regulated by SOS and LexA. The role of Pol IV in *E. coli* mutagenesis is not currently clear. Other genes induced by SOS repair are involved in long-patch excision repair, BER repair, site-specific recombination, and RecF-dependent recombination.

An additional role for the UmuDC proteins in DNA damage tolerance besides catalyzing translesion synthesis has been proposed. Uncleaved UmuD protein and UmuC appear to provide resistance to UV killing by regulating growth after cells have experienced DNA damage in the stationary phase. In starved cells treated with UV, the uncleaved UmuD protein and UmuC delay recovery of DNA replication and cell growth, thereby allowing additional time for accurate repair synthesis (error-proof) to remove or process damage before replication is attempted. RecA-mediated cleavage of UmuD to UmuD' then acts as a molecular switch, permitting UmuD₂C to carry out translesion synthesis at unrepaired lesions that continue to block replication. The strategy affords the cell ample opportunity to repair damage without resorting to mutagenesis.

To Live or Let Die. It should be stressed that many of the mutations that occur following mutagenic treatments are the result of the error-prone *recA*⁺-dependent SOS repair systems. Thus, it may seem contradictory that a *recA*⁺ *uvrA*⁻ strain with its high mutation rate survives better than a *recA*⁻ mutant that does not produce mutations. This apparent paradox can be explained by the adage "it is better to live with mutations than to die with dignity." In *recA*⁻ *uvrA*⁻ cells, the lack of error-prone repair systems leads to a decreased viability following DNA damage (mutagenic treatments) because of the inability to replicate past gaps in the chromosome—a failure that destroys chromosome integrity. The mutation rate is lower in *recA*⁻ *uvrA*⁻ mutants that survive DNA damage because these cells lack the error-prone *recA*⁺-dependent SOS system normally required to synthesize past gaps in DNA. Without the SOS system, incorrect bases are not introduced.

Weigle Reactivation and Weigle Mutagenesis. The experiments that first and most clearly indicated the presence of an inducible repair system involved the infection of UV-irradiated cells with UV-irradiated bacteriophage. Preirradiation of *E. coli* with a low dose of UV was found to greatly increase the survival and mutation rate of UV-irradiated λ phage. This indicated a repair system that also produced mutations was induced in *E. coli* following UV irradiation. These findings by Weigle, which are now known to be part of the SOS phenomenon, have been termed Weigle reactivation and Weigle mutagenesis.

Replication Restart

In bacterial cells, replication forks often encounter DNA damage that can inactivate, not just stall, the fork. Replication forks are routinely inactivated under normal growth conditions as well as following environmentally induced DNA damage. In fact, most, if not all, forks originating from *oriC* encounter DNA damage under normal growth conditions. Many of these encounters cause an enzymatic train wreck, leading to replication fork demise. Replication restart involves recombination functions and a primosome distinct from that used at *oriC*.

DNA polymerase II appears to play a pivotal role in rapid replication restart, a process whereby reinitiation of DNA synthesis on UV-damaged DNA allows lesion bypass to occur in an error-free repair pathway. The gene encoding DNA polymerase II (*polB*) is induced as part of the SOS system. One model for how error-free lesion bypass can occur holds that a RecA filament formed downstream of a blocked Pol III holoenzyme triggers a recombination process that allows the blocked strand to associate with the undamaged complementary strand (Fig. 3-34). Pol II then bypasses the lesion by copying the undamaged daughter strand and switches back to the original strand. The PriABC proteins then work to assemble a new primosome that recruits Pol III

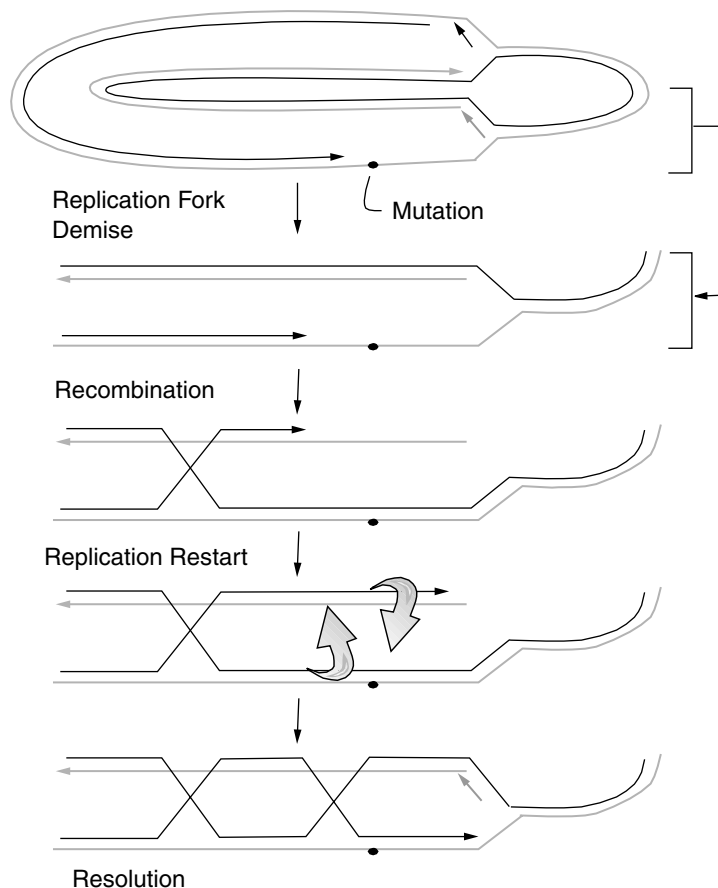


Fig. 3-34. Replication demise and restart.

to take over replication, completing replication restart. Often the result of this process is a concatemer of linked chromosomes requiring resolution by Xer enzymes (see Chapter 2). The repair of stalled replication forks is extremely important for survival and illustrates the intimate relationship between recombination and DNA synthesis. In fact, the most frequent use of bacterial recombination systems is in the rescue of stalled replication forks, not gene exchange.

Adaptive Mutations

In the early 1900s, scientists began to wonder whether mutations occurred spontaneously or were induced or directed when bacteria were exposed to a specific hostile condition. For example, rare phage-resistant variants (mutants) of *E. coli* arise when cells are plated with excess virulent bacteriophage. The question was whether the mutations were caused by the phage or whether the phage simply revealed spontaneous preexisting rare mutants. A classic series of papers published during the 1940s by Luria and Delbruck, as well as Joshua and Esther Lederberg, demonstrated that the phage-resistant mutants were present before exposure to virus. The most elegant proof involved replica-plating *E. coli* to identify which colonies were resistant to the phage. The sibling colony, present on the original master plate, had never been exposed to phage yet also proved to be resistant. These experiments seemed to settle the issue: mutations occurred spontaneously.

However, in 1988, the issue was reopened in a provocative article by John Cairns. Cairns and colleagues argued the flaw in the earlier work was that a lethal selection was used. Consequently, only preexisting mutants could have been found. The early work clearly demonstrated the occurrence of mutations in the absence of selection but did not eliminate the possibility that some mutations might be caused by selection. To support his argument, Cairns examined the ability of a lactose-negative (Lac^-) *E. coli* to revert through mutation to Lac^+ . Lac^+ -revertant colonies will form when Lac^- cells are plated on minimal lactose medium. Each day after plating, the number of Lac^+ colonies that appeared increased. The onset of these mutants could be delayed if they were first plated on medium without lactose and provided with lactose several days later! It seemed many of these late-appearing Lac^+ revertants were being caused (directed) by the presence of lactose!

Additional work from other researchers confirmed this phenomenon with genes other than *lac*. Several models have emerged that seek to explain these results. A popular model proposes that at any given instant during prolonged starvation some fraction of cells in a population might enter into a “hypermutable” state, while the remaining cells remain more-or-less immutable. While in a hypermutable state, mutations can occur randomly in any gene. A cell would exit the hypermutable state and form a colony if any one of those mutations solved the problem—for example, allowed it to grow on lactose. However, if none of the mutations solved the problem, then the cell would die from the accumulation of lethal mutations.

In the Cairns experiments, cells hypermutable before the addition of lactose might still generate lac^+ mutations, but because there is no lactose present they will remain stressed and hypermutable, generating secondary lethal mutations. Thus, they will not form colonies once lactose is added. Only cells that have not yet entered the hypermutable state will still be viable when lactose is introduced and thus have potential for rescuing themselves. This model offers a mechanism with an underlying random

basis that does not invoke true directed mutations. In part, this is why these types of mutations are now called adaptive rather than directed.

In some circumstances, we should realize that the antimutagenic (error-proof) repair mechanisms are not advantageous. Environmental stress responses are often characterized by the potential for genetic diversification in cell populations by mutation. The hypermutable state would be an example. An alternative model is that growth of cells on carbohydrates like glucose limits expression of error-prone repair pathways via catabolite repression. Thus, growth on carbohydrates that do not repress the error-prone systems will increase mutation rate and the chance that the defect will be corrected. Whatever the true mechanism, we may no longer be able to view mutations and selections as entirely separate processes.

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CHAPTER 4

MICROBIAL PHYSIOLOGY IN THE GENOMIC ERA: A REVOLUTIONARY TALE

The field of microbial physiology is undergoing a dramatic revolution. This reformation is the direct result of three technological achievements: the personal computer, the Internet, and rapid DNA sequencing techniques. In the past, examining the physiology of a microorganism was a long, painstaking process. Even now, 60 years after initiating the analysis of the common intestinal microorganism *Escherichia coli*, we are still far from having a completely, integrated picture of its biochemistry and genetics.

Today, however, many organisms that have proven almost intractable to scientific inquiry have had much of their genetics and physiology laid bare. Examples include *Rickettsia prowazekii* and *Mycobacterium leprae*, neither of which can grow outside of a living host. We now know with some certainty what biochemical pathways they have, how they make energy, what possible virulence proteins are in their pathogenic arsenal, and how they relate evolutionarily to other bacteria! All this has come about by modern sequencing techniques that have allowed entire genomes to be decoded in relatively short periods of time and because powerful sequence analysis software has been created that identifies open reading frames [an open reading frame (ORF) is a DNA sequence predicted to encode a protein] and promoter sequences, and compares new sequences with those already known. Comparing the amino acid sequence of an ORF of unknown function with all other known protein sequences (called a homology search) identifies those proteins with sequences and motifs similar to what is present in the ORF. This analysis provides considerable insight as to the possible function of the predicted ORF in the cell without ever having conducted a single, true biochemical experiment! The personal computer and Internet have fueled this renaissance of microbial physiology by allowing unfettered sharing of this information among scientists around the world.

This chapter briefly describes the basic tools molecular biologists use to examine DNA sequences, gene expression, DNA–protein interactions, and protein–protein interactions. Initially, however, we discuss how the sequence of a complete genome is obtained and the means by which that knowledge can be used to study microbial physiology and evolution.

GENOMIC AND PROTEOMIC TOOLS

Cloning a Genome

Every quest to sequence a new genome begins by deconstructing the bacterial chromosome into small (<2 kb) overlapping fragments. This is accomplished through techniques first described in Chapter 3. First, a plasmid or phage vector is chosen and cut with a restriction enzyme. Chromosomal DNA from the subject organism is extracted, digested with the same restriction enzyme, and cloned into the cut vector. Following ligation with DNA ligase, the clones are transformed into *E. coli* (a restrictionless strain). This is known as “shotgun” cloning as opposed to a directed cloning strategy that requires the careful identification of individual DNA fragments encoding a specific gene of interest. The result of shotgun cloning is a random clone library that can be quickly sequenced (some microbial genomes only take days to completely sequence using automated methods).

DNA Sequencing

Once a random clone library is constructed, DNA sequencing efforts can begin. As illustrated in Figure 4-1, a single strand from each clone is used as a template to synthesize a series of fragments, each of which is only one base longer than the previous fragment. The trick is to label the last base synthesized on each fragment (the 3' end) with a fluorescent tag and to do it in such a way that a different colored tag is used for each of the four nucleotides.

The first step is to use a short oligonucleotide primer (about 20–25 bases) that binds to a specific sequence in the vector molecule. Remember that DNA-dependent DNA

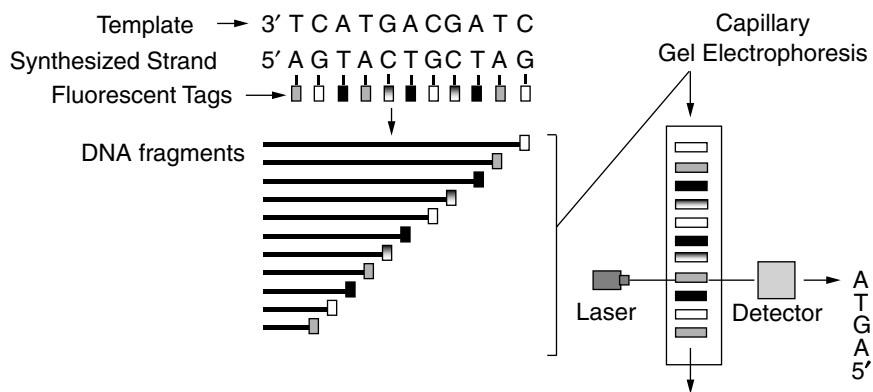


Fig. 4-1. DNA sequencing. A DNA template is sequenced using a DNA oligonucleotide as a primer, DNA polymerase, deoxynucleotides, and a low percentage of dideoxynucleotides. Each dideoxynucleotide is tagged with a different fluorescent tag (differently shaded boxes in figure). Each newly synthesized DNA fragment has only one dideoxynucleotide, and that molecule is incorporated at the 3'-terminal base. As a result, each fragment is labeled with a fluorescent dye that signals the identity of the last base. To read the sequence, the mixture of fragments undergoes electrophoresis through a capillary tube with polyacrylamide that separates the fragments by size. A laser detector positioned near the bottom of the gel reads the fluorescent dye tag of each fragment as it passes.

polymerases only synthesize DNA by adding a new nucleotide to a preexisting 3' hydroxyl group of a growing strand. The primer molecule allows DNA sequencing to begin at a predetermined, fixed point in each clone. Generating different-sized fragments while specifically labeling the last nucleotide incorporated into each fragment is accomplished by including small amounts of fluorescently tagged, dideoxy versions of each nucleotide to the DNA synthesis reaction mix, which also contains normal deoxynucleotides. The key to the method is that dideoxy nucleotides lack both the 2' and 3' hydroxyl groups. So, if DNA polymerase chooses to add a dideoxy nucleotide to an elongating chain, the chain cannot extend any further because there is no 3' hydroxyl primer. Because each dideoxy base is tagged differently, the last base in that fragment, and, thus, the fragment itself, is specifically labeled.

To read the sequence, the mixture of different-sized and labeled DNA fragments are separated by size via polyacrylamide gel electrophoresis, usually conducted in a capillary tube. The overall negative charge on DNA causes all fragments to travel toward the positive pole, and the sieving action of the polyacrylamide matrix allows smaller DNA fragments to travel faster than larger fragments. The result is a parade of fragments progressively moving down the gel, with each fragment being exactly one base longer than the one before it. An optical laser positioned toward the end of the gel will excite the fluorescent tag at the end of each fragment, and the color emitted is read as a specific base.

Because shotgun cloning makes fragments with overlapping ends, computer analysis of the many DNA sequences will identify and assemble the overlapping fragments into **contigs** (a continuous, uninterrupted DNA sequence). With some finishing gap filling, the contigs are assembled into a complete genome ready for annotation. **Annotation** is the process whereby potential genes, known as open reading frames (ORFs), are identified by looking for telltale sequence signatures such as translational start codons in proximity to potential ribosome-binding sites. Then the ORF sequence and the predicted protein product are compared to the existing database to look for similarities with known genes and proteins. Based on what is found, the ORF may be annotated as having a potential function.

The Institute for Genomic Research (TIGR) published the first complete microbial genomic sequence, that of *Haemophilus influenzae*, in 1995. Since then, over 50 genomes have been sequenced and many more are in progress (<http://www.tigr.org/>; click on the TIGR comprehensive microbial resource).

Web Science: Internet Tools for DNA Sequence Analysis

Once you get the DNA sequence for a gene you are studying, what do you do with it? The first stop these days is the World Wide Web, where you can gain amazing insight into the possible function of your gene, even if it has never been sequenced before. How is this level of prognostication possible? It all boils down to hard work and evolution. Before complete genome sequencing efforts became formalized, scientists had already identified the sequences of a large number of genes as well as the functions associated with them. These sequence-function databases have become indispensable in biology because they can be used to assign signature sequences that predict functional motifs. Processes such as gene duplications within an organism, horizontal gene transfers between species, and the mutational divergence of genes are considered to have played major roles in shaping evolution, contributing to the rapid diversification of

enzymatically catalyzed reactions and providing material for the invention of new enzymatic activities. These evolutionary relationships are why genomics is possible and serve as the basis for genomic terminology.

Two genes that appear **homologous** in sequence have likely evolved from a single ancestral sequence. Two terms have been used to classify the types of homology. **Orthologous** genes (or proteins) are homologs in two different organisms (species) that evolved from a common ancestral gene. They may or may not retain ancestral function. So, the gene encoding σ -70 in *E. coli* is orthologous to the σ -70 gene in other species, even species not closely related to *E. coli* (e.g., *Staphylococcus*). **Paralogous** genes (or proteins) are those whose evolution reflects gene duplications. For example, the gene encoding σ -38 is a paralog of the gene encoding σ -70. One gene arose as a duplication of the other, followed by mutational divergence to change—in this case, promoter selectivity.

As noted, the first step once a sequence is in hand is the Web. Major Internet repositories for gene sequences include EMBL (European) and GenBank (USA). These databases can be accessed via a number of Web sites such as through the National Institutes of Health home page (<http://www.ncbi.nlm.nih.gov/>) or the European Bioinformatics Institute home page (<http://www.ebi.ac.uk/Tools/index.html>). Upon sequencing part or all of a gene, the researcher can enter these sites, and, through programs such as BLAST (Basic Local Alignment Search Tool) or FASTA, can compare their query DNA sequence, or deduced protein sequence, to all the known DNA or protein sequences. These programs use statistical calculations to identify significant sequence matches between the query sequence and the sequences in the database. If homology is found over a major portion of a known gene/gene product in another organism and if the function of that homolog is known, the putative function of the query gene can be cautiously predicted. Many times only a portion of the query sequence is homologous to a known gene. This may be due to the presence of a common cofactor-binding site such as ATP. This can be analyzed with another program found on the Internet that screens sequences for known motifs (Table 4-1).

Virtual expeditions to the Internet sites listed in Table 4-1 will reveal a wealth of computer analysis tools including those that predict isoelectric points, molecular weights, three-dimensional structures, and proteolytic peptide patterns of deduced proteins, as well as those that reveal potential DNA regulatory sites, promoters, and so on. A particularly intriguing reference is the KEGG (Kyoto Encyclopedia of Genes and Genomes) site (<http://www.genome.ad.jp/kegg/>), which graphically illustrates the biochemical systems predicted to be present in any microorganism whose genome has been sequenced. To see how this works, go to this site, open KEGG, and under “Pathway Information” click on “Metabolic Pathways.” Click on the metabolic system of interest. The reference pathway for this system will appear. In the “Go to” window, select the organism you want to query (e.g., *Staphylococcus aureus*). The reference pathway will reappear but with colored boxes highlighting the enzymes predicted to be present in that organism based on the genome sequence. Clicking on a box reveals the enzyme name with amino acid and DNA sequences from that organism. A short tour around this site will underscore how extensively genomics and the Internet have transformed the field of microbial physiology.

Genomics has provided some interesting insights into niche selection and species survival. For example, two pathogens, *H. influenzae* and *Mycoplasma pneumoniae*, both infect the respiratory tract, yet their strategies for acquiring solutes are distinct.

TABLE 4-1. Web Site Resources for Genomics and Proteomics

Web Address	Description
http://web.bham.ac.uk/bcm4ght6/genome.html	A database collating research on <i>E. coli</i> genes whose products have been characterized subsequent to in silico predictions from the completed genome sequence
http://cgsc.biology.yale.edu/	<i>E. coli</i> genetic stock center
http://www.ucalgary.ca/%7Ekesander/	Salmonella genetic stock center
http://genome.wustl.edu/gsc/bacterial/salmonella.shtml	Salmonella genome
http://ecocyc.PangeaSystems.com:1555/server.html	Ask specifics about different biochemical pathways in <i>E. coli</i> ; must acquire a password
http://www.genome.ad.jp/kegg/	Ask specifics about biochemical pathways in many sequenced organisms
http://bomi.ou.edu/faculty/tconway/global.html	<i>E. coli</i> functional genomics
http://susi.bio.uni-giessen.de/ecdc/ecdc.html	<i>E. coli</i> database collection (very useful)
http://www2.ebi.ac.uk/fasta33/	Homology searches
http://www.motif.genome.ad.jp/	Motif searches within proteins
http://www.expasy.ch/	Many tools for sequence analysis
http://www.expasy.ch/cgi-bin/map1	Two-dimensional gel profiles
http://www2.ebi.ac.uk/clustalw/	Nucleotide and protein sequence alignment tools
http://www.ncbi.nlm.nih.gov/PubMed/	Pub Med literature searches
http://www.tigr.org/tdb/mdb/mdbcomplete.html	TIGR microbial genome database
http://bmbsgi11.leeds.ac.uk/bmbknd/DNA/genomic.html	Genome databases
http://www.biology.ucsd.edu/~msaier/transport/	Information on transport systems

Analysis of the *H. influenzae* genome indicates that the organism has transporters for 13 different amino acids and small peptides. In contrast, the *M. pneumoniae* genome only encodes three potential transporters with specificity for amino acids. This species, which may have evolved through reductive evolution from a gram-positive ancestor, has discarded most of the transporter genes, and, instead, utilizes a smaller set of porters possessing broad substrate specificity. The two species appear to occupy similar niches but use different approaches to solve the problem of solute acquisition. This example illustrates that an ability to compare, annotate, and design experiments across multiple genomes via computational methods will have a profound influence on microbiological experimentation.

However, be aware that computer analysis of a DNA sequence can be seductive. Many students think that just because a sequence in a database has been annotated as having a potential function, that *is* the function. This is truly fuzzy logic. Until a predicted gene function has been confirmed by genetic or biochemical means, it should only be referred to as a *potential* function. This pitfall reveals that continued sequence annotation by experimental work in any organism is important to increase the accuracy of computer-based theoretical predictions. This is true regardless of whether the genome for the organism under study has been completed.

Gene Replacement

One step toward confirming a predicted function is to mutate the gene in the organism by what used to be called reverse genetics—that is, identify the gene, then make the mutation rather than the other way around. A variety of techniques can be used, but most methods rely on delivering a mutated gene into the organism in such a way that easily selects recombination and replacement of the resident gene. One method, **suicide mutagenesis**, involves deleting an internal portion of the gene you wish to mutate and placing the deleted gene into a **suicide vector** (Fig. 4-2). In this example, a drug cassette encoding chloramphenicol resistance has replaced the deleted section of gene. A suicide vector is an antibiotic-resistant plasmid that only replicates in cells containing a specific replication protein. The only way the plasmid can convey drug resistance on a cell missing the plasmid-specific replication protein is if the plasmid recombines

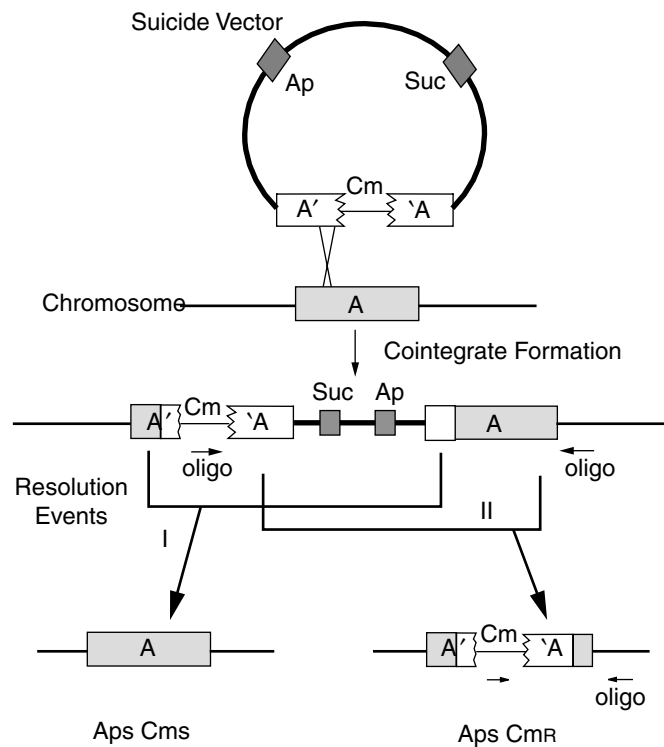


Fig. 4-2. Suicide mutagenesis. A suicide vector plasmid contains the target gene with an internal fragment replaced by a drug resistance marker (chloramphenicol in this case). The plasmid is introduced into a nonpermissive host and Cm resistance is selected. This will only occur if homologous recombination between the mutated gene on the plasmid and the intact copy on the host chromosome cause integration of the plasmid. This plasmid also contains a gene encoding sucrose, an enzyme that confers sensitivity to sucrose on the host cell (*E. coli*). Subsequent selection for sucrose resistance will select for two secondary recombination events (labeled I and II). Both events lead to loss of plasmid-encoded Ap resistance. Resolution event I leaves an intact copy of the gene in the chromosome (loss of Cm resistance) while resolution event II removes the intact gene, leaving the gene A::Cm insertion (The two colons signify the Cm gene is inserted with gene A.)

into the host chromosome. As shown in the figure, recombination occurs at regions of homology determined by gene sequences flanking the deletion. The insertion actually forms a duplication that can be resolved through a second recombinational event. The resolution event can be scored as loss of the plasmid drug marker and retention of the drug marker inserted into the gene (event II in Fig. 4-2). The mutant can now be analyzed for phenotypic characters based on the computer-predicted function.

Gene Arrays

Knowing how the cell coordinates *all* of its biochemical activities will be a major step toward understanding what constitutes life. Unfortunately, a method that would simultaneously examine all the biochemical and structural facets of cell physiology does not yet exist. However, there are powerful techniques available such as gene arrays and two-dimensional separation of cell proteins that provide global macromolecular “snapshots” of the cell. These procedures allow us to view how tweaking one biochemical system impacts the synthesis of all the other systems in the cell.

The first technique, gene array technology, allows us to view gene expression at the whole genome level. The expression of every gene can be monitored simultaneously using a **DNA microchip**. The basic idea is to attach DNA sequences derived from each gene in an organism to a solid support surface in a way that will allow for rapid hybridization to a fluorescently labeled pool of DNA (or cDNA; see below). DNA sequences from individual genes occupy separate spots arranged as a grid on the chip (Fig. 4-3). The entire chip is only about 1 or 2 square inches. Next, the RNA from a cell culture is rapidly extracted and converted into **complimentary DNA (cDNA)** using reverse transcriptase (RNA-dependent DNA polymerase). By incorporating fluorescently tagged nucleotides into the reaction mix, each cDNA molecule is labeled. Individual cDNA molecules will bind to specific locations on the chip that contain their original encoding gene. Laser-scanning and fluorescent-detection devices read the chip surface, and computer analysis reveals which genes were expressed in the original sample and which were not.

Direct comparison of gene expression patterns produced by one strain grown under two *different* conditions can be made using gene array analysis. This is achieved by labeling the cDNA derived from cells grown under different conditions with different fluorescent markers (Fig. 4-3), which is done during the synthesis of cDNA by incorporating different fluorescently tagged dTTP solutions into the reaction mixes. A single array is then used to probe a mixture of the two differentially labeled cDNA preparations. If a specific gene is expressed under only one condition, the corresponding spot on the array will fluoresce one color (e.g., red) because only the RNA extract from cells grown in that condition will have that species of RNA. If a second gene is expressed only when cells are grown under the other condition, its spot on the array will exhibit the other color (e.g., green). When a composite image of the two scans is made via computer, spots corresponding to genes expressed equally under both conditions will exhibit a third color (yellow or orange), which is a blend of the first two colors. This type of analysis can be used to examine the global effects of growth conditions, stresses, or mutations on gene expression.

Proteomics

The proteome encompasses all the expressed proteins of an organism. In contrast to the genome, which is static in composition, the proteome can change moment

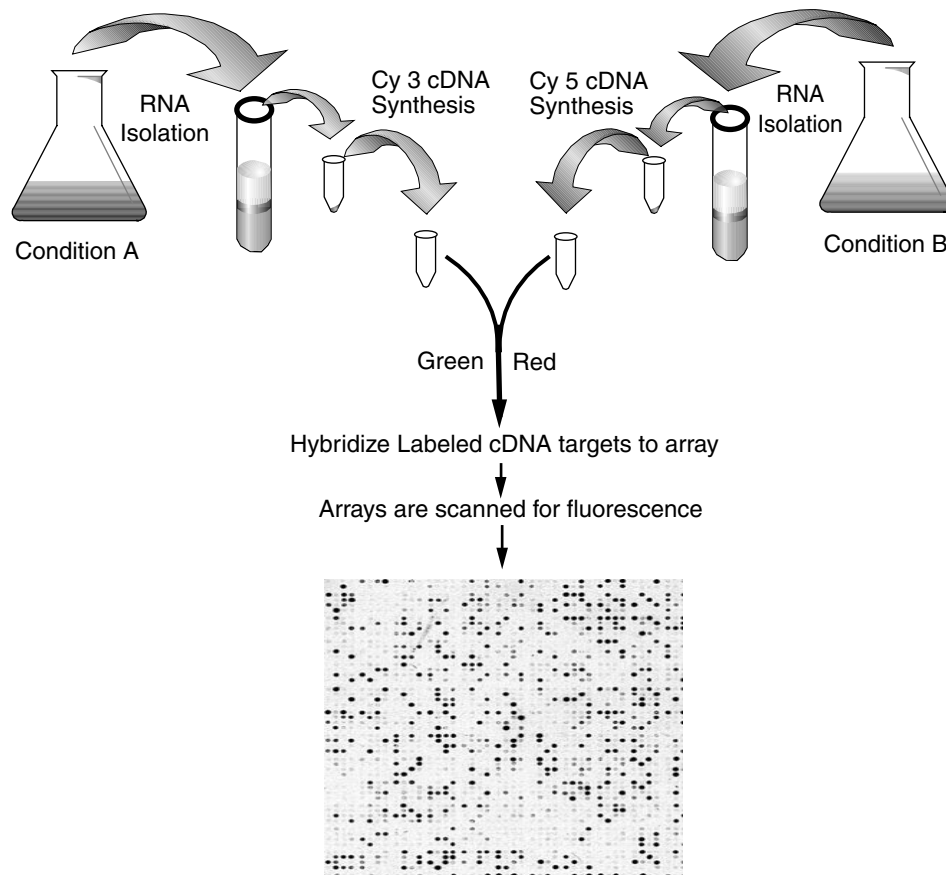


Fig. 4-3. Gene array technology. Dark spots on the array indicate cDNA molecules binding to their corresponding oligos.

by moment in response to changes in the cytoplasmic or extracytoplasmic environments. This is most evident when a cell encounters a stress such as acid shock or heat shock. In response to impending danger, there is a major reorganization of the proteome. Many new proteins are expressed in response to the stress while the levels of other proteins diminish. While gene array technology can predict proteome changes due to transcriptional fluctuations, it fails to predict alterations in the proteome resulting from translational or posttranslational forces—that is, a change in growth condition might not change transcription of a given gene, but the translation of its mRNA might be profoundly affected.

The proteome is typically viewed using two-dimensional polyacrylamide gel electrophoresis (Fig. 4-4). A cell extract containing a mixture of 1000 to 2000 different proteins is subjected first to isoelectric focusing (IEF), a process that separates proteins based on their net charges. In IEF, the proteins are applied to an immobilized pH gradient gel and subjected to a current. Each protein contains many amino acid side groups (e.g., carboxyl or amino groups) that can exhibit a charge depending on whether the group is protonated. A side group changes its protonation state (and, thus, charge)

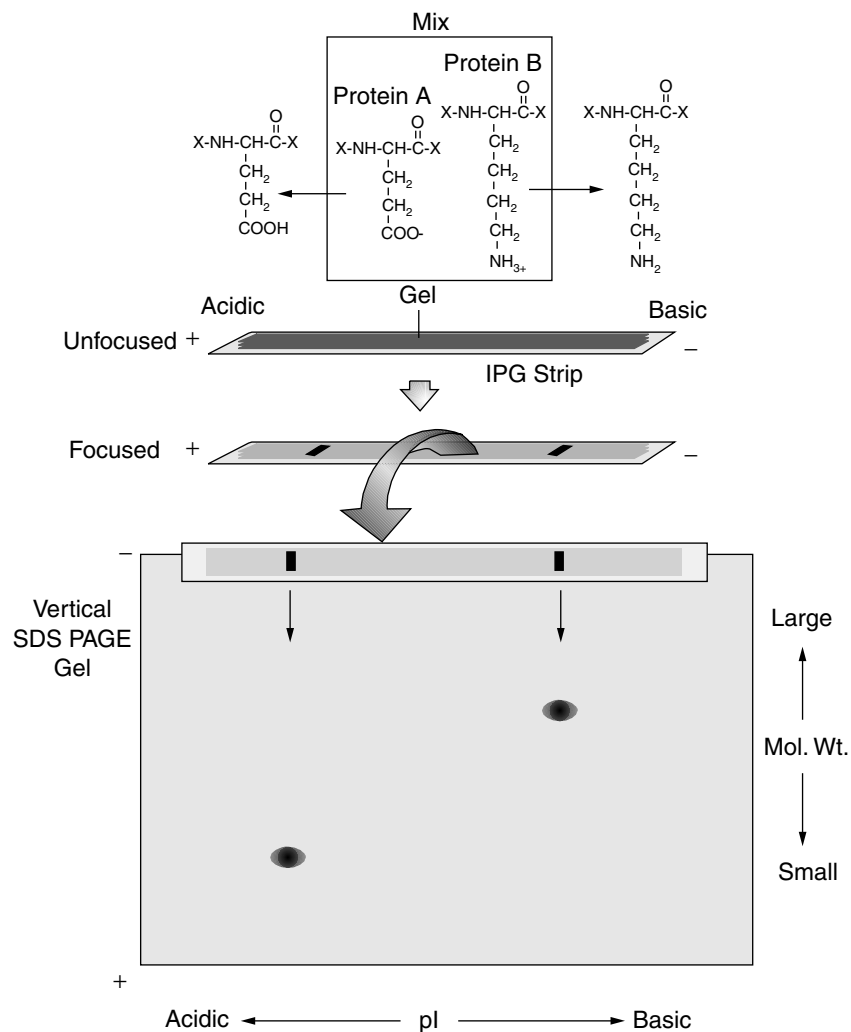


Fig. 4-4. Two-dimensional polyacrylamide gel electrophoresis (2D PAGE). Two proteins with different isoelectric points are depicted in the top box. Horizontal arrows show the direction that each protein travels in the IPG strip.

based on the pKa of the side group(s) and the pH of the surrounding environment. As a rule, side groups will be protonated at pH values below their pKa value.

Side chains of the glutamate and lysine residues in the example have pKa values of 4.3 and 9.8, respectively. So, at pH 7, the carboxyl group of glutamate will be negatively charged (unprotonated) while the amino group of lysine will be positively charged (protonated). The **net charge** of a protein depends on the sum total of the side-group charges. The two proteins in the illustration will begin to move toward opposite poles of the IEF gel because of their initial charges, but, as they move, each protein is exposed to a pH gradient. At some point (called the isoelectric point), the pH encountered at a specific location in the gel results in an equal number of positive and negative charges on the protein, so the net charge is zero. Because there is no

charge, the protein stops moving and focuses at that position. Proteins with different isoelectric points focus to different positions along the IEF gel. IEF separation is the first dimension of two-dimensional gel electrophoresis.

As good as this initial separation process is, proteins of very different molecular weights can and do have identical isoelectric points. As a result, one band on an IEF gel could contain 10 or 15 different proteins. To get a clearer picture of the proteome, a second technique must be used that separates proteins based on their molecular weight. The second dimension begins with the IPG strip from the first dimension being soaked in a sodium dodecyl sulfate (SDS) solution. SDS is a negatively charged molecule that will coat all proteins and impart a negative charge on them. The strip is then placed at the top of a vertical slab gel containing a polyacrylamide matrix and again subjected to an electric current (Fig. 4-4). The proteins begin to move through spaces in the matrix toward the positive pole. The smaller proteins (i.e., lower molecular weight) travel easiest through the spaces while the larger proteins travel more slowly. This sieving effect separates the proteins by molecular weight. Combining the two techniques arranges the proteins in a pattern illustrated in Figure 4-5.

The proteins can be visualized either by radioactively labeling the proteins with ^{35}S -methionine during growth or by staining with conventional coomassie blue dyes

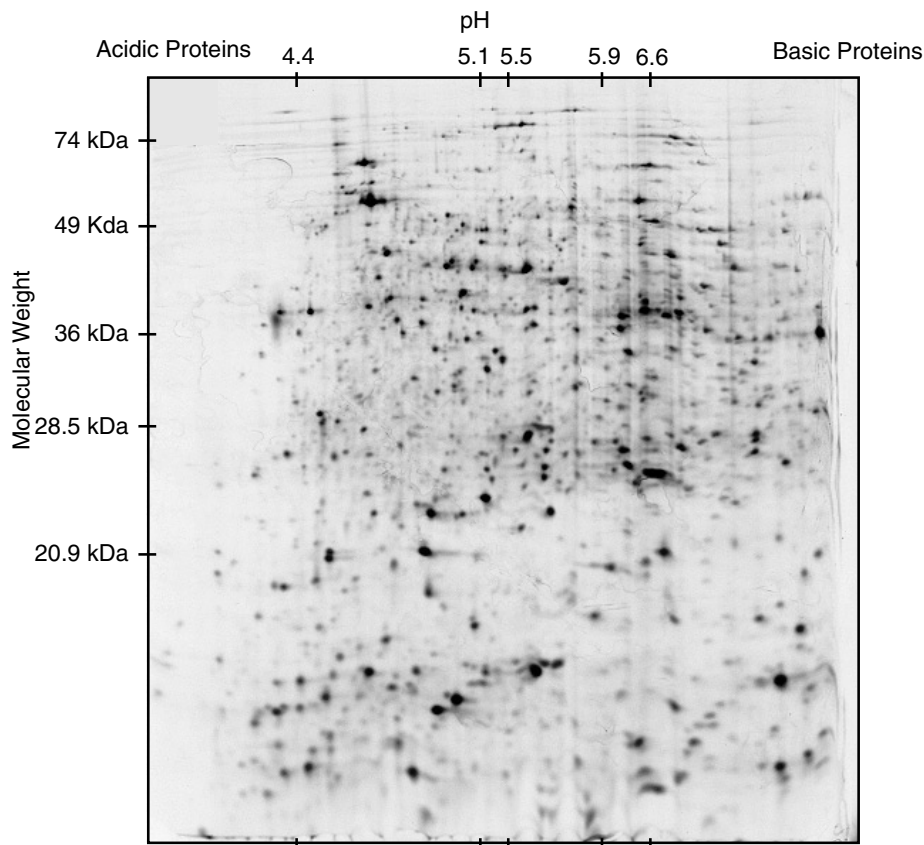


Fig. 4-5. Two-dimensional gel of the *Salmonella typhimurium* proteome.

or the newer fluorescent tagged dyes. Computer analysis can be used to compare the proteomes produced in response to different growth conditions. Once a protein is earmarked for further study, it must be identified. Here, the combined power of genomics and proteomics is most apparent. When the genome of the organism or a closely related organism has already been sequenced, the protein spot of interest can be excised from the gel and the precise mass of the protein can be determined by mass spectrometry. In this technique, a protein is digested with a protease such as trypsin and the fragments are introduced into the mass spectrometer, which precisely measures the mass of each fragment. This can be used to unambiguously assign the identity of the protein when the entire genome of the organism has already been sequenced and the exact molecular weights of all potential peptides have been calculated.

If the genome of the organism has not been sequenced, the spot can be excised and subjected to N-terminal amino acid sequence analysis or cleaved with a protease, such as trypsin, and the fragments separated by HPLC and sequenced. The peptide sequence(s) obtained is then used to query various databases to obtain matches. Alternatively, tandem mass spectrometry, or MS/MS, can be used to derive partial sequence information for a peptide. One of the many tryptic peptides separated by the initial round of MS is selected for further mass analysis. Following further fragmentation, the mass spectrum derived from the overlapping fragments is used to identify positions of specific amino acids in the peptide. The resulting amino acid sequence can be used to query known databases for matches.

TRADITIONAL TOOLS

Mutant Hunts

The classical approach used to expose the details of a biochemical pathway requires the presence of a selectable phenotype, such as an ability to grow on a carbohydrate. Mutants that have lost this phenotype are then sought. Once a mutant is found, the biochemical step in the pathway can be explored and the encoding gene can be mapped, identified, cloned, and sequenced.

It is not unusual for a mutant hunt to screen 10,000 individual mutagenized colonies and only find a handful of mutants in the particular system under study. This is why the selection phenotype is so important. If there is a **positive selection** phenotype [such as resistance to an amino acid analog or a fermentable (red colony) versus nonfermentable (white colony) phenotype on a MacConkey media], thousands of mutants can be screened on a single plate. In the analog example, only mutants that fail to transport or use the amino acid analog will grow and form colonies on a plate containing that analog. In looking for mutants defective in carbohydrate fermentation, mutant colonies that fail to ferment the test sugar will appear white among many nonmutated red colonies.

In the absence of positive selection, the investigator must resort to **negative selection** screens that are more labor intensive. For example, the phenotype of an *E. coli* mutant that cannot make the amino acid alanine will only grow on a defined medium if that medium contains alanine. To screen for *ala* mutants, 10,000 random mutants are inoculated onto 200 rich medium plates in grids of 50 mutants per plate. Each master plate is then replicated to two agar plates containing defined media, one with and one without alanine. Replicating colonies from master to test plates involves making an

imprint of the colonies from the master plate onto sterile velveteen pads and using that pad to “stamp” the new plates. The new plates are inspected after allowing the patches time to grow. An *ala* mutant will be seen as a colony patch that grows on defined media containing alanine but will not grow on the same media lacking alanine.

More sophisticated mutant hunts in which individual cells rather than whole colonies can be screened are now possible with the discovery of fluorescent-activated cell sorting (FACS). To appreciate this technique, imagine making a gene fusion between a gene encoding green fluorescent protein (*gfp*) and a gene that is only expressed under conditions where the cell will *not* grow (an acid pH-inducible gene at pH 4.5) and cannot form colonies. You would like to find mutants that do not express this gene at pH 4.5. To find these mutants, the *gfp* fusion strain can be mutated, a pool of mutants grown at normal pH, adjusted to pH 4.5 to allow induction, and a sample taken and passed through a FACS machine. The cells are passed single file through an orifice where instrumentation reads whether a cell is fluorescing. Simultaneously, a charge is placed on the fluorescing cells. Then charged and uncharged cells can be separated, and the mutant cells that do not express the test gene at pH 4.5 are collected and used for further study. This type of single-cell mutant hunt strategy has dramatically expanded the capability of microbial geneticists.

Transcriptional and Translational Gene Fusions (Reporter Genes)

A very powerful tool for analyzing various aspects of gene expression involves fusing easily assayed reporter genes such as *lacZ* (β -galactosidase) or *gfp* (green fluorescent protein) to host target gene promoters. Two types of fusions are typically used (Fig. 4-6). Operon or transcriptional fusions are used when a promoterless *lacZ* reporter gene is inserted within a target gene in an orientation that places *lacZ* under the control of the target gene's promoter. Thus, whatever factors control expression from the target gene promoter will also control the production of β -galactosidase. Since the *lacZ* message resulting from this fusion still contains its ribosome-binding site, the regulation observed is typically, although not always, due to transcriptional control. The second type of fusion (called a gene or protein fusion) involves inserting a reporter (e.g., *lacZ*) into a target gene that is missing both its promoter and ribosome-binding site. Not only will the messages from the target and reporter gene be fused but the truncated target gene peptide and the reporter gene peptide will be fused when inserted in the proper reading frame. In this case, anything controlling the transcription or translation of the target gene will also control β -galactosidase levels.

Construction of these fusions sometimes involves using a genetically engineered μ phage (Chapter 6) to randomly insert a reporter *lacZ* gene into the *E. coli* or *Salmonella* genome. Alternatively, the fusion can be constructed in vitro and transferred into recipient cells in a manner that will promote allelic exchange of the reporter fusion for the resident gene. Once constructed, these fusions allow for monitoring of how different growth conditions or known regulators influence the transcription or translation of the gene. In addition, these fusions provide convenient phenotypes (e.g., lactose fermentation or fluorescence) that can be exploited in mutant hunts to screen for new regulators.

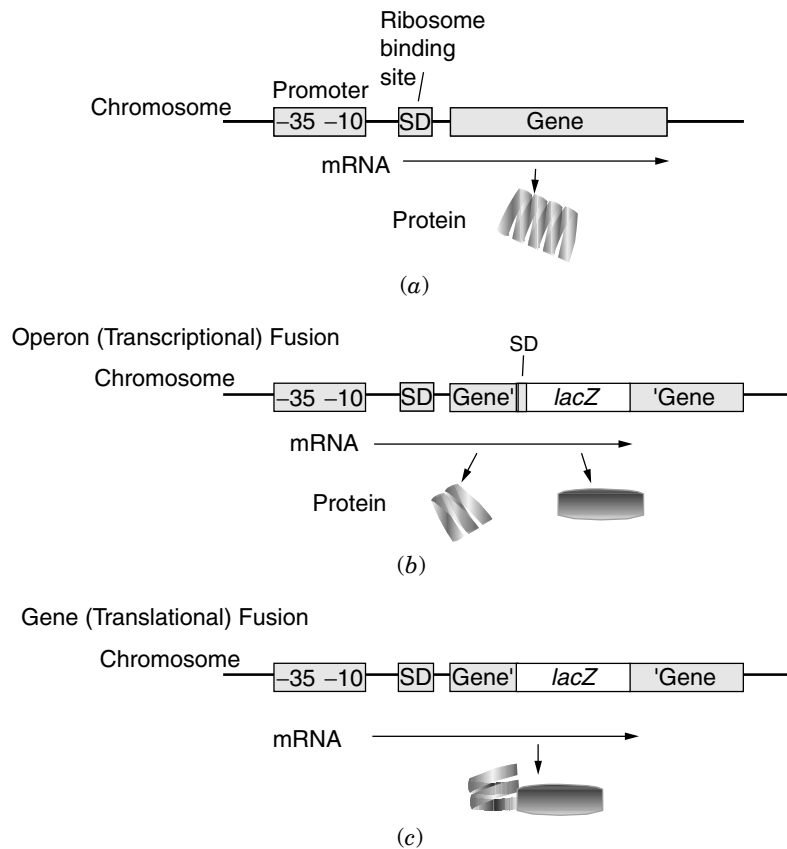


Fig. 4-6. Engineering of operon and gene fusions with *lacZ*.

Polymerase Chain Reaction

In the past, obtaining large concentrations of purified DNA used to be a frustrating rate-limiting step in molecular biology. However, in the late 1980s, the discovery of a relatively simple process called the polymerase chain reaction solved the problem and revolutionized molecular biology. Kary Mullis, a scientist and avid surfer from California, conceived the idea that won him the 1993 Nobel prize in chemistry not through painstaking experimentation, as you might expect, but while cruising California Highway 128 in a Honda Civic.

The basic polymerase chain reaction (PCR) process, outlined in Figure 4-7, can produce a $>10^6$ amplification of target DNA. The figure depicts the use of specific oligonucleotide primers (20–30 bp) that anneal to sequences flanking the target DNA. These primers are used by a thermostable DNA polymerase to replicate the target DNA. For the technique to amplify a specific-size DNA, thermostable DNA polymerases must be used because the reaction mixture must be subjected to repeated cycles of heating to 95°C (to separate DNA strands), cooling to 55°C (to allow primer reannealing), and heating to 72°C (the optimum reaction temperature for the polymerase). The polymerase *Taq*, from the thermophile *Thermus aquaticus*, is commonly used for this purpose.

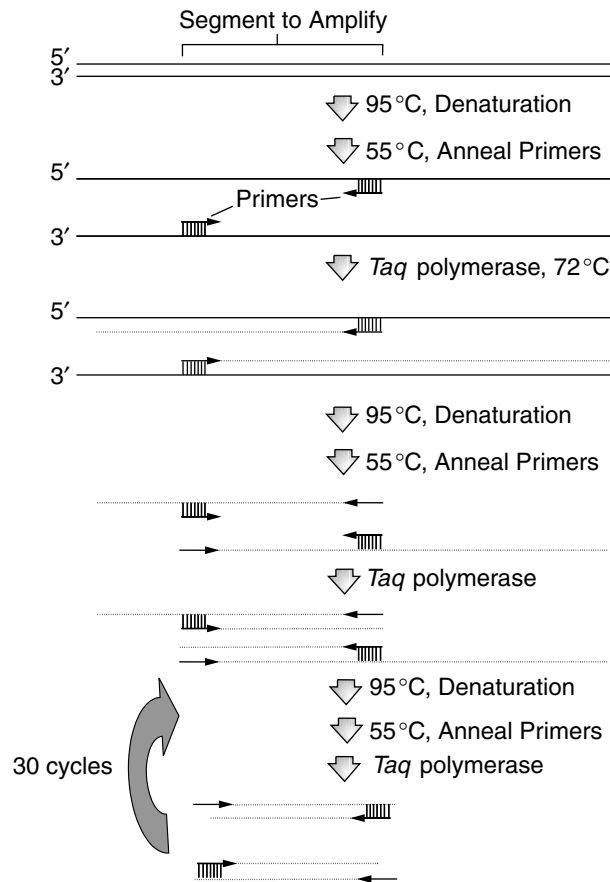


Fig. 4-7. Polymerase chain reaction (PCR).

Since amplification requires 25 to 30 heating and cooling cycles, a machine called a thermocycler is used to reproducibly and quickly deliver these cycles. This basic PCR technique has been modified to serve many different purposes. Primers can be engineered to contain specific restriction sites that simplify subsequent cloning, or if the primers are highly specific for a gene present in only one species of microorganism, PCR can be used to detect the presence of that organism in a complex environment (e.g., the presence of the pathogen *E. coli* O157:H7 in hamburger).

PCR can also be used to engineer specific mutations in a given gene (**site-directed mutagenesis**). For example, computer analysis of a gene reveals a potential functional motif (e.g., Zn^{2+} finger). One step toward determining whether the motif is important to the function of the gene product would be to change one of the key amino acid codons by site-directed mutagenesis and examine the mutant phenotype. The example in Figure 4-8 uses four primers. The two outside primers (#2 and #4) amplify the entire gene. The two inside primers (#1 and #3) overlap the target region and are engineered to contain the desired mutation. There are three PCR reactions required. The first two reactions amplify each end of the gene and include the mutation. To reconstruct the complete gene, the products from the initial reactions are mixed, denatured at 95°C,



sequence for the mutagenic primers (primers 1 and 3). Short arrow points to the target base.

and reannealed to form the hybrid structure shown. Subsequent PCR with the outside primers regenerates the complete gene but with the site-directed mutation.

DNA Mobility Shifts (Gel Shifts and Supershifts)

Once a gene encoding a putative transcriptional regulatory protein is identified, it is imperative to test whether that protein directly binds to the target DNA sequence. One approach is to take purified protein, add it to the putative target DNA fragment, and see if the protein causes a shift in the electrophoretic mobility of that fragment.

An example is illustrated in Figure 4-9. Linear DNA undergoing electrophoresis in an agarose or polyacrylamide gel matrix will travel at a rate based on the size of the DNA molecule. Larger molecules travel slower than smaller molecules. If a protein binds to that DNA molecule, the complex is larger than the DNA alone and will travel more slowly in the gel. This is called a mobility shift or gel shift.

You can determine if a suspected DNA-binding protein is actually part of a mixture of proteins (e.g., a cell extract) that cause a gel shift by performing a **supershift** experiment. A supershift of the target fragment will occur when an antibody directed against the putative DNA-binding protein is incorporated into the binding reaction. The complex of anti-DNA-binding protein, DNA-binding protein, and DNA is even larger than the binding protein–DNA complex, causing an even greater gel shift (supershift). If a supershift does not occur, the suspected protein is not the one causing the original shift.

Finding Transcriptional Starts by Primer Extension

As discussed in Chapter 5, many genes are driven by multiple promoters, each of which produces a different transcript. Documenting whether a gene produces multiple

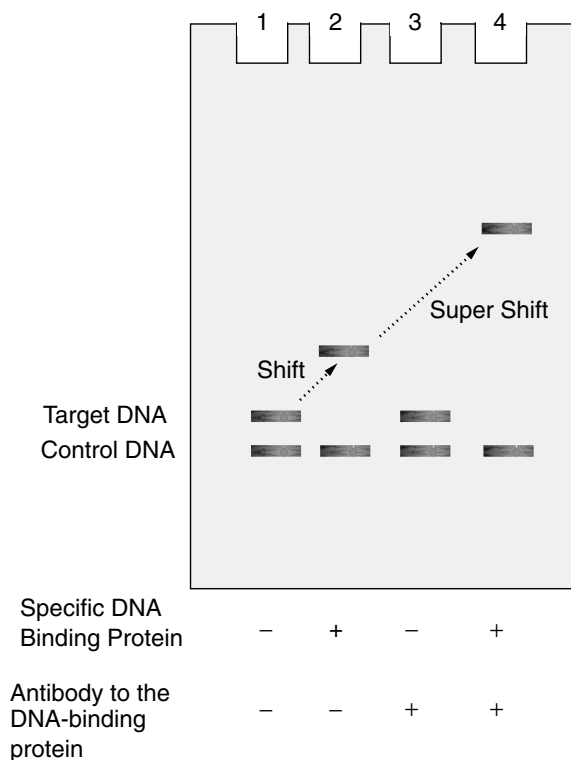


Fig. 4-9. Electrophoretic mobility shift and supershift. Polyacrylamide gel. All lanes contain a target DNA fragment and a control fragment. DNA-binding protein and antibody against the DNA-binding protein are added where indicated *before* running the sample on the gel. The binding of protein to the target fragment (lane 2) makes a larger complex that travels more slowly than the unbound DNA fragment. The addition of DNA-binding protein and antibody makes mobility slower still.

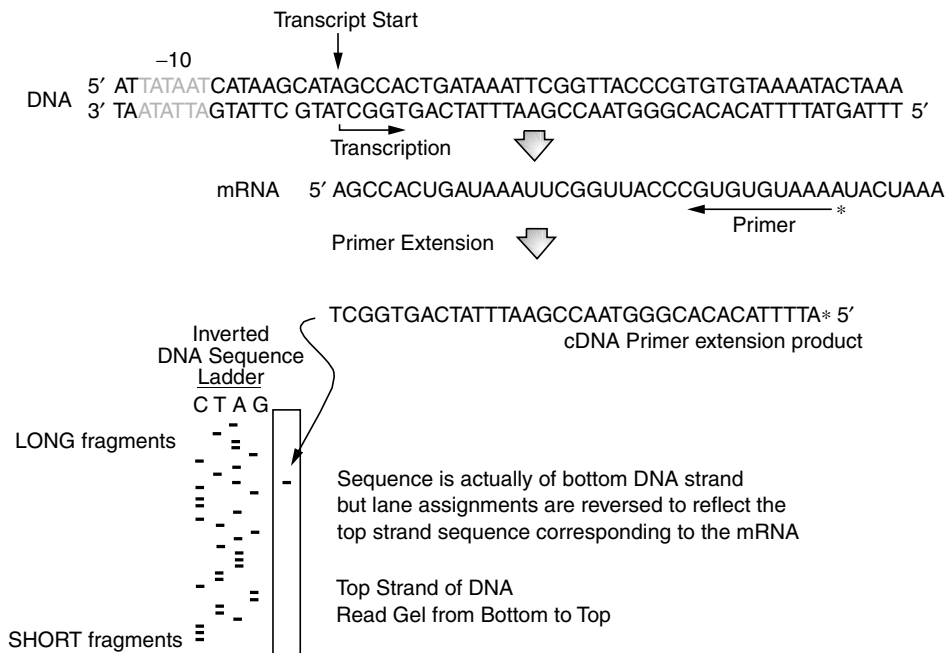


Fig. 4-10. Primer extension. The bottom DNA strand is actually read by RNA polymerase. The leftward arrow below mRNA represents the primer used for primer extension and DNA sequencing. The asterisk indicates the fragment is radiolabeled. Each lane of the DNA sequence gel represented in the lower panel detects a different base as indicated above each lane. Sequence starts after the end of the primer.

transcripts includes defining where each transcript begins. Primer extension is one method that achieves this result (Fig. 4-10). A DNA primer that anneals to the mRNA transcript under investigation is used in a reverse transcription reaction that extends the primer to the 5' end of the message. This produces a cDNA of a precise length that is radiolabeled due to the use of radioactive nucleotides in the reaction mix. The key to this procedure is that the same primer is used in a DNA sequencing reaction with the template DNA. The primer extension fragment is then run alongside a DNA sequencing ladder on a polyacrylamide gel. The size of the primer extension fragment will be identical to one of the “rungs” on the sequencing ladder. That rung represents a base in the sequence, and that base correlates to the transcriptional start site. The RNA from genes that have multiple promoters will produce different-sized primer extension fragments that comigrate with different sequencing fragments.

Detecting DNA, RNA, Protein, and DNA-Binding Proteins by Southern, Northern, Western, and Southwestern Blots

Methods to detect specific DNA, RNA, and protein molecules are critically important tools used to probe gene expression strategies. A **Southern blot**, so named for Ed Southern who developed the technique in 1975, begins with cutting a DNA molecule with one or more restriction endonucleases. The molecules are separated by agarose electrophoresis and then transferred from the agarose gel to a nitrocellulose

membrane overlayed onto the gel. The transfer process involves capillary action in which buffer is drawn first through the agarose gel and then through the facing membrane. DNA traveling with the buffer is deposited on the membrane in exactly the same pattern as it is displayed in the gel. Once DNA fragments are affixed to the membrane, the strands are separated chemically and the membrane is probed with a denatured, tagged DNA fragment under hybridizing conditions that allow annealing between the probe and matching DNA fragment. The probe can be labeled with radioactivity, a fluorescent dye, or biotin. The biotin is detected later by a chemiluminescent enzyme assay. DNA fragment bands that have annealed to the probe are visualized by **autoradiography**—that is, by exposing the membrane to X-ray film and photographically developing the film.

Northern blots are used to analyze the presence, size, and processing of a specific RNA molecule in a cell extract. RNA is extracted from the cell using special precautions to avoid contaminating the preparation with RNases that are ubiquitous in the laboratory environment. The RNA is fractionated by electrophoresis on an agarose-formaldehyde gel and the fragments are transferred onto a nylon or nitrocellulose membrane by capillary action as with Southern gels. Once transferred, the RNA is fixed to the membrane either by baking (nitrocellulose) or UV cross-linking (nylon). The membrane can then be probed with a small radiolabeled DNA probe specific for a given mRNA. After hybridization, the membrane is autoradiographed or subjected to phosphorimaging (a **phosphorimager** records the energy emanated as light or radioactivity from gels or membranes). Visualized bands can be sized by comparing their location with standard RNA-molecular-weight markers loaded in a parallel lane. This technique is useful to determine the presence and turnover of a given message. By estimating size of the transcript, the method can also be used to evaluate whether two adjacent genes are transcribed as an operon.

Western blots are used to detect specific proteins within a complex mixture of heterologous proteins. In Figure 4-11, the protein extract is subjected to SDS PAGE and the protein bands are electrophoretically transferred from the gel to a PVDF membrane. The membrane is probed with a primary antibody directed against the specific protein. Depending on the source of the antibody (mouse, rabbit, other), the membrane is probed with a secondary antibody that specifically binds to the primary antibody (e.g., anti-mouse IgG antibody). This secondary antibody is tagged with horseradish peroxidase, for example. The result is an antibody “sandwich” where the primary antibody binds the target protein and the secondary antibody binds the primary antibody. Adding hydrogen peroxide and luminol sets off a luminescent reaction wherever an antibody sandwich has assembled. The light emitted can then be detected by autoradiography or phosphorimaging. Only protein bands that bind to the primary antibody will be detected. The technique is used to determine if, and at what levels, specific proteins are present in a cell extract. It can also be used to detect the processing of a protein into smaller fragments (Fig. 4-11).

The **Southwestern blot** is one method that can be used to identify proteins that bind specific DNA molecules. Proteins from the cell are separated using standard SDS polyacrylamide gel electrophoresis. The fractionated protein bands are transferred to a nitrocellulose membrane and the membrane is probed with radiolabeled oligonucleotides. Autoradiography will reveal if a specific protein band binds the labeled DNA. The protein can be excised and identified by a variety of techniques.

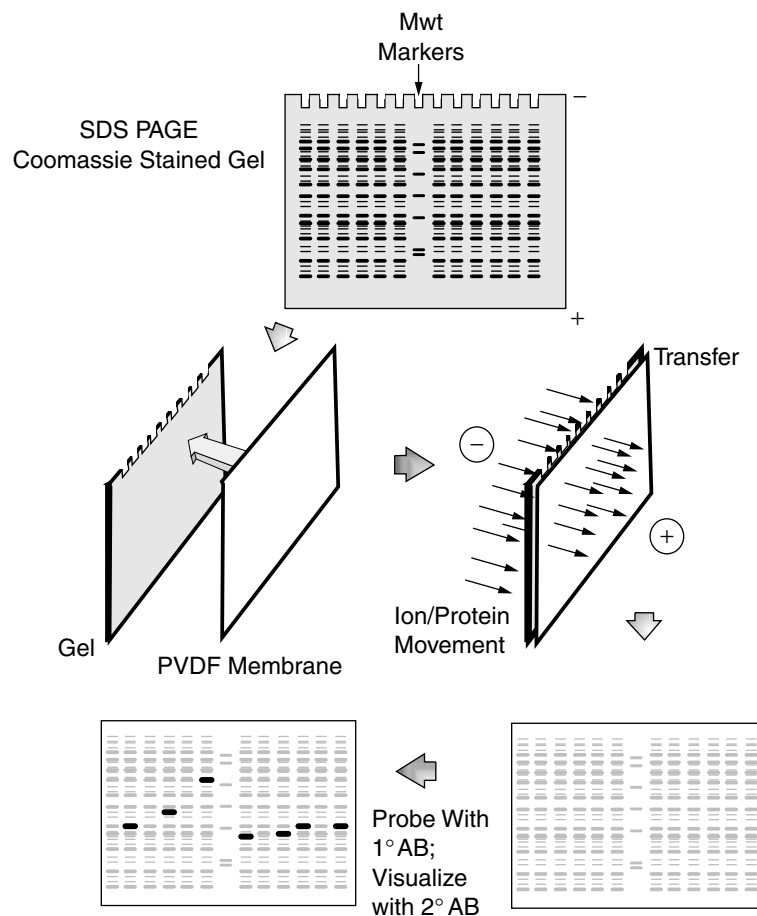


Fig. 4-11. Western blot. Mixtures of proteins from different cultures are loaded onto separate lanes of the SDS polyacrylamide gel (top gel). Standard molecular-weight markers are loaded in the center lane. After electrophoresis, this gel can be stained with Coomassie blue or fluorescent dyes to give the pattern shown. To identify which band in each lane is the protein of interest, the unstained protein bands are electrophoretically transferred to a PVDF membrane as shown. The proteins are deposited on the membrane (the bands are not usually visible but are shown here in gray highlight for convenience). The membrane is then probed with primary antibody directed against the protein of interest. Bands to which these antibodies have bound are then visualized by adding a secondary antibody directed against the first antibody. The secondary antibody is tagged with the enzyme horseradish peroxidase (HRP). After binding of the secondary antibody, hydrogen peroxide and luminol are added to the membrane. HRP, in the presence of H_2O_2 , oxidizes luminol, which enters an excited state that decays and emits light. Thus, areas where the primary antibody bind to the gel (detecting the target protein) will emit light that can be detected with autoradiography (bottom left-hand membrane). Bands corresponding to the target protein are in bold. Note that different extracts exhibit different-sized bands. This could be the result of processing of the protein to smaller sizes under different conditions or crossreacting bands that share antigenic epitopes with the protein of interest.

Two-Hybrid Analysis

Protein–protein interactions are an important part of many regulatory circuits, from anti- σ factors to multisubunit enzyme complexes. It is valuable to probe these interactions in vitro but also in vivo where the proteins actually work. Two-hybrid systems are cleverly designed tools that not only measure in vivo protein–protein interaction but can be used to mine for unknown “prey” proteins that interact with a known “bait” protein.

While there are many types of two-hybrid techniques, the principle can be illustrated with two examples (Fig. 4-12). The now classic method is the **yeast two-hybrid system**. This strategy involves fusing genes encoding two potentially interacting proteins to the separate components of the yeast GAL4 transcriptional activator (Fig. 4-12A). These are the DNA-binding domain fragment and the transcriptional activation domain fragment. Both components of the GAL4 activator must interact to induce expression of a yeast chromosomal *GAL1-lacZ* fusion, although other reporter genes can be used. The *lacZ* gene encodes β -galactosidase, whose enzymatic activity is easily measured. Normally, both domains are part of one GAL4 protein. However, when separated, both domains can still interact if each domain is fused to one of a pair of other interacting proteins.

When the two resulting hybrid proteins interact via the two target protein domains, the GAL4 DNA-binding and activation domains are brought together. Thus assembled, the two-hybrid GAL4 complex will bind and activate transcription of *GAL1-lacZ*. Expression of *GAL1-lacZ* is then visualized on agar plates containing a chromophoric substrate of β -galactosidase called X-gal, or it can be directly assayed in vitro. The power of this technique is that it can also be used to find unknown proteins that interact with a known protein. The known protein is fused to one of the GAL4 domains and used as bait in which cells containing the bait fusion are transformed with a plasmid pool containing randomly cloned prey genes fused to the other GAL4 domain. After plating, colonies that express β -galactosidase are the result of protein–protein interactions occurring between the bait and prey fusion proteins. The gene encoding the prey protein can then be identified by sequencing the DNA insertion.

A novel **bacterial two-hybrid system** utilizes N-terminal (T25) and C-terminal (T18) halves of *Bordetella pertussis* adenylate cyclase (CyaA) produced on separate plasmids. Genes encoding suspected interacting proteins are cloned into the two plasmids, creating gene fusions between each tester gene and the *cyaA* moieties (Fig. 4-12B). Both plasmids are introduced into a *cya* mutant of *E. coli*, which is phenotypically Lac[−] due to the absence of cAMP (see Chapter 5). If the CyaA hybrid proteins interact in vivo, the two CyaA subunits will be brought into close proximity, forming an active adenylate cyclase. As a result, the cells will make cAMP and the lactose operon will be expressed.

Although bacterial protein–protein interactions have been examined using both systems, there is an advantage to using bacterial two-hybrid systems over yeast systems. For example, if a bacterial-motivated protein modification is required for an interaction to occur, the modification may not happen in the yeast system, so the interaction might be missed.

Using the Two-Hybrid System to Identify a New Regulator. *Helicobacter pylori*, a remarkable gram-negative microbe that can inhabit the human stomach and cause gastric ulcers, provides an interesting example in which a two-hybrid

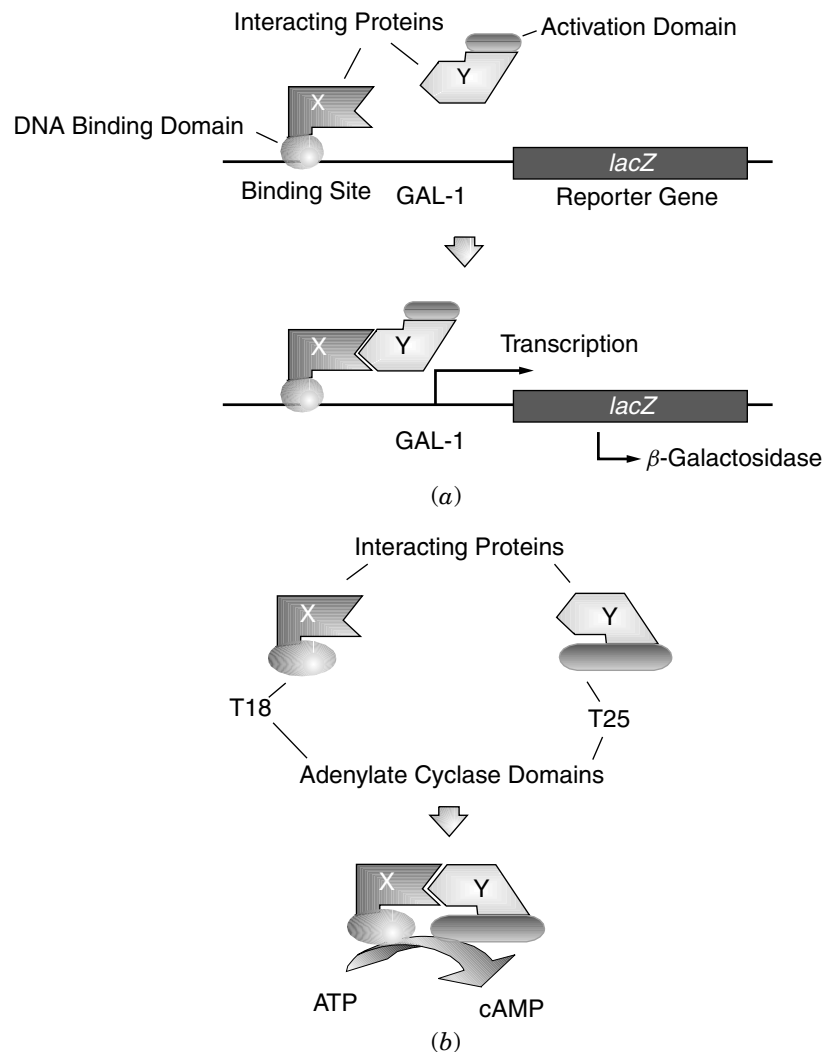


Fig. 4-12. Two-hybrid systems to detect protein–protein interactions. (a) Yeast two-hybrid system. Protein–protein interaction between X and Y fusion proteins bring together the DNA-binding and activation domains of GAL4, which activates transcription of a *GAL1-lacZ* fusion. (b) Bacterial two-hybrid system involving *Bordetella pertussis* adenylate cyclase. Interaction between the X and Y fusion proteins brings together the T18 and T25 moieties of *B. pertussis* adenylate cyclase. The resulting production of cAMP can be measured through expression of a cAMP-dependent gene (e.g., *lacZ*.)

approach was used in conjunction with proteomic and genomic strategies to reveal a new family of regulators (Colland et al., 2001). Computer analysis of the completed helicobacter genome indicated that the organism possesses a σ -28 ortholog responsible for controlling a cascade of flagellar genes. In *E. coli*, the activity of σ -28 is kept in check by a specific anti- σ factor, FlgM (see “Anti- σ ” in Chapter 5). However, scrutiny of the helicobacter genome failed to reveal a similar gene. Convinced that this

organism must contain a FlgM-like molecule, Colland and colleagues used a yeast two-hybrid system to fish out a gene whose product binds to *H. pylori* σ -28 in vivo. DNA sequence analysis of this gene exposed its precise location in the *H. pylori* genome. Once confirmed as an anti- σ factor, BLAST comparison of the deduced amino acid sequence of the *H. pylori* FlgM to known FlgM homologs revealed a carboxy terminal amino acid motif common to other known σ -28 molecules. With this knowledge in hand, the authors used computer analysis to identify potential FlgM orthologs in three other microorganisms. This is a wonderful example of how genomics has transformed the field of microbial physiology.

SUMMARY

There are numerous commonly used molecular tools for studying microbial physiology. This chapter presents only a few key methods chosen to underscore the impact that genomic and proteomic strategies have had on this field. Other techniques such as DNA footprinting (where a DNA-binding protein is shown to protect its target DNA from in vitro nuclease digestion) or reverse transcriptase-mediated PCR (where low levels of an RNA molecule can be visualized using reverse transcriptase to make cDNA and PCR primers to amplify the cDNA to visible levels) have not been discussed. Details of these and many other tools of the trade are readily available in other publications such as *Applied Molecular Genetics* by Roger L. Miesfeld (Wiley-Liss).

As discussions progress in this textbook, you should begin to understand how many of the techniques covered in this chapter are used to answer important biochemical and genetic questions. Perhaps the most striking observation that can be made when comparing the present state of science to that of only 10 or 15 years ago is that many investigations that took decades to complete can now be accomplished in a mere fraction of that time using the current tools of genomics, molecular biology, and biochemistry.

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CHAPTER 5

REGULATION OF PROKARYOTIC GENE EXPRESSION

The cardinal rule of existence for any organism is economy. A cell need not waste energy by simultaneously synthesizing 20 different carbohydrate utilization systems if only one carbohydrate is available. Likewise, it is wasteful to produce all the enzymes required to synthesize an amino acid if that amino acid is already available in the growth medium. Extravagant practices such as these will jeopardize survival of a bacterial species by making it less competitive with the more efficient members of its microbial microcosm. However, there are many instances when economy must be ignored. Microorganisms in nature more often than not find themselves in suboptimal environments (stationary phase) or under chemical (e.g., hydrogen peroxide) or physical (e.g., ultraviolet irradiation) attack. Consequently, in any given ecological niche, the successful microbe not only needs regulatory systems designed to maximize the efficiency of gene expression during times of affluence but must also sense danger and suspend those safeguards in favor of emergency systems that will remove the threat or minimize the ensuing damage. Economy is important, but backup plans ensure survival of the species.

Prokaryotic gene expression is classically viewed as being controlled at two basic levels: DNA transcription and RNA translation. However, it will become apparent that mRNA degradation, modification of protein activity, and protein degradation also play important regulatory roles. This chapter deals with the basics of gene expression, describes some of the well-characterized regulatory systems, and illustrates how these control systems integrate and impact cell physiology. The strategies used by bacteria to avoid or survive environmental stresses are considered in Chapter 18.

TRANSCRIPTIONAL CONTROL

The most obvious place to regulate transcription is at or around the promoter region of a gene. By controlling the ability of RNA polymerase to bind to the promoter, or,

once bound, to transcribe through to the structural gene, the cell can modulate the amount of message being produced and hence the amount of gene product eventually synthesized. The sequences adjacent to the actual coding region (**structural gene**) involved in this control are called regulatory regions. These regions are composed of the promoter, where transcription initiates, and an operator region, where a diffusible regulatory protein binds. Regulatory proteins may either prevent transcription (**negative control**) or increase transcription (**positive control**). The regulatory proteins may also require bound effector molecules such as sugars or amino acids for activity (see “The *lac* Operon” in this chapter). Repressor proteins produce negative control while activator proteins are associated with positive control. Transcription initiation requires three steps: RNA polymerase binding, isomerization of a few nucleotides, and escape of RNA polymerase from the promoter region, allowing elongation of the message. Negative regulators usually block binding while activators interact with RNA polymerase, making one or more steps, often transition from closed to open complexes, more likely to occur.

An **operon** is several distinct genes situated in tandem, all controlled by a common regulatory region (e.g., *lac* operon). The message produced from an operon is **polycistronic** in that the information for all of the structural genes will reside on one mRNA molecule. Regulation of these genes is coordinate, since their transcription depends on a common regulatory region—that is, transcriptions of all components of the operon either increase or decrease together. Often genes that are components in a specific biochemical pathway do not reside in an operon but are scattered around the bacterial chromosome. Nevertheless, they may be controlled coordinately by virtue of the fact that they all respond to a common regulatory protein (e.g., the arginine regulon). Systems involving coordinately regulated, yet scattered, genes are referred to as **regulons**.

Figure 5-1 presents a schematic representation of negative-control versus positive-control regulatory circuits. Whether an operon is under negative or positive control, it can be referred to as **inducible** if the presence of some secondary effector molecule is required to achieve an increased expression of the structural genes. Likewise, an operon is **repressible** if an effector molecule must bind to the regulatory protein before it will inhibit transcription of the structural genes.

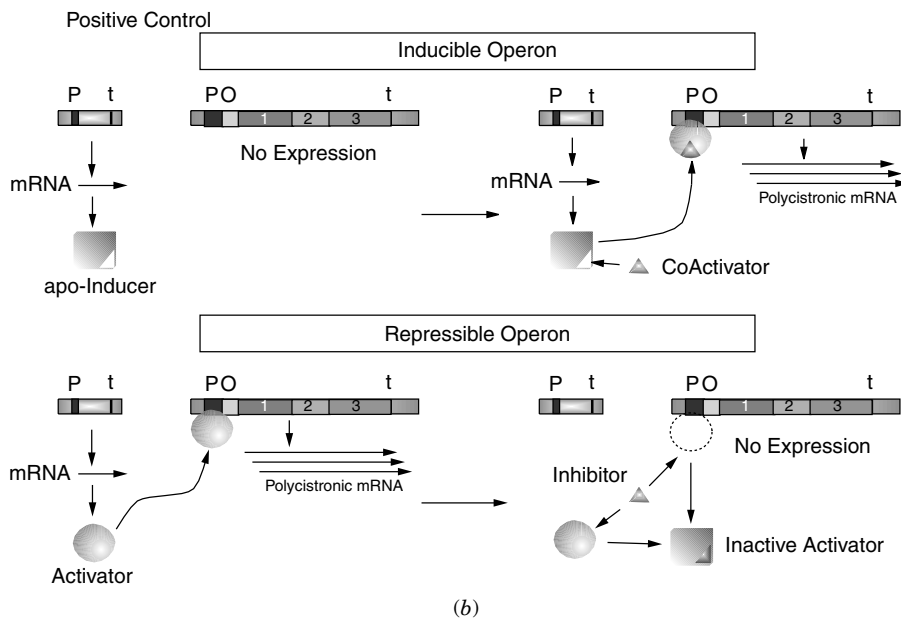
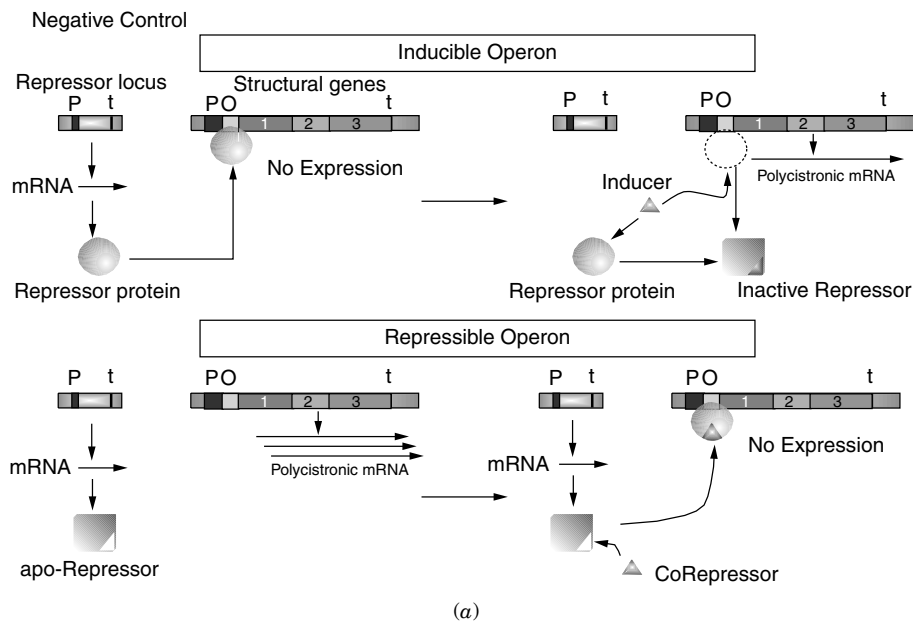
DNA-Binding Proteins

It is evident from Figure 5-1 that regulation at the transcriptional level relies heavily on DNA-binding proteins. Studies conducted on many different regulatory proteins have revealed groups based on common structural features. The following are brief descriptions of several common families.

Helix-turn-helix (HTH) DNA recognition motifs were the first discovered (see “ λ Phage” in Chapter 6). They include the repressors of the *lac*, *trp*, *gal* systems; λ Cro and CI repressors; and CRP. The proteins consist of an α helix, a turn, and a second α helix. A glycine in the turn is highly conserved as is the presence of several hydrophobic amino acids. A diagram of how an HTH motif interacts with DNA can be found in Figure 6-14. It should be noted that, unlike some DNA-binding motifs, the HTH motif is not a separate, stable domain but rather is embedded in the rest of the protein. Commonalities among HTH proteins are (1) repressors usually bind as dimers, each monomer recognizing a half-site at a DNA-binding site; (2) the operator

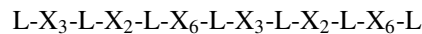
sites are B-form DNA; (3) side chains of the HTH units make site-specific contacts with groups in the major groove.

Zinc fingers were first discovered in the *Xenopus* transcription factor IIIA. Proteins in this family usually contain tandem repeats of the 30 residue zinc-finger motif (Tyr/Phe-x-Cys-X₂ or 4-Cys-X₁₂-His-X₃₋₅-His-X₃-Lys). They contain an antiparallel β sheet and an α helix. Two cysteines, near the turn in the β sheet, and two histidines in the α helix, coordinate a central zinc iron and hold these secondary structures together,



forming a compact globular domain. The zinc finger binds in the major groove of β -DNA, and because several fingers are usually connected in tandem, they are long enough to wrap partway around the double helix, like a finger.

Leucine zipper DNA-binding domains generally contain 60 to 80 residues with two distinct subdomains: the leucine zipper region mediates dimerization of the protein while a basic region contacts the DNA. The leucine zipper sequences are characterized by a heptad repeat of leucines over 30 to 40 amino acids:

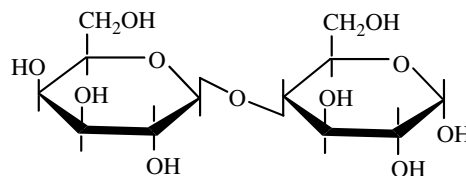


β -sheet DNA-binding proteins bind as a dimer with antiparallel β helices filling the major groove (e.g., MetJ). The arrangement is a 7-residue β sheet, a 14–16-residue α helix, and a 15-residue α helix. The β sheet enters the major groove and side chains on the exposed face of the protein contact the base pairs. IHF and HU proteins may use β -sheet regions for DNA binding.

The *lac* Operon: A Paradigm of Gene Expression

The operon responsible for the utilization of lactose as a carbon source, the *lac* operon, has been studied extensively and is of classical importance. Its examination led Jacob and Monod to develop the basic operon model of gene expression nearly half a century ago. Even now, understanding the many facets of its control underscores the economics of gene expression and reveals that no single operon acts in a metaphorical vacuum. It will become apparent that aspects of cell physiology beyond the simple notion of lactose availability govern the expression of this operon.

Lactose is a disaccharide composed of glucose and galactose:



Lactose (4-D-glucose- β -D-galactopyranoside)

Fig. 5-1. Types of genetic regulatory mechanisms. (a) Negative control regulators (repressors) can govern inducible or repressible operons. Negatively controlled inducible systems utilize a repressor molecule that prevents transcription of structural genes unless an inducer molecule binds to and inactivates that repressor. Negatively controlled repressible systems also involve a repressor protein, but in this instance the repressor is inactive unless a corepressor molecule binds to and activates the repressor protein. Thus, negatively controlled inducible operons are normally “turned off” while repressible operons are normally “turned on” unless a secondary molecule interacts with the respective regulatory proteins. (b) Positive control mechanisms can also regulate inducible or repressible operons. Inducible systems utilize an activator protein that requires the presence of a coactivator molecule in order to enhance transcription. Positively controlled repressible systems produce an activator protein that will enhance the transcription of target structural genes unless an inhibitor molecule is present. t, terminator; P, promoter; O, operator.

The product of the *lacZ* gene, β -galactosidase, cleaves the β -1,4 linkage of lactose, releasing the free monosaccharides. The enzyme is a tetramer of four identical subunits, each with a molecular weight of 116,400. Entrance of lactose into the cell requires the *lac* permease (46,500), the product of the *lacY* gene. The permease is hydrophobic and probably functions as a dimer. Mutations in either the *lacZ* or *lacY* genes are phenotypically Lac^- —that is, the mutants cannot grow on lactose as a sole carbon source. The *lacA* locus is the structural gene for thiogalactoside transacetylase (mwt 30,000) for which no definitive role has been assigned. The promoter and operator for the *lac* operon are *lacP* and *lacO*, respectively. The *lacI* gene codes for the repressor protein, and in this system is situated next to the *lac* operon. Often, regulatory loci encoding for diffusible regulator proteins map some distance from the operons they regulate. The *lacI* gene product (mwt 38,000) functions as a tetramer.

The *lac* operator is 28 bp in length and is adjacent to the β -galactosidase structural gene (*lacZ*). Figure 5-2 presents the complete sequence of the *lacP-O* region, including the C terminus of *lacI* (the repressor gene) and the N terminus of *lacZ*. The operator overlaps the promoter in that the *lac* repressor, when bound to the *lac* operator in vitro, will protect part of the promoter region from nuclease digestion. The mechanism of repression remains somewhat controversial. There is evidence that under some in vitro conditions (e.g., low salt), RNA polymerase can bind to the promoter in the presence of *lac* repressor. Thus, one proposal is that the binding of repressor to the operator region situated between the promoter and *lacZ* physically blocks transcription by preventing the release of RNA polymerase from the promoter and its movement into the structural gene. Other evidence indicates that RNA polymerase and *lac* repressor cannot bind simultaneously, leading to a model whereby *lac* repressor simply competes with RNA polymerase for binding in the promoter/operator region.

Whatever the mechanism, induction of the *lac* operon, outlined in Figure 5-3, occurs when cells are placed in the presence of lactose (however, see “Catabolite control”). The low level of β -galactosidase that is constitutively present in the cell will convert lactose to **allolactose** (the galactosyl residue is present on the 6 rather than the 4 position of glucose). Allolactose is the actual inducer molecule. The *lac* repressor is an allosteric molecule with distinct binding sites for DNA and the inducer. The binding of the inducer to the tetrameric repressor occurs whether the repressor is free in the cytoplasm or bound to DNA. Binding of the inducer to the repressor, however, allosterically alters the repressor, lowering its affinity for *lacO* DNA. Once the repressor is removed from *lacO*, transcription of *lacZYA* can proceed. Thus, the *lac* operon is a negatively controlled inducible system. It should be noted that experiments designed to examine induction of the *lac* operon usually take advantage of a **gratuitous inducer** such as isopropyl- β -thiogalactoside (IPTG)—a molecule that will bind the repressor and inactivate it but is not itself a substrate for β -galactosidase. This eliminates any secondary effects that the catabolism of lactose may have on *lac* expression (see “Catabolite control”).

There are several regulatory mutations that have been identified in the *lac* operon that serve to illustrate general concepts of gene expression. Mutations in the *lacI* locus can give rise to three observable phenotypes. The first and most obvious class of mutations results in an absence of, or a nonfunctional, repressor. This will lead to a constant synthesis of *lacZYA* message regardless of whether the inducer is present. This is referred to as **constitutive expression** of the *lac* operon. The second class of *lacI* mutations, *lacI^s*, produces a **superrepressor** with increased operator binding

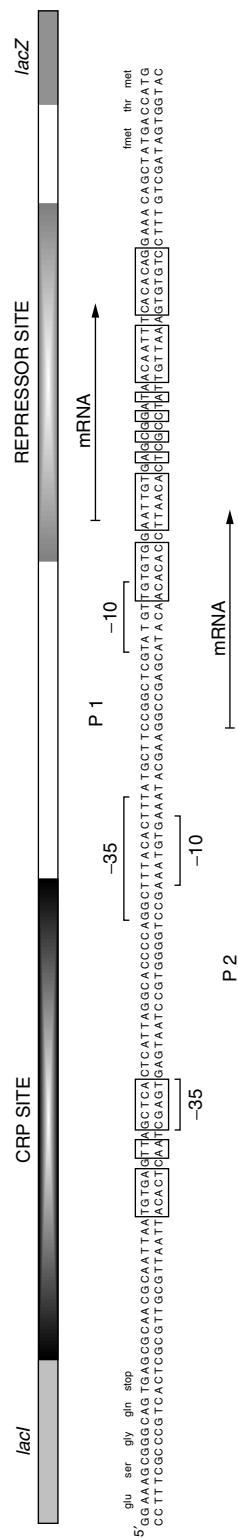


Fig. 5-2. The intergenic region (122 bp) between the *lacI* repressor and *lacZ* genes. This sequence represents the control site for the entire *lacZYA* operon. It includes a CRP-binding site at positions -84 to -55; an RNA polymerase-binding site (promoter; positions -55 to 1) and operator region (1-28). Symmetrical regions in the CRP and operator are boxed. The -10 (Pribnow box) and -35 regions that constitute the sequences recognized by RNA polymerase are indicated by brackets. The symmetrical element of the operator is shown as overlapping the promoter.

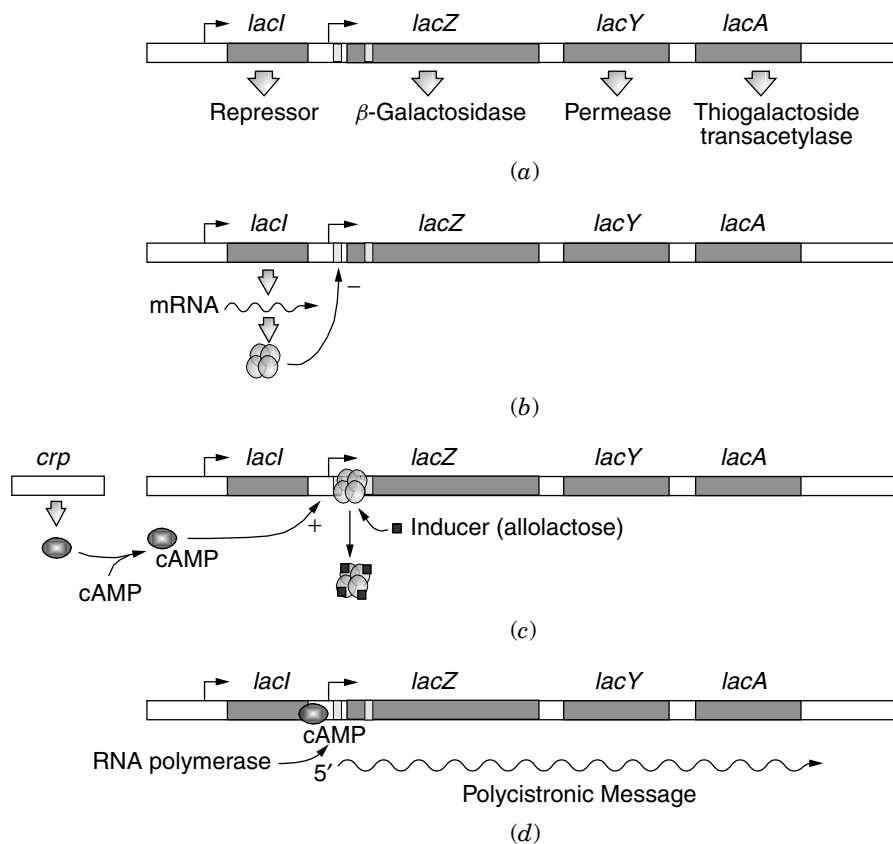


Fig. 5-3. Regulation of the lactose operon. (a) Genetic organization and products of the *lac* operon. (b) Transcription and translation. The *lacI* gene produces the repressor protein, which, as a tetramer, binds to the *lac* operator region and prevents transcription of the *lac* structural genes. (c, d) In the presence of inducer (allolactose), the inducer binds to and allosterically changes the conformation of the LacI repressor such that it will no longer bind to the operator. This allows transcription to proceed. Under conditions of high cAMP concentrations, cAMP will bind to cAMP receptor protein (CRP). The CRP–cAMP complex binds to a specific CRP site on *lacP*. This facilitates polymerase binding and, through interactions with the α subunit, stimulates open complex formation and transcription of the *lac* structural genes.

and/or diminished inducer (IPTG)-binding properties. These mutations result in an uninducible *lac* operon. The *lacI* locus contains its own promoter, *lacI^P*, distinct from *lacP*. The third class of *lacI* mutations occurs in its promoter. Mutations in the *lacI* promoter, called *lacI^q*, have been identified as promoter-up mutations that increase the level of *lacI* transcription. The region of the promoter affected by *lacI^q* mutations is in the -35 region. Increased transcription due to the *lacI^q* mutation occurs by virtue of facilitated RNA polymerase binding. The net result is a 10–50-fold increase in the *lacI* gene product (*lac* repressor). This increased production of LacI repression leads to an almost total absence of basal expression of *lacZYA* yet still allows induction in the presence of the inducer. Certain mutations in the *lac* operator (*lacO^c*) can also result in constitutive expression of the *lac* operon by causing decreased affinity for the repressor.

DNA Looping. Although Figure 5-3 suggests the presence of a single DNA-binding site for the tetrameric LacI repressor, there are actually three DNA-binding sites, all of which are involved in repression. This introduces the concept of **DNA looping** in the control of transcription. Besides the *lacO* site already introduced, the two other binding sites are O_I at -100 bp (placing it within *lacI*) and O_Z at $+400$ bp (placing it within *lacZ*). Binding of LacI to these sites tethers them together, forming competitive loops, meaning either an O_I loop or an O_Z loop can form. The O_I loop represses initiation while the O_Z loop inhibits mRNA synthesis, both directly by preventing transcription elongation and indirectly by stabilizing repressor binding at the primary operator, O . These auxiliary operators may also simply increase the relative concentration of LacI around the principal operator. This is important because there are only 10 tetrameric LacI molecules in the cell. The binding of one dimer of a tetrameric LacI to O_I leaves the other dimer free to search for O . Since O_Z and O are 400 bp apart, this tethering effectively increases the concentration of LacI in the vicinity of O about 200-fold.

“Priming the Pump”. If the *lac* repressor prevents expression of the *lac* operon, then how does the cell allow for production of the small amount of β -galactosidase needed to make allolactose inducer and enough LacP permease to enable full induction when the opportunity arises? A reexamination of Figure 5-2 reveals the presence of a second *lac* promoter called P2. This promoter binds RNA polymerase tightly but initiates transcription poorly. Once initiation occurs from P2, RNA polymerase can transcribe past the LacI-bound operator and into the structural *lacZYA* genes. Note also that the message produced from P2 will contain a large palindromic region corresponding to the operator-binding site. The stem-loop structure that forms will sequester the *lacZ* SD sequence, allowing only minimal production of β -galactosidase. This palindromic sequence is not produced from P1.

Catabolite Control: Sensing Energy Status

The *lac* operon has an additional, positive regulatory control system. The purpose of this control circuit is to avoid wasting energy by synthesizing lactose-utilizing proteins when there is an ample supply of glucose available. Glucose is the most efficient carbon source for *E. coli*. Since the enzymes for glucose utilization in *E. coli* are constitutively synthesized, it would be pointless for the cell to also make the enzymes for lactose catabolism when glucose and lactose are both present in the culture medium. The phenomenon can be visualized in Figure 5-4. This classic experiment shows *E. coli* initially growing on succinate with IPTG as an inducer of β -galactosidase activity (*lacZ* gene product). The early part of the graph shows an increase in β -galactosidase activity due to the induction. Then, at the point indicated, glucose is added to one culture. A dramatic cessation of further β -galactosidase synthesis is observed (**transient repression**) followed by a partial resumption of synthesis (**permanent repression**). It was presumed that a catabolite of glucose was causing this phenomenon, hence the term catabolite repression. However, adding cyclic AMP simultaneously with glucose reduces catabolite repression.

The phenomenon of catabolite repression is partly based on the fact that when *E. coli* is grown on glucose, intracellular cAMP levels decrease, but when grown on an alternate carbon source such as succinate, cAMP levels increase. The enzyme responsible for converting ATP to cAMP is adenylate cyclase, the product of the *cya*

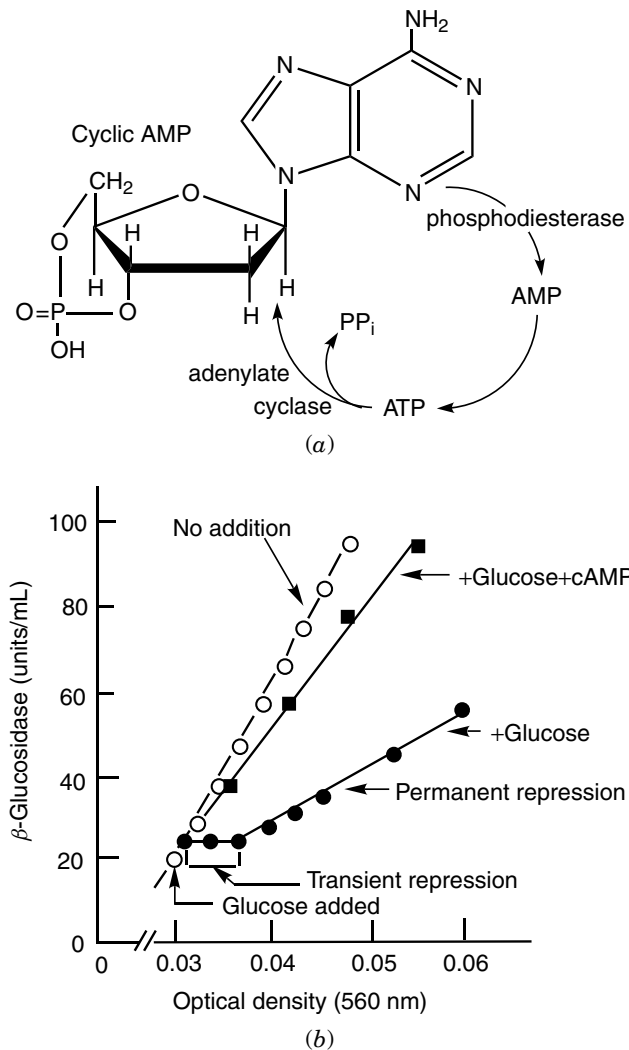


Fig. 5-4. Cyclic AMP formation and reversal of catabolite repression of β -galactosidase synthesis by cyclic AMP. (a) Formation of cyclic AMP by adenylate cyclase and its degradation by cAMP phosphodiesterase. (b) Catabolite repression of β -galactosidase synthesis and its reversal by cyclic AMP. Glucose, with or without cAMP, was added to a culture of *E. coli* growing on succinate in the presence of an inducer of the enzyme, isopropyl- β -o-thiogalactoside (IPTG). The addition of cAMP overcomes both the transient (complete) and the permanent (partial) repression by glucose. The brief lag before repression reflects the completion and the translation of the already initiated messenger.

locus. How does cAMP increase transcription of the *lac* operon? The evidence indicates that cAMP binds to the product of the *crp* locus termed the cAMP receptor protein (CRP, sometimes referred to as catabolite activator protein, CAP). The CRP-cAMP complex then binds to the CRP-binding site on the *lac* promoter. CRP represses transcription from *P*₂ but activates *P*₁. (Fig. 5-2). Several studies have shown that

CRP causes DNA bending of around 90° or more. The bend enables CRP to directly interact with RNA polymerase (see below). CRP interaction with RNP increases RNP promoter binding as well as allows RNA polymerase to escape from the promoter and proceed through elongation.

Many operons are affected by cAMP and CRP in a similar manner. These include the *gal*, *ara*, and *pts* operons involved with carbohydrate utilization. The genes controlled by CRP are referred to as members of the **carbon/energy regulon**. Other operons not directly related to carbohydrate catabolism are positively regulated by cAMP as well. The CRP–cAMP complex can also act as a negative effector for several genes including the *cya* and *ompA* loci.

What, then, is the mechanism by which glucose reduces cAMP levels? What is the catabolite responsible? Actually, it is not a catabolite per se but rather the inducible component of the phosphotransferase system, IIA^{glu} (the product of the PTS *crr* gene), which is at the center of this control system. IIA^{glu} controls the activity of preexisting adenylate cyclase rather than its synthesis. The model is presented in Figure 5-5. The PTS system employs several proteins, some of which are specific for a given sugar, to

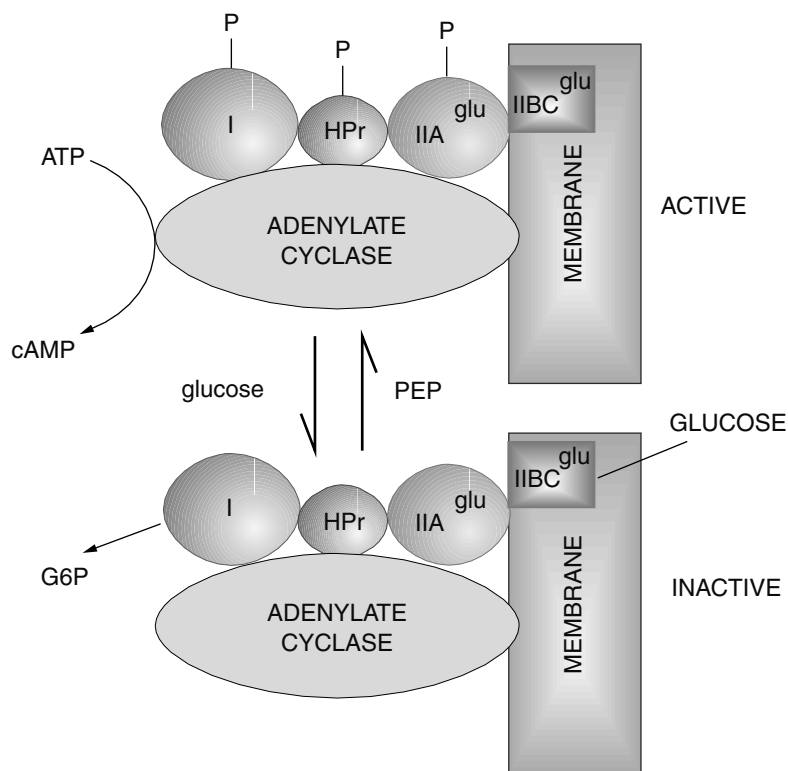


Fig. 5-5. Regulation of adenylate cyclase activity and catabolite repression. Circles and squares represent components of the phosphotransferase system; IIA, inducible component of phosphotransferase system. **Top:** Disuse of the PTS will lead to the accumulation of phosphorylated- IIA^{glu} and thus active adenylate cyclase. **Bottom:** Transport of glucose through the PTS will deplete phosphorylated III^{glu} . The dephosphorylated form of III^{glu} will inhibit adenylate cyclase activity.

transfer a phosphate from PEP to a carbohydrate during transport of that carbohydrate across the membrane. The phosphate group is transferred from protein to protein until, in the case of glucose, it reaches enzyme IIA^{glu} . In the absence of glucose, IIA^{glu} remains phosphorylated. It then interacts with and activates adenylate cyclase in the presence of inorganic phosphate. The result is a dramatic increase in cAMP levels.

However, when glucose is available and transported across the membrane, the phosphate on IIA^{glu} is transferred to the sugar forming glucose-6-phosphate. The dephosphorylated IIA^{glu} binds and inhibits adenylate cyclase activity, causing intracellular cAMP levels to diminish. So, when cells are grown on non-PTS sugars (e.g., glycerol), cAMP levels are high and the CRP–cAMP complex forms. This complex then acts at the transcriptional level to alter the expression of several operons either positively or negatively, depending on the operon. There is also evidence that when CRP–cAMP levels get too high, phosphatases dephosphorylate $\text{IIA}^{\text{glu}}\text{-P}$, which will downregulate adenylate cyclase by decreasing $\text{IIA}^{\text{glu}}\text{-P}$ activator.

CRP–cAMP actually plays two different roles in controlling the *lac* operon. Aside from activating *lacZYA* expression when lactose is present, it actually appears to increase the affinity of LacI to *lacO* by cooperative binding! So, when lactose is present, CRP–cAMP activates *lacZYA* transcription, but when lactose is absent, CRP–cAMP increases repression by LacI.

Another property of catabolite repression is **diauxie growth**. When *E. coli* is given a choice of glucose or lactose in the medium, the organism preferentially uses the glucose. The use of lactose is prevented until the glucose is used up. While part of the reason for this is due to the cAMP effect, the major cause of diauxie is inducer exclusion in which nonphosphorylated IIA^{glu} plays a role. In inducer exclusion, glucose prevents the uptake of lactose, the inducer for the *lac* operon. It appears that nonphosphorylated IIA^{glu} inhibits the accumulation of carbohydrates via non-PTS uptake systems (e.g., lactose). It has been shown that a direct interaction occurs between IIA^{glu} and the lactose carrier (LacP permease), the result being the inhibition of lactose transport.

Class I and Class II CRP-Dependent Genes

CRP, acting as a dimer, regulates more than 100 genes. The cAMP–CRP complex binds to target 22 bp sequences located near or within CRP-dependent promoters. At these promoters, the cAMP–CRP complex activates transcription by making direct protein–protein interactions with RNA polymerase. Simple CRP-dependent promoters contain a single CRP-binding site and are grouped into two classes. Class I CRP-dependent promoters have the CRP-binding site located upstream of the RNAP-binding site (e.g., *lacPI*, –61 bp from transcriptional start). Class II CRP-dependent promoters (e.g., *galPI*) overlap the RNAP-binding site. At class I promoters, CRP activates transcription by contacting RNAP via a surface-exposed β turn on the downstream CRP subunit. The contact patch on CRP, called activating region 1 (AR1), makes contact with the C-terminal domain to the RNAP α subunit (α -CTD). This serves to recruit RNAP to the promoter (Fig. 5-6).

The process of transcription activation at class II promoters is more complex. RNAP binds to promoter regions both upstream and downstream of the DNA-bound CRP, making multiple interactions with CRP. In this situation, AR1 in the upstream CRP subunit interacts with α -CTD situated upstream of CRP. The second activating region (AR2) is a positively charged surface on the downstream CRP subunit. AR2 interacts with the N-terminal domain of the α subunit (α -NTD) but only at class II promoters.

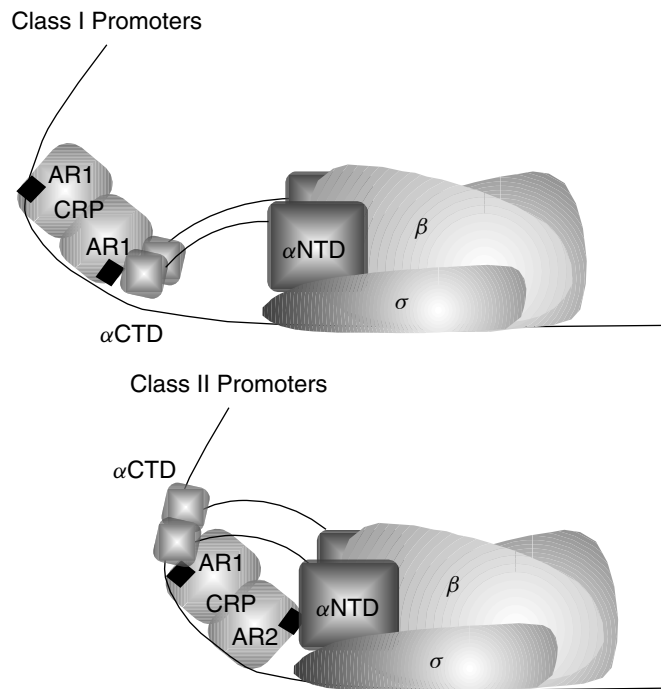


Fig. 5-6. Classes of CRP-dependent promoters. The position of the CRP-binding site on DNA relative to the RNAP-binding site influences which regions of CRP are available to interact with the α -subunit C-terminal domain. The patches on CRP that can bind the α -CTD are called activating regions (AR1 and AR2).

The Catabolite Repressor/Activator Protein Cra

The cAMP–CRP story only partly explains the extent by which catabolite repression affects cell physiology. The rest of the story involves another global regulator called Cra (catabolite repressor/activator) protein. The physiological significance of this regulator is the control of so-called peripheral pathways that allow growth on substrates that are not intermediates in a central pathway such as glycolysis. These peripheral pathways can take intermediates in the TCA cycle, such as malate, and convert them to pyruvate and back up to F-6-P (a reverse glycolysis). This process provides the carbon compounds needed for the synthesis of critical cell components (e.g., nicotinamide coenzymes). Details can be found in Chapter 8.

The Cra regulator controls numerous genes, some positively, others negatively, in response to fructose-6-phosphate or fructose-1,6-bisphosphate. Following transport of glucose, the resulting G-6-P is converted first to F-6-P and then to F-1,6-bP through glycolysis. The presence of a Cra-binding site [consensus TGAA(A/T)C•(C/G)NT(A/C)(A/C)(A/T)] upstream of the RNA polymerase-binding site is found in genes activated by Cra [PEP synthase (*ppsA*), PEP carboxykinase (*pckA*), isocitrate dehydrogenase (*icd*), fructose-1,6-bisphosphatase (*fdp*)]. Genes repressed by Cra have Cra-binding sites that overlap or follow the RNAP-binding site [Entner-Doudoroff enzymes (*edd-eda*), phosphofructokinase (*pfk*), pyruvate kinase (*pykF*)]. Cra is displaced from binding sites by binding F-1-P or F-1,6-bP. Thus, as

glucose is used, F-6-P accumulates, which then causes derepression of Cra-repressed genes such as *pfk* and *pykF* and deactivation of Cra-activated genes, PEP synthetase, and PEP carboxykinase. As a consequence, when cells are growing on glucose, gluconeogenesis genes are switched off, but genes involved in glucose utilization (e.g., Entner-Doudoroff or Embden-Meyerhof; see Chapter 8) are turned on. Figure 5-7 illustrates some of the relationships between the PTS system, adenylate cyclase, and Cra.

Catabolite Control: The Gram-Positive Paradigm

Bacillus subtilis and other gram-positive microorganisms utilize a different system for catabolite repression. A protein called CcpA (catabolite-control protein) is able to differentially bind the phosphorylated forms of HPr. As part of the PTS transport system (see Chapter 9), HPr is phosphorylated at residue His-15 in the absence of carbohydrate transport. When a suitable carbohydrate is transported through the PTS system, HPr donates the phosphate at His-15 in a cascade leading to phosphorylation of the carbohydrate (Fig 5-5). Thus, the presence of phosphorylated HPr indicates the absence of a PTS sugar. In contrast, when glucose is present, the level of phosphorylated HPr is very low. Glucose, however, activates an HPr kinase (PtsK) that phosphorylates a different HPr residue, Ser-46. Phosphorylation of HPr at Ser-46 indicates the presence of glucose. The CcpA protein binds to HPr (ser-P). The complex then acts as a transcriptional regulator of the catabolite-sensitive genes.

The *gal* Operon: DNA Looping with a Little Help from Hu

The *gal* operon of *E. coli* consists of three structural genes, *galE*, *galT*, and *galK*, transcribed from two overlapping promoters upstream from *galE*, P_{G1} and P_{G2} (Fig. 5-8). Regulation of this operon is complex because aside from being involved with the utilization of galactose as a carbon source, in the absence of galactose the *galE* gene product (UDP-galactose epimerase) is required to convert UDP-glucose to UDP-galactose, a direct precursor for cell wall biosynthesis. While transcription from both promoters is inducible by galactose, it is imperative that a constant basal level of *galE* gene product be maintained even in the absence of galactose. The *gal* operon is also a catabolite-repressible operon. When cAMP levels are high the CRP-cAMP complex binds to the -35 region (*cat* site), promoting P_{G1} transcription but inhibiting transcription from P_{G2} . When grown on glucose, however, where cAMP levels are low, transcription can occur from P_{G2} assuring a basal level of *gal* enzymes (Fig. 5-8).

Both of the *gal* promoters are negatively controlled by the *galR* gene product, a member of the LacI family of repressors. The *galR* locus is unlinked to the *gal* operon. There are two operator regions to which the repressor binds: an extragenic operator (O_E), located at -60 from the P_{G1} transcription start site ($S1$), and an intragenic operator (O_I), located at position $+55$ from $S1$ start. A *galR* repressor dimer binds independently to O_E and O_I . For repression to occur, the two DNA-bound GalR dimers must interact with each other, forming a tetramer (Fig. 5-8c) that engineers the DNA into a repression loop. However, GalR dimers only weakly interact. In order to stabilize that interaction, the histone-like protein Hu binds to a specific site (*hbs*) between the two operators. The DNA loop sequesters P_{G1} and P_{G2} from RNA polymerase access. Some access must occur, however, since, as mentioned above, basal levels of the *gal* enzymes are synthesized even under repressed conditions.

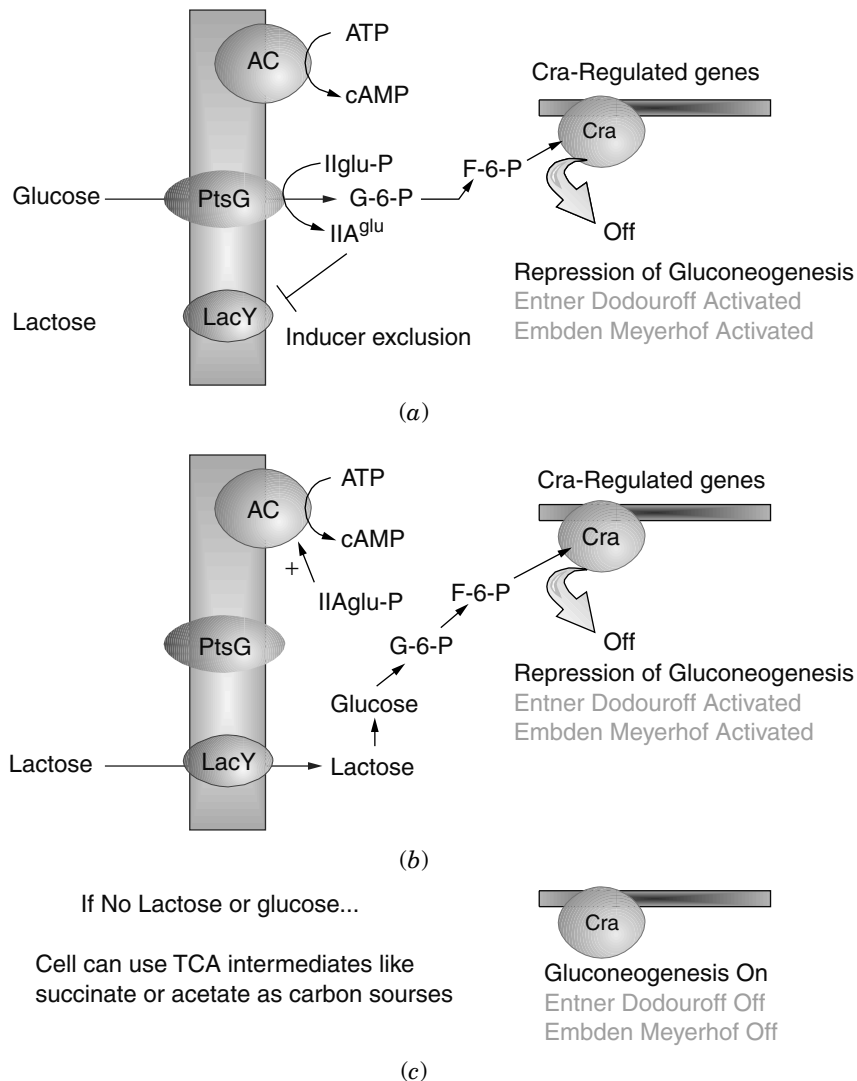


Fig. 5-7. Regulation of carbohydrate metabolism in *E. coli* by Cra. (a) Growth on glucose. The cell will want to inhibit gluconeogenesis, stimulate glycolysis, and prevent use of other carbon sources such as lactose. Consequently, the synthesis of cAMP is low, limiting expression of other carbohydrate operons; the PTS protein IIA^{glu} accumulates, inhibiting lactose transport; and F-6-P is produced. F-6-P binds Cra, releasing it from target genes. The results are repression of gluconeogenesis, which is not required because glucose is plentiful, and an increase in glycolytic enzymes required to use glucose. (b) When lactose is present (no glucose), IIA^{glu}-P accumulates, stimulating cAMP synthesis, which increases *lac* operon expression. The catabolism of lactose also produces significant levels of F-6-P, so, as with glucose catabolism, gluconeogenesis is low while glycolysis is high. (c) If no carbohydrates are available, the cell can use TCA cycle intermediates as carbon sources. This is made possible because F-6-P is not made at high levels, allowing Cra to bind to its target genes. This stimulates the gluconeogenesis pathways needed to use the TCA intermediates and limits glycolysis.

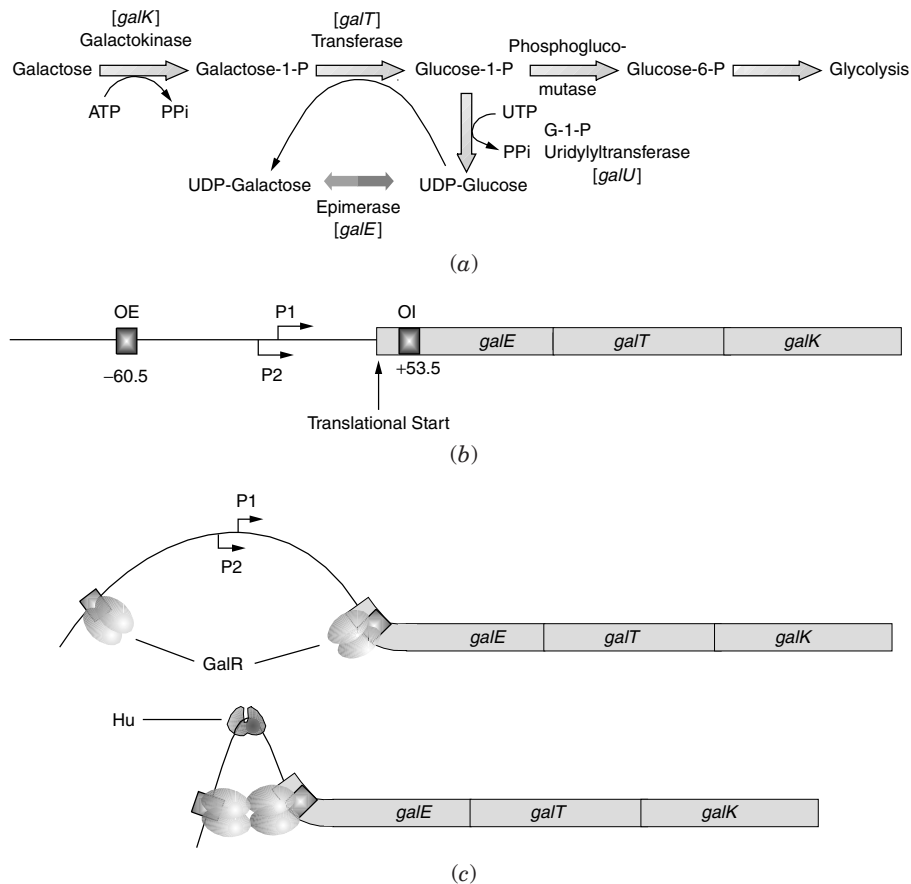


Fig. 5-8. Regulation of the galactose operon. (a) Biochemical pathway. (b) Genetic organization of the *galETK* operon including transcription and translation starts. (c) Architecture of the GalR–Hu *gal* DNA complex. Repression of the *gal* operon by the *galR* product binding to the internal and external operator regions and Hu binding between the two operators.

The Arabinose Operon: One Regulator, Two Functions

Catabolism of the carbohydrate L-arabinose by *E. coli* involves three enzymes encoded by the contiguous *araA*, *araB*, and *araD* genes (Fig. 5-9). The product, xylulose-5-phosphate (xl-5-p), can be catabolized further by entering the pentose phosphate cycle as outlined in Figure 1-10 (see also Chapter 8). The transcriptional activity of these genes is coordinately regulated by a fourth locus, *araC*. The *araC* locus and the *araBAD* operon are divergently transcribed from a central promoter region (Fig. 5-10b). Both the *araC* promoter (P_C) and the *araBAD* promoter (P_{BAD}) are stimulated by cAMP and CRP. Regulation by *araC* is unusual, however, in that the *araC* product activates transcription of *araP_{BAD}* in the presence of arabinose (inducing condition) yet represses transcription of both promoters in the absence of arabinose (repressive condition). Thus, *araC* protein can assume two conformations: repressor and activator (Fig. 5-10). The *araC* product also regulates its own expression in the presence or absence of arabinose. The arabinose operon was one of the first examples where looped DNA structures were shown to regulate transcription initiation.

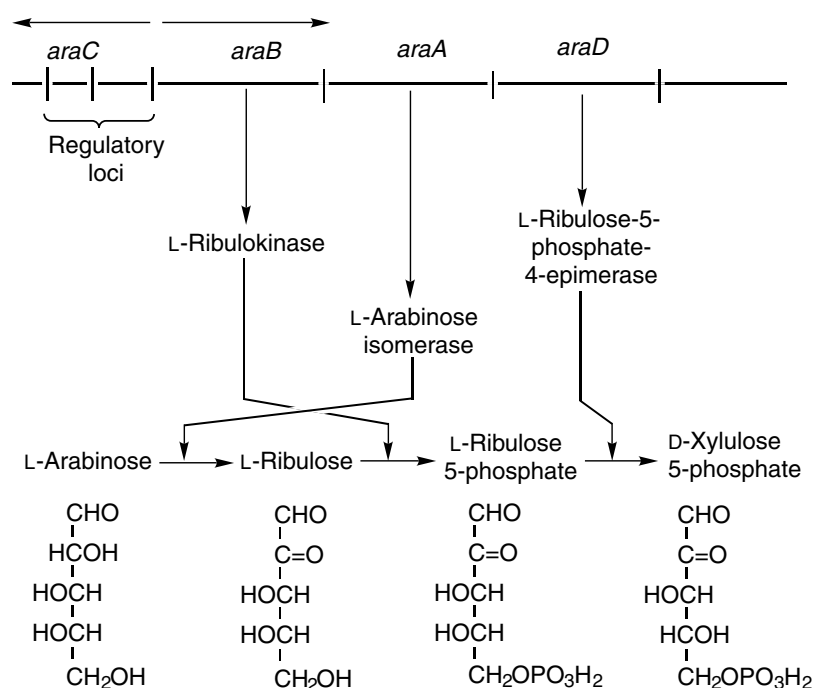


Fig. 5-9. The arabinose operon.

AraC functions as a dimer in vivo. Figure 5-10a presents the basic domain structure of an AraC monomer and two alternative dimer arrangements. The protein consists of a C-terminal DNA-binding domain connected by a flexible linker to an N-terminal dimerization domain. The AraC molecule ends with an N-terminal arm that can bind either the C-terminal domain of the same molecule or the N-terminal domain of a partner AraC molecule depending on whether arabinose is present. Without arabinose, the N-terminal arm binds to the C terminus, which, combined with the linker, forms a very rigid structure. Dimerization of this form results in the rather long complex shown in Figure 5-10a. In the presence of arabinose, the N-terminal arm preferentially binds the N-terminal domain of the partner molecule in an AraC dimer. As a result, only the flexible linker connects the main domains, which allows the dimer to attain the third, more compact, structure in Figure 5-10a. Understanding these very different structures is the key to understanding regulation of this system.

Figure 5-10b presents the *ara* operator and promoter regions. The AraC protein can bind to four operators: O_1 , O_2 , I_1 , and I_2 (I for induction). In the absence of arabinose, the rigid AraC dimer can only bind to O_2 and I_1 , forming a negative repression loop that prevents *araBAD* expression (Fig. 5-10c). Upon binding to arabinose, it becomes energetically more favorable for the N-terminal arm to bind to the dimerization domain of the AraC partner than to the DNA-binding domain. The dimer is now flexible enough to bind to I_1 and I_2 . Occupancy of the I_2 site places AraC near the RNP-binding site, allowing interactions to occur between these two proteins. It is important to note that AraC can sample arabinose while remaining bound to DNA. Having AraC already

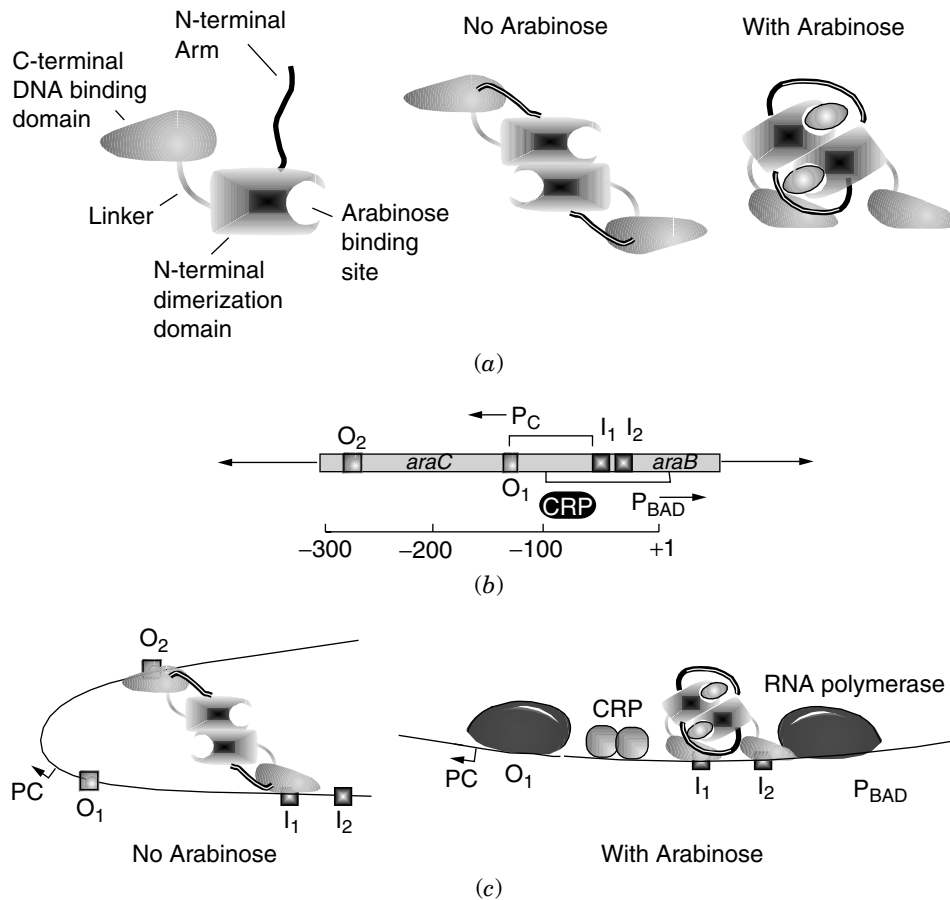


Fig. 5-10. Model for control of the *araBAD* and *araC* promoters. (a) Alternative structures of an AraC dimer formed in the presence or absence of arabinose. (b) The divergent promoter region of the *ara* operon. *O₁*, *O₂*, *I₁*, and *I₂* are binding sites for AraC. *P_C* and *P_{BAD}* represent the promoter regions for *araC* and *araBAD*, respectively. The CRP box marks the binding region for CRP. (c) Alternative AraC binding architectures that allow differential regulation of this divergently promoted operon.

bound to the regulatory region in the absence of arabinose allows it to respond very quickly to the addition of arabinose, since the potentially slow DNA-binding step has already taken place.

Transcription from *P_{BAD}* is also stimulated by cAMP–CRP through protein–protein interactions with RNA polymerase. The α -CTD of RNA polymerase α subunit contacts a third activating region of CRP called activation region 3 (AR3). The combination of arabinose binding AraC and binding of cAMP–CRP to its recognition site causes a dramatic increase in *araBAD* expression.

There are now over 100 transcriptional regulators that contain regions homologous to the helix-turn-helix DNA-binding domain of AraC. These proteins are classified as the **AraC family** of regulators.

ATTENUATION CONTROLS

Transcriptional Attenuation Mechanisms

The *trp* Operon. The genes that code for the enzymes of the tryptophan biosynthetic pathway in *E. coli* are assembled in an operon (Fig. 15-17). There are two regulatory mechanisms that control the expression of this operon at the transcriptional level: transcriptional end-product repression and transcriptional attenuation. Repression involves the product of the *trpR* gene (repressor) binding tryptophan. This increases the affinity of the repressor for the *trp* operator region, thus blocking transcription. However, there is a second level of regulation placed on this and many other amino acid biosynthetic operons.

This control mechanism, called **attenuation**, depends on the coupling between transcription and translation as illustrated in Figure 1-8. Between the *trpO* and *trpE* genes is the attenuator region. This region codes for a small (14 amino acid) nonfunctional polypeptide called the leader polypeptide. The key to attenuation is understanding the competitive stem-loop structures that can form in the leader mRNA. Figure 5-11a presents the sequence of the *trp* leader mRNA along with the amino acid sequence of the leader polypeptide. The brackets above the mRNA sequence show regions of mRNA capable of base pairing with each other, leading to transcription termination. Region 1 can form a stem structure (Fig. 5-11) with region 2 (1:2 stem), and region 3 can form a stem structure with region 4 (3:4 stem). The 3:4 stem (called the **attenuator stem-loop**) principally acts as a transcription termination signal. If it is allowed to form, transcription terminates prior to the *trpE* gene. The brackets below the message show regions that can base pair in an **antiterminator** configuration (2:3 stem-loop). The formation of a 2:3 loop prevents the formation of the 3:4 terminator stem; thus, transcription is allowed to proceed into the *trp* structural genes.

How, then, does the cell regulate the formation of the 2:3 or 3:4 secondary structures? This is accomplished by varying the rate of translation of the leader polypeptide relative to the rate of transcription (Fig. 5-12). Notice the two adjacent Trp codons in the leader sequence. If tryptophan levels within the cell are low, then the levels of charged tRNA^{Trp} will also diminish. As the ribosome translates the leader polypeptide, it will stall over the region containing the Trp codons (Fig. 5-12c), overlapping and thereby preventing region 1 from binding to region 2 and allowing RNA polymerase to synthesize more of the leader transcript. With the 1:2 stem unable to form and once RNA polymerase has transcribed past region 3, the 2:3 stem-loop (called the **antiterminator loop**) can form, which will prevent formation of the 3:4 terminator. Consequently, RNA polymerase can proceed into the *trp* structural genes, ultimately leading to an increase of tryptophan levels. (There is another Shine-Dalgarno sequence in *trpE*, allowing for ribosome binding).

Once intracellular tryptophan (and, therefore, charged tRNA^{Trp}) levels are high, the ribosome will proceed to the translational termination codon present in the leader (Fig. 5-12d). In this situation, the ribosome overlaps region 2 and prevents the formation of a stable 2:3 loop. Consequently, when RNA polymerase transcribes past region 4, the 3:4 transcription terminator stem forms and halts transcription prior to the *trp* structural genes. Attenuation is a common regulatory mechanism for amino acid operons. Figure 5-11b illustrates the attenuator region and alternative stem-loop structures for the *his* operon.

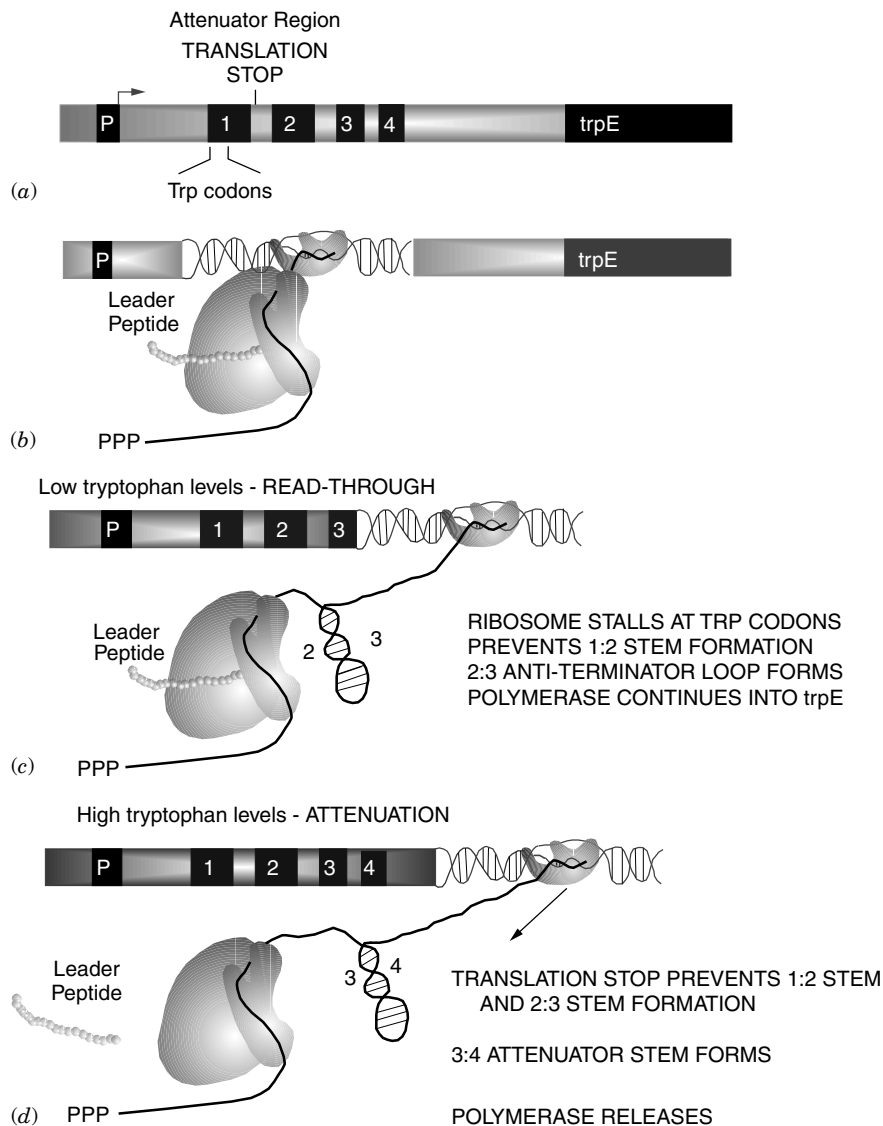


Fig. 5-12. Model for attenuation of the tryptophan operon. (a) The attenuator region illustrating positions of promoter (P), tryptophan codons (vertical lines), and regions capable of forming stem-loop structures (1, 2, 3, 4). (b) Coupled transcription and translation of leader polypeptide. (c) Under conditions where internal tryptophan levels are low, the ribosome stalls over the *trp* codons in the message due to a depletion of charged tryptophanyl tRNA. This prevents 1:2 stem formation but allows the 2:3 stem to form. Consequently, the 3:4 attenuator loop cannot form and will permit RNA polymerase to read through into the *trp* structural genes. (d) At high levels of internal tryptophan, the ribosome encounters the translational stop codon, preventing the formation of the 1:2 and 2:3 stem structures. RNA polymerase continues, allowing 3:4 stem formation, which is a transcriptional stop signal. Transcription does not proceed into the *trp* structural genes.

Attenuation by TRAP in *Bacillus subtilis*. In the discussion of the *E. coli* tryptophan operon, the concept of attenuation involving ribosome stalling was introduced. However, there are other ways a cell can utilize attenuation that do not involve ribosome stalling. The first is an RNA-binding protein-dependent attenuation system that regulates the formation of competing mRNA stem structures in a 5' untranslated region. In *Bacillus*, a protein called TRAP (tryptophan RNA-binding attenuation protein) forms an 11-member ring structure that can bind tryptophan. Tryptophan-activated TRAP binds to 11 closely spaced (G/U)AG repeats present in the nascent *trpEDCFBA* leader transcript. TRAP binding prevents formation of an antiterminator structure in the leader region, which leaves an overlapping Rho-independent terminator free to form. Thus, TRAP binding promotes transcription termination before RNA polymerase reaches the structural genes. In the absence of TRAP, the antiterminator structure prevents formation of the terminator, resulting in transcriptional readthrough into the *trp* structural genes.

The *pyrBI* Strategy. In *E. coli* or *Salmonella enterica*, the de novo synthesis of UMP, the precursor of all pyrimidine nucleotides, is catalyzed by six enzymes encoded by six unlinked genes and operons (see Chapter 16). One operon within this system is *pyrBI*, which codes for the catalytic (PyrB) and regulatory (PyrI) subunits of aspartate transcarbamylase. The operon is negatively regulated over 100-fold by intracellular UTP concentration. Attenuation in this system involves RNA polymerase pausing rather than ribosome stalling. The basic model is presented in Figure 5-13. As with the tryptophan operon, regulation depends on the degree of coupling between transcription and translation. Within the *pyrBI* regulatory region there is an RNA polymerase pause site flanked by U-rich (T-rich in the DNA) areas (Fig. 5-13a). When cellular UTP levels are low (meaning the cell will want to synthesize more UMP), RNA polymerase undergoes a lengthy pause as it tries to find UTPs to extend the message (Fig. 5-13c). This strong pause allows the translating ribosome to translate up to the stalled RNA polymerase. When the polymerase finally escapes the pause region and reaches the transcriptional attenuator region, the ρ -independent terminator hairpin cannot form due to the close proximity of the ribosome. This permits RNA polymerase to transcribe into the *pyrBI* structural genes. However, when UTP levels are high, RNA polymerase undergoes only a weak pause, then quickly releases to transcribe the attenuator before the ribosome has caught up (Fig. 5-13b). Therefore, the transcription termination loop forms, terminating transcription before RNA polymerase has entered structural genes.

It may seem that attenuation is a very costly and inefficient regulatory system for the cell. Why would nature provide a regulatory mechanism that involves the synthesis of nonfunctional polypeptides or transcripts? Remember that all regulatory systems require information and so of necessity require the expenditure of energy. The energetic cost of attenuation may be less than that required to synthesize a larger trans-acting regulatory protein. A clear advantage of attenuation is that it uses information present in the nascent RNA transcript. This is a strategy that quickly and directly translates what is sensed (i.e., low tryptophan levels) into action (i.e., increased transcription).

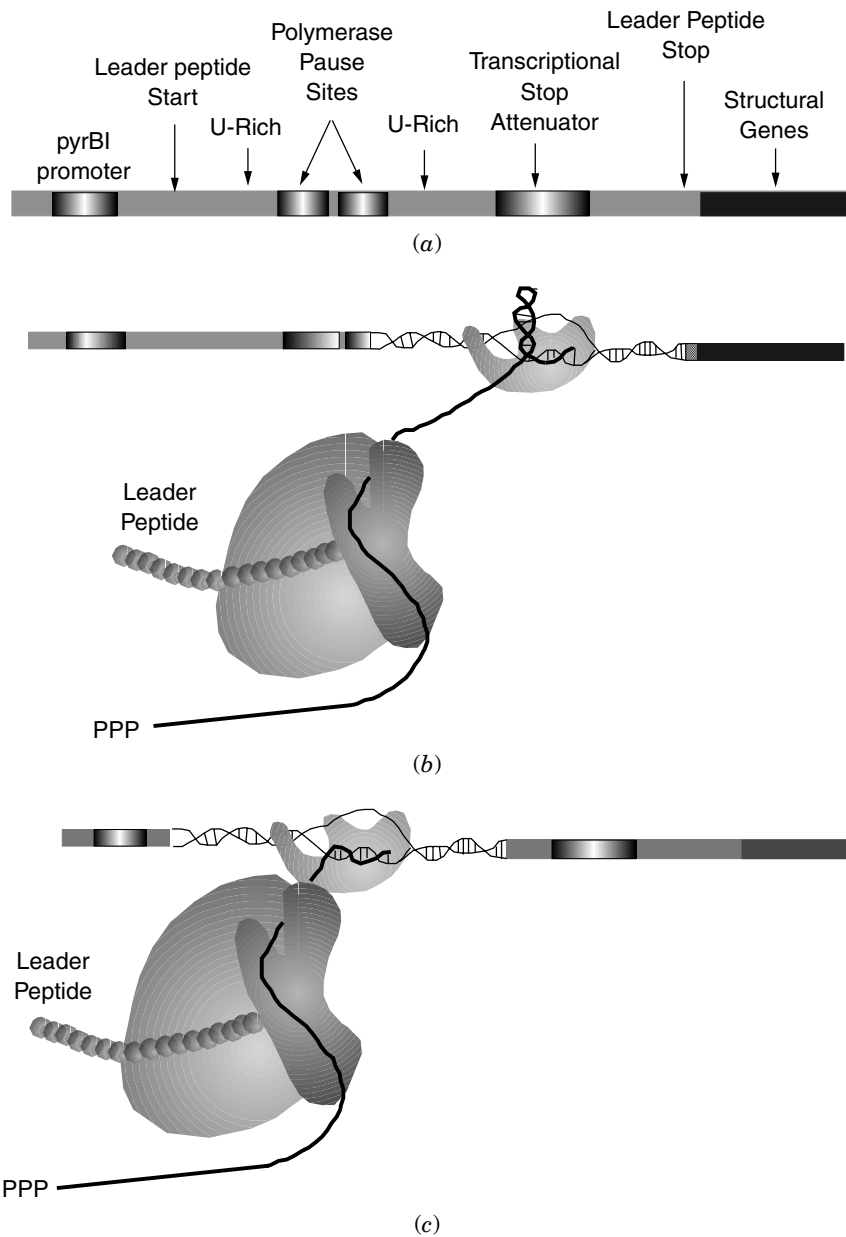


Fig. 5-13. Transcriptional attenuation of the *pyrBI* operon.

Translational Attenuation Control: The *pyrC* Strategy

The *pyrC* locus encodes the pyrimidine biosynthetic enzyme dihydroorotase (see Chapter 16). Expression of this enzyme varies over 10-fold depending on pyrimidine

availability. Regulation is not due to the modulation of mRNA levels but is the result of an elegant translational control mechanism. The promoter region for *pyrC* is shown in Figure 5-14. Transcription starts at one of the bases indicated by an asterisk and depends on pyrimidine availability. When grown under pyrimidine excess, transcription starts predominantly at position C₂. Transcripts started at this site can form a hairpin loop due to base pairing between regions indicated by arrows. This hairpin blocks ribosome binding by sequestering a Shine-Dalgarno sequence and lowers production of the *pyrC* product.

Alternatively, when pyrimidine availability is low, transcripts preferentially start at G₄. This shorter transcript cannot form the hairpin and thus will have an SD sequence available for ribosome binding. This translational control of *pyrC* is a striking example of a regulatory mechanism employing only the basic components of the transcriptional and translational machinery. RNA polymerase directly senses nucleotide availability and by differential transcription initiation will control translational initiation of the transcript.

Membrane-Mediated Regulation: The *put* System

Salmonella enterica can degrade proline for use as a carbon and/or nitrogen source. The process requires two enzymes: a specific transport system encoded by *putP* and the *putA*-coded bifunctional membrane-bound proline oxidase that converts proline

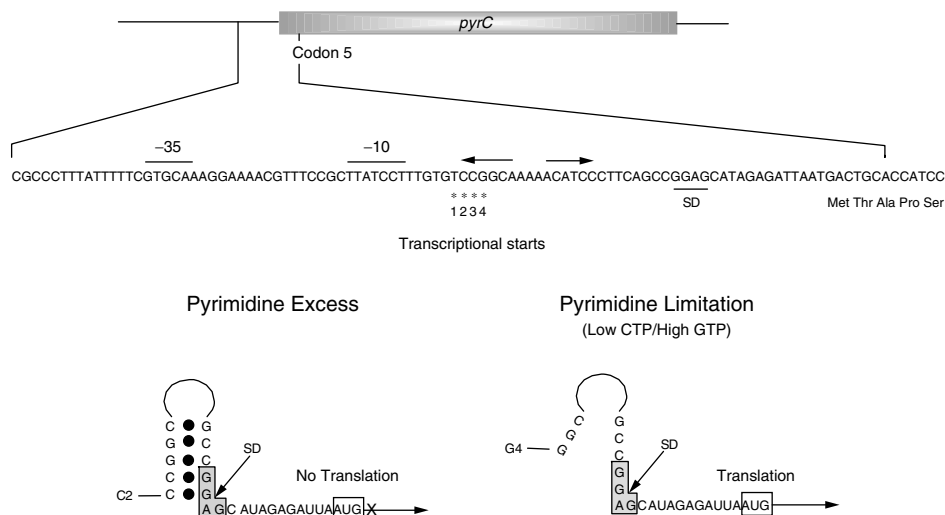


Fig. 5-14. Translational control of *pyrC* expression. The top figure shows the *pyrC* locus with the promoter region expanded. Arrows above the sequence indicate the areas capable of base pairing in the mRNA. The numbered asterisks indicate potential transcriptional start sites. The choice of start site varies with the ratio of CTP to GTP in the cell. The bottom pair of illustrations indicate the mRNA stem-loop structure that forms during pyrimidine excess but fails to form during pyrimidine limitation due to the shorter transcript. This stem-loop structure will sequester a ribosome-binding site and prevent synthesis of the *pyrC* product.

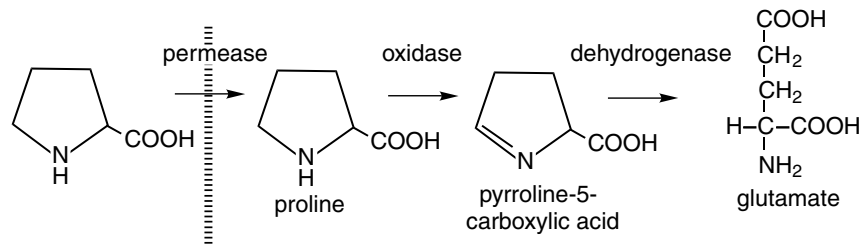


Fig. 5-15. Proline degradation pathway. P-5-C, pyrroline-5-carboxylic acid.

to glutamate (Fig. 5-15). The two genes are adjacent but are transcribed in opposite directions from a common controlling region. The *putA* gene product also possesses regulatory properties in that it represses the expression of both *putA* and *putP*. The regulatory and enzymatic properties of *putA* are separable by mutation—that is, some *putA* point mutations result in the constitutive expression of *putAP* yet do not affect proline oxidase activity.

The PutA protein is a flavoprotein that, when bound to proline, exists as a membrane-bound enzyme associated with the electron transport chain. When associated with the membrane, PutA will not repress *putAP* expression. The regulatory switch centers on the redox state of the FAD prosthetic group, which in turn depends on the presence of proline. Proline interaction with PutA reduces the FAD. PutA containing reduced FAD has a greater affinity for membrane lipids. Thus, when proline is present, PutA is sequestered away from *putAP* operator sequences and the system is induced. As the levels of free proline drop, the PutA FAD moiety oxidizes. This form of PutA is released from the membrane and accumulates in the cytoplasm where it will bind *putAP* and repress expression. Other examples of redox-sensitive genetic switches include Fnr (anaerobic control) and OxyR (oxidative stress response, See Chap 18).

Recombinational Regulation of Gene Expression (Flagellar Phase Variation)

Salmonella periodically alters the antigenic type of the polypeptide flagellin, the major structural component of bacterial flagella. It does this as a form of misdirection designed to hinder mammalian immune systems from getting a “fix” on the organism. As the initial antibody response begins to attack, the invader has changed its appearance. Two genes, *fliC* (formerly H1) and *fliB* (formerly H2), encode different forms of the flagellin protein, but only one gene is expressed in the cell at any one time. The organism can switch from the expression of one gene to the other at frequencies from 10^{-3} to 10^{-5} per cell per generation. The two flagellin genes do not map near each other, with *fliC* and *fliB* mapping at 40 and 56 min, respectively. Figure 5-16 presents the model for the flagellar phase variation switch.

The principle of the control involves the protein-mediated inversion of a 993 bp region of DNA called the H region. The H region is flanked by two 26 bp sequences designated *hixL* (repeat left) and *hixR* (repeat right). Each 26 bp sequence is composed of two imperfect 13 bp inverted repeats. Note in Figure 5-16 that the promoter for the *fliB* and *fliA* (the *fliC* repressor) operons resides within the H region. Thus, when this region occurs in one orientation (Fig. 5-16a), the promoter is positioned to transcribe

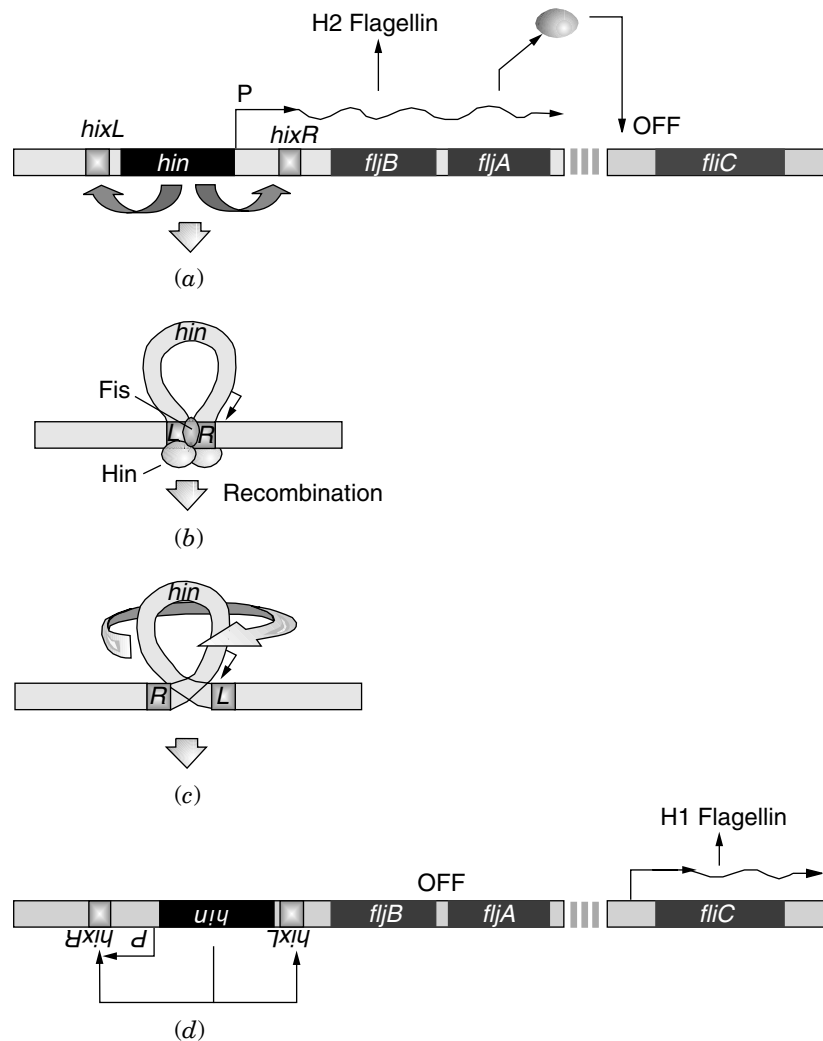


Fig. 5-16. Molecular mechanism of flagellar phase variation in *Salmonella typhimurium*. P, promoter regions.

fljB and *fljA*. Consequently, the H1 locus is repressed by the *fljA* product (also known as rH1) and H2 flagellin is synthesized. The *hin* gene product (22 kDa) is the recombinase that mediates the inversion of this region. When the region between *hixL* and *hixR* is inverted (Fig. 5-16d), the promoter is reoriented and now reads away from the H2 operon, preventing the synthesis of H2 flagellin and FljA repressor. Without FljA, the *fliC* gene can be expressed, producing H1 flagellin.

Within the H region are **enhancer sequences** that stimulate the inversion process 100-fold. Enhancers are cis-acting DNA sequences that magnify the expression of a gene. They do not require any specific orientation relative to the gene and can function at great distances from it. The stages of inversion include Hin binding to the *hix* sites and Fis (a 12 kDa dimeric protein) binding to the enhancer. The *hix* sites are brought

Translational Repression

Fig. 5-17. Organization and regulation of genes contained within the *str-spc* and *rif* regions of the bacterial chromosome. Genes are represented by the protein product. (→) For each operon, the direction of transcription from the promoter (P). It has been shown that the L11 and β operons are probably a single operon. The L11 promoter functions as the major promoter for all genes contained within the L11 and β operons in exponentially growing cells. Regulatory ribosome proteins are indicated by boxes. Effects of the boxed proteins on the in vitro and in vivo synthesis of proteins from the same operon are shown. L10 can function as a repressor in a complex with L12. + indicates specific inhibition of synthesis; −, no significant effect on synthesis; $\pm$, weak inhibition of synthesis; (+), inhibition presumed to occur in vivo; ND, not determined. It has not been established how the regulation of the synthesis of ribosomal proteins S12, L14, and L24 is achieved.

										$\xleftarrow{\boxed{L1}}$	L11	P	<u>L11 OPERON</u>			
										+	+		in vitro			
										(+)	+		in vivo			
										$\xleftarrow{\beta'}$	β	L7/12	$\boxed{L10}$	P	<u>β OPERON</u>	
										-	-	+	+		in vitro	
										-	-	+	(+)		in vivo	
										$\xleftarrow{EF-Tu}$	EF-G	$\boxed{S7}$	S12	P	<u>str OPERON</u>	
										-	+	+	-		in vitro	
										-	+	(+)	-		in vivo	
S17	L29	L16	S3	(S19, L22)	L2	L23	$\boxed{L4}$	L3	S10	P	<u>S10 OPERON</u>					
-	-	-	-	-	$\pm$	+	+	+	+		in vitro					
+	+	+	+	+	+	+	(+)	+	+		in vivo					
$\xleftarrow{L15}$	L30	S5	L18	L6	$\boxed{S8}$	S14	L5	L24	L14	P	<u>spc OPERON</u>					
ND	ND	-	-	-	-	+	+	-	-		in vitro					
+	+	+	+	+	(+)	+	+	-	-		in vivo					
										$\xleftarrow{L17}$	α	$\boxed{S4}$	S11	S13	P	<u>α OPERON</u>
										-	-	+	+	+		in vitro
										+	$\pm$	(+)	+	+		in vivo

protein synthesis states that the translation of a group of ribosomal protein genes encoded on a polycistronic message is inhibited by one of the proteins encoded within that operon. It appears, for example, that L1 of the L11 operon can bind to its own message, preventing its translation.

Many of the proteins that serve as translational repressors also bind specifically to either 16S or 23S rRNA during ribosome assembly. Figure 5-18 illustrates the similarities in the secondary structure of the S4 and S8 protein-binding sites on rRNA versus mRNA. Thus, if growth slows, leading to a decrease in rRNA, the translational repressors will bind instead to their own message. Homologies have been found between the binding sites on the rRNA and the target site on the mRNA for several operons. From a single target site on a polycistronic message, a repressor can affect the translation of all the sensitive downstream cistrons through sequential translation—that is, translation of downstream cistrons can be dependent on the translation or termination of the upstream target cistron. Notice that in each operon presented in Figure 5-17 not all of the cistrons in a polycistronic message are affected.

Anti- σ Regulation by Molecular Hijacking

Since σ factors contain the information needed to recognize promoters, cells will often use alternative σ factors to coordinately transcribe related families of genes in response to environmental or cellular signals. Because different σ factors bind competitively to core RNA polymerase, gene expression can be modulated by regulating the availability of the various σ factors. The cellular levels of many σ factors are controlled by transcription, translation, or proteolysis, but other alternative σ factors are synthesized and maintained in an inactive state until they are actually needed. Proteins that inhibit σ -factor activity are known as **anti- σ factors**. These negative regulatory proteins interact directly with their cognate σ factors, preventing them from stably associating with core RNAP. Three σ /anti- σ examples are considered: σ^F /SpoIIAB involved in the *Bacillus* sporulation cascade; σ^{70} and the *E. coli* phage T4 protein AsiA (see Chapter 6, T4 phage, middle gene expression); and σ^{28} /FlgM involved with flagellar assembly in *Salmonella enterica*.

As noted in Chapter 2, σ factors contain four major conserved regions designated 1 through 4. The SpoIIAB protein of *Bacillus* binds to conserved σ^F regions 2.1, 3.1, and 4.1, sterically hindering the binding of σ^F to core RNAP. The T4 product AsiA, however, binds to region 4.2 of σ^{70} , which recognizes the -35 promoter region. AsiA does not hinder σ^S association with core RNAP but will prevent the σ from recognizing a target promoter. Details of how these regulatory mechanisms integrate into sporulation and phage development are discussed in Chapters 6 and 19, respectively.

FlhA. Flagellar biogenesis is an extraordinarily complex process involving many different gene products. Separate systems are involved in assembling the basal body, motor, and the flagellum. Each system must be expressed at the appropriate time and in the correct order. Attempting to synthesize flagella before the basal body is ready won't work. Consequently, the genes for flagellar biogenesis in *Salmonella* are organized into a transcriptional hierarchy composed of three classes. Class 1 consists of the master regulators FlhD that, together with σ^{70} , transcribe the Class 2 genes. The Class 2 genes encode proteins required for the assembly and structure of the hook–basal

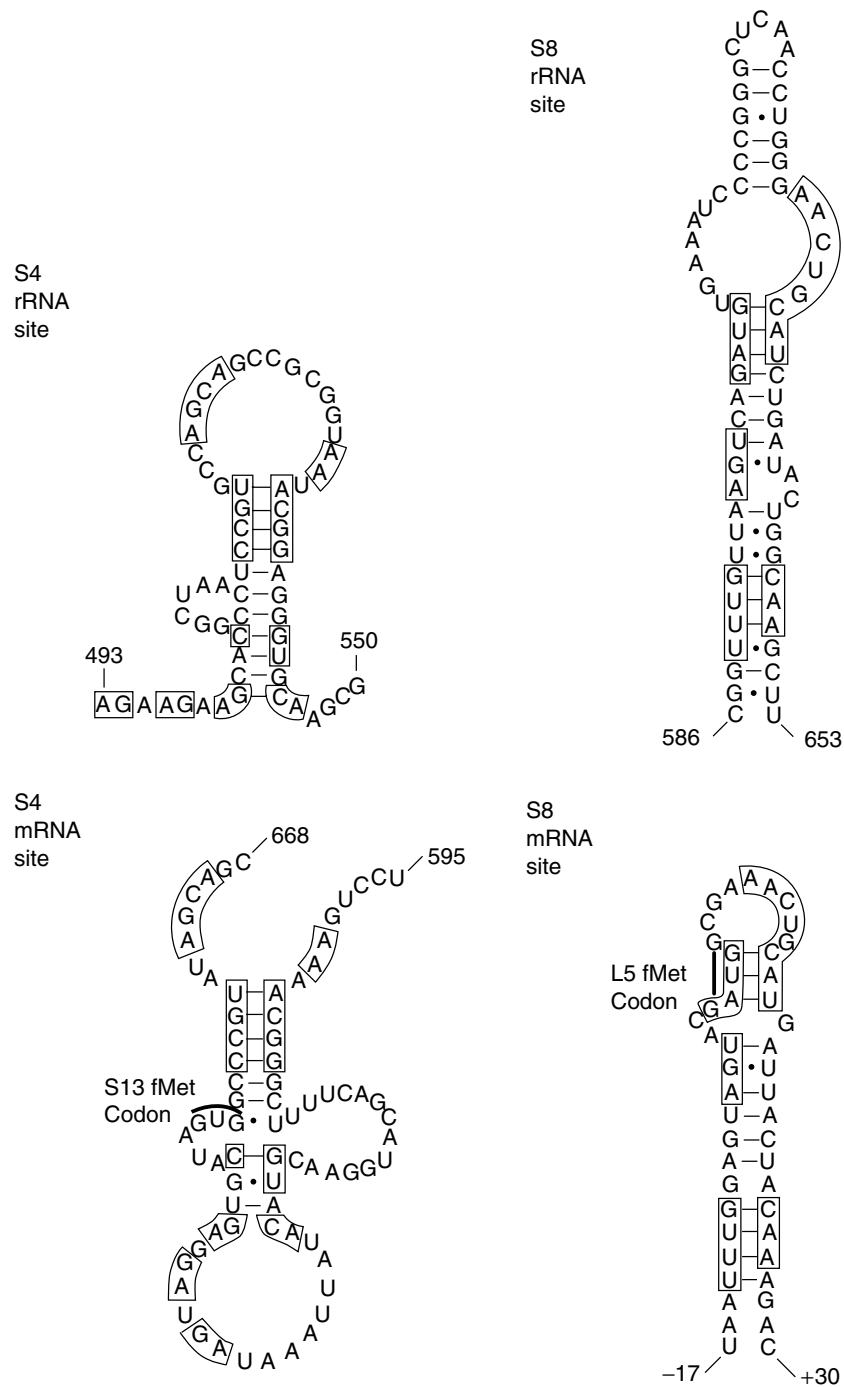


Fig. 5-18. Predicted secondary structures of S4- and S8-binding sites on 16S rRNA and their respective mRNAs. The binding site of S4 on the mRNA overlaps the S13 initiation region; the mRNA site for S8 overlaps the L5 initiation region. Boxed sequences indicate homologies.

body (see Chapter 7 for more details) as well as for the FliA σ (σ^{28}) and its anti- σ FlgM. The Class 3 genes encoding the flagellar filament, motor, and chemotaxis proteins are transcribed exclusively by σ^{28} . Although σ^{28} is produced simultaneously with the hook–basal body proteins, expression of Class 3 genes does not occur until the basal body structure is complete. The reason is that FlgM keeps σ^{28} in check during the assembly process. Once assembly is complete, however, the basal body structure secretes FlgM into the surrounding medium through its core channel. This lowers the intracellular concentration of the anti- σ to a level that permits σ^{28} -dependent transcription of the Class 3 genes.

Titration of a Posttranscriptional Regulator: The CsrA/CsrB Carbon Storage Regulatory Team

During the transition from exponential phase growth into the stationary phase, nonsporulating bacteria readjust their physiology from allowing robust metabolism to

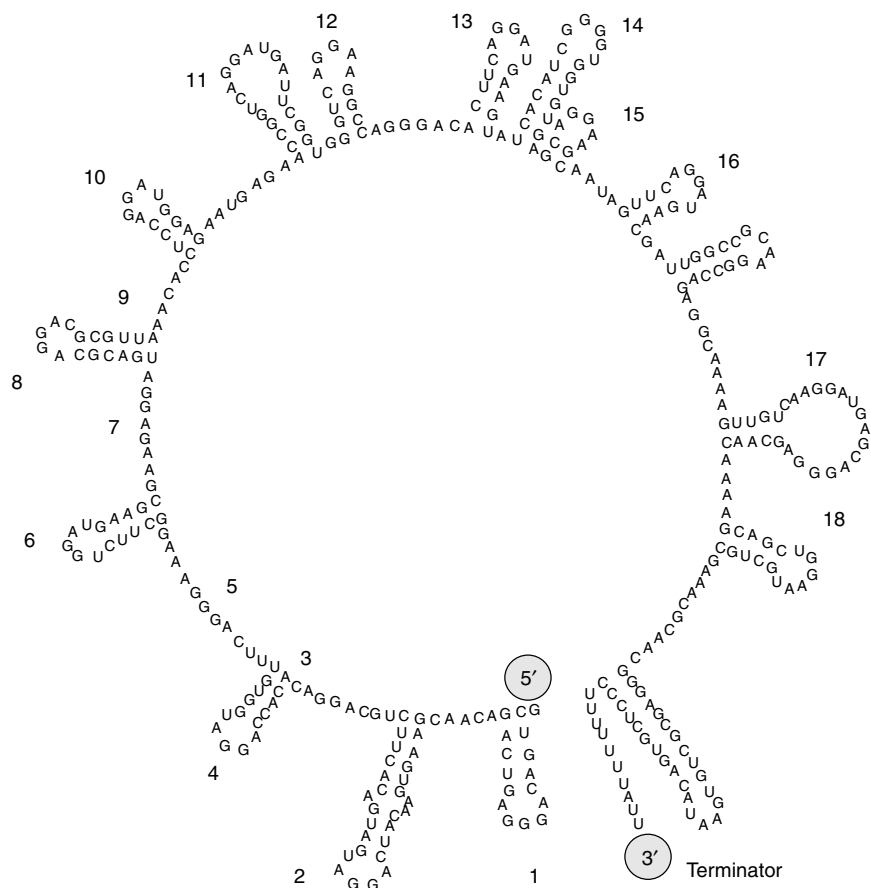


Fig. 5-19. CsrB untranslated RNA. There are 18 copies of the consensus sequence 5'-CAGGA(U,C,A)G-3' embedded in the 350 bp RNA (copies are numbered). These sequences bind 10 molecules of CsrA protein.

providing greater stress resistance and an enhanced ability to scavenge substrates from the medium. One induced pathway is the glycogen (*glg*) biosynthesis pathway, which may provide carbon and energy needed to survive the rigors of the stationary phase (see Chapter 8). The novel regulatory mechanism that in large part controls the expression of glycogen synthesis genes involves a 61 amino acid protein called CsrA and a 350 bp RNA called CsrB. As shown in Figure 5-19, CsrB contains 18 copies of the sequence 5'-CAGGA(U,C,A)G-3' that appear to bind 18 molecules of CsrA protein. CsrB RNA, in effect, serves as a CsrA “sink,” lowering the concentration of free CsrA. CsrA is itself a negative regulator of glycogen synthesis genes. Free CsrA binds to regions around the ribosome-binding sites of target RNAs and seems to facilitate message degradation. Since the stationary phase increases *glg* expression, entry into it might increase CsrB RNA levels to “soak up” more CsrA. Alternatively, CsrA protein levels could themselves decrease. Either way, the prediction is that the ratio of CsrA to CsrB is what controls CsrA activity, although this has not been tested to date.

CsrA also activates some glycolysis genes (e.g., *pfkA*, phosphofructokinase) and represses expression of some genes associated with gluconeogenesis (e.g., *fbp*, fructose-1,6-bis-phosphate phosphatase). Therefore, CsrA has an effect on these carbon utilization pathways opposite that of Cra (see above). How CsrA activates the expression of a gene is not clear but may involve CsrA negatively regulating a negative regulator of the actual target. In this case, the loss of CsrA through mutation would increase production of the negative regulator, which would decrease expression of the ultimate target. This is in contrast to the direct negative control of *glg* where loss of CsrA increases *glg* mRNA stability.

GLOBAL CONTROL NETWORKS

Escherichia coli has the ability to grow in or survive numerous environmental stress conditions including (1) shifts from rich to minimal media, (2) changes in carbon source, (3) amino acid limitations, (4) shifts between aerobic and anaerobic environments, (5) heat shock, (6) acid shock, and (7) numerous starvations such as phosphate, nitrogen, and carbon. Through all of these stresses, the cell modulates DNA replication, cell growth, and division in order to remain viable. The cell must coordinate many different regulatory circuits that control seemingly disparate aspects of cellular physiology. This integrated response to environmental stress is referred to as **global control**. Global regulatory networks include sets of operons and regulons with seemingly unrelated functions and positions scattered around the chromosome, yet they all are coordinately controlled in response to a particular stress.

Some attention should be given to the terminology used to describe regulatory networks. A **regulon** is a network of operons controlled by a single regulatory protein (e.g., the arginine regulon). A **modulon** refers to all the operons and regulons under the control of a common regulatory protein (e.g., the CRP modulon). The simplest way for all of the operons in a modulon to respond to a stress is through the production of a signal molecule (**alarmone**) that accumulates during the stress. Alarmones are often small nucleotides such as cyclic AMP or guanosine tetraphosphate (ppGpp). The term **stimulon** is used to describe all of the genes, operons, regulons, and modulons that respond to a common environmental stimulus. This word is useful when describing all of the coinduced or corepressed proteins that form a cellular response without knowing the mechanism of the response.

Communication with the Environment: Two-Component Regulatory Systems

Bacteria constantly sense and adapt to their environment in order to optimize their ability to survive and multiply. We have covered a variety of regulatory systems, each of which utilizes a regulatory protein (repressor/activator) as a sensor of the internal environment. We now discuss how bacteria sense and interpret changes in the external environment. The general theme here is that bacteria sense and respond to changes in the outside world through a network of two-component signal transduction mechanisms.

Stimuli sensed by two-component systems include pH, osmolarity, temperature, the presence of repellents and attractants, nutrient deprivation, nitrogen and phosphate availability, plant wound exudates, and others. In two-component systems, a protein spanning the cytoplasmic membrane (**sensor/kinase component**) will sense an environmental signal and then transmit a signal through phosphorylation/dephosphorylation reactions to regulatory protein components (**response regulators**) located in the cytoplasm. The phosphorylated cytoplasmic component then regulates target gene expression (Fig. 5-20). Although each system only responds to specific stimuli, they are functionally similar. This conclusion is based on the conservation of amino acid sequences identified within key domains of these proteins (Fig. 5-20).

As illustrated in Figure 5-20, the sensor kinase (or transmitter) protein is usually, but not always, a transmembrane protein. Typically, the extracytoplasmic domain (sensor) senses a change in the environment, which triggers an intrinsic histidine

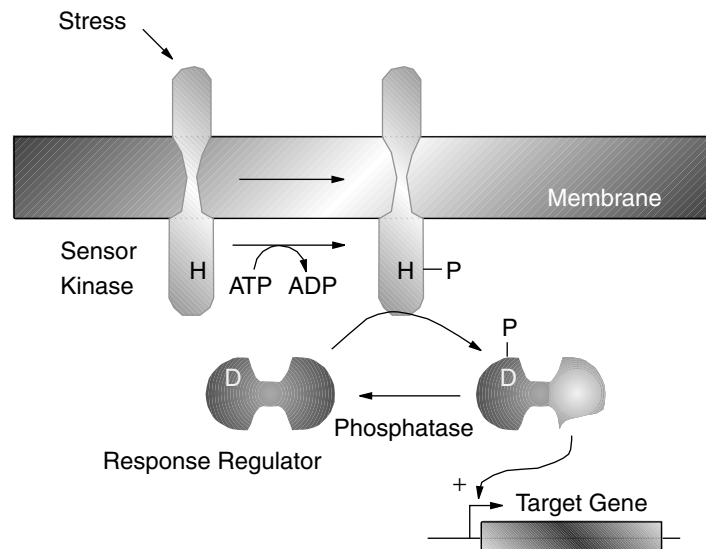


Fig. 5-20. General model for two-component signal transduction. An external stimulus interacts with the periplasmic (usually N-terminus) portion of the sensor/transmitter protein. This activates an autokinase activity in the C-terminal cytoplasmic region, which phosphorylates a conserved histidine residue. The phosphoryl group is then transferred from the sensor kinase histidine to a conserved aspartate residue in the receiver module of the response regulator protein. This phosphorylation induces a conformational change in the output domain (usually DNA binding), activating its DNA-binding function.

kinase activity that phosphorylates a conserved histidine residue in the cytoplasmic domain (**transmitter module**). The sensor/kinase transmitter domain then transfers the phosphate to a conserved aspartate residue in the amino-terminal end (**receiver domain**) of the regulator protein. Phosphorylation of the response regulator receiver domain induces a conformational change in the regulator or **output domain** located at the carboxyl end of the protein. The output domain of most of the known response regulators typically binds target DNA sequences, although some output domains are involved with protein–protein interactions (see “Chemotaxis” in Chapter 7).

Most, if not all, two-component systems are downregulated by specific phosphatases that dephosphorylate the aspartyl phosphate in the response regulator. Phosphatase activity can reside in a protein separate from the signal transduction proteins (e.g., SixA) or may be part of the sensor kinase itself (e.g., EnvZ). Thus, the regulatory switch for two-component signal transduction systems equates to changes in the ratio of kinase activity to phosphatase activity brought about by sensing an environmental signal. The result is a change in phosphorylated versus dephosphorylated levels of the cognate response regulator.

It is also important to recognize that phosphorylation of the response regulator does not necessarily lead to a downstream response, as might be assumed. The role of a given histidine kinase may be to maintain the response regulator in its phosphorylated state, which prevents it from triggering a response. The dephosphorylated regulator might, in fact, be the species that promotes subsequent events.

Sensor Kinases. Several examples of two-component signal transduction systems are listed in Table 5-1. The sensor kinase proteins can be grossly lumped into two classes with respect to membrane location. The integral membrane proteins show typical transmembrane topology. These include, among many others, EnvZ, which is discussed in detail in “Osmotic Control of Gene Expression” in Chapter 18. The second group of sensor kinase proteins are cytoplasmically located with no obvious transmembrane domains. NtrB and CheA are examples that are discussed in this chapter under “Regulation of Nitrogen Assimilation and Nitrogen Fixation” and in Chapter 7 under “Chemotaxis.”

In addition to membrane location, sensor kinases can be grouped by their domain structure. There are classic sensor kinases such as EnvZ that possess a sensor or **input domain** at the amino terminus and a **transmitter domain** at the carboxyl terminus containing the conserved histidine phosphorylation residue. Another group is referred to as the “hybrid” sensor kinase proteins. These resemble the classic sensor kinases except they also contain a response regulator–type receiver domain (Fig. 5-21). The hybrid sensor kinases undergo a sequential series of reactions starting with phosphorylation of the conserved histidine in the transmitter domain, which can either directly transfer the phosphate extrinsically to a cognate response regulator aspartate residue or undergo intrinsic phosphotransfer to a receiver module aspartate residue within the sensor kinase itself. A subsequent phosphotransfer must be made to a histidine in a Hpt domain (histidine phosphotransfer) that is either located at the carboxyl terminus of the sensor kinase (e.g., ArcA; see Chapter 18, oxygen regulation) or is a separate protein. In either case, a final phosphotransfer takes place from the histidine in the Hpt domain to the aspartate residue in the response regulator protein. Examples of classic and hybrid systems are shown in Figure 5-21. A useful Web site for signal transduction histidine kinases is <http://wit.mcs.anl.gov/WIT2/Sentra/HTML/sentra.html>.

TABLE 5.1. Some Properties of Various Two-Component Systems

Organism	Sensor/ Transmitter	Regulator/ Receiver	Response or Regulated Gene	Environmental Stimulus
<i>E. coli</i> / <i>S. typhimurium</i>	EnvZ	OmpR	<i>ompC</i> , <i>ompF</i>	Changes in osmolarity
<i>E. coli</i>	PhoR	PhoB	<i>phoA</i> , <i>phoE</i> , etc.	Phosphate limitation
<i>E. coli</i>	PhoM	PhoM-ORF2	<i>phoA</i> , etc.	Extracellular glucose
<i>E. coli</i>	CpxA	CpxR	<i>traJ</i> , etc.	Dyes and toxic chemicals
<i>E. coli</i>	UhpB	UphA	<i>uhpT</i> , etc.	Glucose-6-phosphate
<i>E. coli</i>	?	UrvC-ORF2	Unknown	Unknown
<i>E. coli</i>	NarX	NarL	<i>narGHJI</i>	Nitrate concentration
<i>E. coli</i> / <i>S. typhimurium</i>	NtrB(NRII)	NtrC(NRI)	<i>glnA</i> , etc.	Ammonia limitation
<i>E. coli</i> / <i>S. typhimurium</i>	CheA	CheB/CheY	Chemotaxis	Repellents and attractants
<i>S. typhimurium</i>	PhoQ	PhoP	<i>phoN</i> , etc., virulence	Magnesium, pH
<i>Klebsiella pneumoniae</i> / <i>Bradyrhizobium parasponiae</i>	NtrB	NtrC	<i>nifLA</i>	Nitrogen limitation
<i>Rizobium meliloti</i>	FixL	FixJ	<i>nifLA</i> , <i>fixN</i>	Nitrogen limitation
<i>R. leguminosarum</i>	DctB	DctD	<i>dctA</i>	C4-dicarboxylic acids
<i>Pseudomonas putida</i>	?	XylR	<i>xylCAB</i> , <i>xylS</i>	m-xylene or m-methylbenzyl alcohol
<i>Pseudomonas aeruginosa</i>	?	AlgR	<i>algD</i>	Not known
<i>Bordetella pertussis</i>	BvgC	BvgA/BvgC	<i>ptx</i> , <i>fha</i> , <i>cya</i> , etc., virulence	Temperature, nicotinic acid, MgSO ₄
<i>Agrobacterium tumefaciens</i>	VirA	VirG	<i>virB</i> , <i>virC</i> , etc., virulence	Plant wound exudate
<i>Bacillus subtilis</i>	?	SpoOA/SpoOF	Control of sporulation	Nutrient deprivation
<i>B. subtilis</i>	DegS	DegU	Regulation of degradative enzymes	Not known

Response Regulators. The second components (receiver/regulator proteins) of the sensory transduction systems can be grouped into at least four families. First, there are typical DNA-binding transcriptional activator proteins such as OmpR from *E. coli* (see “Osmotic Control of Gene Expression” in Chapter 18). A second family has, in addition to an N-terminal receiver module with conserved regions to the OmpR family, a conserved region in the C terminus (presumably involved with DNA binding) that is different from OmpR. Examples include BvgA of *Bordetella pertussis* and NarL and UhpA of *E. coli*. The third family of the receiver group, exemplified by DNA-binding proteins NtrC and NifA, has C-terminal regulator regions similar to each other but distinct from the other families. A fourth family, including CheY from *S. typhimurium* and SpoOF from *B. subtilis* only, has a receiver module. CheY is not a DNA-binding protein but interacts with other proteins at the flagellar motor (see “Chemotaxis” in Chapter 7).

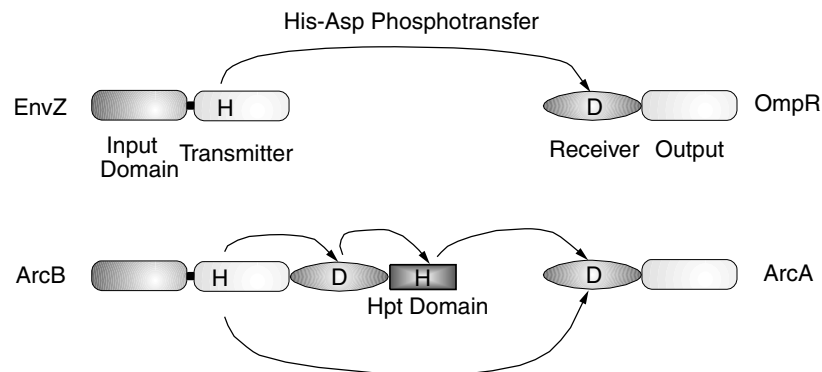


Fig. 5-21. Phosphotransfer reactions in classic (top) and hybrid (bottom) two-component systems. EnvZ and ArcB are sensor kinases. Their cognate response regulators are OmpR and ArcA, respectively. H and D indicate conserved histidine and aspartate phosphorylation target residues.

A one-component sensory transduction system has also been characterized that combines the features of the two-component system into one protein. The ToxR system from *Vibrio cholerae* mediates regulation of several virulence components for this intestinal pathogen in response to environmental stimuli. ToxR is a transmembrane sensor that also binds specifically to the promoter regions of virulence-factor genes through its cytoplasmic domain. Because it is a single protein, it does not require transmitter or receiver modules. There is homology at the DNA-binding cytoplasmic N-terminus region with the C terminus of the OmpR-like regulators. Therefore, it is classified as a member of the OmpR family.

Cross-Talk. Even though signal transduction proteins respond to different stimuli, they still exhibit a high degree of amino acid sequence conservation within key domains. Thus, you can gaze at the completed genome sequence and predict with some confidence the number of potential two-component systems contained in an organism's sensory array. *E. coli*, for example, has 32 potential two-component systems based on sequence analysis. The relatively high degree of sequence conservation between these different systems leads one to wonder about their fidelity. Can different systems end up “cross-talking” to each other? Under normal in vivo circumstances each system is able to specifically transduce its own signal. However, if a system is disturbed by mutations, cross-talk between systems can be observed whereby a component of one system can, at a low level, substitute for the missing component of another. It seems likely, then, that the various two-component systems present in the same bacterial cell may communicate with each other and form a complex and highly balanced regulatory network.

Regulation of Nitrogen Assimilation and Nitrogen Fixation: Examples of Integrated Biochemical and Genetic Controls

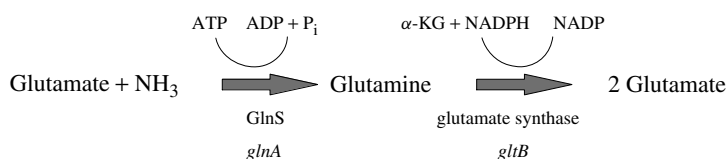
Discussion of several specific two-component regulatory systems are discussed later in the book where they are viewed in the context of their physiological role (see “Chemotaxis” in Chapter 7; Chapters 18 and 19). However, two basic examples are

presented here. The acquisition of carbon, nitrogen, and phosphate are major nutritional challenges for any organism. We have already discussed some regulatory aspects of carbon source utilization (e.g., CRP above) and now examine regulatory features governing the utilization of nitrogen and phosphate sources, both of which rely heavily on two-component signal transduction mechanisms.

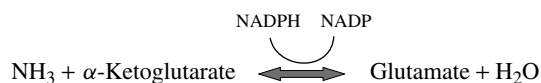
The preferred nitrogen source for enteric bacteria is ammonia, with the principal product of ammonia assimilation being glutamate. Glutamic acid, in turn, is the precursor of several amino acids and ultimately furnishes the amino group of numerous amino acids via transamination (see Chapters 1 and 14). Glutamate is also the precursor of glutamine, which is involved in the synthesis of other amino acids as well as purines and pyrimidines. A total of 85% of the cellular nitrogen is derived from the amino nitrogen of glutamate and 15% from the amide nitrogen of glutamine.

In the absence of ammonia, other nitrogen-containing compounds such as histidine or proline can serve as alternate sources of nitrogen. Rapid synthesis (induction) of these alternate nitrogen assimilatory pathways occurs during ammonia deprivation. This chapter deals with the regulatory mechanisms controlling various pathways of nitrogen assimilation and nitrogen fixation. Because of the ingenious way that *E. coli* entwines the biochemical and genetic controls of nitrogen metabolism, both forms of regulation are discussed here.

Ammonia Assimilation Pathways. Glutamine synthetase (GlnS) is centrally important to cells growing in medium containing less than 1 mM ammonia (NH_4^+). The pathway is illustrated below:



Under conditions where the NH_4^+ concentration is above 1 mM, the GlnS/GS system is inactive, but glutamate dehydrogenase (*gdh*) can be used for ammonia assimilation via the following reaction:



Left unabated, GlnS converts all of the cellular glutamate to glutamine, which is detrimental to the cell. As a result, GlnS activity and synthesis must be highly regulated. The precise regulation of GlnS is exerted by regulating expression of its structural gene, *glnA*, and regulating GlnS activity by reversible covalent adenylation. The link between genetic and biochemical controls is a protein called PII encoded by *glnB*. PII communicates with the glutamine synthetase adenylation enzyme that modulates the biochemical activity of GlnS. PII also communicates with the sensor kinase of a two-component signal transduction system that regulates *glnA* expression.

Biochemical Regulation. PII activity is regulated by small molecules that signal carbon and nitrogen status. The nitrogen signal molecule is glutamine while the carbon

signal is α -ketoglutarate. The sensing of both molecules keeps the assimilation of nitrogen in balance with the rate of carbon assimilation. The nitrogen signal (glutamine) is sensed by the GlnE adenylyltransferase (ATase) and the bifunctional GlnD uridylyl transferase/uridylyl-removing enzyme (Utase/UR). As illustrated in the center of Figure 5-22, Low levels of glutamine stimulate GlnD UTase activity, which uridylylates PII. PII-UMP binds to GlnE ATase and stimulates the deadenylylation of GlnS-AMP. This activates glutamine synthetase activity and increases ammonia assimilation.

High levels of glutamine do the opposite by stimulating GlnD uridylyl removal from PII. The unmodified form of PII activates the GlnE ATase, which adenylylates and inactivates GlnS, decreasing ammonia assimilation. However, when α -KG levels are high (meaning a large sink is still available to assimilate more ammonia), the PII activation of ATase activity is overridden even if glutamine levels are high. As a result, GlnS remains active and ammonia assimilation continues. This maintains a balance between available nitrogen and the available carbon sink for nitrogen (α -KG).

Genetic Control. As stated above, PII protein links the biochemical control of nitrogen assimilation to the genetic regulatory system, governing synthesis of the assimilating enzymes. The *glnA* (GlnS), *ntrB*, and *ntrC* loci form an operon as shown in Figure 5-23. Note that expression of the *glnA ntrB ntrC* operon is driven by three promoters: *glnAp1* (transcriptional start 187 bp upstream of *glnA* translational start), *glnAp2* (73 bp upstream of *glnA* translational start), and *ntrBp* (also known in the literature as *glnLP*; transcriptional start 256 bp downstream of *glnA* translational termination and 33 bp upstream of *ntrB* translational start). The *glnAp1* and *ntrBp* promoters both utilize σ -70 for initiation and have the appropriate -10 and -35 regions. They are not strong promoters but allow the cell to maintain a low level of glutamine synthetase as well as low levels of the nitrogen regulators NtrC (NRI) and NtrB (NRII) during growth in high nitrogen. In contrast, the *glnAp2* promoter is a

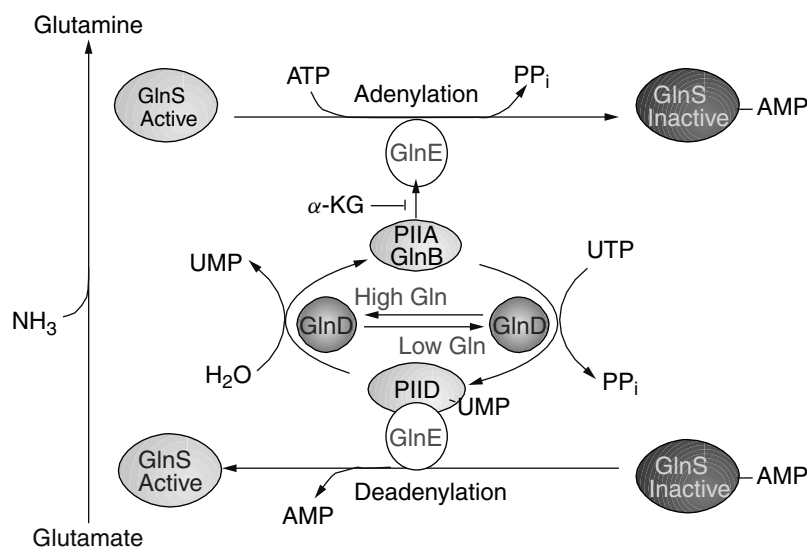


Fig. 5-22. Biochemical regulatory circuit controlling nitrogen assimilation.

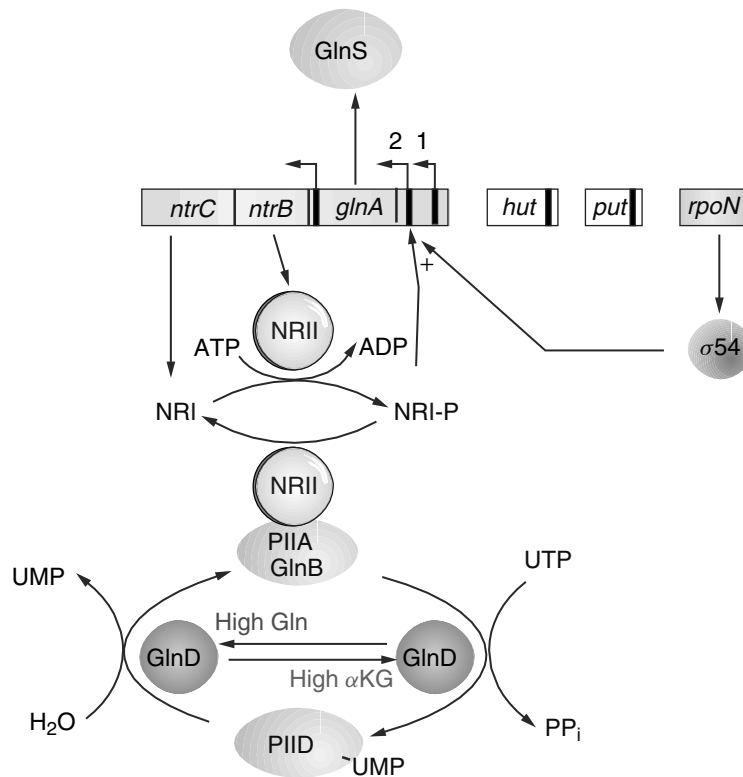


Fig. 5-23. Genetic regulation of nitrogen assimilation under high or low intracellular glutamine concentrations.

strong promoter that utilizes an alternate σ factor, σ -54, the product of *rpoN*. It contains the -10 consensus sequence TTGGCACAN₄TCGCT common among other nitrogen-regulated promoters. Nitrogen and carbon regulation of *glnA* expression centers on the activation of this promoter.

The NtrB and NtrC proteins, called NRII and NRI, respectively, form a two-component signal transduction system that regulates the expression of the *glnA* operon as well as other operons encoding nitrogen assimilation proteins (e.g., histidine or proline utilization). The NtrB (NRII) sensor kinase phosphorylates and activates the NtrC (NRI) response regulator when glutamine levels are low. The resulting NtrC-P is a transcriptional activator of the *glnA* operon. Increasing expression of this operon, of course, increases glutamine synthetase production and, thus, glutamine. When glutamine levels rise too high, the intrinsic phosphatase activity of NRII removes the phosphate from NRI and activation of *glnA* transcription stops. Regulation of this system hinges on changing the relative kinase and phosphatase activities of NRII in response to glutamine and carbon flux.

The regulatory cascade leading to NtrC phosphorylation once again begins with GlnB (PII). Along with interacting with the ATase as described, PII will modulate the kinase/phosphatase activities of the NRII sensor kinase, providing an elegant link between biochemical control of GlnS activity and the transcriptional activation of the *glnAntrBC* operon. Figures 5-23 and 5-24 illustrate this regulatory network.

How is the system shut off? As the level of glutamine rises in the cell, the UTase activity of GlnD is activated, which converts PII-UMP to PII (GlnB). In addition to *inactivating* GlnS ATase as described earlier, PIIA will activate the phosphatase activity of NRII that removes phosphate from NRI-P (Fig. 5-23). Since dephosphorylated NRI cannot activate the *glnAp2* promoter but will repress *glnAp1* and *glnLp*, the operon is deactivated and returns to background levels of expression.

This is not the whole story, however. As with biochemical control, the level of α -KG (the nitrogen sink) binds to and influences PII effects on the phosphatase/kinase

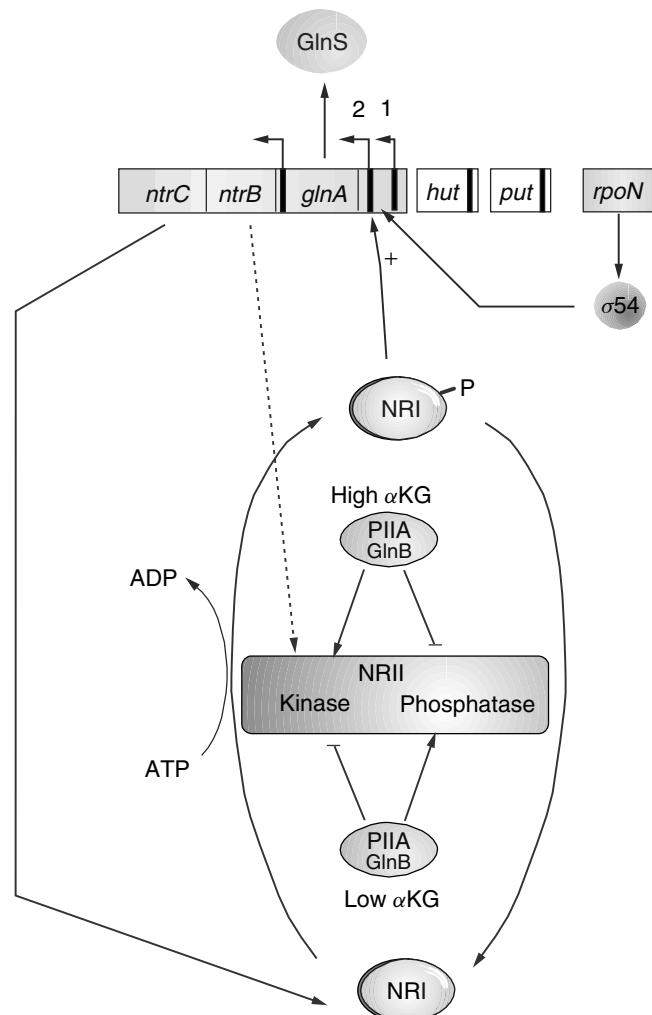


Fig. 5-24. Modulation of nitrogen regulation by α -ketoglutarate.

activities of NRII (Fig. 5-24). Thus, when glutamine levels are high and α -KG levels are low (not much of a nitrogen sink), a PII trimer binds a single α -KG molecule. This complex stimulates NRII phosphatase, inhibits NRII kinase, and, because the resulting NRI-P levels are low, shuts down *glnA* transcription. If the α -KG level remains high even in the presence of high glutamine, PII will bind three α -KG molecules. The PII:3 α -KG complex inhibits NRII phosphatase but stimulates kinase activity. This leaves NRI-P levels high and *glnA* transcription active. Thus, the cell can continue to assimilate ammonia even though intracellular glutamine levels are high.

Enhancer Sequence. It is of considerable importance that the binding site for NRI at *glnAp2* can be moved over 1000 bp away from the original site without diminishing the ability of NRI to activate transcription from *glnAp2*. Thus, the NRI-binding sites do not function as typical operator regions but resemble **enhancer sequences** found in eukaryotic cells. Enhancers work by increasing the local concentration of an important regulator near a regulated gene by tethering it to the region. NtrC(NRI)-P initially binds its DNA target as a dimer but eventually ends up as a large oligomer in which some of the NRI molecules are bound only through protein-protein interactions. Activation of *glnp2* expression does not involve recruitment of $E\sigma^{54}$ by NtrC-P but actually involves DNA-bound NtrC-P interacting with an $E\sigma^{54}$ complex already bound to *glnp2*. This interaction, via a DNA loop, converts the $E\sigma^{54}$ -promoter complex from a closed to an open form, thereby initiating transcription.

The cell has other systems designed to capture nitrogen from amino acids that may be available in the environment. These include the *hut* (histidine utilization) and *put* (proline utilization) operons. Both systems require NRI-P and σ -54 RNA polymerase for their expression and thus are regulated in a manner similar to the *glnA* operon.

Control of Nitrogen Fixation. Along with the general nitrogen assimilation regulatory system (*ntr*) just described, *Klebsiella pneumoniae* can fix atmospheric nitrogen (N_2). Nitrogen fixation in *Klebsiella* involves 17 genes all participating in some way with the synthesis of nitrogenase, the enzyme complex that converts N_2 to NH_3 (see Fig. 14-2). The *nifHDK* transcriptional unit codes for nitrogenase proper while the remaining *nif* genes are responsible for (1) synthesizing the Mo-Fe cofactor, (2) protein maturation, (3) regulation, or (4) unknown functions. The *ntrC* and *ntrB* products will also regulate *nifLA* transcription in a manner analogous to what was described above. The *nifAL* operon is activated by NRI-P, which results in increased intracellular concentration of NifA. NifA can then activate expression of the other *nif* operons. NifA activity is regulated by the NifL product in response to oxygen and probably ammonia, although the mechanism is not known. The *nif* system is active only under anaerobic and low NH_4^+ conditions. For additional discussion of nitrogen fixation and its regulation, see Chapter 14.

Phosphate Uptake: Communication Between Transport and Two-Component Regulatory Systems

Another global regulatory system studied extensively is the phosphate-controlled stimulation of *E. coli*. Expression of the majority of phosphate-regulated loci is specifically **phosphate-starvation inducible** (*psi*). However, other phosphate-controlled loci are **phosphate-starvation repressible**. Many of the *psi* genes encode proteins associated

with the outer membrane or the periplasmic space and function either in the transport of inorganic phosphate across the cell membrane or in scavenging phosphate from organic phosphate esters. Two of these inducible genes, *phoA* (alkaline phosphatase, PhoA) and *phoE* (outer membrane porin protein, PhoE), have been studied and shown to be under the control of a complex regulatory circuit. This circuit involves the two-component regulators *phoR* (sensor kinase) and *phoB* (response regulator), *phoU*, and a high-affinity phosphate transport system.

There are actually two transport systems for inorganic phosphate: the low-affinity PIT (phosphate inorganic transport) and an inducible high-affinity PST (phosphate-specific transport). In addition to *phoU*, *phoR*, and *phoB*, the *pst* operon also plays a role in regulation. This was first evident upon observing that *pst* mutants lacking the PST transport system constitutively synthesize alkaline phosphatase (derepressed) even though they grow well and maintain a high internal phosphate concentration using the PIT system. Clearly, phosphate itself is not the corepressor for this system. The system really works through communication between the PST system and PhoR.

A model for phosphate-controlled gene regulation is outlined in Figure 5-25. PhoR and PhoB are members of a two-component regulatory system. PhoR is an integral membrane-sensing protein that can trigger induction of the Pho regulon under limited phosphate conditions (less than $4 \mu\text{M}$). When the cell senses low P_i levels, the PhoR protein is autophosphorylated. PhoR-P then transphosphorylates PhoB. PhoB-P activates transcription of target genes by binding to the consensus “Pho-box” sequences (CTTTTCATAAACTGTCA) located upstream of *pho* regulon promoters. As extracellular phosphate levels increase, transport through PST increases, and PstB,

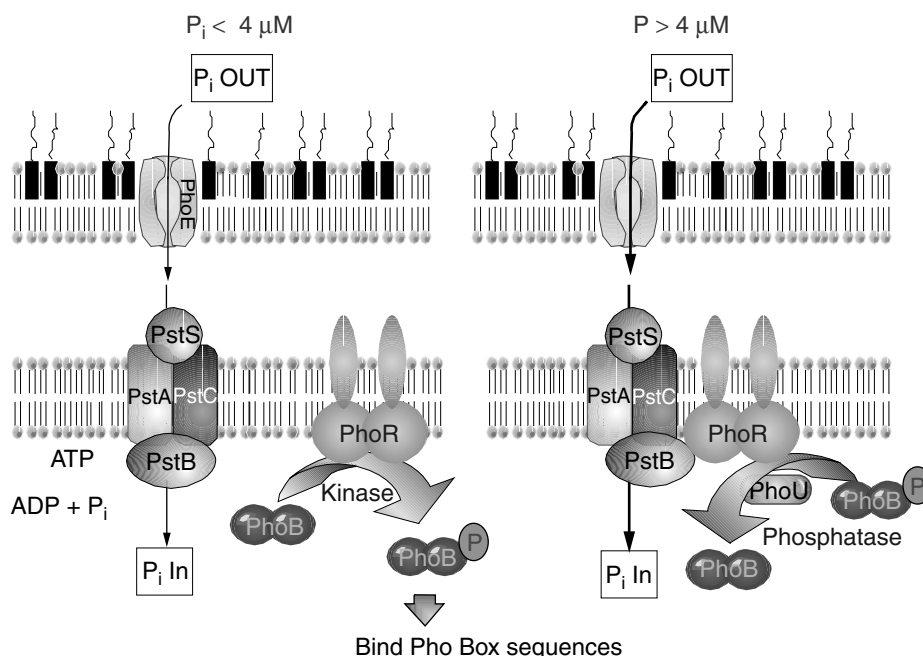


Fig. 5-25. Model for the role of PhoM, PhoR, and PhoB in phosphate-regulated gene expression.

a member of the high-affinity transport system, will interact with PhoR and inhibit its kinase activity. Next, PhoR, together with another protein, PhoU, will dephosphorylate PhoB, thereby turning off transcription of the *pho* regulon. So, this system successfully links the level of inorganic phosphate transport to the expression of phosphate-scavenging systems through protein–protein interactions between a member of the transport system (PstB) and the governing sensor kinase (PhoR).

Quorum Sensing: How Bacteria Talk to Each Other

Thirty years ago, A bacterium was thought to be a self-contained bag of protoplasm that had no sense of community. This concept was shattered with the discovery of a microbial cell-to-cell signaling phenomenon known now as **quorum sensing**. *Vibrio fischeri* is a free-living, bioluminescent bacterium that can become a symbiont of squid, colonizing their light organs. The bacteria only emit light, however, when they are present at a high cell density as can be found in the contained space of the light organ. The gene cluster (*lux*) responsible for bioluminescence is shown in Figure 5-26. LuxI is an enzyme that synthesizes the autoinducer molecule N-(3-oxohexanoyl)-L-homoserine lactone. At low cell densities, there is very little expression of the *luxICDABEG* operon and, thus, very slow production of the autoinducer. As the cell density increases, the autoinducer will accumulate in the medium. As the concentration of autoinducer increases, it will diffuse across the membrane back into the cell and bind the LuxR protein. The LuxR–AI complex activates transcription of the divergent *lux* operon, increasing the production of enzymes responsible for bioluminescence. The organism then emits light.

In contrast to the relatively simple system in *V. fischeri*, *Vibrio harveyi* possesses two autoinducer-response systems that function in parallel to control the density-dependent expression of the luciferase structural operon *luxCDABE* (Fig. 5-27). Autoinducer-1 (AI-1) is a homoserine lactone molecule, but the structure of AI-2 is not fully elucidated. Another difference with the *V. fischeri* system is that the *V. harveyi* *lux* inducer systems utilize complex two-component response regulators to sense levels of extracellular autoinducer rather than simple diffusion of AI across the cell membrane. As is typical of two-component regulatory systems, signals are transmitted via a series of phosphorylation/dephosphorylation reactions to a DNA-binding response regulator protein. However, the two transmembrane histidine kinases that detect each autoinducer (LuxN for AI-1 and LuxQ for AI-2) channel their signal (via phosphorylation) to a

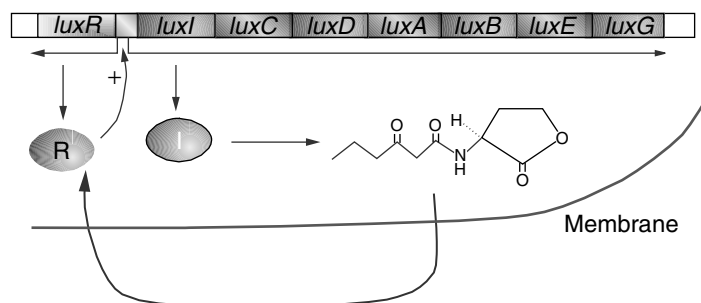


Fig. 5-26. Cell-to-Cell communication. The *lux* operon operon of *Vibrio fischeri*.

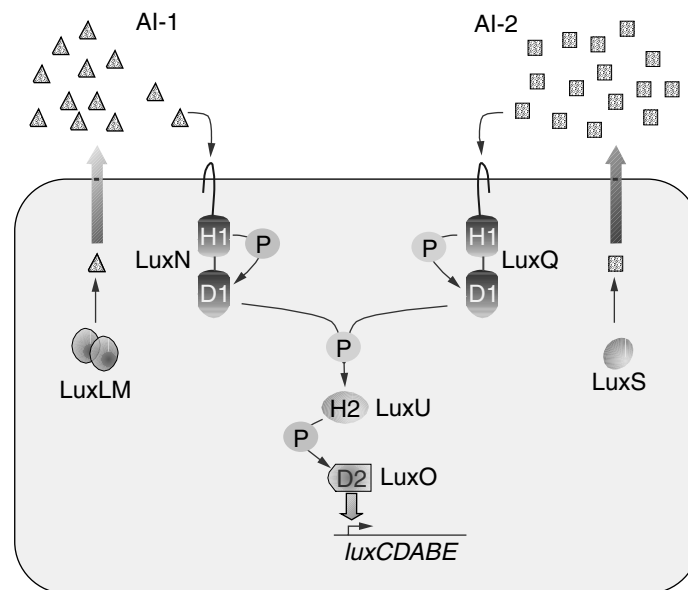


Fig. 5-27. Quorum sensing. The luciferase systems of *Vibrio harveyi*.

shared integrator protein (LuxU). LuxU then transphosphorylates the LuxO response regulator, which in turn activates *luxCDABE* expression. Complex two-component systems generally have a hybrid transmembrane protein with two modules: a sensor kinase that becomes phosphorylated at a conserved histidine residue and a C-terminal response regulator module that becomes phosphorylated at a conserved aspartate residue. The phosphate is then transferred to a conserved histidine on a phosphorelay protein (LuxU in this case) and then finally to a conserved aspartate on a second response regulator (LuxO).

The reason for dual systems appears to involve whether the organism is communicating with its own species or with other species. Homologs of LuxS have been identified in many gram-negative and gram-positive species. Amazingly, the AI-2 autoinducers produced from many of these species can “talk” to *V. harveyi* via the LuxQ AI-2 sensing system. Mutants of *V. harveyi* that do not make AI-2 (*luxS* mutants) will bioluminesce when in the presence of other species making AI-2 molecules! The capacity to respond to both intra- and interspecies signals could allow *V. harveyi* to know not only its own cell density but also its proportion in a mixed population. Although evidence indicates that *E. coli* and *S. enterica* produce AI-2-like molecules, their role in the physiology of these organisms is unclear at present. Recall that other autoinducer systems were discussed in the section on transformation in Chapter 3.

Proteolytic Control

Lon Protease. The *lon* locus of *E. coli* codes for a protease that degrades “abnormal” proteins (see Chapter 2). Mutations in the *lon* locus have pleiotropic effects including an increased UV sensitivity, filamentous growth, and mucoid colonies due to capsule overproduction. UV sensitivity and filamentous growth due to *lon*, explained in

Chapter 3, results from an inability to degrade the SOS-inducible cell division inhibitor SulA. The mucoid phenotype is the result of decreased degradation of RcsA, a positive regulator of capsular polysaccharide synthesis. In both cases, the proposed role of Lon protease is to assure that certain regulatory proteins persist for only a short time. However, if we define a global regulator in terms of initiating a coordinated response of a given set of operons to a specific stimulus, then *lon* does not qualify as such. Rather than initiating a response, Lon acts to limit responses and may be important as a mechanism for maintaining and returning a cell to equilibrium. For a description of the interaction of *lon* protease with the cell division process, see Chapter 17.

The Alternate Sigma Factor σ^S , the ClpXP Protease, and Regulated Proteolysis. As *Salmonella* and *E. coli* enter stationary phase or when exponentially growing cells experience a sudden stress such as acid or high osmolarity, the level of an alternate σ factor, called σ^S (38 kDa, encoded by *rpoS*), dramatically increases and directs RNA polymerase to a variety of stress-inducible gene promoters. While the *rpoS* gene is subject to some transcriptional control, the primary reasons for the stress-induced elevation in σ^S levels involve increased translation of *rpoS* message and decreased proteolysis of σ^S protein. The protease responsible for degrading σ^S is ClpP and its associated ClpX recognition subunit (see Chapter 2). Under rapidly growing conditions, ClpXP rapidly degrades σ^S , but following encounters with stress, degradation slows, which allows σ^S to accumulate.

However, ClpXP alone cannot recognize σ^S . An adaptor recognition protein called RssB (regulation of sigma S, also known in *Salmonella* as *mviA*, mouse virulence gene) is required to present σ^S to ClpXP. Recent evidence indicates that the amino terminus of RssB (*mviA*) binds σ^S while the carboxy terminus binds ClpX. Regulation of σ^S degradation is mediated by RssB, apparently through phosphorylation of a conserved aspartate residue within an amino-terminal domain that bears homology to the receiver domains of two-component response regulators. Once phosphorylated, RssB efficiently presents σ^S to ClpXP. The cognate sensor kinase has not been identified, but acetyl phosphate has been implicated as a phospho donor.

SUMMARY

In this chapter, the basic concepts of how cells regulate gene expression are introduced. It should be more and more apparent that no single system operates independently in the cell. There are several layers of control that monitor and modulate the expression of any given gene. The reason for this overlapping control is clearly the rule of economy. Ensuing chapters discussing the biochemical aspects of cellular metabolism can now be viewed in the overall context of regulatory networks where one metabolic system impacts other seemingly far-removed systems.

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CHAPTER 6

BACTERIOPHAGE GENETICS

Even after nearly 100 years of study, bacterial viruses continue to play an important role in the development of bacterial genetics. Their study has been instrumental in revealing many important biological processes applicable to prokaryotic and eukaryotic organisms alike. For example, T4 and ϕ X174 have been important tools used to reveal cardinal features of the DNA replication fork. Likewise, λ phage was integral to studies dealing with the initiation of DNA replication. Interchangeable σ factors were discovered using T4 while λ proved central to our understanding of transcriptional repressors and activators.

Another phage we discuss, μ , has taught us much regarding DNA rearrangements. As you will see, bacteriophages remain critically important research tools. For example, the λ family of phages has enjoyed a resurgent interest in the genomic era. The availability of complete genome sequences has revealed a pivotal role for members of this phage family in the development of *E. coli* virulence. The genomes of many pathogens contain entire sections, called **pathogenicity islands**, that encode virulence genes with an AT content different from the rest of the chromosome. The DNA flanking these islands often contain remnants of viral genes, like scars marking how the organism may have acquired these genes. (see “Type III secretion” in Chapter 20). In addition to their evolutionary significance, the unique recombination systems of λ have been exploited to simplify prokaryotic and eukaryotic genetic engineering strategies. Even more remarkable is the renewed debate over an old idea: the potential therapeutic use of bacterial viruses as antimicrobial agents. Clearly, phage genetics has shaped modern biology and continues to be a critical factor in its development.

GENERAL CHARACTERISTICS OF BACTERIOPHAGES

The first step toward understanding bacteriophages is to learn something about their classification. However, characterization and classification of the bacteriophage viruses

poses many problems. From a practical point of view, they can be distinguished on the basis of their natural host, host range, and other similar characteristics. In addition, they can be characterized on the basis of their RNA or DNA content. Each phage generally contains only one kind of nucleic acid (the filamentous viruses have been shown to be an exception to this general rule). Structural symmetry and susceptibility or resistance to ether and other solvents and to other chemical agents also provide additional criteria for classification (Table 6-1). There are over 5000 phage types that comprise 13 phage families (Table 6-2).

Bacteriophages range in size from 20 to 300 nm. Large phages are barely within the realm of resolution of the light microscope. For visualization of any details of viral structure, the electron microscope is essential. The nucleic acid may be either double stranded, single stranded, circular, or linear. Morphologically, phages generally display a considerable degree of geometric symmetry. They generally have a head or **capsid** composed of identical protein subunits called **capsomeres**. The nucleic acid **core** is housed within the capsid. Some bacteriophages have a prominent tail structure. The capsid structure may assume an icosahedral form (20 sides with 12 vertices). The entire infectious unit is generally referred to as a **virion**. The smallest phages contain nucleic acid (either DNA or RNA) with a molecular weight on the order of 1×10^6 daltons. This relatively short-chain nucleic acid can obviously only code for a small number of genes. These genes, of necessity, must encode information that governs the formation of the basic viral subunits. The presence of such a small number of genes reflects the degree to which viruses are dependent on the host cell.

Each species of microorganism is susceptible to a limited subset of viruses. Put another way, each phage has a specific (and limited) host range of susceptible microorganisms that it can infect. For example, a hypothetical phage A might only infect one strain of *E. coli* while phage B could infect many different strains of *E. coli*. Phage A is viewed as having a narrow host range relative to that of phage B. Host range depends on the presence of specific viral receptors on the host cell surface. These receptors are usually composed of specific carbohydrate groups on lipopolysaccharides (LPS) of cell surface structures. Some viruses, such as M13, attach to sex pili and are referred to as male-specific phages. The salient point here is that receptors are actually normal cellular proteins with a specific function that are co-opted by the phage.

Two types of infection cycles may occur following the initial infection. One is a **virulent** or **lytic** infection that ends in lysis and death of the host cell. The other type of infection is called **lysogenic**, which may be quite unapparent since the host cell does not die. In fact, the general appearance and activity of the lysogenized host cell may not be altered in any overt manner. A lytic type of phage only undergoes virulent infection. Temperate phages, on the other hand, can undergo either lytic or lysogenic infection.

Bacteriophage infections of bacteria follow a characteristic pattern. The growth curve of bacteriophages, sometimes referred to as the one-step growth curve, is depicted in Figure 6-1. Once a population of bacterial cells has been inoculated with a given number of bacteriophages, the number of detectable infectious particles rapidly decreases. This stage is termed the **eclipse phase** and represents that portion of the time following infection during which bacteriophages cannot be detected either in the culture medium or within the cell. The overall period encompassing adsorption and eclipse is referred to as the **latent period**. After the latent period has been completed (approximately 20 minutes for *E. coli* cells infected with coliphage), infectious viruses

TABLE 6-1. Characteristics of Some Phages of *Escherichia coli*

Phages	Virion Morphology			Growth Cycle at 37 °C			
	Head (nm)	Tail (nm)	Nucleic Acid, Type and Amount (daltons)	Latent Period (min)	Average Yield per Cell	Lysogeny	Peculiarities
T1	Icosahedral 50	10 × 150	DNA, 2.5 × 10 ⁷	13	150	—	Resistant to drying
T2, T4, T6	Prolated icosahedral 65 × 95	25 × 110	DNA, 1.2 × 10 ⁸	21–25	150–400	—	Contain glucosylated HMC
T3, T7	Icosahedral 47	10 × 15	DNA, 2.4 × 10 ⁷	13	300	—	Give semitemperate mutants
T5	Icosahedral 65	10 × 17	DNA, 7.5 × 10 ⁷	40	200	—	DNA injection in two steps
λ, φ80	Icosahedral 54	10 × 140	DNA, 3.3 × 10 ⁷	35	100	+	DNA circularizes, <i>in vitro</i> or <i>in vivo</i>
P1	Icosahedral 65	12 × 150	DNA, 6 × 10 ⁷	45	80	+	General transduction
P2	Icosahedral 50	10 × 150	DNA, 2.2 × 10 ⁷	30	120	+	Multiple chromosomal locations
φX174 S13	Icosahedral 30	None	DNA, 1 strand 1.7 × 10 ⁶	13	180	—	Circular DNA
f2, MS2	Icosahedral 24	None	RNA, 9 × 10 ⁵	22	20,000	—	Male specific attachment to F pili
f1, fd	None	6 × 800	DNA, 1 strand 1.3 × 10 ⁶	30	100–200 (continuous release)	—	Male specific, circular DNA
χ ^a	Icosahedral 67.5	12.5 × 230	DNA	60	200	—	Attaches to motile flagella

^aNot a coliphage; grows on many strains of *Salmonella*.

TABLE 6-2. Families of Bacterial Viruses

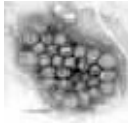
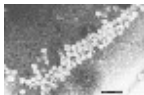
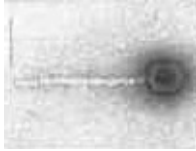
Corticoviridae icosahedral capsid with lipid layer, circular supercoiled dsDNA	
Cystoviridae enveloped, icosahedral capsid, lipids, three molecules of linear dsRNA	
Fuselloviridae pleomorphic, envelope, lipids, no capsid, circular supercoiled dsDNA	
Inoviridae, genus Inovirus long filaments with helical symmetry, circular ssDNA	
Leviviridae quasi-icosahedral capsid, one molecule of linear ssRNA	
Lipothrixviridae enveloped filaments, lipids, linear dsDNA	
Microviridae icosahedral capsid, circular ssDNA	
Myoviridae icosohedral capsid, dsDNA	See T4 and μ Phages
Plasmaviridae pleomorphic, envelope, lipids, no capsid, circular supercoiled dsDNA	(Image not available)
Podoviridae tail short and noncontractile, head isometric	
Rudiviridae helical rods, linear dsDNA	(Image not available)

TABLE 6-2. (Continued)

Siphoviridae, morphotype B1 tail long and noncontractile, head isometric	
Tectiviridae icosahedral capsid with inner lipoprotein vesicle, linear dsDNA, “tail” produced for DNA injection	

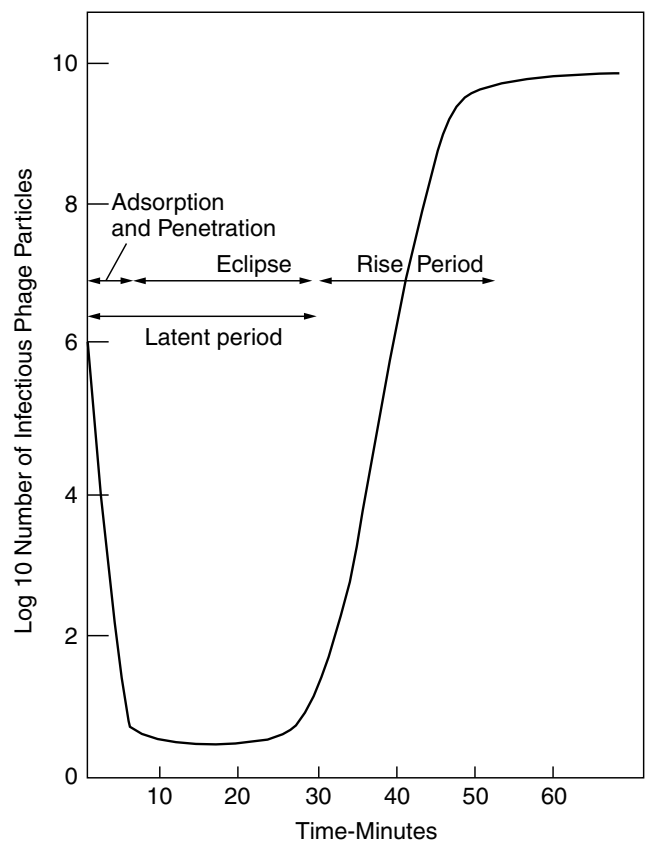


Fig. 6-1. Typical one-step growth curve of bacteriophages.

begin to be released from the cell. The average number of infectious bacteriophage particles released per cell is referred to as the **burst size**. The replication cycle may be depicted as follows:

1. Attachment (see Fig. 6-2).
2. Injection of nucleic acid or penetration of the entire virus into the cell.

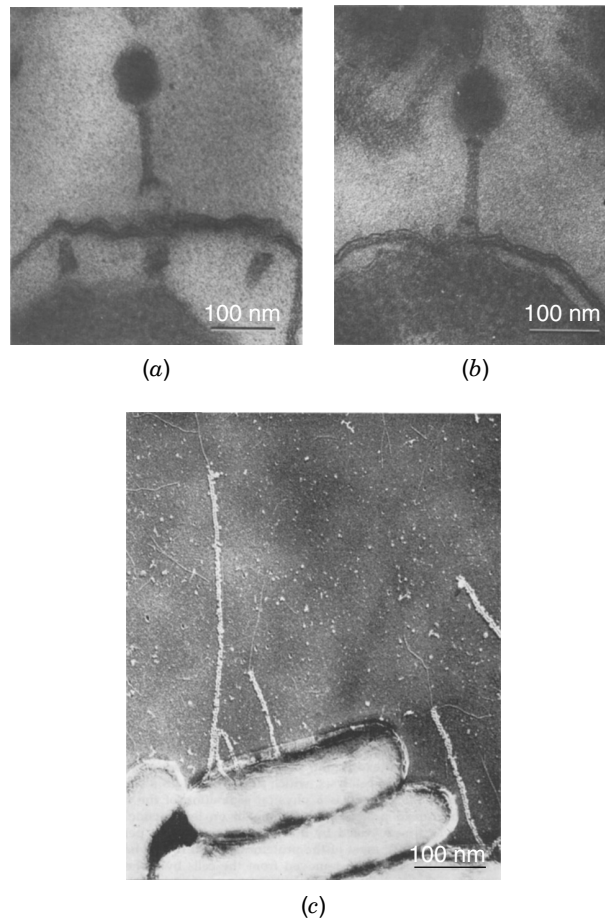


Fig. 6-2. Bacteriophage attachment to cells and pili. (*a, b*). Attachments of T-even virions to points of adhesion between cytoplasmic membrane and outer membrane (cell wall) of *E. coli*. Bars equal 100 nm. (*C*) *E. coli* cells with F pili showing attached particles of MS2 (icosahedral) and M13 (filamentous) phages. (Source: From Luria et al., (1978). General Virology, 3rd Edition, Wiley, NY.

3. Uncoating (i.e., removal of the capsid surrounding the nucleic acid “core”).
4. Intracellular synthesis of viral subunits (capsid and nucleic acid).
5. Assembly of the subunits into complete virus particles (maturation). Complete bacteriophages within an *E. coli* cell are shown in Figure 6-3.
6. Release of viruses (burst or lysis).

Steps 1 through 4 comprise the latent period of which steps 2, 3, and 4 represent eclipse.

Lysogenic infection occurs when the nucleic acid of the virus enters the host cell, becomes aligned with the genome of the cell, and eventually is integrated into the cell genome. In this way, the virus is replicated in unison with replication of the host cell genome. At some later time, information coded within the viral genome may be **induced** to function. This triggers replication of the virus and entry into the lytic stage.

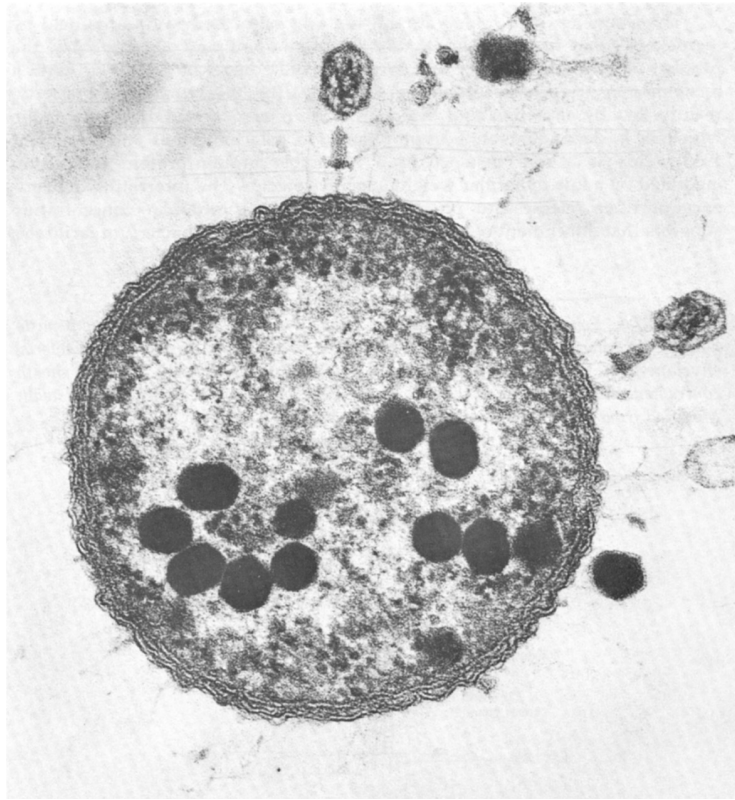


Fig. 6-3. A section of a cell of *E. coli* strain B from a culture infected with bacteriophage T2. Clearly visible are the cell envelopes (wall and cytoplasmic membrane); the clear area of the phage DNA pool containing many condensed phage DNA cores; and attached to the cell surface, three phages, one empty and two still partially filled. The phage at the top shows the long tail fibers and the spikes of the tail plate in contact with the cell wall. The tail sheath is contracted and the tail core has apparently reached the cell surface but has not penetrated it. (Courtesy of Dr. Lee D. Simon.)

Tumor viruses may initiate lysogenic infections of animal cells in a manner comparable to lysogenic infections of bacteria by bacteriophages. Additional discussion regarding the genetics of four specific bacterial viruses, each of which represents a different lifestyle, may be found next.

T4 PHAGE

Structure

T4 phage is one member of the Myoviridae family of T-even (T2, T4, T6) lytic phages of *E. coli*. Figure 6-4 presents the structure of a typical T-even phage of *E. coli* (T4). T4 possesses an oblong head (80×120 nm) and a contractile tail (95×20 nm). Within the capsid is a linear, double-stranded DNA molecule (1.3×10^8 daltons or 1.69×10^5 base pairs). Associated with the DNA are a number of polyamines (putrescine, spermidine, and cadaverine). The base composition of T4 DNA is unusual in that all of

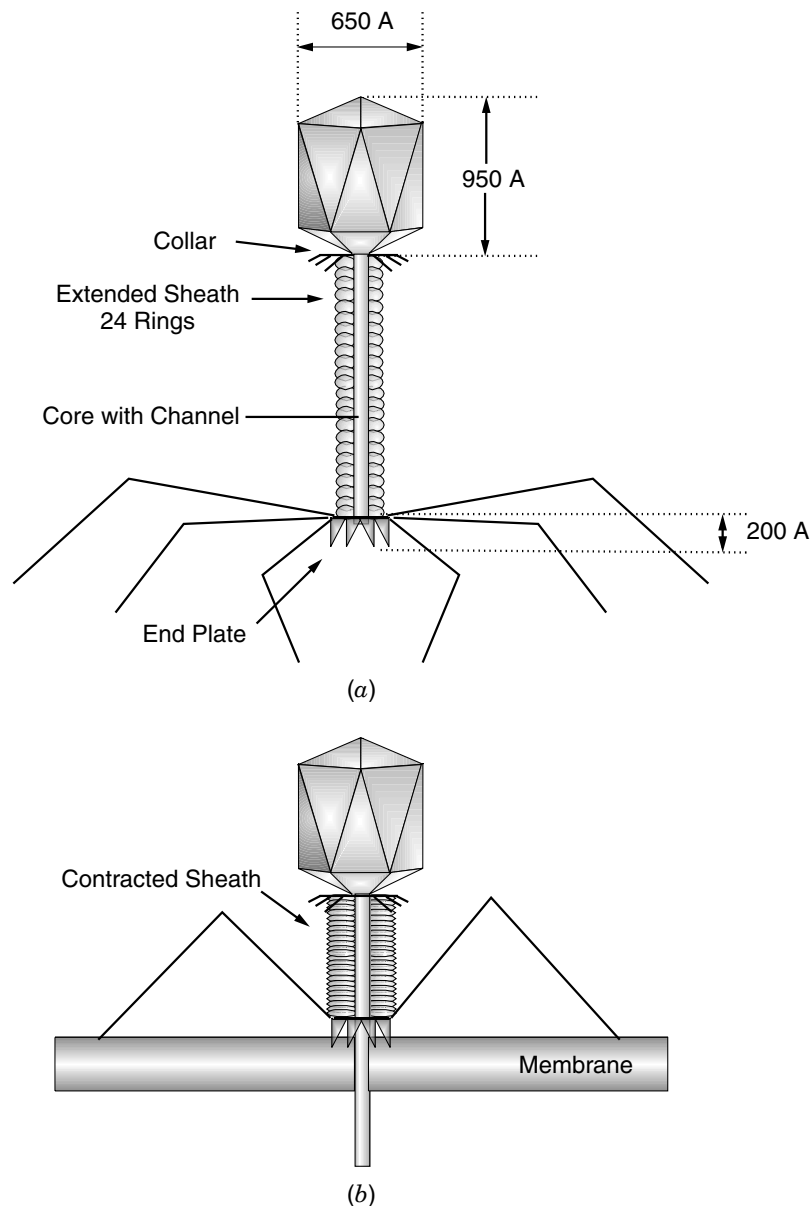


Fig. 6-4. Schematic representation of phage T4 virion, its component parts and penetration of the phage core through the bacterial envelope. (a) Virion with extended tail fibers. (b) Virion with the tail sheath contracted and the spikes of the tail plate pointing against the bacterial cell wall. (Source: Adapted from Luria et al., 1978.)

the cytosine residues have been replaced by a modified base, hydroxymethylcytosine (HMC). The reason for this replacement will become apparent from subsequent discussion. While virions in general are thought not to possess enzymes, a number of enzymatic activities have been found to be associated with the T4 virion. Several of these are listed in Table 6-3 along with their possible roles in T4 infection.

TABLE 6-3. Virus-Specific Enzymatic Activities Associated with T4 Virions

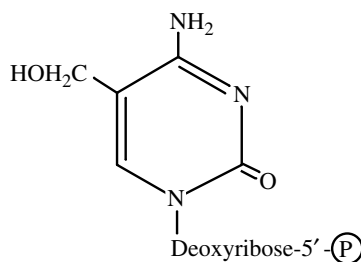
Enzyme	Location and/or Function
Dihydrofolate reductase	Located in base plate, may have role in unfolding tail fibers
Thymidylate synthetase	Found in base plate; necessary for infectivity
Lysozyme	Possible role in penetration through cell wall
Phospholipase	Possible role in host cell lysis
ATPase	Associated with the tail sheath; presumed to be involved with the contractile process
Endonuclease V	Excision repair (see 13.1A) of phage or host DNA
<i>alt</i> function	Alteration of the host RNA polymerase

The one-step growth experiment described earlier was originally developed with T4. The curve reflects a series of events beginning with the tail fibers of T4 binding to specific *E. coli* cell surface receptors (adsorption). Once the pins of the base plate contact the outer surface of the cell, the tail sheath contracts and the tail core penetrates the cell wall (Fig. 6-4). The T4 DNA molecule and associated proteins are then injected into the host. Animations of T4 phage attachment and injection can be found within the phage images section at www.phage.org.

General Pattern of T4 Gene Expression

Early Gene Expression. It is a general strategy of viruses to defer some gene expression to later stages of infection and progeny virus maturation. T4 phage is no exception. Transcription of certain T4 genes begins almost immediately. The **immediate-early genes** are transcribed at 30 seconds postinfection and the **delayed-early** or **middle genes** are transcribed at 2 minutes postinfection (model for T4 infection in Fig. 6-5). The products of these genes establish the infection. Since transcription of the immediate early genes is directed by σ^{70} promoters, normal host RNA polymerase is used. Among the early gene products are nucleases and other T4 proteins that stop host transcription, unfold the host chromosome, and degrade host DNA to nucleotides. Several enzymes are involved with the synthesis of T4 DNA including a T4 DNA polymerase (gene product 43, gp43).

Stealth Technology. A unique feature of T4 DNA is that its DNA contains the unusual base 5-hydroxymethylcytosine (hm-dCTP):



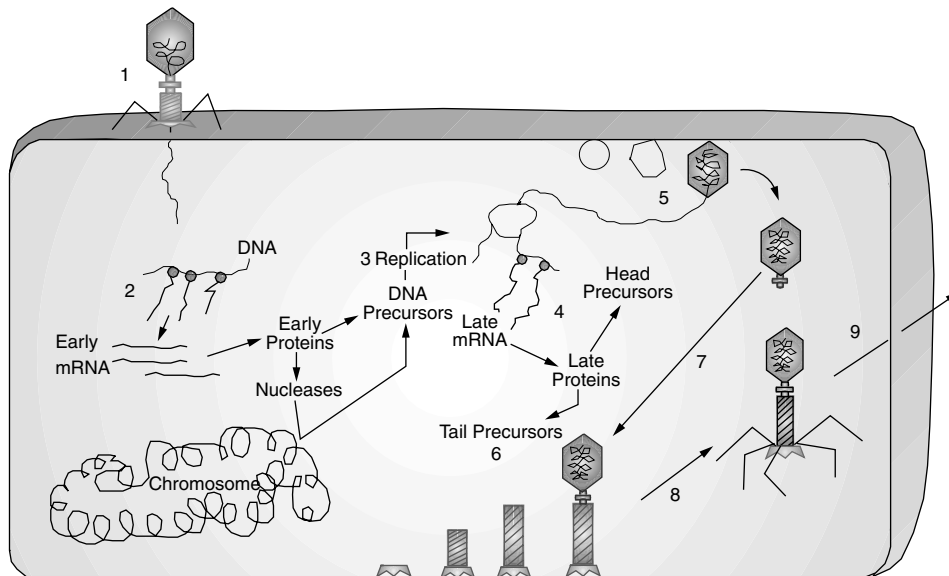


Fig. 6-5. Life cycle of T4 phage. (1) Attachment and injection of DNA; (2) transcription of early genes; (3) replication and concatemer formation; (4) transcription of late genes; (5) head assembly; (6) tail assembly; (7) attachment of head to tail; (8) attachment of tail fibers; (9) cell lysis and release of mature phage.

This base is synthesized as hm-dCMP from dCTP by a T4 enzyme prior to incorporation of the base into T4 DNA. Following incorporation, this nucleotide is glucosylated by a T4-encoded HMC glucosyltransferase. These modifications serve two basic functions: (1) phages with HMC-modified DNA can use cytosine-specific endonucleases to scavenge deoxyribonucleotides from host DNA without damaging their own DNA; and (2) the HMC-modified phages can shuttle back and forth undetected between strains carrying cytosine-specific restriction-modification systems, since the phages do not contain cytosine proper (see “Restriction and Modification” in Chapter 3).

Middle Gene Expression. The transcriptional switch from immediate-early to middle genes is associated with a series of T4-directed modifications as shown in Table 6-4. Alt and Mod proteins catalyze the ADP ribosylation of Arg265 in the RNA polymerase α subunits. This modification lowers the affinity of core polymerase for the housekeeping σ^{70} protein and is part of a strategy to reduce host RNA synthesis. The Alt protein is actually packaged in the T4 capsid and injected with the DNA during infection. Mod, on the other hand, is synthesized after infection. Transcription of middle genes requires the MotA protein, a positive regulator that binds to a TAT/AGCTT DNA sequence 13 bp upstream of a -10 box, converting the region into an active promoter even though the region lacks a -35 box. T4 also produces a 10 kDa anti- σ^{70} protein (AsiA) that prevents transcription of the T4 σ^{70} early gene promoters. AsiA has been shown to bind region 4.2 of σ^{70} (Chapter 2), blocking interaction between σ^{70} and the -35 upstream promoter element of early T4 genes and most *E. coli* genes.

TABLE 6-4. T4-Evoked Changes of *E. coli* RNA Polymerase

Change	Time First Detectable in RNAP (min after infection at 30 °)	Effect or Function
ADP ribosylation of α subunits (Arg-265); encoded by <i>alt</i> (70K)	<0.5, synthesized late, packaged, and injected	Lowers affinity for σ ; participation in host shutoff.
ADP ribosylation of α subunits (Arg-265); encoded by <i>mod</i> (27K)	1.5–2.0	Lowers affinity for σ ; selective shutoff of some early genes around 4 min; participation in host shutoff?
DNA-binding activator of middle transcription (MotA)	<5	
Binding of 10K protein (AsiA)	<5	σ Antagonist, anti- σ -70.
Binding of 15K protein	5	T4 gene 60 codes for a subunit of T4 DNA topoisomerase and may also code for the 15K protein.
Binding of gp33 (12K)	5–10	Positive control of late transcription, helper of σ^{gp55} .
Binding of σ^{gp55} (22K)	5–10	Positive control of late transcription, alternate σ .
Interaction with gp45 (25K)	(Does not copurify with RNA polymerase)	Gp45 is a component of the core of the T4 replisome and is also directly involved in <i>late</i> transcription.

Late Gene Expression. Expression of the early and middle genes ceases about 12 minutes postinfection. However, at 5 minutes postinfection, DNA replication has begun and transcription of the **late genes** begins. Further modifications of RNA polymerase accompany this transition. For example, gp55 is a viral-encoded σ factor that is required for late gene transcription. It essentially hijacks the host RNA polymerase by altering its promoter sequence recognition properties. This works because the molecular structure of T4 late promoters is quite different from other known prokaryotic promoters. It uses the consensus sequence TATAAATACTATT at the –10 region but exhibits no consensus sequence at the –35 region. The alternate σ gp55 and the unusual structure of late gene promoters directs T4 transcription toward the late genes.

Another curiosity of T4 transcription is that expression of late genes is closely tied to DNA replication. One of the proteins involved with late gene transcription (gp33) serves as a link between RNA polymerase- σ^{gp55} and components of the T4 replication machinery. Apparently the T4 replication complex serves as a “mobile enhancer,” although how this works is not clear.

Capsid Assembly and Phage Release. The late gene products are primarily components of viral capsids. Approximately 40 have been identified. In spite of the complexity of the T4 capsid, the assembly of the various components (e.g., heads, tail fibers, base plate) is remarkably efficient. It is fascinating that final assembly of the various phage component parts occurs spontaneously, without guidance. This feature is referred to as self-assembly and is due to the remarkable affinity each component has for its mate. The inner portion of Figure 6-6 illustrates the pathways involved with capsid formation and points out where the genes responsible are positioned on the T4 genome. The entire process results in the production of at least 200 particles per cell by 25 to 30 minutes postinfection (at 37 °C, in rich medium). At the completion of the T4 multiplication cycle the cell will lyse. Lysis is accomplished by a sudden cessation of respiration and requires the phage *t* gene product plus the phage *e* product (T4 lysozyme). The *t* protein is an example of a **holin protein** that assembles in the cell membrane and allows passage of the *e* lysozyme through the inner membrane, allowing *e* lysozyme access to the outer membrane to degrade lipopolysaccharide carbohydrates.

Holin proteins in general have to be regulated; otherwise, the cell will prematurely lyse. One piece of the timing puzzle has finally been exposed. It involves **lysis inhibition**, an intriguing and historically important feature of T4 phage. Over 50 years ago, it was observed that the normal lysis of T4-infected cells could be inhibited by secondary T4 infection occurring at least 3 minutes after the primary infection. Lysis inhibition by superinfecting phages is effective even if the superinfecting DNA does not replicate. T4 mutants in the *rI*, *rII*, and *rIII* genes are not subject to lysis inhibition and exhibit unusual plaque morphology. It now appears that the *r* genes are part of the holin timing mechanism. The key role appears to be *rI* protein, located in the periplasm. Injection of superinfecting DNA appears to activate *rI*, which in turn interacts with the *t* holin, inhibiting its pore forming ability. The rest of the timing mechanism, however, remains a mystery.

T4 Genome

Terminally Redundant and Circularly Permuted. A useful website dedicated to T4 phage can be found at (<http://www.evergreen.edu/user/T4/home.html>). It should be noted at this point that while the DNA of T4 phages is linear both in the virus and within the host cell, the genetic map is circular (i.e., genes situated at opposite ends of the linear structure, which should not map close to each other, nevertheless, upon genetic mapping, are found to be linked). The resolution of this paradox can be found in the realization that the ends of each T4 DNA molecule are **terminally redundant**—that is, a duplication of genetic information occurs at both ends of a single molecule. This phenomenon is the result of the **headful mechanism** utilized for the packaging of T4 DNA into the T4 capsid. After DNA replication (discussed more fully below), replica molecules can recombine with each other at their terminal redundancies, creating very long molecules (concatemers) many genomes in length. Each mature virion is then made by packaging a “headful” of DNA cut from such a T4 concatemer. One headful, however, is 105% of a single genome. Therefore, the packaging process will generate a population of phages carrying all possible **circular permutations** of the T4 genome (Fig. 6-7). Genetic crosses between members of this population will result in a circular map.

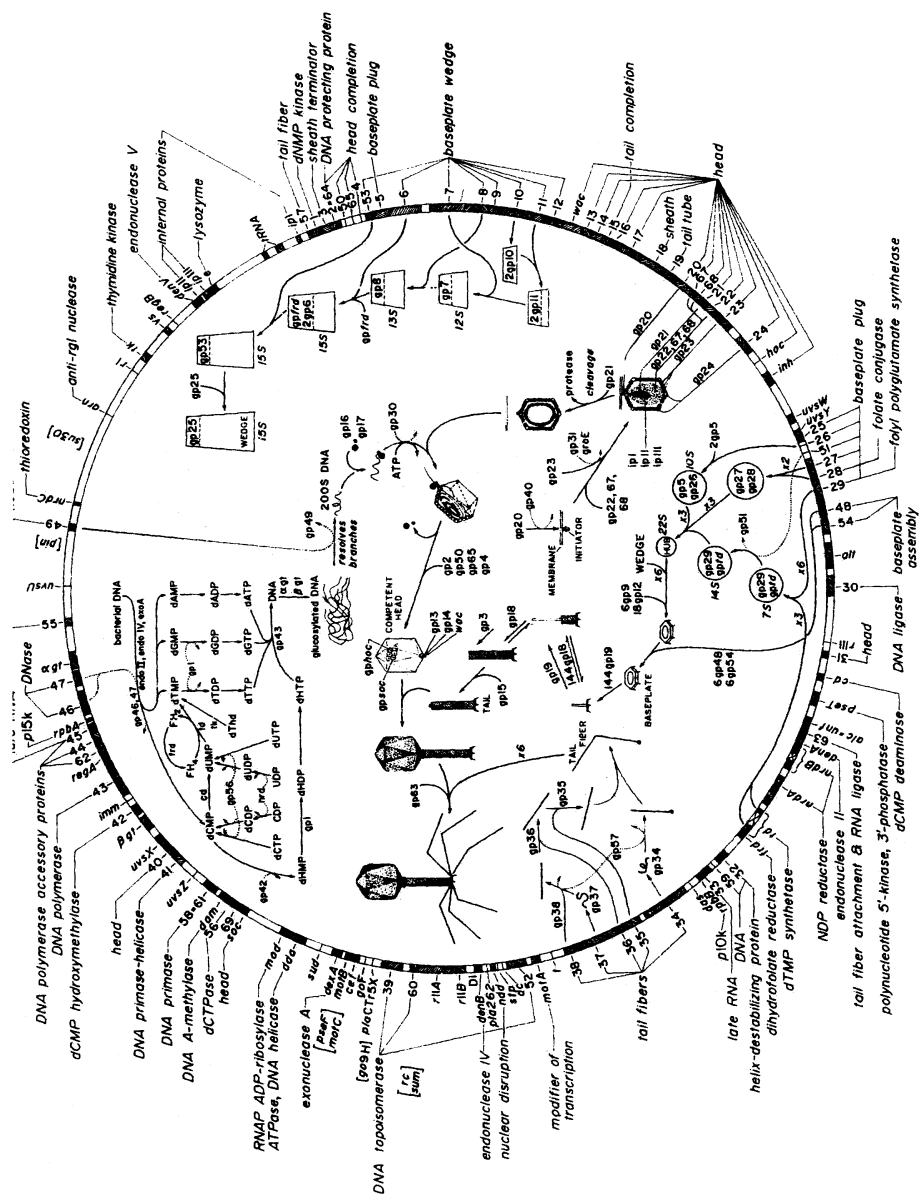


Fig. 6-6. Genomic map of bacteriophage T4. (Courtesy of Drs. B. S. Guttman and E. M. Kutter, Evergreen State College, Olympia, WA.)

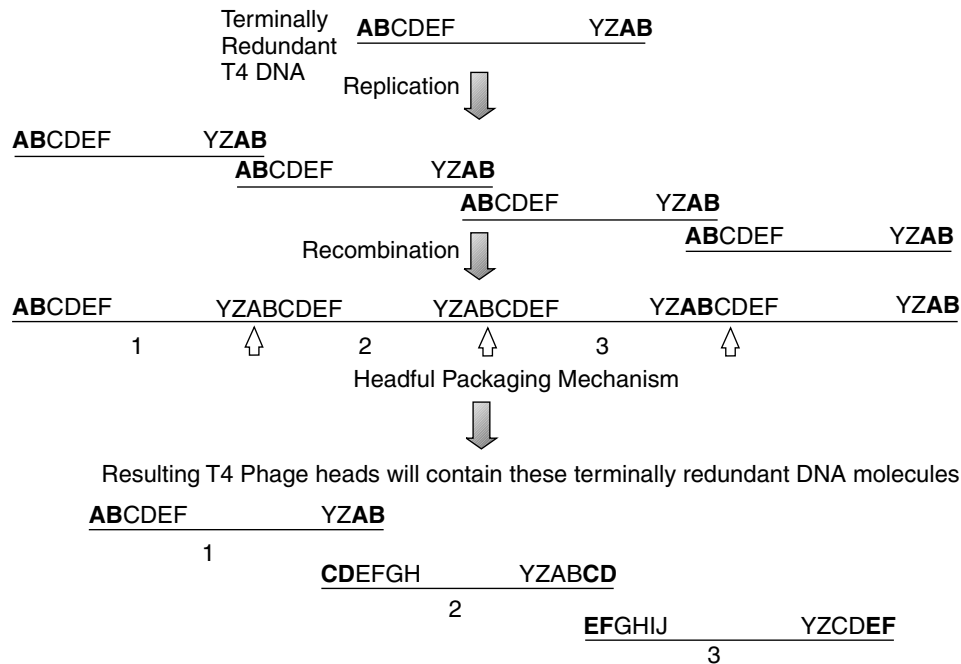


Fig. 6-7. Formation of terminally redundant, circularly permuted T4 DNA.

T4 DNA Replication. There are approximately 30 T4 genes known to be involved with some aspect of T4 DNA synthesis. Many of these are not directly involved with the replication fork mechanism but encode enzymes required for nucleotide precursor synthesis. In this section we deal only with those genes directly involved with the replication process as well as analyze the replication machinery itself.

Because of its rapid infection cycle, T4 uses multiple replication origins and initiation mechanisms. There is an origin-dependent “early” mode of initiation involving multiple origins spaced around the T4 genome. Transcription in these areas is essential to initiating replication. As illustrated in Figure 6-8, transcription at an origin creates a substrate for RNase H. Following cleavage of the RNA in the RNA:DNA hybrid, the remaining RNA serves as a replication primer. One major initiation site occurs between genes *rII* and 42, and there are up to four additional sites at other locations. As with host DNA replication, there is a leading or continuously synthesized strand and a lagging strand. Okazaki fragments in T4 are about 2000 nucleotides long. The major protein responsible for nucleotide synthesis is gp43, the T4 DNA polymerase (mwt 110,000). Accessory proteins include g44, g45, and g62, which collectively form a sliding clamp on the DNA much as DnaB protein does on chromosomal DNA (see Chapter 2).

The gene 41 protein is a DNA helicase that will also serve to unwind dsDNA at the replication fork. A complex of gp41 and gp61 is envisioned to move along the lagging strand in the 5' to 3' direction dependent on nucleotide hydrolysis. The gene 32 product, a helix-destabilizing protein (ssDNA-binding protein), will bind to the single-stranded regions. At intervals of about 2000 nucleotides, the gp41/61 complex stops at specific sites to make pentaribonucleotide primers (pppApCpNpNpNp). The primers probably

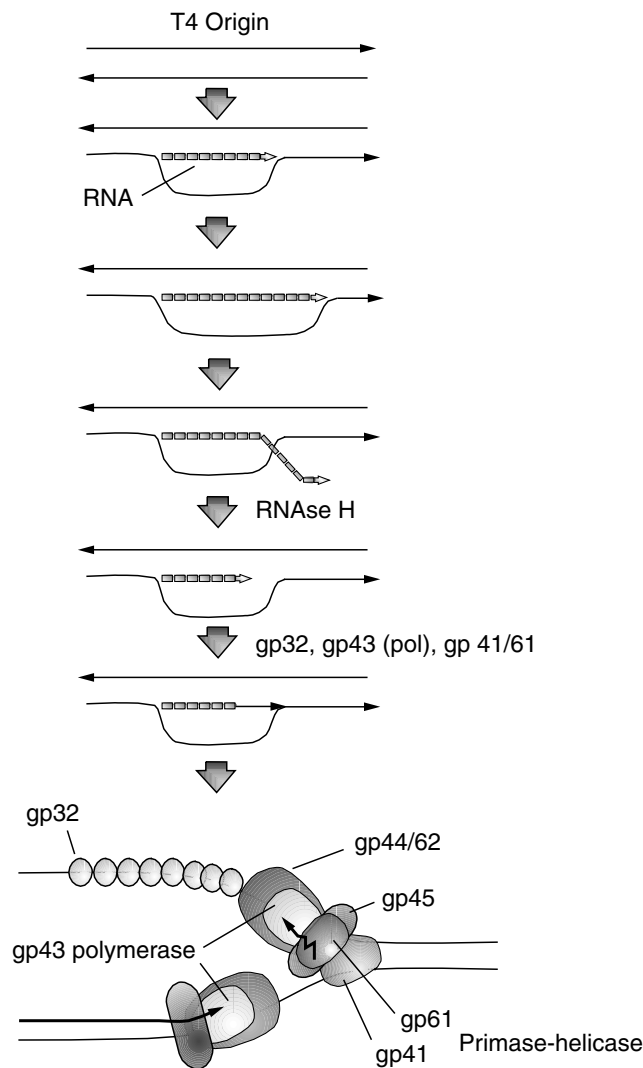


Fig. 6-8. Origin-dependent T4.

remain attached to the priming complex until elongated by T4 DNA polymerase (gene 43 product). Joining the Okazaki fragments to make a continuous DNA chain requires a special DNA repair system that includes T4 RNase H (44 kDa) to degrade the RNA primer, T4 DNA polymerase to replace the RNA with DNA, and T4 DNA ligase to form a phosphodiester bond between the 3' end of each fragment and the 5' end of the preceding one (Fig. 6-8). The gene 44/62 complex and gene 45 proteins appear to keep the polymerase continuously on the growing 3'-OH chain.

After a few rounds of DNA synthesis, the T4 UvsW protein inactivates origin-dependent replication. UvsW is a helicase that unwinds RNA primers from these origins, effectively halting this form of initiation. Subsequently, T4 switches to a recombination-dependent initiation mechanism. This is an essential feature of T4

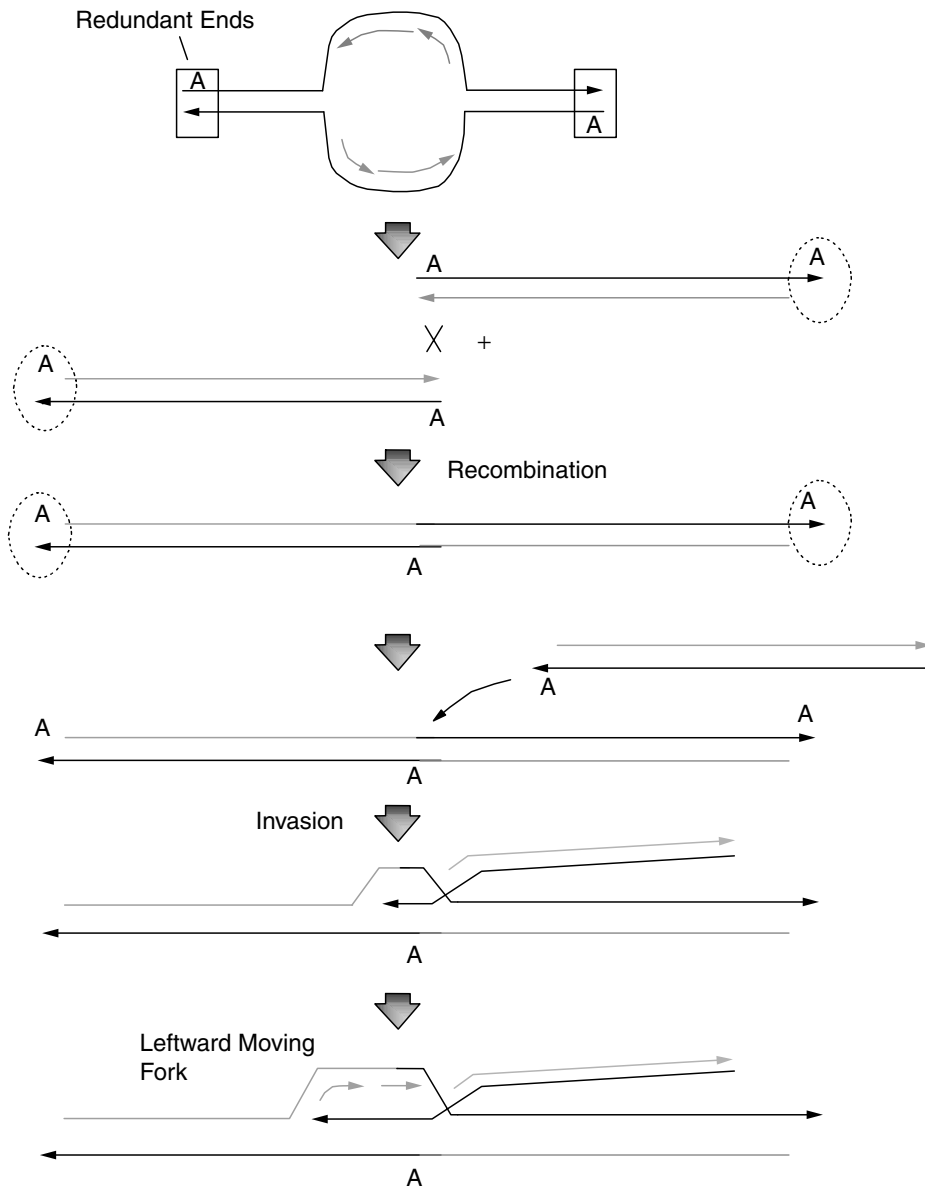


Fig. 6-9. Recombination-dependent initiation of T4 DNA replication. Redundant ends are marked by circles and boxes. Recombination between redundant areas of two molecules creates a concatemer. A third molecule can invade the internal redundancy of the concatemer and generate a leftward-moving replication fork.

development because its genome is linear. No known DNA polymerase can initiate chains *de novo*. As a result, the 3' ends of linear DNA molecules replicated from specific origins will remain unreplicated. The first clue as to how T4 solves this problem was the discovery that its genome is terminally redundant (see above). The redundancy extends 3 to 4 kb. As diagrammed in Figure 6-9, daughter T4 molecules can recombine

at the homologous blunt ends to form a two-copy concatemer. The 3' single-stranded end of another T4 molecule can then invade the homologous region positioned in the middle of the concatemer, forming a new replication fork.

The expression of four of the DNA replication genes (*g43*, *g44*, *g45*, and *g62*) is regulated by controlling the initiation and termination of transcription as well as translation (Fig. 6-10). Although these four genes are transcribed by the host RNP, the T4 *mot* product influences promoter selection by RNAP, stimulating transcription of *g43* and *g45* while inhibiting transcription of *g44*. The gp43 product (T4 DNA polymerase) also competes with Mot for binding, thereby autoregulating gene 43 transcription. Translational control by *regA*, also indicated in Figure 6-10, is explained below.

Regulation of T4 Gene Expression. Regulation of T4 gene expression has been shown to occur both at the transcriptional and translational levels. At the translational level two systems are recognized; autogenous translational repression by the gene 32 product (helix-destabilizing protein) and translational repression of many T4-induced early mRNAs by the phage *regA* product. In the systems of translational repression already discussed (see “Translational repression” in Chapter 5) each protein repressor was specific to only one species of mRNA. This mechanism appears the same for the T4 gene 32 product. However, translational regulation by RegA (12,000 mw) is unique in that it appears to repress the translation of different mRNA species. The mechanism appears to be the simple recognition of and binding to a consensus mRNA

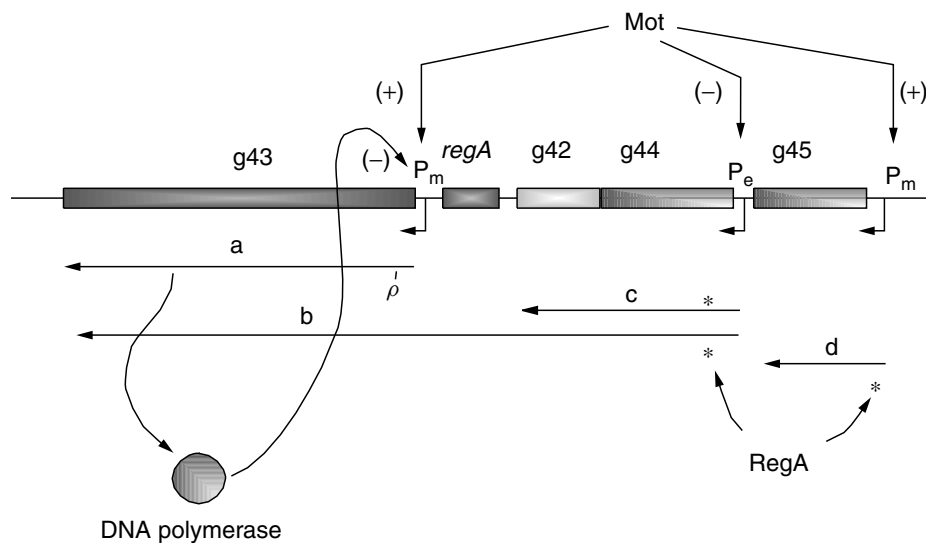


Fig. 6-10. Schematic summary of proposed mechanisms controlling expression of replication genes. P_e indicates an immediate-early promoter recognized by *E. coli* RNA polymerase that is independent of *mot*. P_m indicates a middle-early promoter that is dependent on *mot* for transcription. The wavy lines indicate length and direction of mRNA synthesis. Transcript *a* accounts for 75% of the cellular level of gene 43 mRNA and the *b* transcript contributes 25%. Translation of transcripts *c* and *d* may be controlled by the product of the *regA* gene. ρ indicates a potential site for ρ -mediated termination that the *mot* protein may modulate. (Source: Modified from Spicer and Konigsberg, 1983, *Bacteriophage T4*, in Matthews, et al., 1983. pp. 291–301.

sequence [(AUG)UACAAU-3')] thus blocking ribosome access. Secondary mRNA structures appear to be unnecessary. Although the regulatory significance of *regA* remains a mystery, it would seem to play a role in coordinating the early events in T4 development.

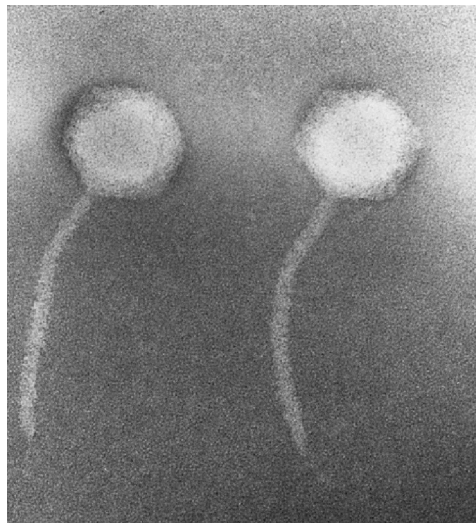
Hyphenated Genes: Introns in T4. Many eukaryotic genes include one or more nontranslated intervening sequences (IVS), or **introns**. These intervening sequences must be excised from the primary transcripts prior to translation. Few examples of introns have been documented in prokaryotes. The first was a 1 kb intron within the thymidylate synthase (*td*) gene of bacteriophage T4. Since then, two more T4 genes, *nrdB* and *sunY*, have been shown to contain introns. The T4 introns are all self-splicing, in which a series of transesterifications, or phosphodiester bond transfers, occurs with the RNA functioning as an enzyme. The introns within *td* and *sunY* are actually mobile introns that can catalyze their transfer to another DNA molecule.

A novel form of gene splicing is found in gene 60, which codes for an 18 kDa subunit of the T4 topoisomerase. Within the message is a 50 nucleotide untranslated region. However, this region is not removed as an intron from the message. It seems a pair of 5 nucleotide direct repeats flanking the IVS can base pair, forming a hairpin that brings codons on either side close together. A ribosome can presumably move through or jump across this structure, ignoring the nucleotides in the loop.

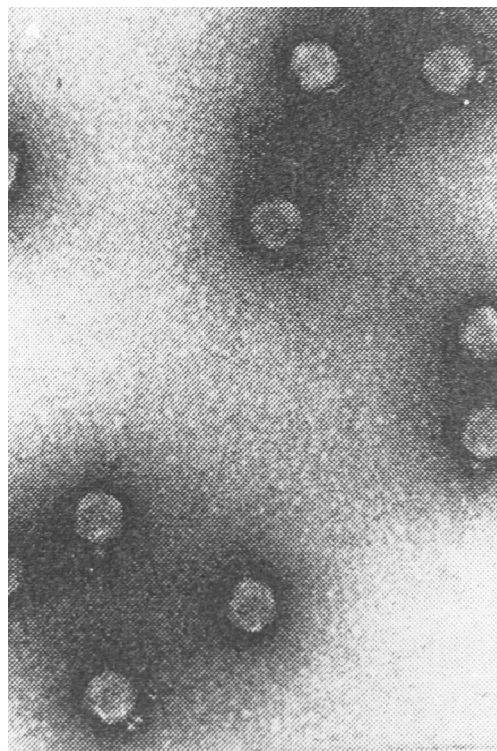
λ PHAGE

The λ (lambda) phage (<http://www.ncbi.nlm.nih.gov/cgi-bin/Entrez/framik?db=Genome&gi=10119>) (Fig. 6-11) of *E. coli* is a *Siphoviridae* and a classic example of a temperate phage—that is, it can undergo either a lytic or a lysogenic cycle upon infecting a sensitive *E. coli*. This property manifests itself in the form of a cloudy plaque when λ is grown on a lawn of *E. coli*. The cloudy center is due to growing bacterial cells that have been lysogenized by the phage DNA and to the fact (discussed below) that these lysogens are resistant to superinfection. Mutations in certain λ genes (*cI*, *cII*, *cIII*) will prevent lysogeny, so the mutant phage will produce clear plaques in which all infected cells are lysed. The λ genome (48.5 kDa) expresses approximately 50 proteins. The genes of this phage can be grouped into four categories based on their roles in either lysis or lysogeny: (1) genes involved in lytic development, (2) genes involved with the development of lysogeny, (3) genes participating in both, and (4) genes with unknown functions. The genes in the second and fourth groups are nonessential for phage growth. Figure 6-12A presents a general map of the λ genome.

The DNA within a λ capsid is linear (48,502 bp) but contains cohesive ends designated *cos* sites. Upon phage attachment and injection, the *cos* ends of λ DNA anneal and are ligated to form a circular molecule. There are two phases of λ replication as the phage undergoes lytic development: an early phase in which the circular molecules replicate bidirectionally and a late phase, approximately 10 minutes postinfection, consisting of a rolling circle mechanism that generates long concatemeric molecules containing multiple copies of the λ genome. This is discussed in detail below. The concatemers are then acted upon by the λ-packaging system and are used to fill preformed capsids with λ DNA. A precise amount of λ DNA is packaged, corresponding to one complete genome, following cleavage by λ terminase at the unique *cos* sites mentioned above (Fig. 6-13).



(a)



(b)

Fig. 6-11. Electron micrographs of some temperate bacteriophages. (a) λ phage negatively stained with potassium phosphotungstate. (Micrograph by E. A. Birge.) (b) Phage P22, also negatively stained. Bar equals 100 nm. (Source: From King and Casjeus, *Nature* (London) **251**:112–119, 1974.)

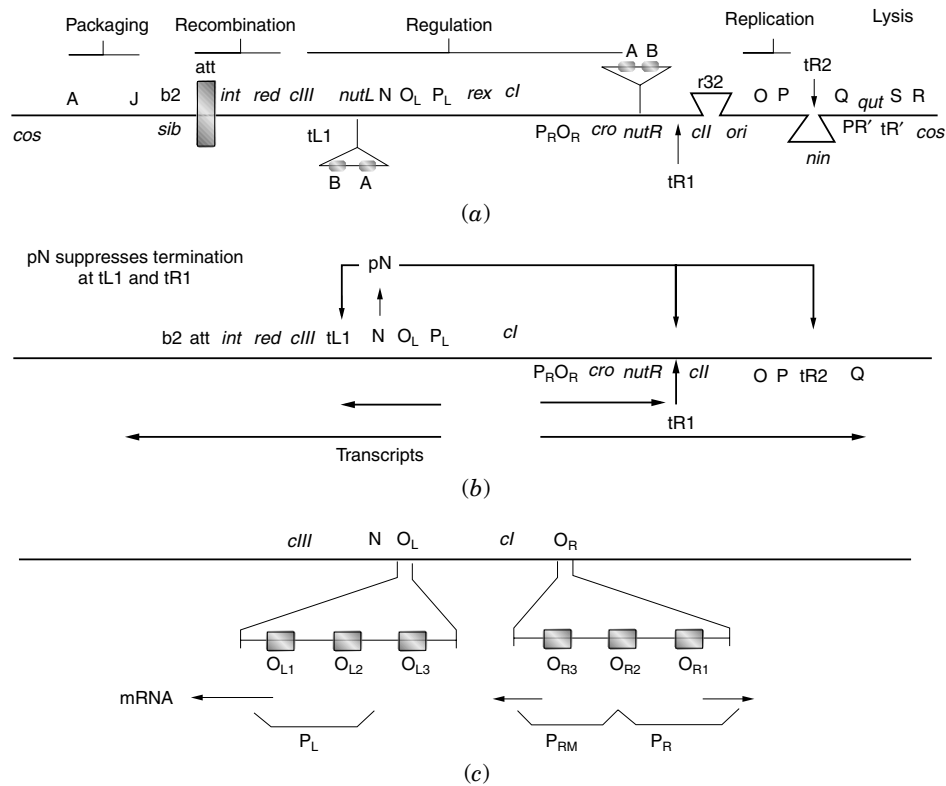


Fig. 6-12. λ phage. (a) Genetic organization. (b) Transcriptional organization. (c) Structure of the leftward and rightward promoter/operator regions.

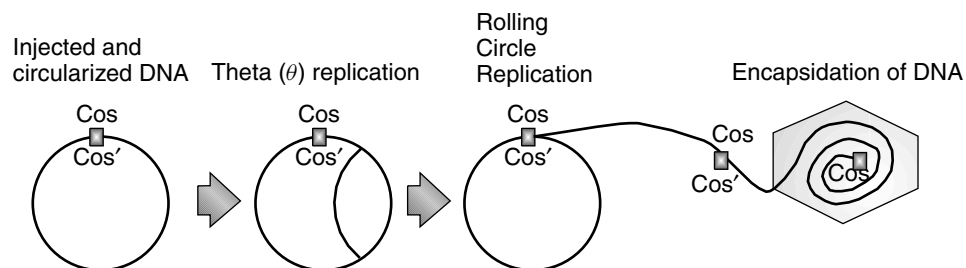


Fig. 6-13. Infective pathway of bacteriophage λ . The phage DNA forms a covalently closed circle through pairing of the cohesive ends, (*cos*), followed by ligation. Early DNA replication by the mode generates circular monomeric molecules. The switch to the rolling circle mode of replication is dependent on the product of the *gam* gene, which inhibits the host *recBC* nuclease. The multimeric DNA is the substrate for encapsidation. (Modified From Murray, in Hendrix et al., 1983.) Lambda II. Cold Spring Harbor Laboratory, Cold Spring Harbor, N.Y.

The Lysis-Lysogeny Decision

Upon infection, λ phage must make a molecular decision to multiply in a lytic manner or “hibernate” as an integrated prophage in the lysogenic state. As will become evident, it is impossible for λ to go down one path (i.e., toward lysis) without partially going down the other (i.e., toward lysogeny). The multiplicity of infection greatly influences the final decision. In general, a high multiplicity of infection leads to lysogeny, whereas a low ratio of phage to cells results in lytic multiplication. This correlation should be kept in mind while reading the following sections.

Transcription

Transcription of λ initiates at P_L (leftward promoter) and P_R (rightward promoter), with the early transcripts terminating at the ρ -dependent terminators at t_{L1} and t_{R1} (Fig. 6-12B). The leftward transcript results in the production of the antiterminator N while the rightward transcript codes for Cro protein. Transcription into genes beyond the terminators requires N protein acting as an antiterminator at t_{L1} and t_{R1} . Six factors produced by *E. coli* are required for effective N utilization: *nusA* (71 min), *nusB* (9 min), *nusC* (*rpoB*, 90 min), *nusD* (*rho*, 84 min), *nusE* (*rpsJ*, 74 min), and *nusG* (90 min). The *nus* acronym stands for N-utilization substance. *E. coli nus* mutants fail to support λ growth by blocking the action of pN rather than its synthesis. The *nusA* and *nusG* gene products and their normal roles in *E. coli* are discussed in Chapter 2 with regard to transcription. As noted above, *nusC*, *nusD*, and *nusE* mutations occurred in the genes for RNA polymerase β subunit, Rho, and ribosomal protein S10 respectively.

The model for N action involves a modification of RNA polymerase at the *nut* sites (see Fig. 6-12A and B). The presence of NusA, NusB, and/or NusG on core polymerase enables pN to modify RNA polymerase when a *nut* site is encountered. The modified RNA polymerase can then transcribe past t_{R1} and t_{L1} , resulting in the synthesis of *CII* and *CIII* message, respectively, and past t_{R2} , yielding another antiterminator pQ. The *nut* site in λ mRNA is divided into three parts: BoxA, a site that binds NusB and S10; a spacer region; and a stem-loop structure called BoxB that binds N and NusA. NusB binds to the BoxA sequence of nascent message and communicates with RNA polymerase through the S10 collaborator. N-antiterminator protein binds to the *nut* BoxB hairpin loop and interacts with RNA polymerase through NusA, an interaction stabilized by NusB–S10 and NusG. NMR studies have revealed that the N–BoxB interaction involves the aromatic ring of a tryptophan residue in N actually stacking with an unpaired adenine in Box B. This multifactorial complex ultimately suppresses the pausing of RNA polymerase at the downstream transcriptional termination signal.

Several minutes after λ infection, the synthesis of early (N and Cro) and delayed early (*CII*, *P*, *O*, *Q* and *CIII*, *red*, *int*) transcripts diminishes due to the action of Cro repressor on P_L and P_R following its binding to O_L and O_R . However, the Q antiterminator will allow a P'_R transcript to extend through the late genes involved with capsid production and lysis. In contrast to N, which binds mRNA, Q binds to a DNA site called *qut*. RNA polymerase initiating at P'_R pauses at +16/+17, which allows Q to bind and modify RNAP, shortening the pause. The P'_R extended transcript eventually terminates at *b* (remember the λ genome is circular at this point). As a general rule, when the multiplicity of infection is low, there is an insufficient quantity of CII or CIII to establish lysogeny. However, enough Q is synthesized to transcribe through the late genes, producing the lytic cycle of λ phage.

Function of Cro Versus CI Repressor and the Structure of O_L and O_R

As opposed to CI, which promotes lysogeny, Cro protein promotes lytic multiplication of λ by preventing the synthesis of CI. Although both Cro and CI bind to the same operators (O_L and O_R), these repressor proteins serve distinct physiological functions due to several unique features of the operators as illustrated in Figure 6-12C. First, within each operator there exist three CI (and Cro)-binding sites. Each binding site (designated O_{L1} , O_{L2} , etc.) is similar in sequence and possesses axes of 2-fold hyphenated symmetry. The two repressors have different relative affinities for each of these binding sites. At low concentrations, dimers of CI bind to O_{R1} and O_{R2} , blocking transcription from P_R . Actually, CI repressor binds cooperatively to O_{R1} and O_{R2} , binding first at O_{R1} , which increases the affinity of O_{R2} for CI, then at O_{R2} . Binding of CI to O_{R2} will activate P_{rm} (promoter for repressor maintenance), ensuring continued production of CI. In contrast, Cro protein binds more tightly to O_{R3} . This prevents transcription from P_{rm} , halting synthesis of CI. Thus, Cro binding to O_R will promote the lytic cycle of development by preventing lysogeny.

The study of both Cro and CI repressors revealed a motif that has proven to be common to many repressor proteins—that is, the helix-turn-helix motif. The DNA-binding domain of CI resides at the amino-terminal end of this protein. In this region are several alpha helices, two of which participate in DNA binding. As shown in Figure 6-14, the two helices interact, forming a helix-turn-helix structure that can fill the major groove of the operator DNA. Both Cro and CI bind the operator as dimers. Figure 6-14 illustrates how two of the HTH regions of a dimer fill the major groove and also shows the importance of the N-terminal arms (last six amino acids) in reaching around the major groove to contact the backside of the operator. The affinity of an armless CI repressor for DNA is reduced about 1000-fold.

Establishment of Repressor Synthesis

If *cI* transcription from P_{rm} requires CI as a positive regulator, how is synthesis of CI established in the first place? After infection, *cI* transcription originates from a different promoter called P_{re} (promotor for repressor establishment). Transcription from P_{re} is positively regulated by the CII protein (Fig. 6-15). CII also activates the integrase gene whose product is required for λ DNA to integrate into the host chromosome (prophage state). Thus, CII coordinately regulates the two transcriptional units required for lysogeny.

One of the main criteria for whether lysis or lysogeny occurs is the amount of CII activator produced upon infection. This amount is dependent on several factors: (1) the level of *cII* mRNA, (2) the rate of *cII* mRNA translation, and (3) the rate of CII processing. The level of *cII* mRNA is controlled by CI and Cro repressors as well as by pN, which is required for transcription past t_{R1} into *cII*. Translational control of *cII* expression involves the host HimA protein. Efficient translation of CII mRNA requires HimA, which appears to allow ribosome access to the sequestered Shine-Dalgarno sequence and translation initiation codon in CII mRNA. There is a potential stem-loop structure found at the 5' end of the CII message that protects these sequences.

In addition to the amount of CII message produced and the efficiency with which it is translated, the amount of active CII produced in the cell is dependent on its processing. The first two amino acids (N-formyl methionine and valine) must be removed in order to form an active CII protein. Also, turnover (or degradation) of CII is very

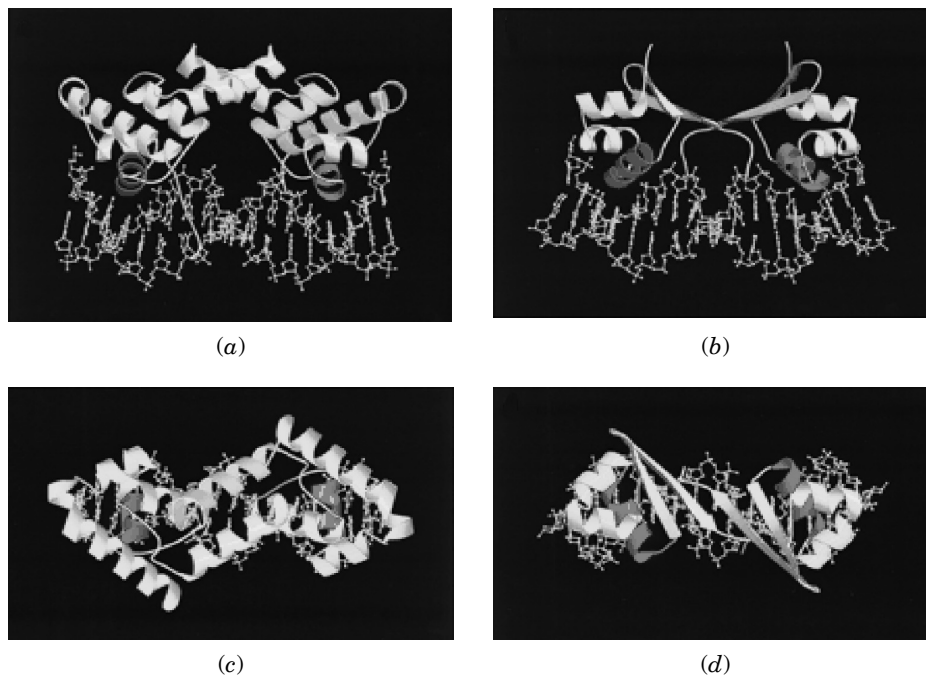


Fig. 6-14. Helix-turn-helix region of the λ CI and Cro repressors. Comparison of the structures of complexes of λ repressor (5, 6) and Cro (unpublished results) with operator DNA. In both cases, the proteins were crystallized with a 19 bp duplex of which the central 17 bp are shown. (a) Headpiece of λ repressor bound to DNA. The consensus half is to the left. As can be seen, there is substantial asymmetry, especially in the location of the amino-terminal arm. The bulk of repressor contacts occur on one side of the DNA, but the N-terminal arms reach around to contact the other side. (b) Cro bound to operator DNA. (c) View down the pseudo-2-fold axis of λ repressor bound to operator DNA. (d) Related view of Cro bound to operator DNA. (Source: From Albright and Mathews, 1998. Copyright, National Academy of Sciences, USA.)

important. The products of the *hflA* and *hflB* loci (94 and 72 centisomes, respectively) are responsible for CII degradation. HflB is actually a membrane-bound protease called FtsH. FtsH is responsible for degrading CII as well as σ 32 (see “Thermal Stress and the Heat Shock Response” in Chapter 18) and the SecY component of the *E. coli* protein secretory system. The *hflA* locus consists of three genes: *hflK*, *hflC*, and *hflX*. HflK and C are membrane proteins (periplasmic side) that interact with the FtsH protease and regulate its activity. It is the apparent role of the CIII protein to inhibit this protease, thus allowing sufficient accumulation of CII to promote transcription from P_{re} (i.e., CI production). Once again, a high multiplicity of infection will result in a large amount of CIII. This inhibits FtsH (HflB) protease from degrading CII. The accumulation of CII promotes the transcription of CI and integrase, which allows λ to integrate and represses all but CI transcription—that is, establishes the lysogenic state.

It is interesting, too, that the overall energy state of the cell affects the process of lysogenization. Under conditions where cAMP levels are low, lysogeny is the preferred developmental route, while the reverse is true where cAMP levels are high. It was discovered that cAMP and CRP appear to inhibit the transcription or activity of HflA. Thus, high cAMP levels lead to lowered HflA protease activity. This allows for a greater

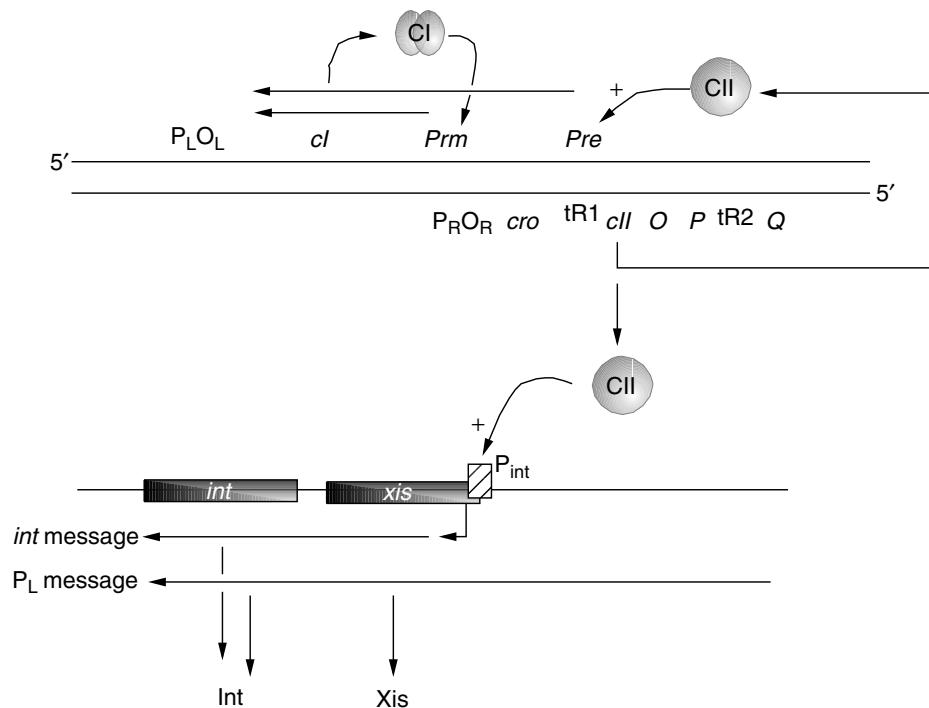


Fig. 6-15. Top: Establishment of lysogeny by λ phage. CII protein stimulates the transcription of *cI* from *pre*, the promoter for repressor establishment. Once CI protein is produced, it will stimulate its own transcription from *prm*, the promoter for repressor maintenance. Note that *prm* overlaps O_R (Fig. 6-12C) such that when CI binds to O_{R1} and O_{R2} transcription is inhibited from P_R but stimulated from *prm*. **Bottom: Regulating integration: the P_{int} promoter region and the mechanism of regulation by CII.** The CII protein recognizes and activates transcription from P_{int} . Positive regulation of *int* (integrase) but not *xis* occurs because the CII-stimulated P_{int} RNA transcript lacks the translation start for Xis.

level of CII, thereby stimulating CI production. However, there is some evidence that contradicts this model, so it is not precisely clear how cAMP affects this system.

Control of Integration and Excision

Once molecular events lead toward the lysogenic state, the circularized genome must integrate into the host chromosome as depicted in Figure 6-16. Integration involves site-specific recombination between the phage *att* and bacterial *att* attachment sites. The integration requires the *int* gene product plus integration host factor (IHF; see below). Prophage attachment normally occurs at the *att* site located between the *gal* and *bio* operons at 17 minutes. Note the permutation of gene order that occurs during the integration—that is, A and R are distal loci in the linear λ DNA present at infection but end up as proximal loci in the prophage.

The *int* and *xis* loci are tightly linked, and their coding regions partially overlap (Fig. 6-15). The *int* locus is controlled by the P_{int} promoter while transcription of *xis* usually results from P_L . However, as illustrated in Figure 6-15, activation of P_{int} by CII results in transcription starting beyond the beginning of *xis*. Thus, an incomplete

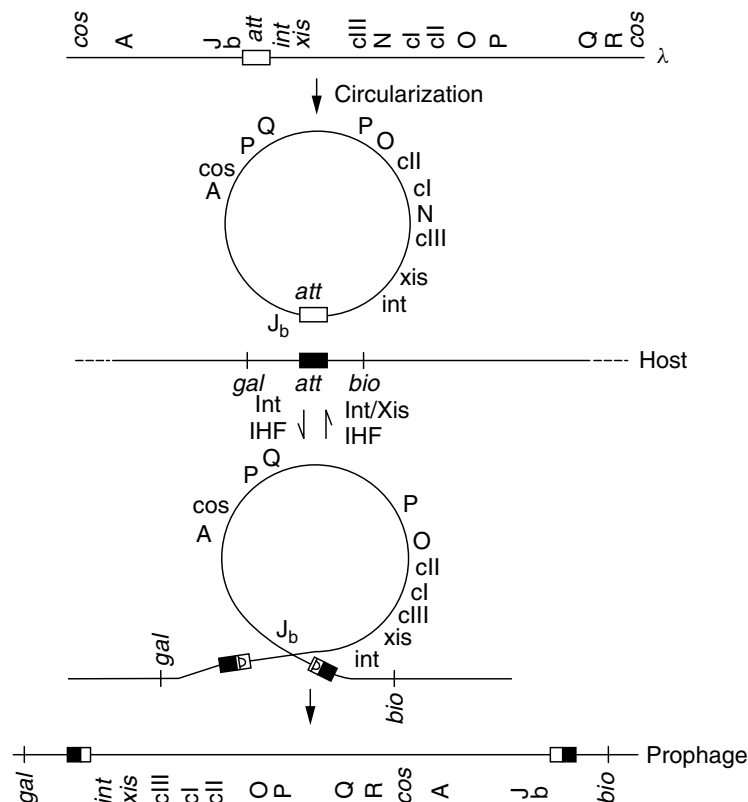


Fig. 6-16. Integration and excision of the λ genome. Note the permutation of gene order as a result of integration (i.e., *b* and *int* are proximal in the vegetative form but distal in the prophage.)

Xis message results from P_{Int} . The end result of CII activation, then, is the preferential synthesis of Int over Xis, which is obviously preferred for integration.

Negative Retroregulation of *int* by *sib*

Upon induction of prophage by UV mutagenesis, it is preferred that more Xis be produced than Int in order to achieve lytic replication. The *sib* locus within the *b* region is responsible for this. Upon exposure to UV light, activation of the host *recA* coprotease results in the autocleavage of CI protein (see “Inducible repair” in Chapter 3). Transcription of λ is allowed to occur from P_L and will proceed through *xis* and *int*. Both Int and Xis are needed for prophage excision (host integration factor is also needed). Examination of Figure 6-16 reveals that in the prophage state *sib* (in region *b*) is removed from its normal proximal location to *int*. However, upon excision, the *sib* region is situated near *int* once again (separated by about 200 bp). At this point, Int is no longer required and could prove detrimental if it causes reintegration of λ in a cell ravaged by UV irradiation.

Regulation by *sib* is posttranscriptional and occurs only in *cis*. The basis for *sib* retroregulation (retro because *sib* is located downstream from *int*) is the formation of a stem-loop structure near the 3' end of *int* message (Fig. 6-17). This stem-loop structure

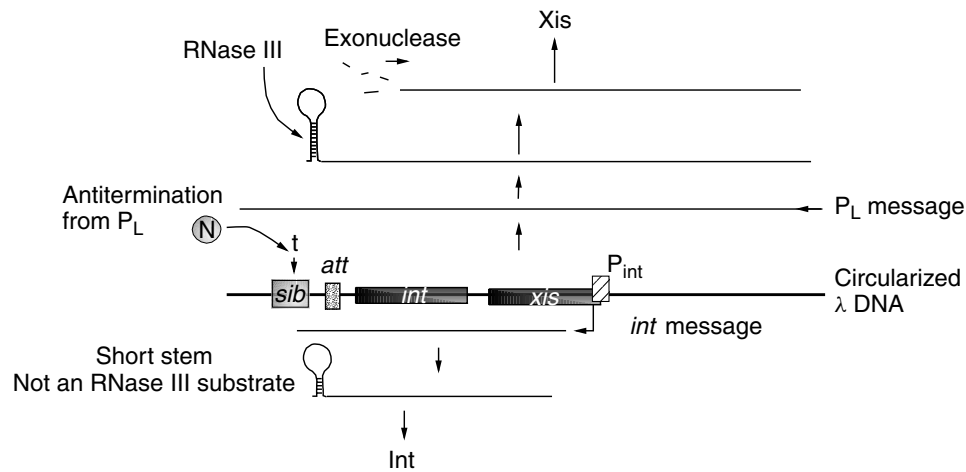


Fig. 6-17. Retroregulation by the *sib* region. Top: Abandoning ship. After excision of λ prophage from the chromosome, the *sib* region is repositioned downstream from *int*. This allows λ to make more Xis than Int, a situation that will prevent reintegration and favor lytic growth. Transcription of the *xis int* region from P_L , which only occurs when CI is inactive, allows antitermination of the downstream terminator by N protein. The longer mRNA includes the *sib* region. The *sib* RNA sequence forms a stem-loop structure that can be degraded by RNase III; which triggers additional degradation of *int* message. Thus, little integrase is produced. Xis mRNA sequence integrity is maintained and XIS is produced. **Bottom:** Setting up camp. This is the situation with a newly infected, circularized λ DNA that made a large amount of CII and enough CI to stop transcription from P_L . When CII initiates transcription of *xis int* at P_{int} , N is not available to antiterminate the *int* terminator. The shorter stem loop is not an RNase III substrate, so integrase is made and integration occurs. Recall from Figure 6-15 that no Xis is made from the P_{int} transcript, making integration the only option.

is a substrate for RNase III; thus, RNase III cleaves at the stem-loop region and the 3' end is degraded by a 3' exonuclease. Retroregulation, however, only regulates the P_L transcript but not the transcript resulting from P_I . The reason is that transcription from P_L is regulated by N protein, which, upon binding RNA polymerase, renders it resistant to most termination signals. Thus, the P_L transcript will proceed through the *sib* region, bypassing a termination signal. The CII-activated P_I transcript terminates within *sib*. The longer stem-loop structure formed from the P_L transcript is probably a more suitable substrate for RNase III allowing retroregulation.

λ -Phage Replication

Upon infection and circularization, early replication is bidirectional, with the origin of replication (*ori*) situated within the O gene. Evidence suggests that the amino-terminal end of the O protein (as four dimers) interacts with *ori* at four iteron sequences as the first step in a chain of interactions that control replication. The result is a nucleosome-like structure called the O-some. The λ P protein's job is to compete with DnaC in the host replication machinery for the DnaB helicase. The λ P-DnaB complex then interacts with the O-some, forming a preprimosome complex. The recruitment of DnaB into the λ -replication complex sequesters DnaB away from host replication machinery, with the consequence of decreasing *E. coli* DNA replication. Activation

of DnaB within the λ *ori* complex is different from its activation at the chromosomal origin. Three *E. coli* heat shock proteins, DnaK, DnaJ, and GrpE, release λ P from the complex, triggering DnaB activation (see “Protein folding and chaperones” in Chapter 2). Following unwinding of λ *ori* by DnaB, DNA replication can proceed upon the addition of primase and DNA polymerase III.

Transcriptional activation of *ori* is also required for replication with transcription through the origin from P_R as a specific requirement. It is thought that DnaB can only unwind DNA in one direction from a preprimosome complex that includes λ O protein. Transcription directed toward the complex on the opposite side dislodges λ O, which is then degraded by the ClpXP protease. Because of this transcriptional requirement, replication is directly inhibited by CI repressor binding to the right operator. This is desirable, of course, when choosing lysogeny as a lifestyle.

As noted previously, early bidirectional replication of λ only occurs for a brief period after which a rolling circle mode of replication, referred to as σ replication, begins to make the long concatemeric DNA required for packaging (Fig. 6-13). Searches for a gene that would mediate the switch from θ - to σ -replication modes have led to the theory that DnaA protein is involved. Transcription from P_R is required for initiation of bidirectional replication, and DnaA protein (see “Initiation of DNA Replication” in Chapter 2) activates transcription from P_R . Since λ DNA contains many DnaA-binding sites, as the λ copy number increases, so, too, does the number of DnaA-binding sites, leading to the hypothesis that the titration of DnaA by these sites is part of the switch to σ replication.

The *gam* and *red* genes are clearly important for the switch to σ replication. The host *recBC* product, exonuclease V, will degrade the tails produced during σ -mode replication. The λ *gam* product inhibits exonuclease V activity, thus allowing concatemers to form. A *gam*⁻ mutant, therefore, will only undergo θ -mode replication. However, concatemers are required for packaging to occur. The *red* product of λ mediates recombination between θ molecules, resulting in a large, circular concatemer also suitable for packaging. The packaging process itself is illustrated in Figure 6-18.

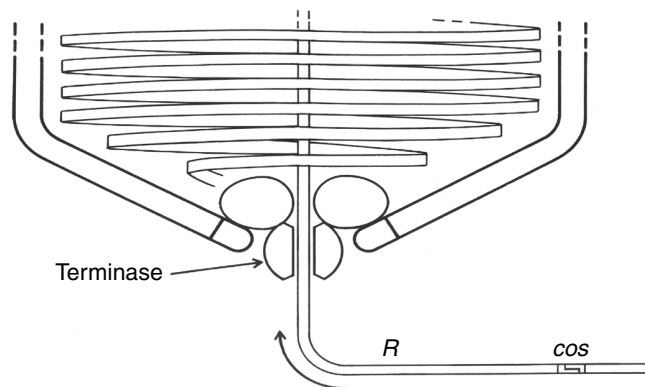


Fig. 6-18. Scanning for the terminal *cos* site, showing part of a nearly filled head with terminase positioned at the connector. The DNA strand being packaged is “scanned” by terminase until the cohesive end symmetry segment of the terminal *cos* site is recognized and cut. Other aspects of packaging, such as the terminase–connector interaction, are purely hypothetical. (Source: From Feiss and Becker, in Hendrix et al., 1983.) Lambda II. Cold Spring Harbor Laboratory, Cold Spring Harbor, N.Y.

The linear DNA molecule is inserted into an empty procapsid until the terminase enzyme (gene A product) identifies a *cos* site, at which point the enzyme introduces a staggered double-strand cut. The filled capsid then has a tail assembly attached and is ready for release from the cell. Release of mature phage particles from an infected cell involves λ lysozyme (the R gene product), which is a 17 K protein with murein transglycosylase activity, and the 11 K S gene product. The S product is believed to interact with the cell membrane, causing pore formation.

μ PHAGE: TRANSPOSITION AS A LIFESTYLE

The genome of μ (mu) phage (<http://www.ncbi.nlm.nih.gov/cgi-bin/Entrez/framik?db=Genome&gi=15122>) is a linear molecule of about 38 kb consisting of 36 kb of μ DNA and 1.5 kb of host DNA at one end and 50 to 150 of host DNA at the other. The attached host sequences reflect the μ life cycle. When μ replicates, it undergoes repeated cycles of transposition in the cell (approximately 100 events per cell during the lytic cycle) and is ultimately packaged along with adjacent host sequences into virus particles. While μ transposition results in the duplication of a 5 bp target sequence, μ is different from other transposons in that the ends of μ are not inverted repeats.

The basic map of μ can be found in Figure 6-19 and Table 6-5. Two gene products are required for the transposition process. The A gene product (mwt 70,000) is the transposase, and the B gene product (mwt 33,000) is an accessory protein necessary for replication and full transpositional activity. During the lytic cycle where this activity is greatest, μ transposes by a cointegrate process. This fact indicates that during replication and cointegrate formation, inversion insertions or reciprocal adjacent deletions will occur (i.e., the host chromosome will be completely rearranged).

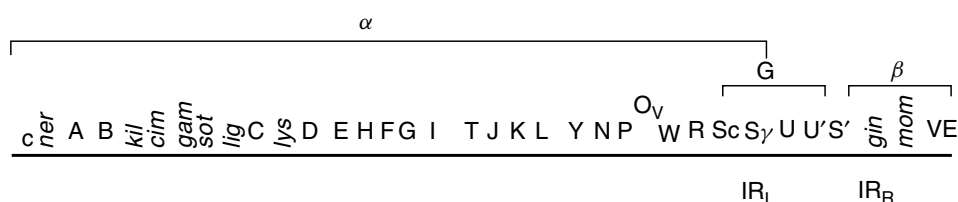


Fig. 6-19. Map of μ DNA. The DNA is 37 kb long. Lengths are shown in kb (not drawn to scale). Mature μ DNA in phage particles is flanked by heterogeneous host sequences that result from packaging of μ inserted into the host chromosome at different sites. α , G, β : Different segments of the μ genome. The α segment contains the early genes and the genes for head and tail morphogenesis. The early region contains the repressor gene *c* and a negative regulator (*ner*). In addition to transposase gene A, the main replicator gene B, and a gene for amplified replication of μ (*arm*), other genes in this region are involved in DNA metabolism. Gene C appears to control the late functions. The G segment contains genes for the μ host range and is flanked by 34 bp inverted repeats. The *gin* protein, whose gene is located to the right of G, acts on the 34 bp inverted repeats to invert the G segment; hence, G is known as the flip-flop segment. The rightmost gene in μ encodes the DNA modification function (*mom*). (Modified From Toussaint and Resibois, 1983.) In, *Mobile genetic elements*, J. A. Shapiro, Ed., Academic Press, New York, pp 105–158.

TABLE 6-5. Characteristics of the μ Genes and Functions

Gene	Localization with Respect to the <i>c</i> End (Kb)	Function	Molecular Weight of Protein
<i>c</i>	0–0.9	Repressor	26,000
<i>ner</i> (negative regulation)		Shutoff of early transcription	6,000
<i>A</i>	1.3–3.3	Transposase	75,000
<i>B</i>	3.3–4.3	Replication, stimulates MuA	33,000
<i>cim</i> (control of immunity)		When inactivated in prophage lacking the S end, no repressor synthesis	7,000
<i>kil</i> (killing)		Killing of the host even in the absence of replication	8,000
<i>gam</i>	5–6	Protects DNA from <i>exoV</i> digestion by binding to DNA	14,000
<i>sot</i> (stimulation of transfection)	6–7	When present and expressed in a bacterium, stimulates efficiency of transfection with μ DNA	
<i>arm</i> (amplified replication of μ)		When deleted, less replication of μ DNA	
<i>lig</i> (ligase)		Expression of an activity that can substitute for <i>E. coli</i> and T4 ligase	
<i>C</i>	9–10	Positive regulator of late gene expression	15,500
<i>lys</i> (lysis)		Lysis of host cell	18,900
<i>D</i>		Head protein	
<i>E</i>		Head protein	
<i>H</i>		Head protein	64,000
<i>F</i> (gp30)			49,400
<i>G</i>			17,200
<i>I</i>		Possible protease	38,900
<i>T</i>		Major head protein	
<i>J</i>		Head protein	
<i>K</i>		Tail protein	
<i>L</i>		Tail protein	55,000
<i>M</i>		Tail protein	12,800
<i>Y</i>		Tail protein	12,500
<i>N</i>		Tail protein	60,000
<i>P</i>		Tail protein	41,800
<i>O</i>		Tail protein	
<i>V</i>		Tail protein	
<i>W</i>		Tail protein	
<i>R</i>		Tail protein	
<i>S</i>	α and G	Tail protein	55,300
<i>S'</i>	α and G	Tail protein	48,000

(continued)

TABLE 6-5. (Continued)

Gene	Localization with Respect to the <i>c</i> End (Kb)	Function	Molecular Weight of Protein
<i>U</i>	In G	Tail protein	22,000
<i>U'</i>	In G	Tail protein	22,000
<i>com</i>		Regulates expression of DNA modification function	7,408
<i>gin</i> (G inversion)	Left of β	Inversion of the G region	21,500
<i>mom</i> (modification of μ)	Middle of β	Modifies μ DNA, protecting against restriction; is not a methylase	28,000
<i>ner</i>		Negative regulator of early transcription	8,500

The μ -phage particle, another myoviridae, is comprised of an icosahedral head 600 Å long bound to a contractile tail that ends in a base plate containing supporting spikes and fibers (Fig. 6-20). Infection with μ involves adsorption of the phage tail to a lipopolysaccharide component of the bacterial outer membrane followed by injection of viral DNA into the host cytoplasm. The G segment of μ is an invertible region that affects host range by changing the type of tail fiber produced. Inversion of the G region occurs via a reciprocal recombination between the short inverted repeats that flank this region. This switching occurs during the prophage state and is controlled by the *gin* product (a recombinase). The switch determines which of two alternative tail fibers are produced in a manner similar to what was described for *S. enterica* flagellar phase variation (see Chapter 3). μ G(+) phages, with Sv and U at the left end of G, will infect *E. coli* K-12 or *S. enterica*. μ G(−) phages, with S'v and U' at the left end of G, will infect such strains as *E. coli* C, *Shigella sonnei*, *Erwinia cloacae*, or other *Erwinia* species.

Following infection, the linear μ DNA circularizes, but the circular DNA is proteinase sensitive, suggesting that circularization may be the result of proteins covalently attached to the DNA. Ten minutes after infection, much of the infecting μ DNA is associated with the chromosome. Transposition to the chromosome from infecting phage involves a nonreplicative simple insertion as opposed to replicative cointegrate formation (see Fig. 6-21). There is a striking difference between μ and other temperate phages in that μ integrates into the host chromosome whether or not it enters the lytic or lysogenic cycle. Integration of μ into the bacterial chromosome occurs at a more-or-less random location, although there are preferred sites for insertion.

μ DNA synthesis via replicative transposition commences 6 to 8 minutes after infection or after induction of prophage. The original prophage copy is not excised prior to the onset of replication. The newly replicated μ copies also remain associated with the chromosome. Each new round of replication involves additional covalent attachment of the ends of μ to a distant target site. In effect, μ replicates by transposition to numerous sites on the bacterial chromosome. Both the *A* and *B* gene products are required for replication and transposition. In addition, transposition

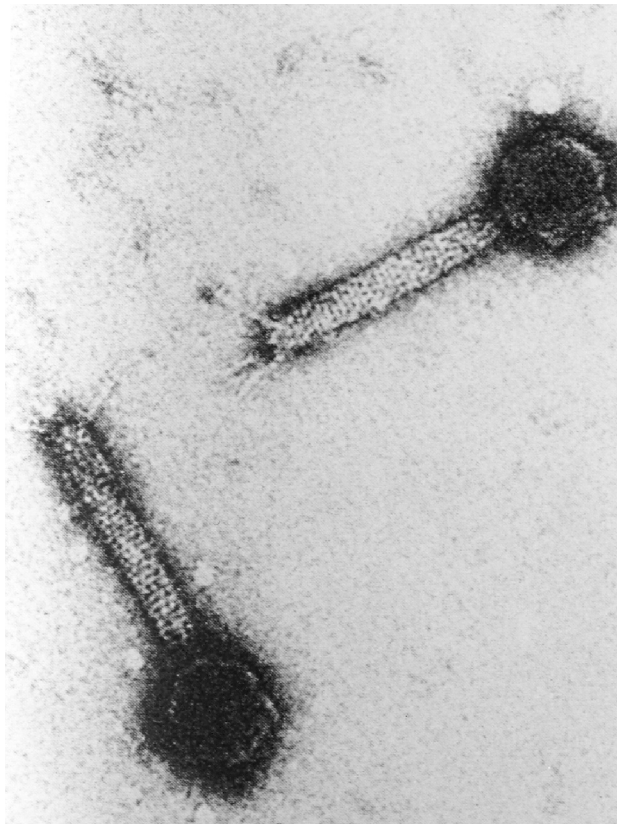


Fig. 6-20. Electron micrographs of the μ -phage particle. (Source: From Toussaint and Resibois, 1983.) In, *Mobile genetic elements*, J. A. Shapiro, Ed., Academic Press, New York, pp 105–158.

requires both the *c* and *S* ends of μ . Two *c* ends or two *S* ends are not suitable substrates for the transposition enzymes. The *A* gene product (MuA) recognizes the ends of the phage genome, assembles as a tetramer, and catalyzes the two basic transposition reactions. The protein introduces single-strand nicks at each end of μ and promotes strand transfer where the 3' ends of the nicked DNA attack a new DNA site. MuA alone, however, only promotes a low level of transposition.

The *B* product (MuB) stimulates transposition by binding to target DNA and to MuA, essentially recruiting target DNA to the transpososome complex. In addition, MuB binding to MuA stimulates the strand transfer activity of MuA and prevents premature disassociation of the MuA tetramer by the ClpX chaperone (see Chapter 2). Recent evidence indicates that MuB is an ATPase and that ATP hydrolysis acts as a switch, regulating the affinity of MuB for target DNA. Multiple rounds of transposition into the same target molecule are actively discouraged, a phenomenon known as *target immunity*. If MuB-ATP binds to target DNA near a preexisting Mu, MuA stimulates MuB to hydrolyze the bound ATP. The resulting MuB-ADP disassociates from the inappropriate target. The presumed purpose of target immunity is to prevent one copy of μ from inserting

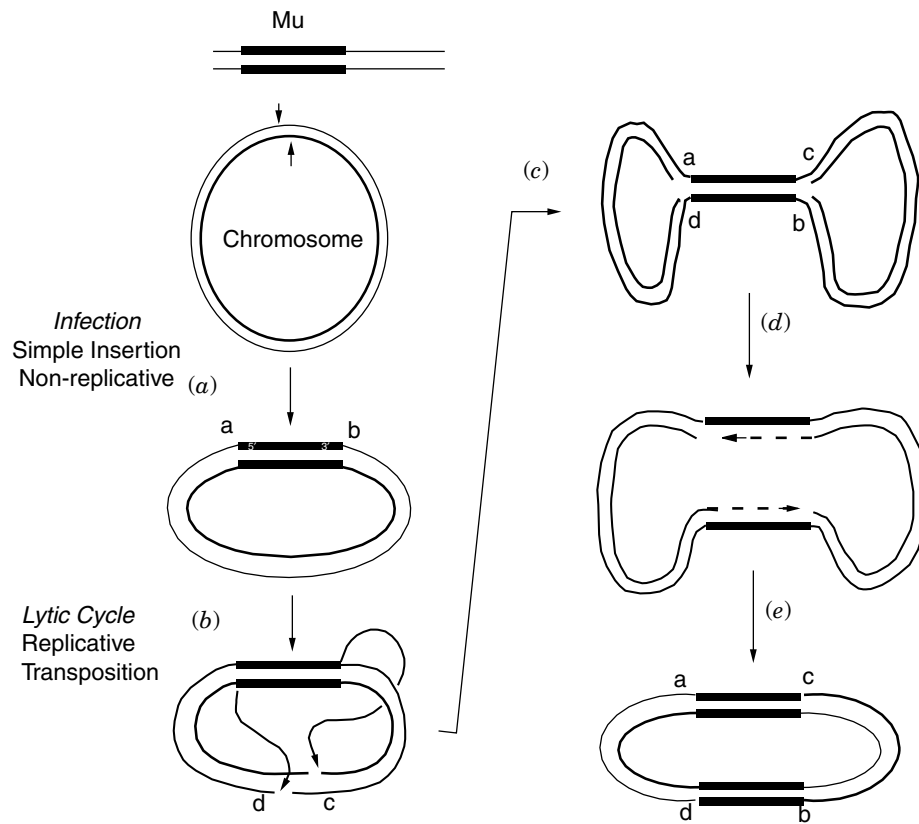


Fig. 6-21. Integration and replication of infecting μ DNA. (a) Integration. The paired filled rectangles represent μ DNA sequences; the lines extending from either side represent variable host DNA that was packaged into the infecting μ capsid. The circle represents the host chromosome; the arrowheads mark the staggered cut at the target site for integration. Integration occurs as a nonreplicative transposition event. (b) Lytic cycle. *a* and *b* represent the 5 bp repeat formed during integration as a result of the staggered cut in the chromosomal DNA. *c* and *d* represent host DNA sequences that flank a target for replicative transposition. Curved arrows indicate the attachment points of specific μ strands at the new target site. (c, d) represent a transposition intermediate and its replication to form two μ insertions. (e) illustrates the final transposition product. Note the chromosomal rearrangement that has occurred with respect to markers *a*, *b*, *c*, and *d*.

into another during the extensive rounds of transposition that occur during lytic growth.

μ transposition is regulated by the Mu *c* repressor protein. Mu repressor binds to an operator region 1 kb from the left end of the phage genome. When bound, μ repressor negatively controls the synthesis of the transposase and other functions involved in transposition, thus maintaining μ in a lysogenic state. Accumulation of a second repressor called Ner can initiate the lytic cycle by binding to another operator that controls production of μ repressor. The μ repressor is also selectively degraded by the ClpXP protease. This will trigger lytic growth by allowing synthesis of the μ A and μ B proteins. ClpX, therefore, has a dual role in controlling transposition. It

limits transposition by disassembling μ A (see above) and can stimulate transposition by targeting μ repressor for degradation.

At the end of the lytic cycle, the various μ -DNA copies scattered around the chromosome are cut and packaged into phage particles. Packaging of μ chromosomes into capsids occurs by a headful mechanism starting from a *pac* site located at the *c* end (see “Transduction” in Chapter 3 for *pac* sites). One cut is made 100 nucleotides outside this end, with the second cut made at a fixed length past the *S* end. Thus, cuts are made in the flanking bacterial DNA, so the DNA from each phage particle contains different terminal sequences. These particles are subsequently released into the environment to seek new hosts.

ΦX174

The Φ X174 phage (<http://www.ncbi.nlm.nih.gov/cgi-bin/Entrez/fram?db=Genome-&gi=10126>) is an example of single-stranded DNA phages (Microviridae) that also include M13 and G4. Φ X174 has been used extensively in studies designed to unravel the mysteries of how the *E. coli* chromosome, 1000 times its size, is replicated. The viral strand of Φ X174 contains 5386 bases and is packaged within an icosahedral capsid with a knob at each vertex (Fig. 6-22). The viral strand is designated (+) because it

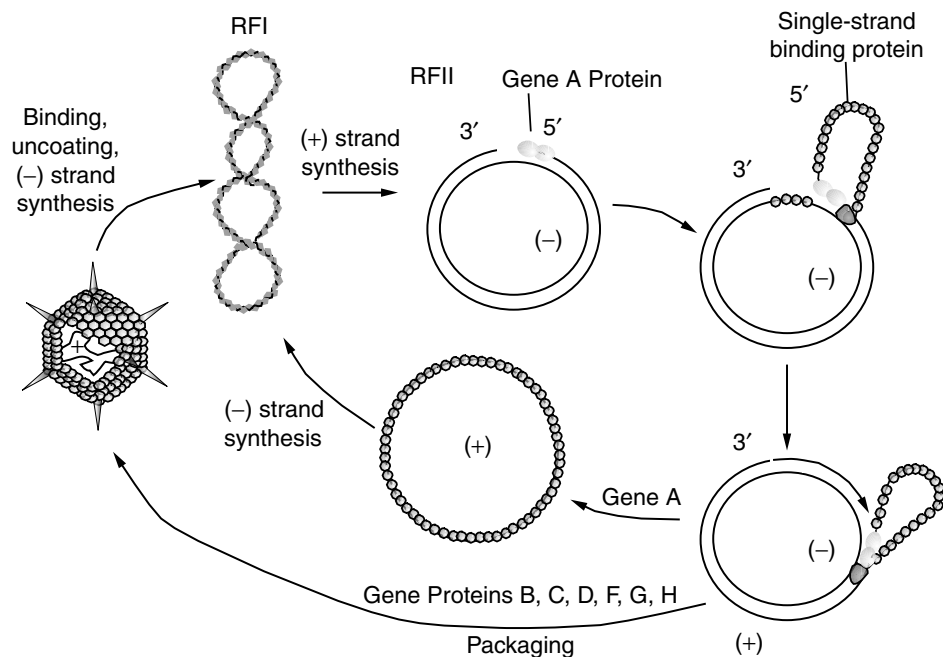


Fig. 6-22. Life cycle of $\phi 174$. Continuous replication initiated by gene A protein cleavage generates viral (+) circles, and discontinuous replication of the viral circles by the SS $\rightarrow$ RF system (host replication machinery) produces RF. In the presence of phage-encoded maturation and capsid proteins, viral circles are encapsulated rather than replicated, DNA making it unavailable for (–) strand synthesis. This strategy only produces new + strands, which are used in the new progeny phage. Following morphogenesis of the phage particle (40 min postinfection), cell lysis ensues.

is the template strand and can be transcribed directly by RNA polymerase. The strand contains genes for 11 gene products. However, the combined molecular weights of these proteins exceed the limit imposed by the available DNA. How is this possible? The apparent incongruity is explained by **overlapping genes** where one gene sequence encodes more than one protein.

The genetic map of Φ X174 is shown in Figure 6-23 and Table 6-6. The region originally assigned to gene A is now known to also contain all of gene B. This phenomenon of overlapping genes is possible because gene B is translated in a different reading frame from gene A. In a similar fashion, gene E is encoded in its entirety within gene D. Another translational control mechanism further expands the use of gene A. The 37 kDa gene A* protein is formed by reinitiating translation at an internal AUG codon within the gene A message. The same translational phase is used, but the functions of the two proteins are distinct. The manner in which this virus extends the one gene–one protein hypothesis demonstrates a frugal usage of genetic information in small phage genomes.

During infection (Fig. 6-22), the phage adsorbs irreversibly to the lipopolysaccharide in the outer membrane of rough strains of *E. coli* and *S. enterica*. Rough strains lack the O antigen outer chains but retain intact core and lipid A (see Chapter 7). Attachment is accomplished via the spikes on the capsid. Following attachment, the virus enters the

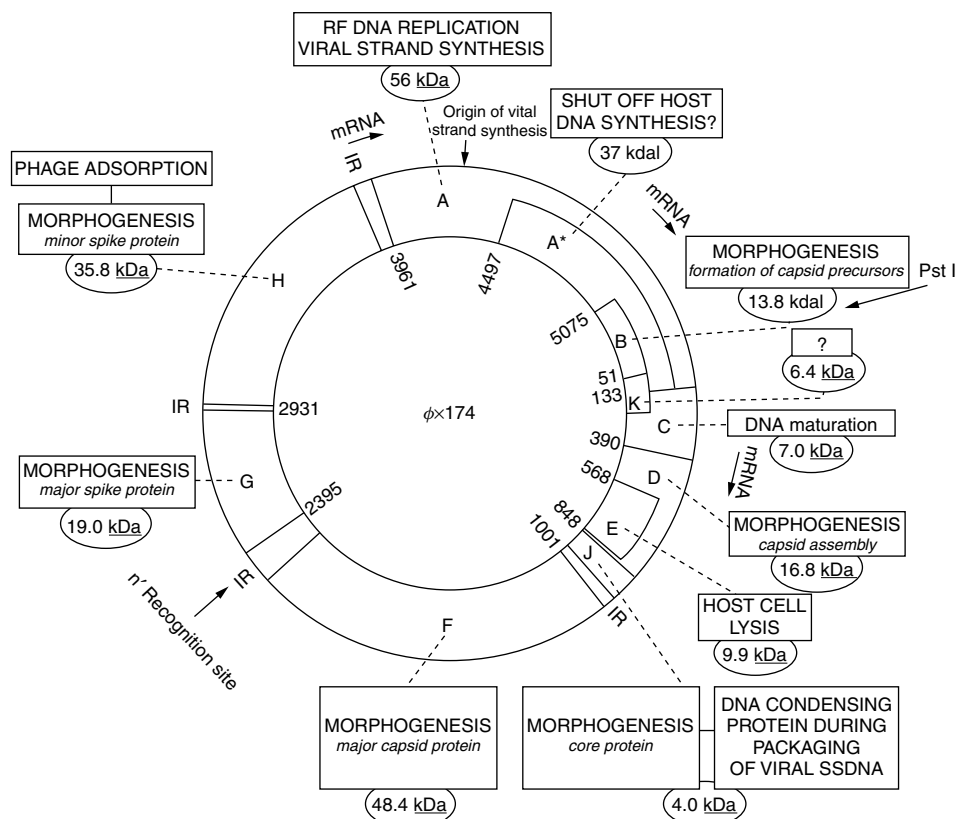


Fig. 6-23. Genetic map of bacteriophage ϕ X174.